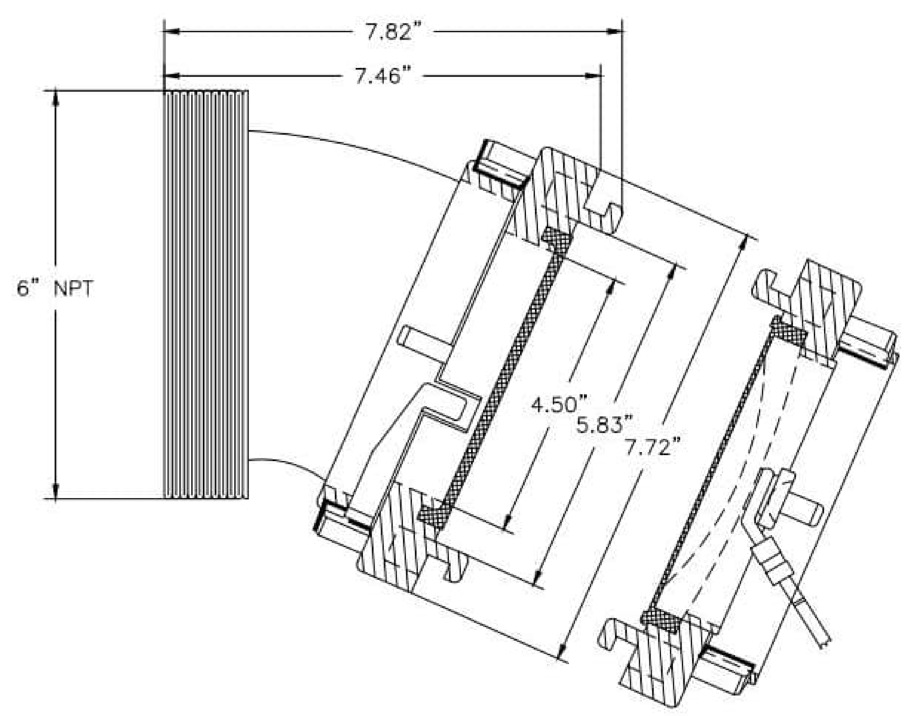
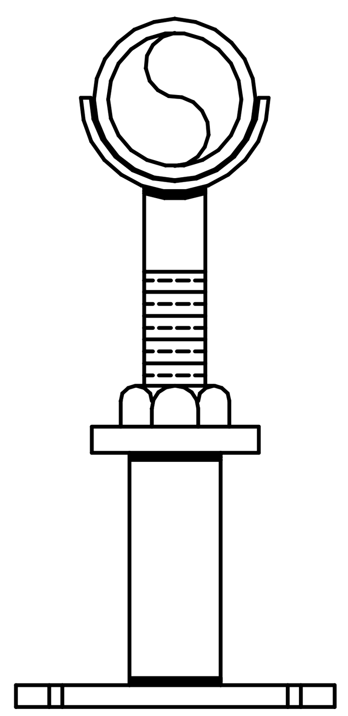
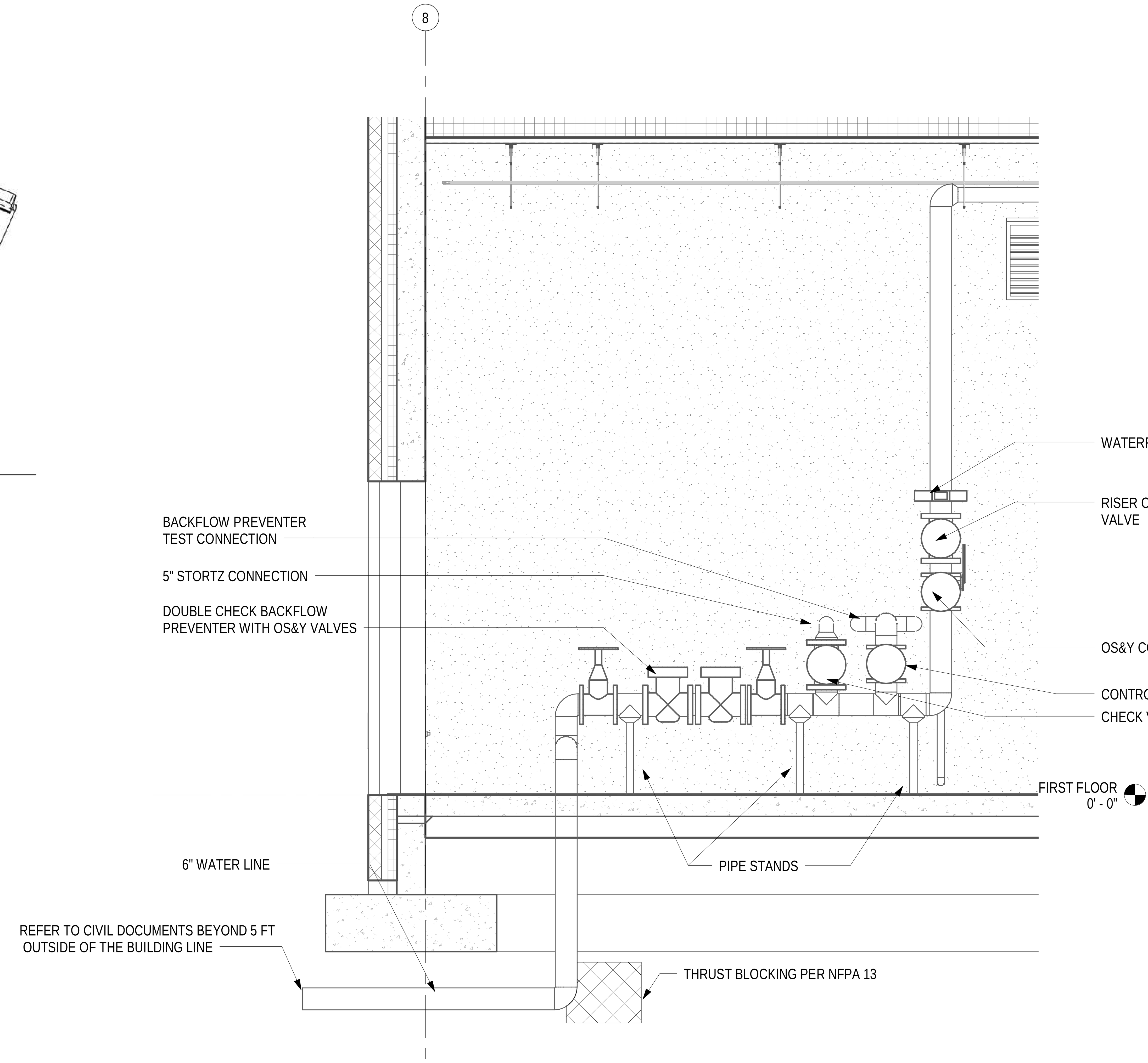


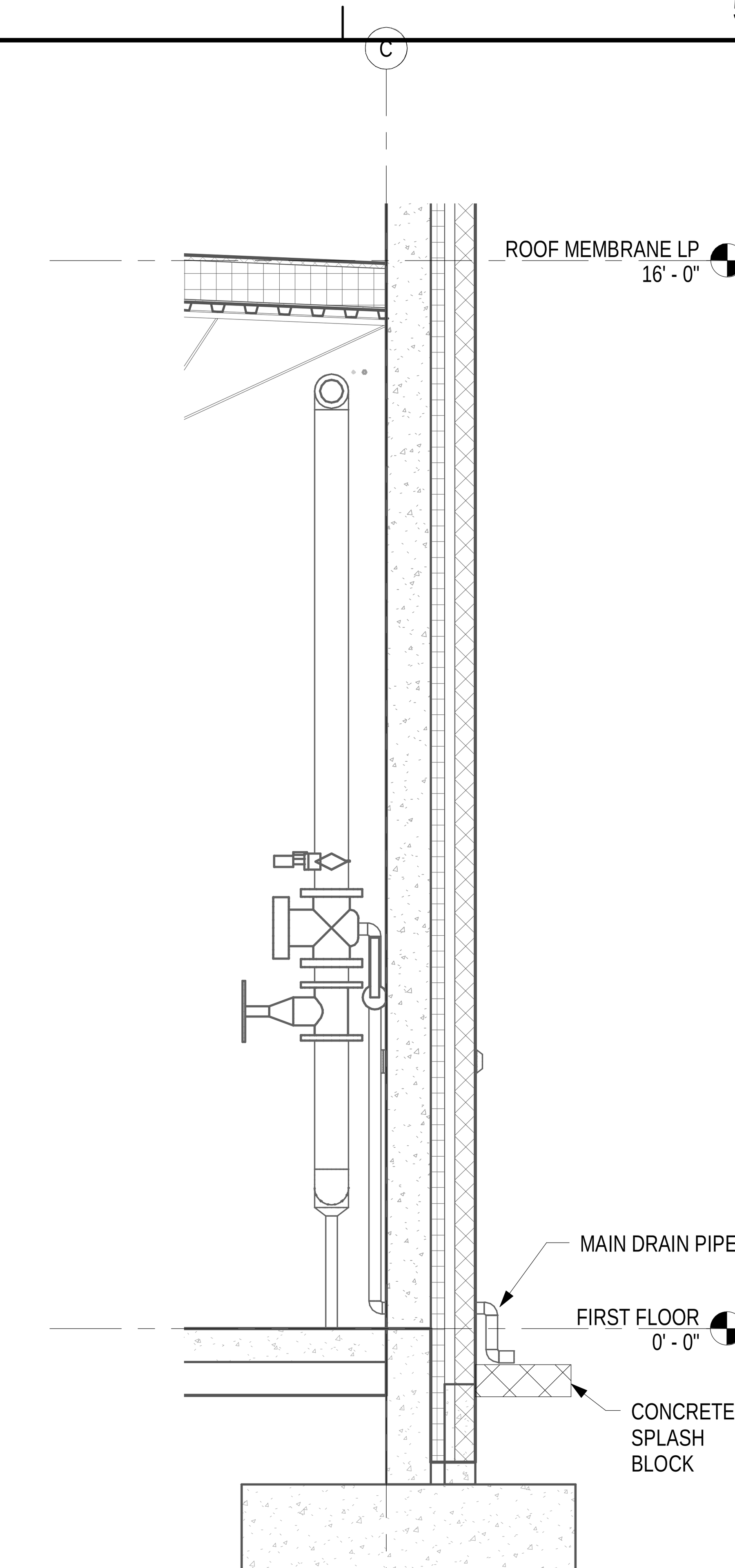
C1 STORZ CONNECTION
NTS



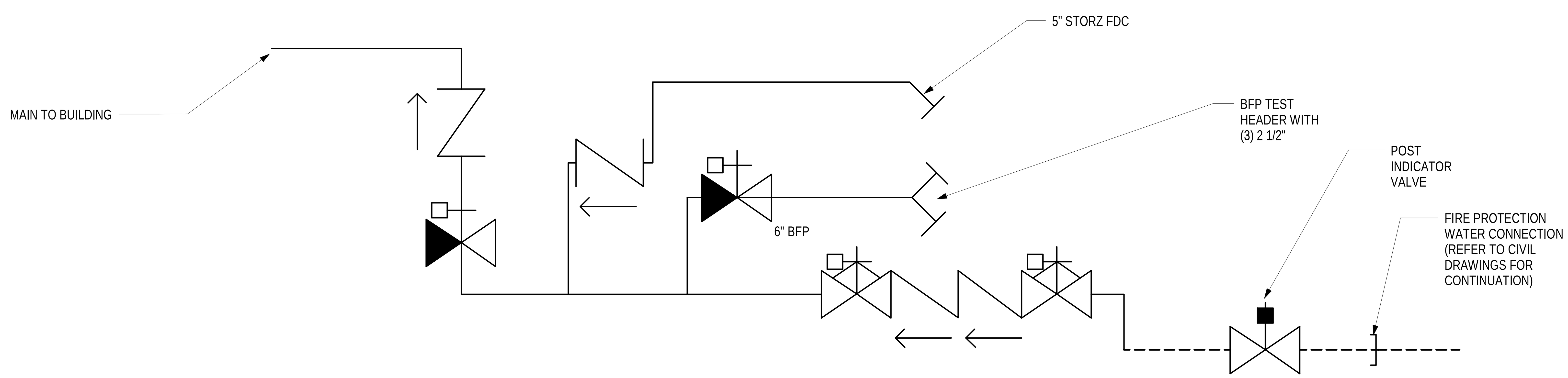
B1 PIPE STAND
6" = 1'-0"



B2 FIRE SUPPRESSION RISER DETAIL 2
1/2" = 1'-0"



B4 FIRE SUPPRESSION RISER DETAIL 1
1/2" = 1'-0"



A1 FIRE PROTECTION RISER DIAGRAM
NTS

0 1" 2" 4" 6"
6" = 1'-0"

0 1' 2' 4' 6'
1/2" = 1'-0"

SYM DESCRIPTION DATE APPR

NAVFAC

STATE OF CONNECTICUT
JEFFREY C. McLAUGHLIN
No. 25368
LICENSED PROFESSIONAL ENGINEER
6/23/2025

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	JM	DRW	CL	CHK	JB
PM/DM	SMS/CWJ				
BRANCH MANAGER	HPN				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
FIRE SUPPRESSION RISER AND DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894002
SHEET 52 OF 140
FX501

6/20/2025 11:38:50 AM FX501
\\dc1vs01\fedpr\FDZE1628\FPRO\BLDG-1667770-F-BIM_v21.rvt

GENERAL SHEET NOTES

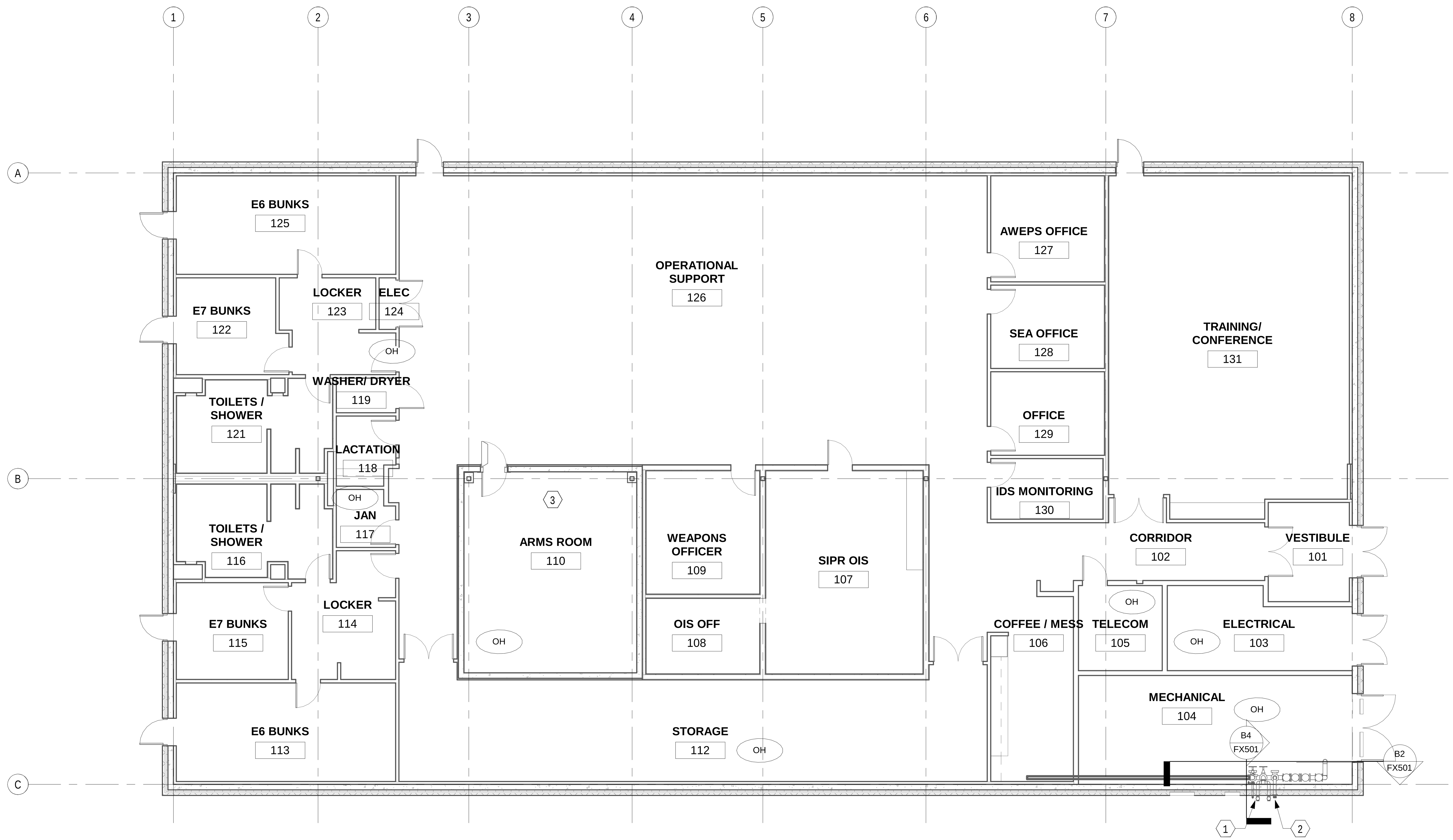
A. REFER TO SHEET FX001 FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS.

B. ALL AREAS LIGHT HAZARD UNLESS OTHERWISE NOTED.

C. SPRINKLER PIPE MUST NOT BE LOCATED ABOVE ELECTRICAL EQUIPMENT (PANEL BOARDS, SWITCH BOARDS, SWITCH GEAR MOTOR CONTROLLERS, ETC.)

SHEET KEYNOTES

1. BACKFLOW PREVENTER TEST HEADER.
2. 5" STORTZ FIRE DEPARTMENT CONNECTION.
3. PROVIDE A SINGLE PIPE PENETRATION FOR SPRINKLERS PROTECTING THE ARMORY.



B1 FIRST FLOOR PLAN - FIRE SUPPRESSION
1/8" = 1'-0"

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

NAVFAC

STATE OF CONNECTICUT JEFFREY C. McLAUGHLIN No. 25368 LICENSED PROFESSIONAL ENGINEER 6/23/2025

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	JM	DRW	CL	CHK	JB
PM/DM	SMS/CWJ				
BRANCH MANAGER	HPN				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
FIRE SUPPRESSION FLOOR PLAN

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894001
SHEET 51 OF 140

FX101

6/20/2025 11:38:47 AM FX101
\\dc1vs01\edgr\jFD2E16289\FPRO\BLDG-1667770-F-BIM_v21.rvt

GENERAL NOTES AND SYMBOLS

(ALL SYMBOLS SHOWN ARE NOT NECESSARILY USED ON THE DRAWINGS)

FIRE SPRINKLER SYMBOLS

OH — ORDINARY HAZARD

WET SYSTEM RISER

WALL-MOUNTED STORZ FIRE DEPARTMENT CONNECTION

INSPECTOR'S TEST CONNECTION/AUXILIARY DRAIN

FIRE SPRINKLER ZONE/ FLOOR CONTROL VALVE ASSEMBLY

PIPING SYMBOLS

ELBOW UP

ELBOW DOWN

VALVE IN DROP

VALVE IN CENTER DROP

VALVE IN RISE

DIRECTION OF FLOW

DIRECTION OF SLOPE DOWN

CONCENTRIC REDUCER

ECCENTRIC REDUCER

TEE OUTLET UP

TEE OUTLET DOWN

UNION

STRAINER WITH BLOWDOWN VALVE

GATE VALVE

CHECK VALVE

PRESSURE CONTROL VALVE

RELIEF VALVE

PRESSURE GAUGE WITH GAUGE COCK

TEST PORT

SPRINKLER SYMBOLS

PENDENT SPRINKLER, STANDARD TEMPERATURE

UPRIGHT SPRINKLER, STANDARD TEMPERATURE

SIDEWALL SPRINKLER, STANDARD TEMPERATURE

DRY PENDENT SPRINKLER, STANDARD TEMPERATURE

PENDENT SPRINKLER, HIGH TEMPERATURE

UPRIGHT SPRINKLER, HIGH TEMPERATURE

MISCELLANEOUS

POINT OF NEW CONNECTION TO EXISTING PIPING

ABBREVIATIONS

AFF	ABOVE FINISHED FLOOR
AHJ	AUTHORITY HAVING JURISDICTION
AVG	AVERAGE
BFP	DOUBLE CHECK VALVE BACKFLOW PREVENTER
CFM	CUBIC FEET PER MINUTE
CO	CONTRACTING OFFICER
COTR	CONTRACTING OFFICER'S TECHNICAL REPRESENTATIVE
ELEV	ELEVATION
FD	FLOOR DRAIN
FDC	FIRE DEPARTMENT CONNECTION
FHV	FIRE HOSE VALVE
FPDOR	FIRE PROTECTION DESIGNER OF RECORD
FVC	FIRE VALVE CABINET
GALS	GALLONS
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
GV	GATE VALVE
MIN	MINIMUM
NC	NONCOMBUSTIBLE
NIC	NOT IN CONTRACT
NP	NOT PERMITTED
OS&Y	OUTSIDE SCREW & YOKE
PSIG	POUNDS PER SQUARE INCH GAUGE
QFPE	QUALIFIED FIRE PROTECTION ENGINEER
SOV	SOLENOID VALVE
SQFT	SQUARE FEET
UH	UNIT HEATER
VLV	VALVE

FIRE SPRINKLER NOTES

1. PROVIDE A COMPLETE AUTOMATIC WET-PIPE FIRE SUPPRESSION SPRINKLER SYSTEM THROUGHOUT THE FACILITY, DESIGNED AND PROVIDED IN ACCORDANCE WITH APPLICABLE CODES:
   - INTERNATIONAL BUILDING CODE (IBC) (2021 EDITION)
   - NFPA 13, STANDARD FOR THE INSTALLATION OF SPRINKLER SYSTEMS (2022 EDITION)
2. VERIFY LOCATIONS AND MEASUREMENTS AS CONDITIONS MAY VARY THROUGHOUT CONSTRUCTION. CHANGES FROM THE ORIGINAL DESIGN MUST BE APPROVED BY THE FPDOR.
3. PROVIDE TAMPER AND FLOW SWITCHES. CONNECTIONS TO THE FIRE ALARM SYSTEM'S SIGNALING LINE CIRCUIT MUST BE MADE UNDER SPECIFICATION DIVISION 28 31 76.
4. WATER FLOW TEST DATA IS PROVIDED TO SHOW HYDRAULIC CALCULATIONS AND DETERMINE PIPE SIZING. WATER FLOW TEST TO BE USED FOR INFORMATION ONLY. CONTRACTOR TO PROVIDE WATER FLOW TEST.
5. OBTAIN REQUIRED PERMITS AND APPROVALS FROM THE AHJ PRIOR TO COMMENCING WORK.
6. HYDRAULICALLY CALCULATE THE SPRINKLER SYSTEM PER NFPA 13 DENSITY AND SPACING LISTED IN THE SPRINKLER DESIGN CRITERIA TABLE BELOW. FIELD VERIFY SYSTEM DESIGN AND LAYOUT PRIOR TO FABRICATION OR INSTALLATION.
7. PROVIDE PIPE HANGERS ON FIRE SPRINKLER SYSTEM PIPING AND EQUIPMENT PER NFPA 13.
8. SEE ARCHITECTURAL DRAWINGS FOR ROOM FINISH SCHEDULE AND ROOM USE. SEE ARCHITECTURAL, ELECTRICAL, MECHANICAL, AND TELECOMMUNICATIONS PLANS FOR THE LOCATION OF CEILING-MOUNTED DEVICES. FIELD VERIFY AND COORDINATE THE LOCATIONS OF SYSTEM COMPONENTS INCLUDING PIPING, ALARMS, DRAINS, TEST POINTS, ETC. WITH ARCHITECTURAL, STRUCTURAL, MECHANICAL, AND ELECTRICAL EQUIPMENT WITHIN THE BUILDING.
9. PROVIDE AUXILIARY DRAINAGE WITH A DRAIN VALVE AND PLUG FOR TRAPPED SECTION OF PIPE, PER NFPA 13.
10. THESE DOCUMENTS DEPICT A PERFORMANCE LEVEL ENGINEERING DESIGN LAYOUT AS GUIDANCE FOR THE PLANNING OF THE FIRE SPRINKLER SYSTEM. PROVIDE COMPLETE DOCUMENTS FOR REVIEW AND APPROVAL FROM THE FIRE PROTECTION DESIGN OF RECORD AND AHJ PRIOR TO INSTALLATION. INCLUDE IN THE SHOP DRAWINGS AND CALCULATIONS FOR ADDITIONAL EQUIPMENT NECESSARY TO PROVIDE A COMPLETE INSTALLATION.
11. PIPE MAIN DRAIN TO OUTSIDE OF BUILDING. PROVIDE SPLASH BLOCK.
12. ALL SPRINKLERS INSTALLED IN FINISHED CEILINGS MUST BE CONCEALED TYPE.

WATERFLOW TEST DATA

TEST DATA: 9/28/2022 BY: JACOBS FPE

STATIC PRESSURE:	99 PSI
RESIDUAL PRESSURE:	93 PSI
PITOT PRESSURE:	62 PSI
DISCHARGE FLOW:	1320 GPM
GAUGE HYDRANT #:	46
FLOW HYDRANT #:	47

NOTE: REFER TO G-103 FOR SITE PLAN SHOWING FIRE HYDRANT LOCATIONS AND WATER DISTRIBUTION SYSTEM PIPING.

SPRINKLER DESIGN CRITERIA

GENERAL ROOM/AREA CLASSIFICATION	OCCUPANCY CLASSIFICATION	DENSITY REQUIREMENTS	DURATION OF SUPPLY	MAXIMUM SPRINKLER PROTECTION AREA	SPRINKLER TEMPERATURE	HOSE ALLOWANCE
OFFICES, RESTROOMS, CORRIDORS, BREAKROOM, BUNK ROOMS	LIGHT HAZARD	0.10 GPM/SQFT 1,500 SQFT SPRINKLER: K5.6	60 MINUTES	225 SQFT	ORDINARY	250 GPM
MECHANICAL ROOM, TELECOM ROOM, STORAGE, ARMORY	ORDINARY HAZARD	0.20 GPM/SQFT 2,500 SQFT SPRINKLER: K8.0	60 MINUTES	130 SQFT	ORDINARY	250 GPM

SYM DESCRIPTION DATE APPR

NAVFAC

STATE OF CONNECTICUT JEFFREY C. McLAUGHLIN No. 25368 LICENSED PROFESSIONAL ENGINEER 6/23/2025 SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY: APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE 05/27/2025

DES JM DRW CL CHK JB

PM/DM SMS/CWJ

BRANCH MANAGER HPN

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
MID-ATLANTIC
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
FIRE SUPPRESSION SYMBOLS, LEGEND, ABBREVIATIONS AND NOTES

SCALE: AS NOTED

EPROJECT NO.: 1667770

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12894000

SHEET 50 OF 140

FX001

FIRE ALARM/MASS NOTIFICATION SYMBOLS

FMCU	FIRE ALARM / MASS NOTIFICATION PANEL
ANN	ANNUNCIATOR PANEL
RAT	RADIO ALARM TRANSMITTER - FIRE ALARM
NAC	NOTIFICATION APPLIANCE CONTROL PANEL
LOC	LOCAL OPERATING CONSOLE
MM	MONITOR MODULE
CM	CONTROL MODULE
F	MANUAL PULL STATION
P	PHOTOELECTRIC SMOKE DETECTOR
S / R	DUCT SMOKE DETECTOR (S - SUPPLY; R - RETURN)
CO	CARBON MONOXIDE DETECTOR
	WALL-MOUNTED FIRE ALARM SPEAKER
	WALL-MOUNTED FIRE ALARM SPEAKER STROBE
XX	COMBINATION FIRE ALARM / MASS NOTIFICATION SPEAKER STROBE WITH CLEAR LENS (CEILING MOUNTED) (XX DENOTES MINIMUM CANDELA RATING)
	REMOTE ANTENNA
VTD	VISUAL TEXT DISPLAY
"WP"	DESIGNATES WEATHERPROOF BACKBOX AND DEVICE
KB	KNOX BOX
C TS	FIRE SUPPRESSION CONTROL VALVE TAMPER SWITCH
C WF	WATER FLOW SWITCH
SPD	SURGE PROTECTIVE DEVICE
VTDC	VISUAL TEXT DISPLAY CONTROLLER

FIRE ALARM / MASS NOTIFICATION ABBREVIATIONS

AFF	ABOVE FINISHED FLOOR
AHJ	AUTHORITY HAVING JURISDICTION
ACU	AUTONOMOUS CONTROL UNIT
AVG	AVERAGE
CO	CONTRACTING OFFICER
COR	CONTRACTING OFFICER REPRESENTATIVE
COTR	CONTRACTING OFFICER'S TECHNICAL REPRESENTATIVE
DFPE	DESIGNATED FIRE PROTECTION ENGINEER
ELEV	ELEVATION
EOL	END OF LINE RESISTOR
FPDOR	FIRE PROTECTION DESIGNER OF RECORD
IDC	INITIATING DEVICE CIRCUIT
MAX	MAXIMUM
MIN	MINIMUM
NAC	NOTIFICATION APPLIANCE CIRCUIT
NIC	NOT IN CONTRACT
NP	NOT PERMITTED
NC	NONCOMBUSTIBLE
QFPE	QUALIFIED FIRE PROTECTION ENGINEER
SLC	SIGNALING LINE CIRCUIT
SQFT	SQUARE FEET
TYP	TYPICAL
WP	WEATHERPROOF

FIRE ALARM / MASS NOTIFICATION SYSTEM GENERAL NOTES

1. DESIGN, PROVIDE, AND COMMISSION THE FIRE ALARM/ MASS NOTIFICATION SYSTEM IN ACCORDANCE WITH APPLICABLE CODES:
   - NFPA 72, NATIONAL FIRE ALARM AND SIGNALING CODE (2021 EDITION)
   - INTERNATIONAL BUILDING CODE (IBC) (2021 EDITION)
   - UFC 3-600-01, FIRE PROTECTION ENGINEERING FOR FACILITIES (CHANGE 6, MAY 2021)
   - UFC 4-010-01, DOD MINIMUM ANTITERRORISM STANDARDS FOR BUILDINGS (CHANGE 2, 30 JULY 2022)
   - UFC 4-021-01, DESIGN AND O&M: MASS NOTIFICATION SYSTEMS (CHANGE 1, JANUARY 2010)
2. THESE DOCUMENTS DEPICT A PERFORMANCE LEVEL ENGINEERING DESIGN LAYOUT AS GUIDANCE FOR THE PLANNING OF THE SYSTEM. PROVIDE COMPLETE DOCUMENTS FOR REVIEW AND APPROVAL FROM THE ENGINEER OR RECORD AND THE AUTHORITY HAVING JURISDICTION PRIOR TO INSTALLATION. INCLUDE IN THE SHOP DRAWINGS AND CALCULATIONS FOR ADDITIONAL EQUIPMENT NECESSARY, INCLUDING INITIATING DEVICES AND NOTIFICATION APPLIANCES, TO PROVIDE A COMPLETE INSTALLATION AND COMPLY WITH LOCAL CODES, AMENDMENTS AND BASE STANDARDS.
3. FOR COMPLETE SYSTEM CONTROL, REFER TO THE MATRIX OF OPERATION WITHIN THIS PACKAGE.
4. PROVIDE AN ADDRESSABLE FIRE ALARM/MASS NOTIFICATION SYSTEM CONTROL PANEL WITH RADIO ALARM TRANSMITTER TO TRANSMIT ALARM, TROUBLE, AND SUPERVISORY SIGNALS TO THE BASE FIRE ALARM REPORTING SYSTEM. PROVIDE ADDITIONAL MASS NOTIFICATION INTERFACE PANEL TO ACCEPT SIGNALS TO BROADCAST VOICE MESSAGES WITHIN THE FACILITY. MASS NOTIFICATION INTERFACE PANEL MUST BE COMPATIBLE WITH THE EXISTING BASEWIDE MASS NOTIFICATION SYSTEM.
5. FINAL FIRE ALARM/MASS NOTIFICATION SYSTEM TEST AND PRE-TEST MUST BE WITNESSED BY THE QFPE. FILE A "RECORD OF COMPLETION" PREPARED IN ACCORDANCE WITH NFPA 72, AND THE FIRE ALARM TEST REPORT WITH THE FPDOR AND AHJ.
6. THE FLASH RATE OF VISIBLE NOTIFICATION APPLIANCES MUST NOT EXCEED 2 FLASHES PER SECOND, NOR BE LESS THAN 1 FLASH EVERY SECOND, IN ACCORDANCE WITH NFPA 72. A MAXIMUM PULSE DURATION MUST BE 0.2 SECONDS WITH A MAXIMUM DUTY CYCLE OF 40%, IN ACCORDANCE WITH NFPA 72. VISIBLE NOTIFICATION APPLIANCES MUST BE SPACED IN ACCORDANCE WITH NFPA 72. STROBES MUST BE SYNCHRONIZED.
7. AUDIBLE FIRE ALARM SOUND LEVELS MUST BE AT LEAST 15dBA ABOVE THE AVERAGE AMBIENT SOUND LEVEL OR 5dBA ABOVE THE MAXIMUM SOUND LEVEL HAVING A DURATION OF AT LEAST 60 SECONDS, WHICHEVER IS GREATER, MEASURED 5 FEET ABOVE THE FLOOR IN THE OCCUPIABLE AREA, BUT NOT GREATER THAN 120 dBA AT THE MINIMUM HEARING DISTANCE FROM THE AUDIBLE APPLIANCE. PROVIDE NOTIFICATION APPLIANCES THROUGHOUT THE FACILITY TO MEET THE NFPA 72 REQUIREMENTS.
8. SPRINKLER SYSTEM TAMPER AND FLOW SWITCHES PROVIDED UNDER SPECIFICATION SECTION 21 13 13. CONNECT DEVICES TO THE FIRE ALARM SIGNALING LINE CIRCUITS.
9. OBTAIN APPROVALS FROM THE FPDOR AND AHJ PRIOR TO COMMENCING WORK.
10. DO NOT LOCATE DETECTORS IN DIRECT AIR FLOW OR CLOSER THAN 3 FEET FROM AN AIR SUPPLY OR RETURN DIFFUSER.
11. MOUNT WALL-MOUNTED VISUAL AND VISUAL/AUDIBLE NOTIFICATION APPLIANCES WITH THE ENTIRE STROBE LENS BETWEEN 80" TO 96" ABOVE THE FINISHED FLOOR.
12. FIRE ALARM SIGNALING LINE CIRCUITS (SLC) AND NOTIFICATION APPLIANCE CIRCUITS (NAC) ARE CLASS B. T-TAPS AND WIRE NUTS ARE NOT PERMITTED.
13. THE INTENT OF THE SPEAKER SPACING IS TO PROVIDE A COMMON INTELLIGIBILITY SCALE (CIS) SCORE TO MEET THE MINIMUM INTENT OF UFC 4-021-01. THE INSTALLING CONTRACTOR MUST VERIFY THEIR DESIGN MEETS THIS INTENT EITHER BY PROVIDING SPEECH INTELLIGIBILITY CALCULATION BASED ON THE PROVIDED PRODUCTS' THEORETICAL Q AND DIRECTIVITY INDEX OR BY ADDING SPEAKERS THROUGH TESTING. MINIMUM CIS VALUE IN EACH AREA WHERE OCCUPANTS ARE LOCATED IS 0.7
14. PROVIDE SURGE SUPPRESSION ON FIRE ALARM AND MASS NOTIFICATION CIRCUITS LEAVING THE BUILDING ENCLOSURE AND ON 120VAC CIRCUITS TO CONTROL PANELS, SUBPANELS, TRANSMITTERS, AMPLIFIER PANELS, AND BOOSTER PANELS IN ACCORDANCE WITH UL 1449 AND UL 1283. SURGE SUPPRESSION IS NOT REQUIRED FOR EXTERIOR WALL MOUNTED DEVICES.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

STATE OF CONNECTICUT
JEFFREY C. McLAUGHLIN
No. 25368
LICENSED PROFESSIONAL ENGINEER
6/23/2025
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES JM	DRW CL	CHK JB
PM/DM		SMS/CWJ
BRANCH MANAGER		HPN
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA

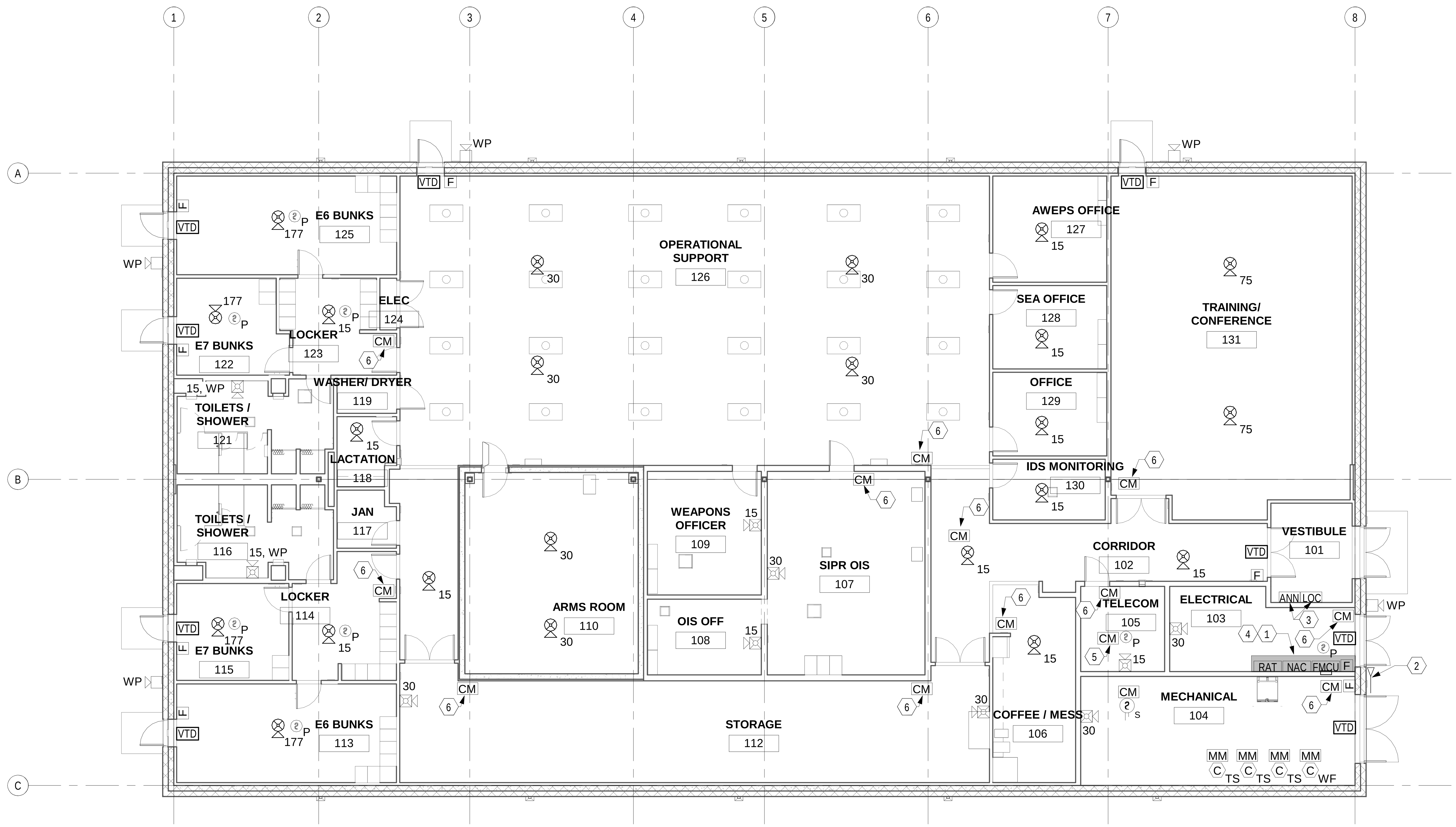
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS

FIRE ALARM AND MASS NOTIFICATION SYMBOLS, LEGEND, ABBREVIATIONS AND NOTES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894003
SHEET 53 OF 140

FA001

6/20/2025 11:37:51 AM
\\dc1vs01\fedprj\FDZE1628\FPRO\BLDG-1667770-F-BIM_v21.rvt
FA001



B1 FIRST FLOOR PLAN - FIRE ALARM / MASS NOTIFICATION SYSTEM

1/8" = 1'-0"

GENERAL SHEET NOTES

A. REFER TO FA001 FOR GENERAL NOTES, SYMBOLS AND ABBREVIATIONS.

SHEET KEYNOTES

1. DESIGNATED SPACE FOR FIRE ALARM EQUIPMENT.
2. RADIO ALARM TRANSMITTER ANTENNA MOUNTED AT ROOF LEVEL.
3. RECESSED LOC AND REMOTE ANNUNCIATOR MAY BE COMBINED INTO A SINGLE UNIT PROVIDED THE MICROPHONE IS ACCESSIBLE TO BUILDING OCCUPANTS WHILE THE FIRE ALARM FUNCTIONS REMAIN INACCESSIBLE.
4. GREY SHADING TO SHOW FIRE PROTECTION BLOCKOUT.
5. CONTROL MODULE FOR PUBLIC ADDRESS SYSTEM. REFER TO T- 201 FOR LOCATION OF PA SYSTEM CONTROLS
6. CONTROL MODULE FOR LIGHTING CONTROLS.

PLAN NORTH

0 4' 8' 16' 24'
1/8" = 1'-0"

SYM	DESCRIPTION	DATE	APPR

NAVFAC

STATE OF CONNECTICUT — JEFFREY C. McLAUGHLIN — No. 25368 — LICENSED PROFESSIONAL ENGINEER — 6/23/2025

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE 05/27/2025

DES	JM	DRW	CL	CHK	JB
PM/DM	SMS/CWJ				
BRANCH MANAGER	HPN				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA

NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS

FIRE ALARM AND MASS NOTIFICATION FLOOR PLAN

SCALE:	AS NOTED
EPROJECT NO.:	1667770
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12894004
SHEET	54 OF 140

FA101

6/20/2025 11:37:54 AM
FA101
\\dc1vs01\edgr\jFDZE1628\FPRO\BLDG-1667770-F-BIM_v21.rvt

MATRIX OF OPERATIONS

● INDICATES ACTION WILL CAUSE A SPECIFIC RESULT

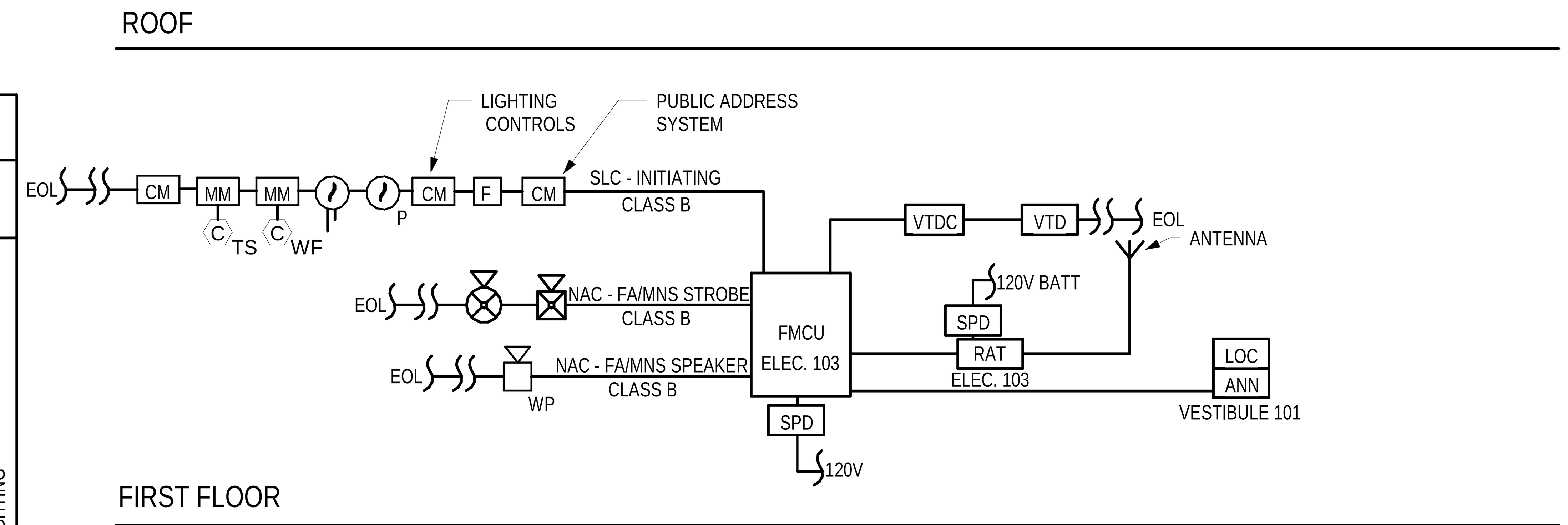
INDICATES ACTION WILL NOT CAUSE A SPECIFIC RESULT

SYSTEM OUTPUTS:
A. GENERAL EVACUATION FIRE ALARM
B. MASS NOTIFICATION GENERAL ALARM
C. OUTPUT TO EMERGENCY RECIEVER
D. AUDIBLE & VISUAL ALARM SIGNAL AT FACP
E. AUDIBLE & VISUAL SUPERVISORY SIGNAL AT FACP
F. AUDIBLE & VISUAL TROUBLE SIGNAL AT FACP
G. RECORD EVENTS IN MEMORY
H. OUTPUT SIGNAL TO SHUT DOWN AIR HANDLING UNIT
I. OVERRIDE GENERAL FIRE ALARM
J. DISPLAY DEVICE ON ANNUCIATOR PANEL
K. ACTIVATE EXTERIOR STROBE AT FD RESPONSE POINT
L. OVERRIDE PUBLIC ADDRESS SYSTEM
M. ACTIVATE EGRESS LIGHTING

	SYSTEM INPUTS	A	B	C	D	E	F	G	H	I	J	K	L	M
1.	AREA/ ROOM SMOKE DETECTOR	●		●	●			●			●	●	●	●
2.	DUCT SMOKE DETECTOR			●		●		●	●		●			
3.	WATER FLOW SWITCH	●		●	●			●			●	●	●	●
4.	MANUAL PULL STATION	●		●	●			●			●	●	●	●
5.	MONITOR SWITCH (FIRE SUPPRESSION VALVE TAMPER SWITCH)			●		●		●			●			
6.	BUNK ROOM SMOKE DETECTOR	●		●	●			●			●	●		●
7.	MASS NOTIFICATION IN- BUILDING EVENT		●	●	●			●		●		●	●	
8.	MASS NOTIFICATION - AHU SHUTDOWN SWITCH (AT FMCU/ACU AND LOC)		●	●		●		●		●			●	●
9.	MASS NOTIFICATION - INPUT SIGNAL FROM BASE SYSTEM		●	●		●		●		●			●	
		A	B	C	D	E	F	G	H	I	J	K	L	M

B1 FIRE ALARM MATRIX OF OPERATION

NTS



NOTE: RISER DIAGRAM IS A SCHEMATIC REPRESENTATION OF THE FIRE ALARM/ MASS NOTIFICATION SYSTEM. REFER TO THE FIRE ALARM/ MASS NOTIFICATION PLANS FOR LOCATIONS AND QUANTITIES OF DEVICES AND APPLIANCES.

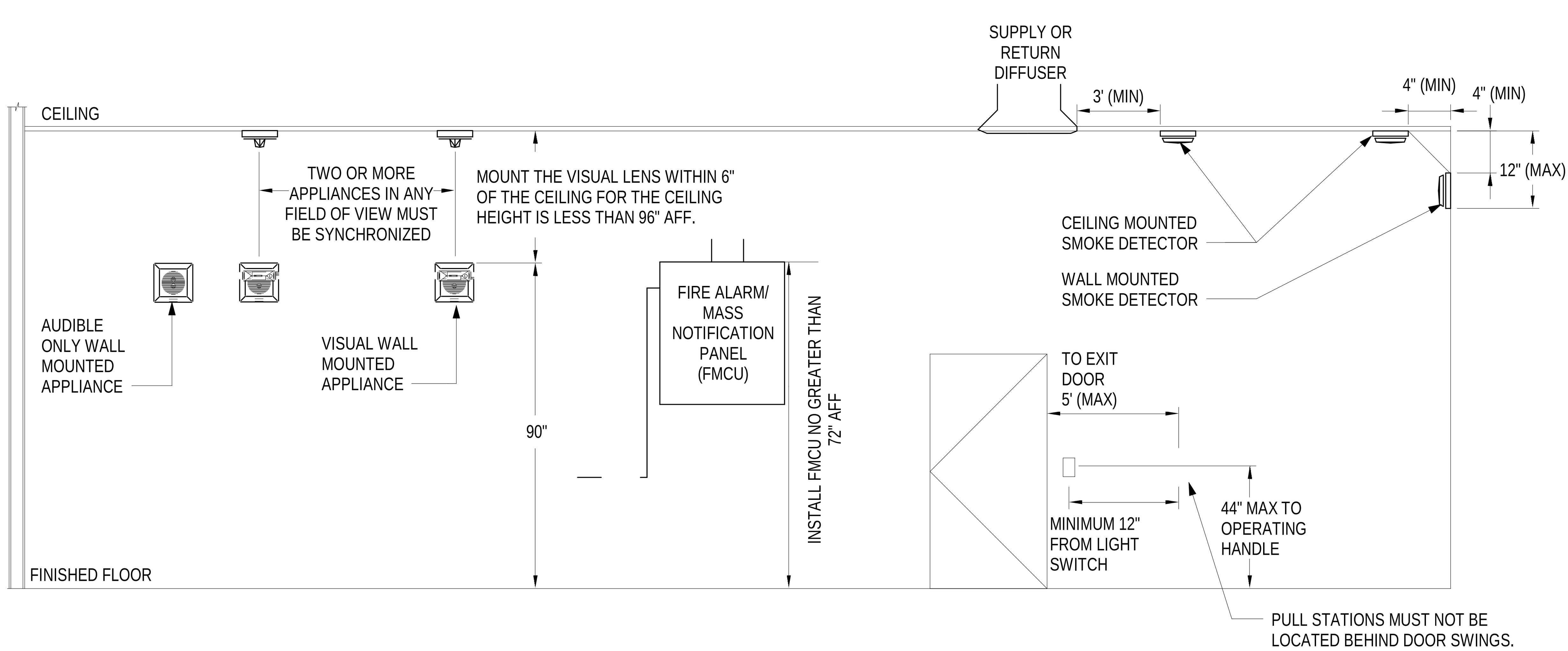
C2 FIRE ALARM RISER DIAGRAM

NTS

GENERAL SHEET NOTES

A. THE DESIGN SHOWN IS A PERFORMANCE DESIGN ILLUSTRATING GENERAL DESIGN CRITERIA ONLY. PROVIDE ALL ADDITIONAL EQUIPMENT REQUIRED TO COMPLETE THE SYSTEM FOR COMPLIANCE WITH ALL APPLICABLE CODES.

B. ALL CIRCUITS MUST BE IN MINIMUM 3/4" CONDUIT.



A1 DETAIL - TYPICAL DEVICE MOUNTING DETAIL

NTS

PLAN NORTH

SYM	DESCRIPTION	DATE	APPR

NAVFAC

STATE OF CONNECTICUT JEFFREY C. McLAUGHLIN No. 25368 LICENSED PROFESSIONAL ENGINEER 6/23/2025 SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY: APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE 05/27/2025

DES JM DRW CL CHK JB

PM/DM SMS/CWJ

BRANCH MANAGER HPN

CHIEF ENG/ARCH JWJ

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
FIRE ALARM AND MASS NOTIFICATION RISER DIAGRAM, MATRIX, AND DETAILS

SCALE: AS NOTED

EPROJECT NO.: 1667770

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12894005

SHEET 55 OF 140

FA501

PLUMBING LEGEND

	2-WAY VALVE
	3-WAY VALVE
	AUTOMATIC AIR VENT
	AUTOMATIC BALANCING VALVE
	BACKFLOW PREVENTER
	BALL VALVE
	BUTTERFLY VALVE
	CHECK VALVE
	DOWNWARD SLOPE
	DRAIN VALVE
	CONCENTRIC REDUCER
	ECCENTRIC REDUCER
	FLANGE
FCO	FLOOR CLEAN OUT
FD	FLOOR DRAIN
M	GAS OR WATER METER
	GATE VALVE
	GLOBE VALVE
	ISOLATION VALVE
	MANUAL AIR VENT
	MANUAL FLOW CONTROL BALANCING VALVE
	PIPE CAP
	PIPE DOWN
	PIPE UP
	PRESSURE GAUGE WITH SHUTOFF COCK
	PRESSURE REDUCING VALVE
	RELIEF VALVE
	SOLENOID VALVE
	SQAURE HEAD BALANCING VALVE
	STRAINER WITH BLOWDOWN VALVE
	TEMPERATURE PRESSURE RELIEF VALVE
	UNION
	WALL CLEAN OUT
W	WATER METER
	WATER PUMP
	WYE STRAINER
	POINT OF DISCONNECTION
	POINT OF CONNECTION
(ICE) (REF)	PLUMBING FIXTURES OR EQUIPMENT BY OTHERS

DCW	DOMESTIC COLD WATER PIPING
DHW	DOMESTIC HOT WATER PIPNG
DHWR	DOMESTIC HOT WATER RETURN PIPING
SS	SOIL, WASTE OR SANITARY SEWER PIPING
R	RADON MITIGATION PIPING
SD	RAIN LEADER - PRIMARY STORM DRAINAGE
OFD	RAIN LEADER - OVERFLOW STORM DRAINAGE
V	VENT PIPING
CA	COMPRESSED AIR PIPING
	THIN LINE WEIGHT INDICATES EXISTING ITEM TO REMAIN
	HEAVY LINE WEIGHT INDICATES NEW WORK TO BE PROVIDED
	HEAVY DASHED LINE WEIGHT INDICATES EXISTING ITEM TO REMOVE, REFER TO DEMOLITION NOTES

NOTE:
NOT ALL ABBREVIATIONS & SYMBOLS MAY BE USED.

DETAIL/SECTION TITLE	DETAIL CALLOUT SYMBOL	SECTION CUT SYMBOL
SECTION, DETAIL OR ELEVATION IDENTIFICATION MARK; SECTION, DETAIL OR ELEVATION TITLE; XX / X-XXX; PLATE NUMBER WHERE SECTION, DETAIL OR ELEVATION IS TAKEN	DETAIL NUMBER; XX / X-XXX; PLATE NUMBER WHERE DETAIL IS DRAWN	SECTION LETTER; PLATE NUMBER WHERE SECTION IS DRAWN; XX / X-XXX; VIEW SECTION

PLUMBING ABBREVIATIONS

ABA	ARCHITECTURAL BARRIERS ACT
ADA	AMERICANS WITH DISABILITIES ACT
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AMI	ADVANCED METERING INFRASTRUCTURE
BF	BELOW FLOOR
BFF	BELOW FINISHED FLOOR
BTUH	BRITISH THERMAL UNIT PER HOUR
BV	BALANCING VALVE
CLG	CEILING
CO	CLEANOUT
COR	CONTRACTING OFFICER'S REPRESENTATIVE
CV	CHECK VALVE
DCP	DOMESTIC HOT WATER CIRCULATING PUMP
DCW	DOMESTIC COLD WATER
DFU	DRAINAGE FIXTURE UNITS
DHW	DOMESTIC HOT WATER
DHWH	DOMESTIC HOT WATER HEATER
DHWR	DOMESTIC HOT WATER RECIRCULATION
DIAM.	DIAMETER
DN	DN
DWV	DRAIN, WASTE, VENT
EA	EACH
ELEV	ELEVATION
ET	EXPANSION TANK
EWC	ELECTRIC WATER COOLER
FCO	FLOOR CLEANOUT
FD	FLOOR DRAIN
FLA	FULL LOAD AMPS
FT	FEET
GAL	GALLON
GFCI	GOVERNMENT FURNISHED, CONTRACTOR INSTALLED
GPH	GALLONS PER HOUR
GPM	GALLONS PER MINUTE
H x W	HEIGHT AND WIDTH
HB	HOSE BIBB
HP	HORSEPOWER
I.E.	INVERT ELEVATION
IM	ICE MAKER
IN.	INCHES
IN. WC	INCHES OF WATER COLUMN
KS	KITCHEN SINK
KW	KILOWATTS
LAV OR L	LAVATORY
MAX	MAXIMUM
MIN	MINIMUM
MS	MOP SINK
N.C.	NORMALLY CLOSED
N.O.	NORMALLY OPEN
NFHB	NON-FREEZE HOSE BIBB
PH	PHASE
PSI	POUNDS PER SQUARE INCH
R	RADON MITIGATION PIPING
REF	REFRIGERATOR
RF	RADON FAN
RL	ROOF LEADER
RPM	REVOLUTIONS PER MINUTE
RPZ	REDUCED PRESSURE ZONE BACKFLOW PREVENTER
SAN OR SS	SANITARY
SF	SQUARE FEET
SH	SHOWER
SK	SINK
SOV	LINE SIZE, FULL OPEN PORT SHUTOFF VALVE
SS	SERVICE SINK
TAB	TESTING, ADJUSTING, BALANCING
TMV	THERMOSTATIC MIXING VALVE
TP	TRAP PRIMER
TYP	TYPICAL
U	URINAL
U.O.N.	UNLESS OTHERWISE NOTED
V	VENT
VTR	VENTTHROUGH ROOF
W	WASTE
W/	WITH
WC	WATER CLOSET
WCO	WALL CLEANOUT
WHA	WATER HAMMER ARRESTOR
WM	WATER METER
WSFU	WATER SUPPLY FIXTURE UNITS
°C	DEGREES CELSIUS
°F	DEGREES FAHRENHEIT
Ø	DIAMETER

GENERAL PLUMBING NOTES

1. THE INTENT OF THESE DRAWINGS IS TO PROVIDE COMPLETE AND PROPERLY FUNCTIONING PLUMBING SYSTEMS. LABOR, MATERIALS, AND EQUIPMENT REQUIRED TO COMPLETE THIS WORK MUST BE PROVIDED.
2. PLUMBING DRAWINGS ARE IN PART DIAGRAMMATIC AND INTENDED TO SHOW THE SCOPE AND GENERAL ARRANGEMENT OF THE WORK UNDER THIS CONTRACT. FOLLOW THESE DRAWINGS FOR SIZES OF THE EQUIPMENT, PIPING AND GENERAL PIPE ROUTING. DRAWINGS ARE NOT INTENDED TO BE SCALED FOR ROUGH-IN MEASUREMENTS NOR TO SERVE AS SHOP DRAWINGS. WHERE JOB CONDITIONS REQUIRE CHANGES OR ADJUSTMENTS IN THE INDICATED LOCATIONS OR ARRANGEMENT OF THE WORK, SUCH CHANGES MUST BE MADE WITHOUT CHANGE IN THE CONTRACT AMOUNT.
3. PLUMBING WORK ON THIS PROJECT MUST BE PROVIDED IN ACCORDANCE WITH THE 2021 INTERNATIONAL PLUMBING CODES AND APPLICABLE LOCAL CODES.
4. COORDINATE PLUMBING WORK WITH SITE CONDITIONS AND THE WORK OF OTHER TRADES.
5. VERIFY SLOPES AND INVERT ELEVATIONS PRIOR TO COMMENCEMENT OF WORK.
6. THE WORK REQUIRED MUST INCLUDE PROVIDING MATERIALS, CUTTING, PATCHING, ADAPTING AND OTHER INCIDENTAL WORK, TOGETHER WITH SUCH TEMPORARY CONNECTIONS AS MAY BE REQUIRED.
7. HORIZONTAL VENT PIPING MUST RUN ABOVE CEILING AND HORIZONTAL SANITARY PIPING MUST RUN BELOW SLAB UNLESS OTHERWISE NOTED.
8. SLOPE ALL HORIZONTAL SANITARY WASTE PIPING 2" AND SMALLER AT 1/4" PER FOOT AND SLOPE ALL HORIZONTAL SANITARY WASTE PIPING 3" AND LARGER AT 1/8" PER FOOT UNLESS OTHERWISE SPECIFIED ON PLANS.
9. PROVIDE FIRE-STOPPING AT PENETRATIONS THROUGH RATED PARTITIONS. THIS INCLUDE PENETRATIONS THROUGH RATED WALLS, BARRIERS, CEILINGS, ROOFS, AND FLOORS. PROVIDE MANUFACTURER'S STANDARD FIRE-STOPPING SEALANT, WITH ACCESSORY MATERIALS, HAVING A MINIMUM FIRE RESISTANCE RATING EQUAL TO THE FIRE-RATED BARRIER THAT IS PENETRATED. FIRE-STOPPING SYSTEMS MUST BE UL 1479 LISTED AND TESTED IN ACCORDANCE WITH ASTM E814.
10. PROVIDE ACCESS PANELS FOR VALVES, TRAP PRIMERS AND OTHER PLUMBING SPECIALTIES LOCATED INSIDE CHASES, BEHIND WALLS OR ABOVE INACCESSIBLE CEILINGS.
11. SHUT OFF VALVES MUST BE PROVIDED AT THE BEGINNING OF BRANCH LINES AND AT EACH INDIVIDUAL WATER LINE SERVING WALL HYDRANTS OR HOSE BIBBS. INDIVIDUAL SHUT OFF VALVES (ANGLE STOP VALVES) MUST BE PROVIDED AT WATER SUPPLY LINES SERVING THE PLUMBING FIXTURES. REFER TO RISER DIAGRAMS FOR LOCATIONS OF SHUT OFF VALVES.
12. DOMESTIC HOT WATER RECIRCULATION SYSTEM MUST BE PROVIDED WITH BALANCING VALVES AND CHECK VALVES AT EACH RECIRCULATION BRANCH.
13. PAINT VENT THROUGH ROOF (VTR) TO MATCH THE ADJACENT ROOF.
14. SANITARY AND STORM PIPE INVERTS EXITING THE BUILDING MUST BE COORDINATED WITH BUILDING FOOTERS.
15. FLOOR DRAINS AND FLOOR CLEANOUTS MUST HAVE SQUARE STRAINERS/TOPS IN AREAS WITH TILE FLOOR AND ROUND STRAINERS/TOPS IN OTHER FLOOR TYPES.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/25/2025
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY: APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH | DRW BPT | CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING LEGEND AND GENERAL NOTES

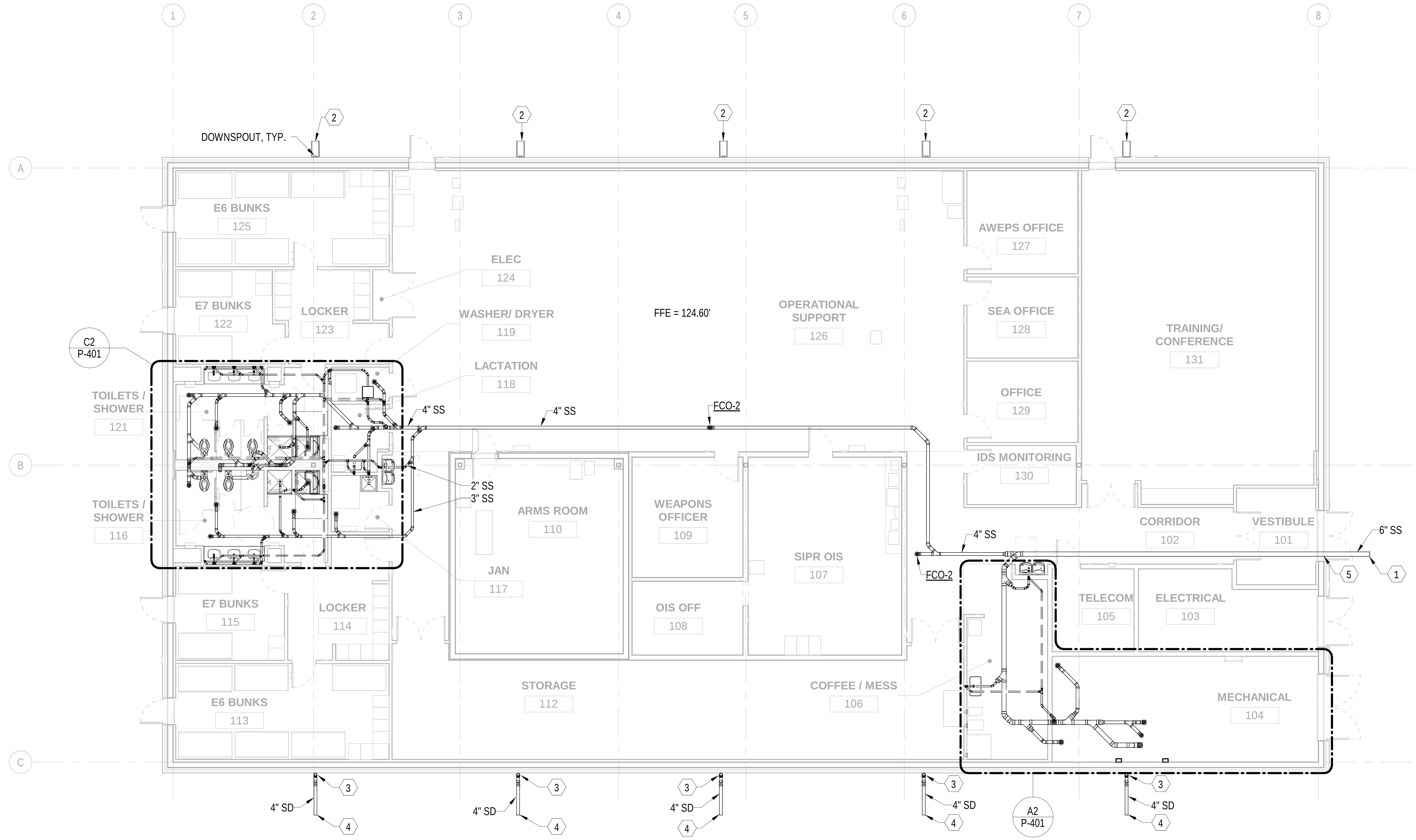
SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894006
SHEET 56 OF 140
P-001

GENERAL SHEET NOTES

A. REFER TO SHEET P-001 FOR LEGEND AND GENERAL NOTES.
B. REFER TO ENLARGED PLANS FOR DETAILED PIPE SIZING AND FIXTURE TAG DESIGNATIONS.
C. PROVIDE UL LISTED FIRE-STOPPING SYSTEMS AT PENETRATIONS THROUGH RATED PARTITIONS.

SHEET KEYNOTES

1. 6" SANITARY TO MAIN SANITARY SEWER, EXTEND 5' FROM EXTERIOR FROM BUILDING. REFER TO CIVIL FOR CONTINUATION. I.E.: 5'-0" BFF.
2. PROVIDE 1' x 2' CONCRETE SPLASH BLOCK TO SERVE DOWNSPOUT FROM ABOVE.
3. DOWNSPOUT FROM ABOVE, REFER TO ARCHITECTURE. PROVIDE CAST IRON DOWNSPOUT BOOT WITH INTEGRAL CLEANOUT PRIOR TO TRANSITION TO STORM PIPE AND PENTRATING BELOW GRADE. REFER TO CIVIL FOR DOWNSPOUT CONNECTION DETAIL.
4. 4" STORM DRAINAGE, EXTEND 5' FROM EXTERIOR OF BUILDING. SEE CIVIL FOR CONTINUATION. I.E.: 2'-0" BFF.
5. REFER TO STRUCTURAL FOR PIPING DETAIL BELOW FOUNDATION.



B1 PLUMBING SANITARY & VENT FLOOR PLAN
1/8" = 1'-0"

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/25/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	CWH	DRW	BPT	CHK	MJL
PM/DM	SMS/CWJ				
BRANCH MANAGER	ALG				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING SANITARY & VENT FLOOR PLAN

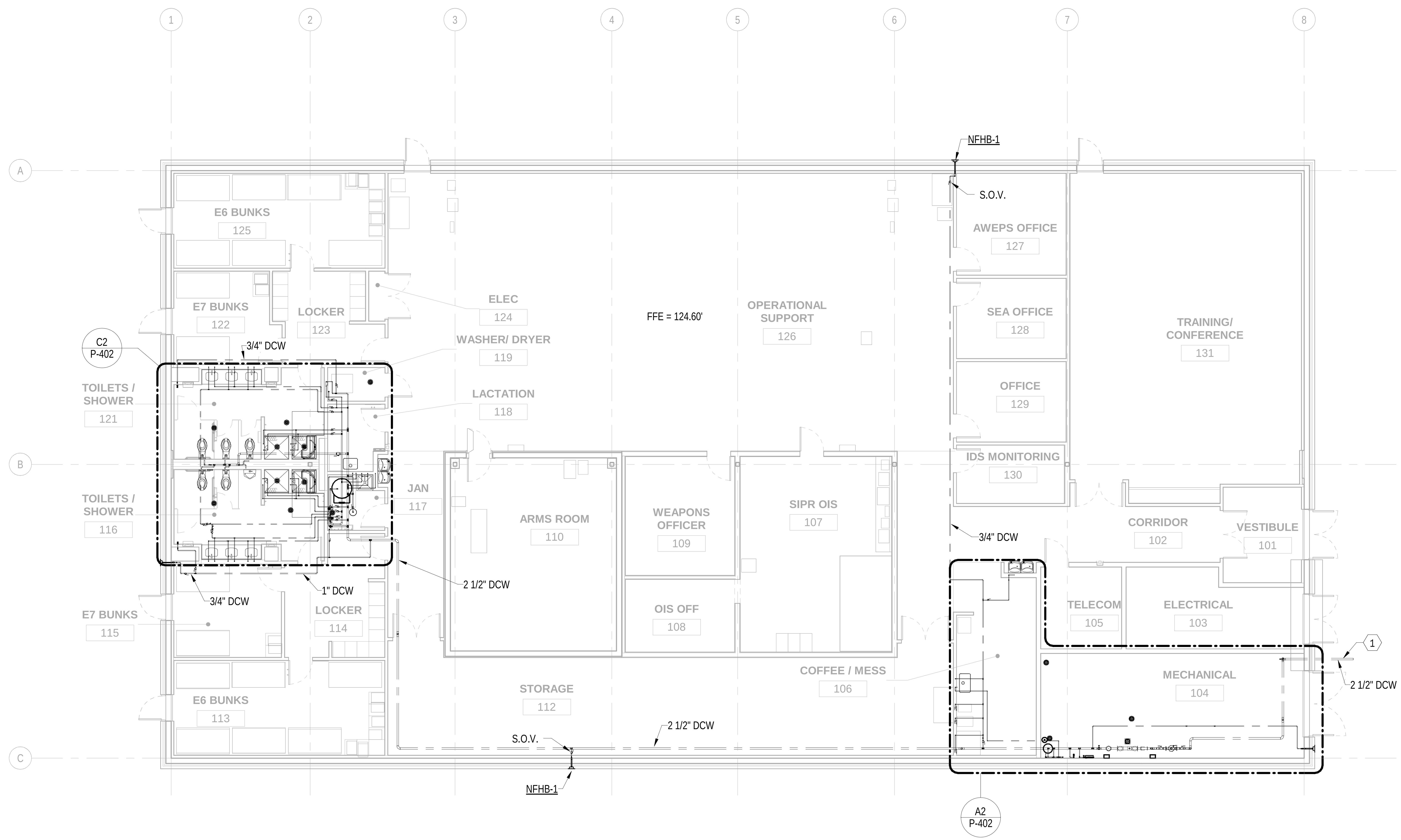
SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894007
SHEET 57 OF 140

P-101

6/20/2025 12:13:00 PM
\\dc1vs01\FEDPRJ\FED\E16\28\PLUMB\BLDG-1667770-P-BIM_v21.rvt
P-101

1 2 3 4 5

D C B A



B1 PLUMBING DOMESTIC WATER FLOOR PLAN

1/8" = 1'-0"

GENERAL SHEET NOTES

A. REFER TO SHEET P-001 FOR LEGEND AND GENERAL NOTES.
B. REFER TO ENLARGED PLANS FOR DETAILED PIPE SIZING AND FIXTURE TAG DESIGNATIONS.
C. PROVIDE UL LISTED FIRE-STOPPING SYSTEMS AT PENETRATIONS THROUGH RATED PARTITIONS.

SHEET KEYNOTES

1 2-1/2" DOMESTIC WATER SERVICE, EXTEND 5' FROM BUILDING. I.E.: 6'-1" BFF. WATER PIPE SIZES BASED ON A MINIMUM WORKING PRESSURE OF 65 PSI AT A FLOW RATE OF 84 GPM AT THE LOCATION WHERE THE MAIN WATER SERVICE ENTERS THE BUILDING. REFER TO CIVIL FOR CONTINUATION.

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/20/2025
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

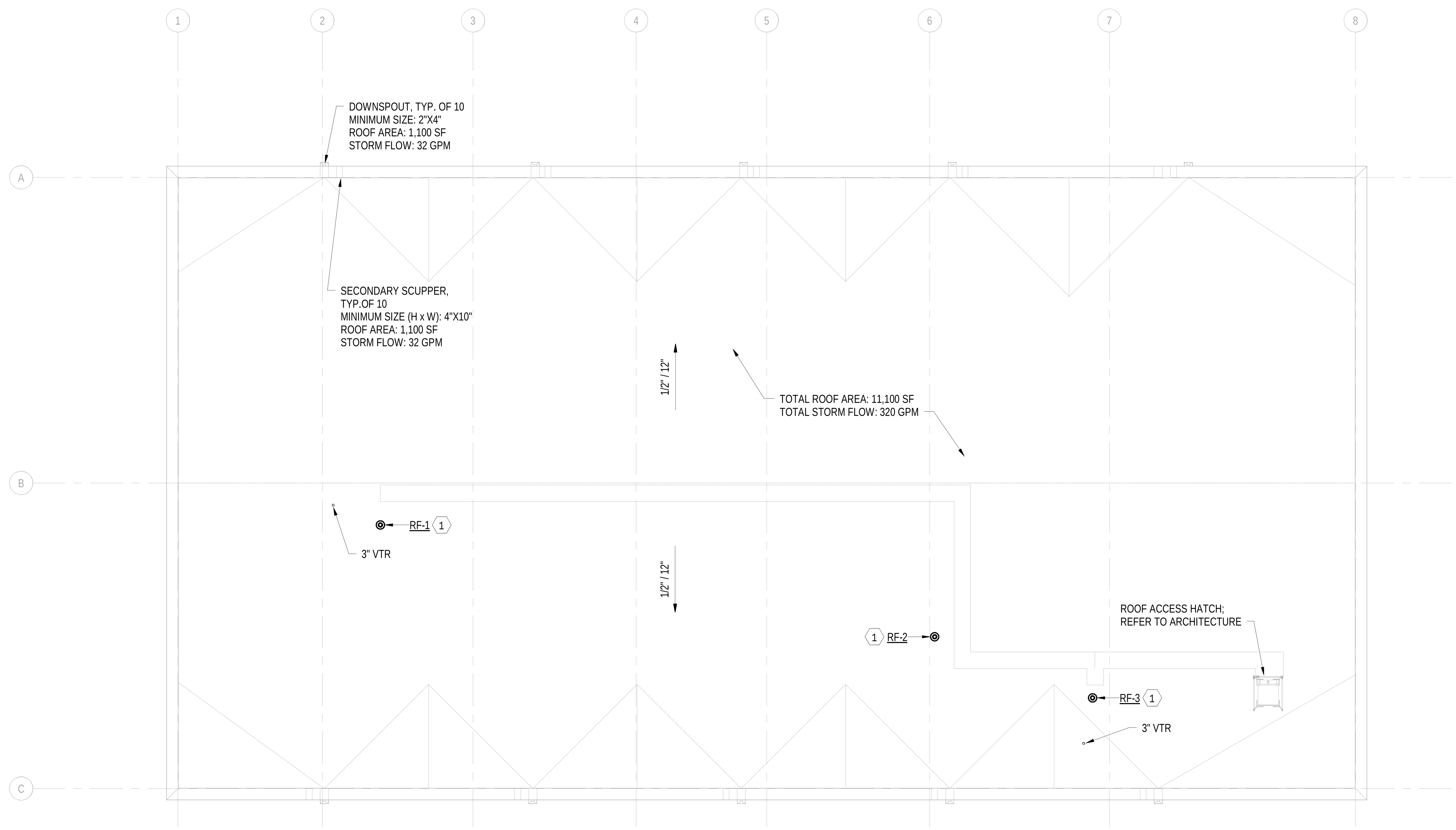
DES	CWH	DRW	BPT	CHK	MJL
PM/DM					SMS/CWJ
BRANCH MANAGER					ALG
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING DOMESTIC WATER FLOOR PLAN

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894008
SHEET 58 OF 140

P-102

6/20/2025 12:13:03 PM P-102
\\dc1vs01\FEDPRJ\FD2E1628\PLUMB\LDG-1667770-P-BIM_v21.rvt



B1 PLUMBING ROOF PLAN

1/8" = 1'-0"

GENERAL SHEET NOTES

A. REFER TO SHEET P-001 FOR LEGEND AND GENERAL NOTES.

B. ROOF DRAINAGE SIZED FOR MAXIMUM RAINFALL RATE OF 2.75"/HOUR AND TABLES 1106.2 AND 1106.3 OF THE 2021 IPC.

C. SCUPPER AND DOWNSPOUT SIZES INDICATED ARE THE MINIMUM REQUIRED SIZES BASED ON TABLES 1106.2 AND 1106.3 OF THE 2021 IPC. REFER TO ARCHITECTURAL DRAWINGS FOR MORE INFORMATION ON THE SCUPPERS AND DOWNSPOUTS.

SHEET KEYNOTES

1 PROVIDE RADON FAN ASSEMBLY AT ROOF. REFER TO RADON FAN ASSEMBLY DETAIL ON SHEET P-502 FOR DETAILED PIPING AND MOUNTING ARRANGEMENT.

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

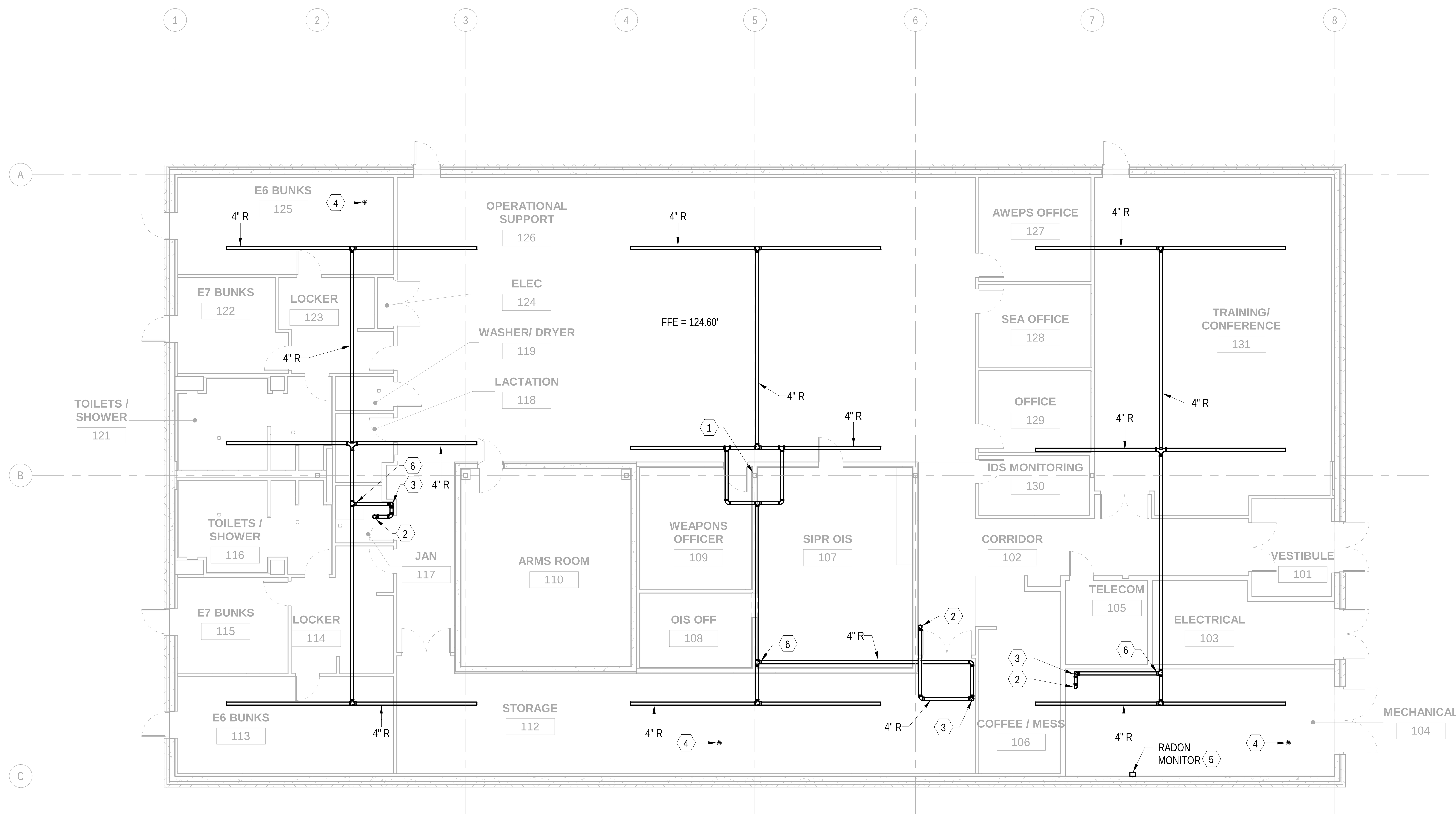
SATISFACTORY TO DATE 05/27/2025

DES	CWH	DRW	BPT	CHK	MJL
PM/DM					SMS/CWJ
BRANCH MANAGER					ALG
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING ROOF PLAN

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894009
SHEET 59 OF 140

P-103



B1 PLUMBING RADON MITIGATION FLOOR PLAN

1/8" = 1'-0"

GENERAL SHEET NOTES

A. REFER TO SHEET P-001 FOR LEGEND AND GENERAL NOTES.
B. BELOW GRADE RADON PIPING MUST BE PERFORATED SCHEDULE 40 PVC. RADON PIPING FROM THE VAPOR EXTRACTION POINT BELOW GRADE TO THE DISCHARGE POINT ON THE ROOF MUST BE SOLID SCHEDULE 40 PVC.
C. RADON PIPING INDICATED ON THIS SHEET IS ROUTED BELOW THE SLAB UNLESS NOTED OTHERWISE.

SHEET KEYNOTES

1. ROUTE RADON PIPING TO AVOID COLUMN FOOTER AT THIS LOCATION; REFER TO STRUCTURAL.
2. 4" RADON UP TO RADON FAN AT ROOF.
3. 4" RADON UP FROM BELOW SLAB. REFER TO RADON RISER FROM SLAB DETAIL ON SHEET P-502.
4. RADON VAPOR MONITORING POINT. REFER TO VAPOR MONITORING POINT DETAIL ON SHEET P-502.
5. PROVIDE INDOOR RADON MONITOR CAPABLE OF AUTOMATICALLY DISPLAYING CURRENT AND LONG TERM RADON MEASUREMENTS. MOUNT 60" ABOVE THE FLOOR.
6. RADON SYSTEM VAPOR EXTRACTION POINT. TRANSITION TO SCHEDULE 40 SOLID WALL PVC FROM THIS POINT TO THE ROOF TERMINATION.

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

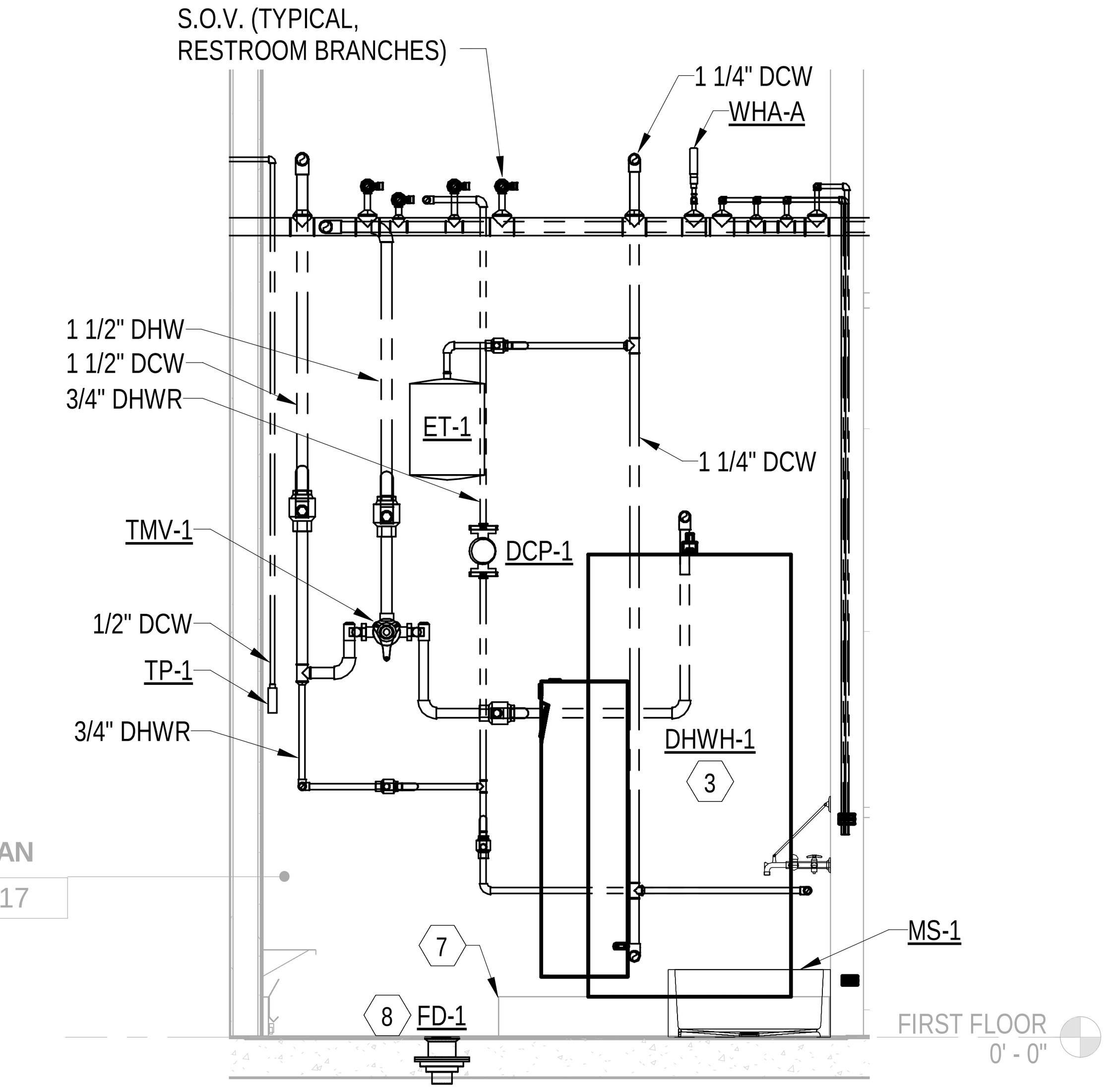
APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH DRW BPT CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING RADON MITIGATION FLOOR PLAN

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894010
SHEET 60 OF 140

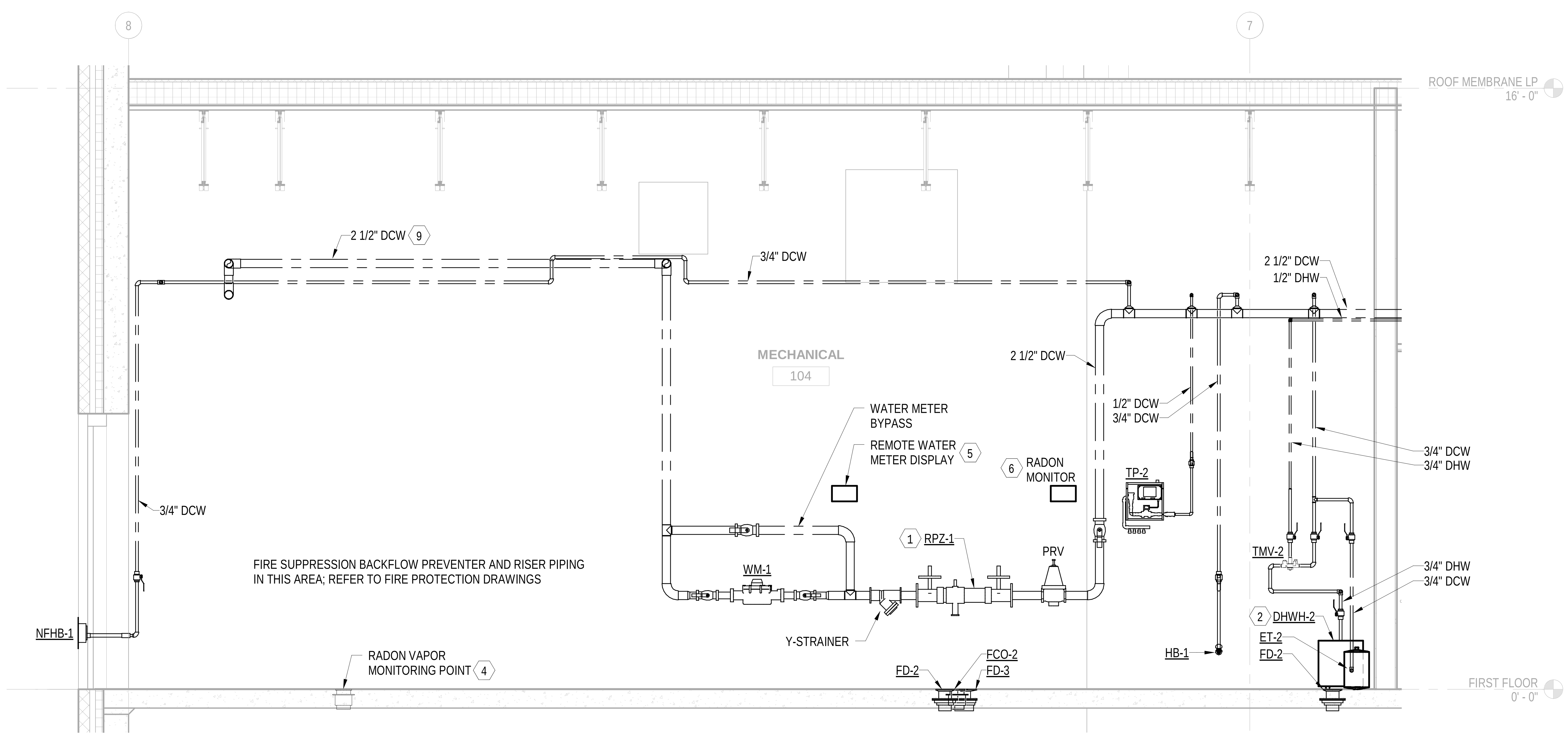
P-104

6/20/2025 12:13:09 PM P-104
\\dc1vs01\FEDPRJ\FD2E1628\PLUMBLDG-1667770-P-BIM_v21.rvt



C3 JANITOR CLOSET SECTION - WEST

1/2" = 1'-0"



A1 MECHANICAL ROOM SECTION - SOUTH

1/2" = 1'-0"

GENERAL SHEET NOTES

A. REFER TO SHEET P-001 FOR LEGEND AND GENERAL NOTES.
B. PROVIDE UL LISTED FIRE-STOPPING SYSTEMS AT PENETRATIONS THROUGH RATED PARTITIONS.

SHEET KEYNOTES

1. REFER TO WATER SERVICE ENTRANCE DETAIL ON SHEET P-501 FOR RPZ PIPING SUPPORT LOCATIONS AND DETAILED PIPING ARRANGEMENT.
2. REFER TO ELECTRIC WATER HEATER DETAIL (MECH ROOM) ON SHEET P-501 FOR DETAILED PIPING ARRANGEMENT.
3. REFER TO ELECTRIC WATER HEATER DETAIL (JANITOR CLOSET) ON SHEET P-501 FOR DETAILED PIPING ARRANGEMENT.
4. RADON VAPOR MONITORING POINT. REFER TO VAPOR MONITORING POINT DETAIL ON SHEET P-502.
5. PROVIDE REMOTE DISPLAY FOR WM-1 MOUNTED 60" ABOVE THE FLOOR.
6. PROVIDE INDOOR RADON MONITOR CAPABLE OF AUTOMATICALLY DISPLAYING CURRENT AND LONG TERM RADON MEASUREMENTS. MOUNT 60" ABOVE THE FLOOR.
7. 6" HOUSEKEEPING PAD, EXTEND 6" FROM EQUIPMENT FOOTPRINT. REFER TO STRUCTURAL DRAWINGS FOR DETAILED DIMENSIONS AND SPECIFICATIONS. COORDINATE EQUIPMENT PAD DIMENSIONS WITH FINAL EQUIPMENT SELECTIONS.
8. COORDINATE FLOOR DRAIN LOCATIONS WITH FOOTPRINTS OF EQUIPMENT HOUSEKEEPING PADS. COORDINATE FLOOR DRAIN LOCATIONS ONCE FINAL EQUIPMENT SUBMITTALS HAVE BEEN APPROVED AND HOUSEKEEPING PAD DIMENSIONS ARE FINAL.
9. ENCASE DCW PIPING WITHIN ALUMINUM JACKETING TO PROTECT THE PIPING FROM UNAUTHORIZED CONNECTIONS PRIOR TO BACKFLOW PREVENTER. PROVIDE LABEL ON PROTECTIVE SLEEVE INDICATING "NO CONNECTIONS ALLOWED" EVERY 5 FEET.

0 1' 2' 4' 6'
1/2" = 1'-0"

PLAN NORTH

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

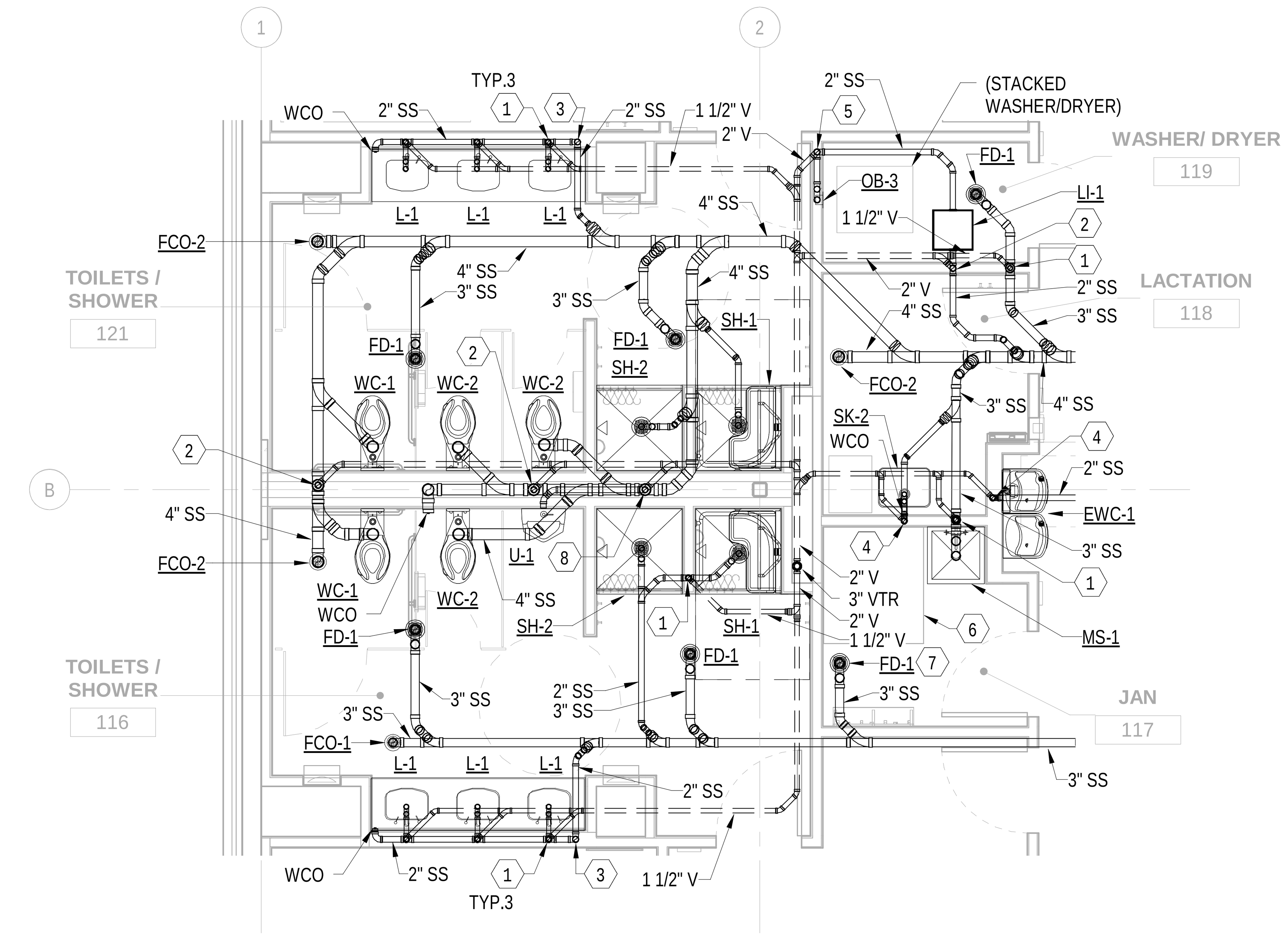
APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	CWH	DRW	BPT	CHK	MJL
PM/DM	SMS/CWJ				
BRANCH MANAGER	ALG				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

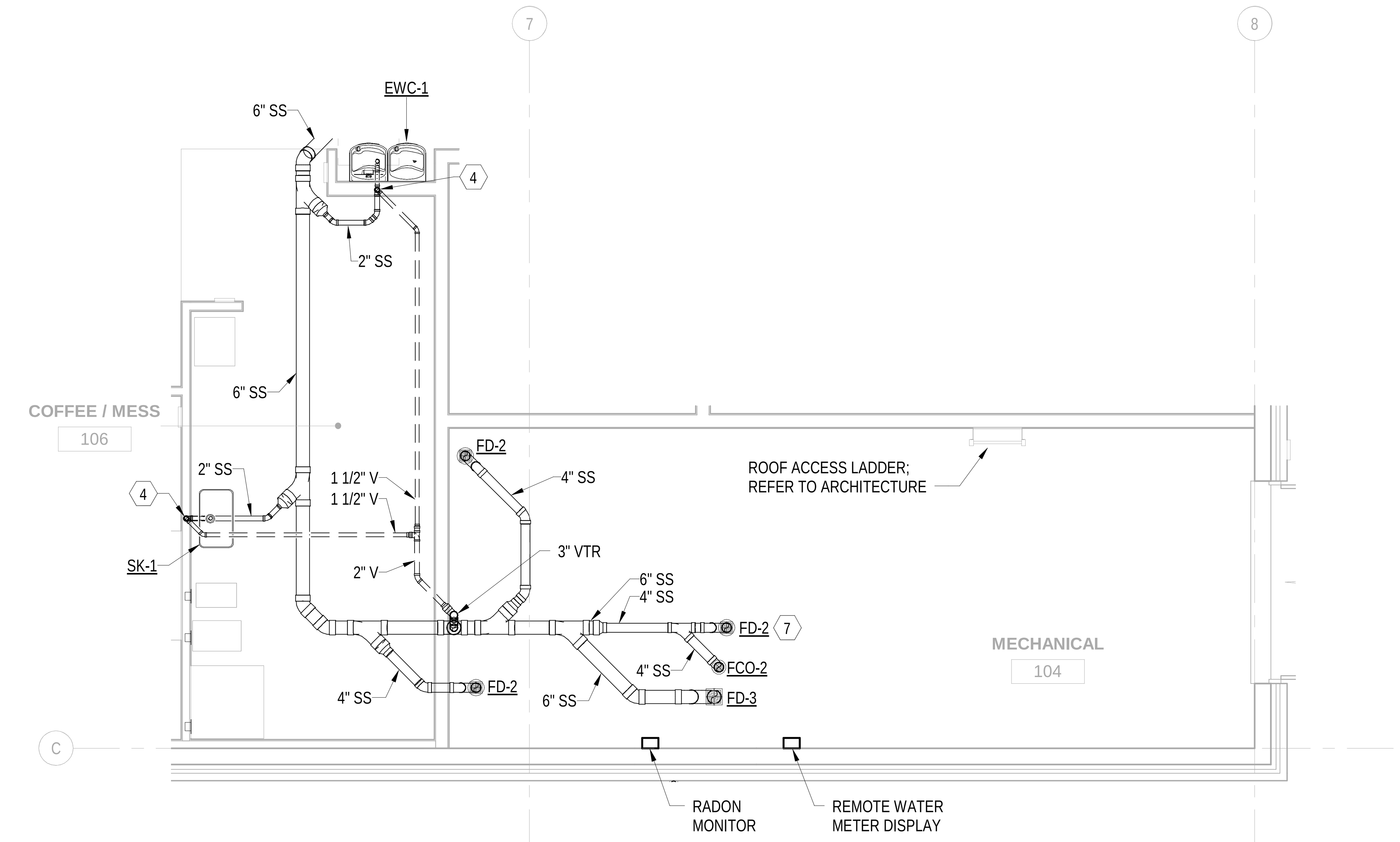
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING SECTIONS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894011
SHEET 61 OF 140

P-301



C2 PLUMBING SANITARY & VENT ENLARGED PLAN - RESTROOMS
1/4" = 1'-0" (P-101)



A2 PLUMBING SANITARY & VENT ENLARGED PLAN - MECHANICAL ROOM
1/4" = 1'-0" (P-101)

GENERAL SHEET NOTES

A. REFER TO SHEET P-001 FOR LEGEND AND GENERAL NOTES.
B. PROVIDE UL LISTED FIRE-STOPPING SYSTEMS AT PENETRATIONS THROUGH RATED PARTITIONS.

SHEET KEYNOTES

1. 1-1/2" V UP.
2. 2" V UP.
3. 2" SS DOWN THROUGH SLAB.
4. 2" SS DOWN THROUGH SLAB, 1-1/2" V UP.
5. 2" SS DOWN THROUGH SLAB, 2" V UP.
6. 6" HOUSEKEEPING PAD, EXTEND 6" FROM EQUIPMENT FOOTPRINT. REFER TO STRUCTURAL DRAWINGS FOR DETAILED DIMENSIONS AND SPECIFICATIONS. COORDINATE EQUIPMENT PAD DIMENSIONS WITH FINAL EQUIPMENT SELECTIONS.
7. COORDINATE FLOOR DRAIN LOCATIONS WITH FOOTPRINTS OF EQUIPMENT HOUSEKEEPING PADS. COORDINATE FLOOR DRAIN LOCATIONS ONCE FINAL EQUIPMENT SUBMITTALS HAVE BEEN APPROVED AND HOUSEKEEPING PAD DIMENSIONS ARE FINAL.
8. 2" SS DOWN THROUGH SLAB, 2" V UP (SERVING U-1).

0 2' 4' 8' 12'
1/4" = 1'-0"

PLAN NORTH

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

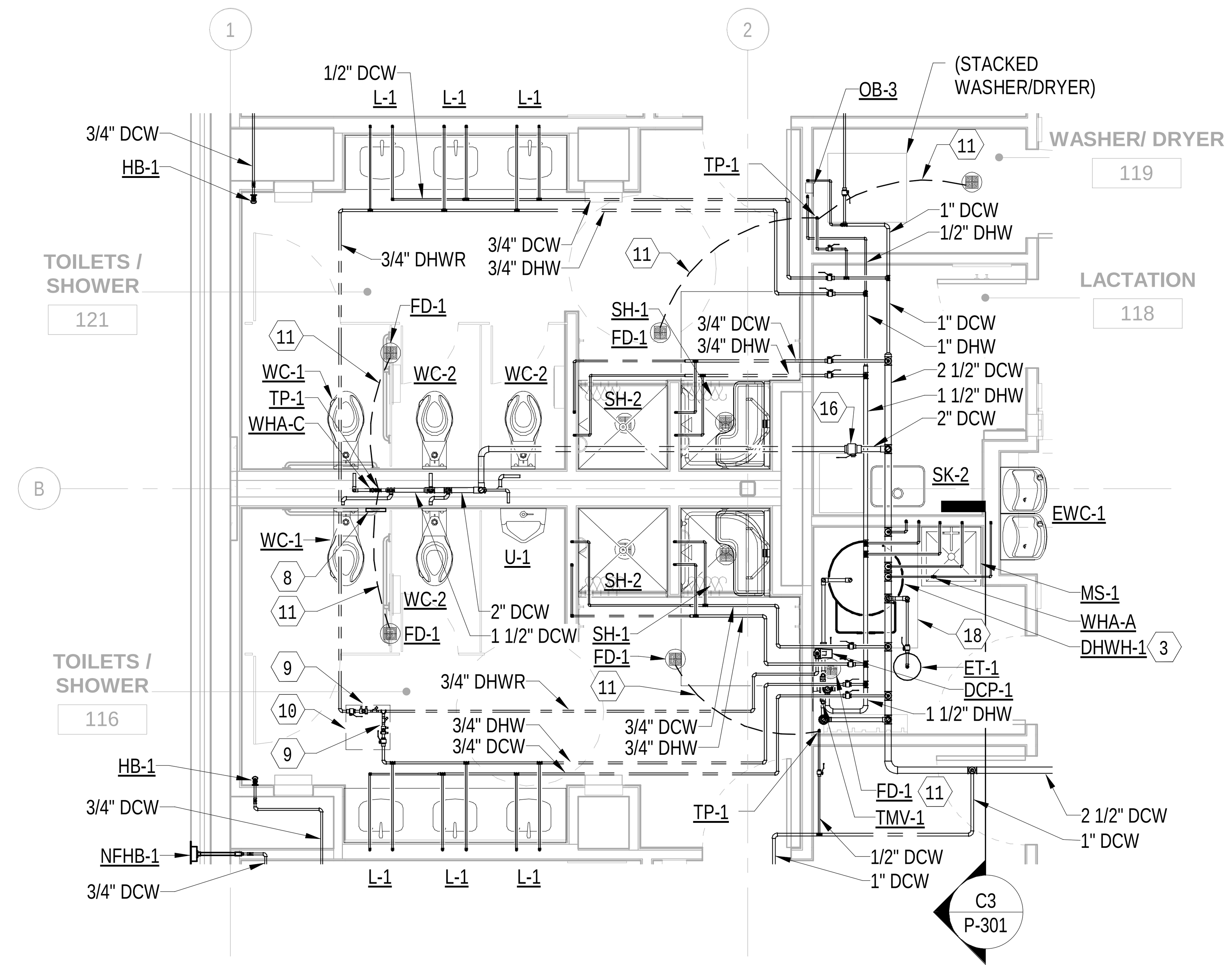
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES CWH	DRW BPT	CHK MJL
PM/DM	SMS/CWJ
BRANCH MANAGER	ALG
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

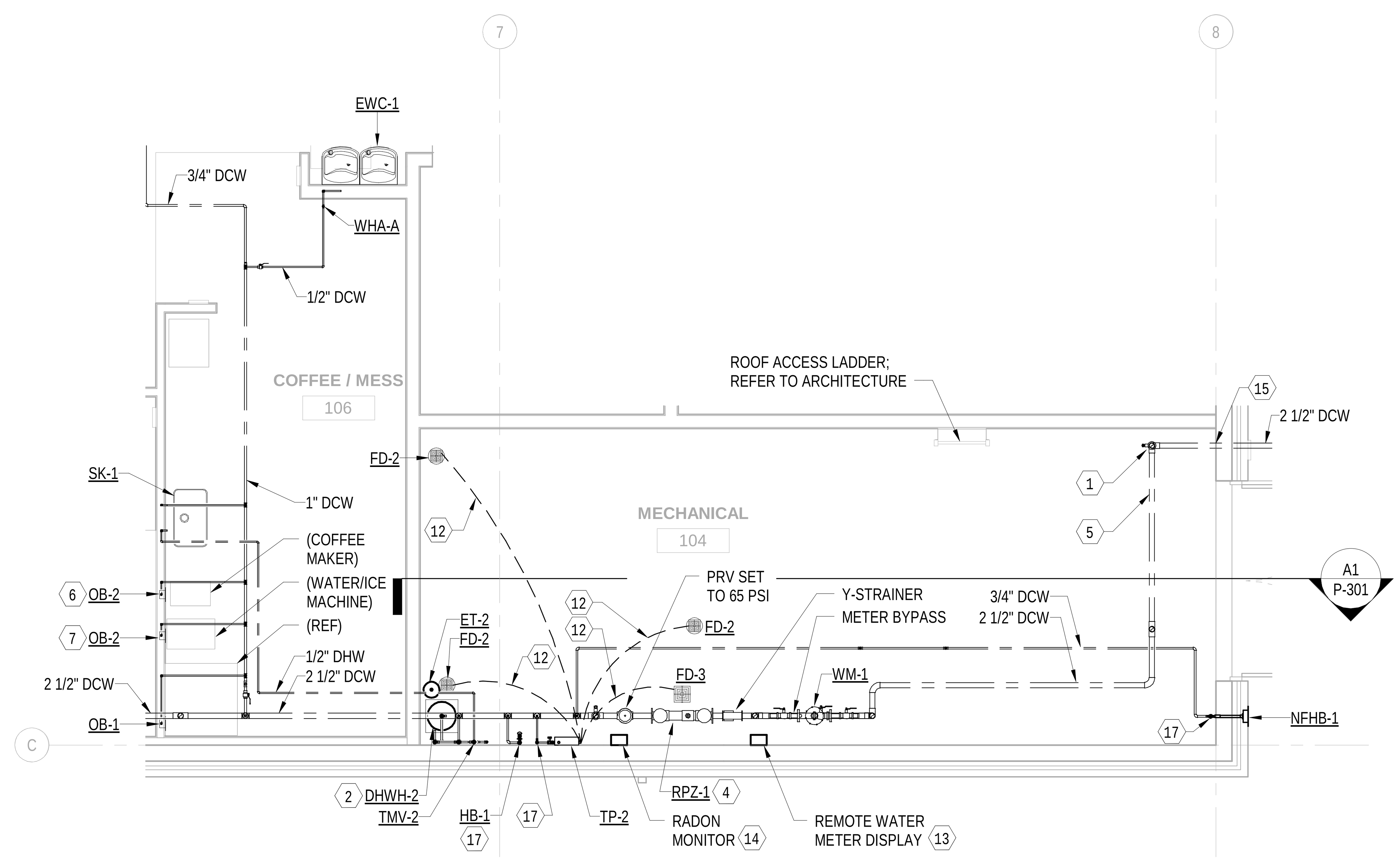
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING ENLARGED PLANS - SANITARY & VENT

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894012
SHEET 62 OF 140
P-401



C2 PLUMBING DOMESTIC WATER ENLARGED PLAN - RESTROOMS

1/4" = 1'-0" (P-102)



A2 PLUMBING DOMESTIC WATER ENLARGED PLAN - MECHANICAL ROOM

1/4" = 1'-0" (P-102)

GENERAL SHEET NOTES

A. REFER TO SHEET P-001 FOR LEGEND AND GENERAL NOTES.
B. PROVIDE UL LISTED FIRE-STOPPING SYSTEMS AT PENETRATIONS THROUGH RATED PARTITIONS.

SHEET KEYNOTES

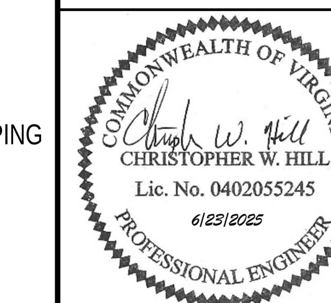
1. 2-1/2" DCW SERVICE UP THROUGH SLAB; PROVIDE LINE SIZE S.O.V. IN VERTICAL.
2. REFER TO ELECTRIC WATER HEATER DETAIL (MECH ROOM) ON SHEET P-501 FOR DETAILED PIPING ARRANGEMENT.
3. REFER TO ELECTRIC WATER HEATER DETAIL (JANITOR CLOSET) ON SHEET P-501 FOR DETAILED PIPING ARRANGEMENT.
4. REFER TO WATER SERVICE ENTRANCE DETAIL ON SHEET P-501 FOR RPZ PIPING SUPPORT LOCATIONS AND DETAILED PIPING ARRANGEMENT.
5. ENCASE DCW PIPING WITHIN ALUMINUM JACKETING TO PROTECT THE PIPING FROM UNAUTHORIZED CONNECTIONS PRIOR TO BACKFLOW PREVENTER. PROVIDE LABEL ON PROTECTIVE SLEEVE INDICATING "NO CONNECTIONS ALLOWED" EVERY 5 FEET.
6. PROVIDE INDIVIDUAL ASSE 1024 DUAL CHECK VALVE ON SUPPLY TO OB-2 FOR COFFEE MAKER.
7. PROVIDE INDIVIDUAL ASSE 1024 DUAL CHECK VALVE ON SUPPLY TO OB-2 FOR WATER/ICE MACHINE.
8. PROVIDE 8" x 8" METAL ACCESS PANEL FOR TRAP PRIMER VALVE & WATER HAMMER ARRESTOR ACCESS.
9. PROVIDE BALANCING VALVE AND CHECK VALVE FOR RECIRCULATION BRANCH. SET BALANCING VALVE TO 0.5 GPM. FINALIZE BALANCING VALVE FLOW DURING TAB.
10. PROVIDE 18" X 18" WHITE POWDERCOATED ACCESS PANEL IN CEILING AT BALANCING VALVES.
11. PROVIDE 1/2" DCW FROM TP-1 TO FLOOR DRAIN.
12. PROVIDE 1/2" DCW FROM TP-2 TO FLOOR DRAIN.
13. PROVIDE REMOTE DISPLAY FOR WM-1 MOUNTED 60" ABOVE THE FLOOR.
14. PROVIDE INDOOR RADON MONITOR CAPABLE OF AUTOMATICALLY DISPLAYING CURRENT AND LONG TERM RADON MEASUREMENTS. MOUNT 60" ABOVE THE FLOOR.
15. REFER TO STRUCTURAL FOR PIPING DETAIL BELOW FOUNDATION.
16. PROVIDE LINE SIZE ISOLATION VALVES ON BRANCH LINES (TYPICAL).
17. PROVIDE LINE SIZE S.O.V. IN VERTICAL.
18. 6" HOUSEKEEPING PAD, EXTEND 6" FROM EQUIPMENT FOOTPRINT. REFER TO STRUCTURAL DRAWINGS FOR DETAILED DIMENSIONS AND SPECIFICATIONS. COORDINATE EQUIPMENT PAD DIMENSIONS WITH FINAL EQUIPMENT SELECTIONS.

0 2' 4' 8' 12'
1/4" = 1'-0"

PLAN NORTH

SYM	DESCRIPTION	DATE	APPR

NAVFAC



SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

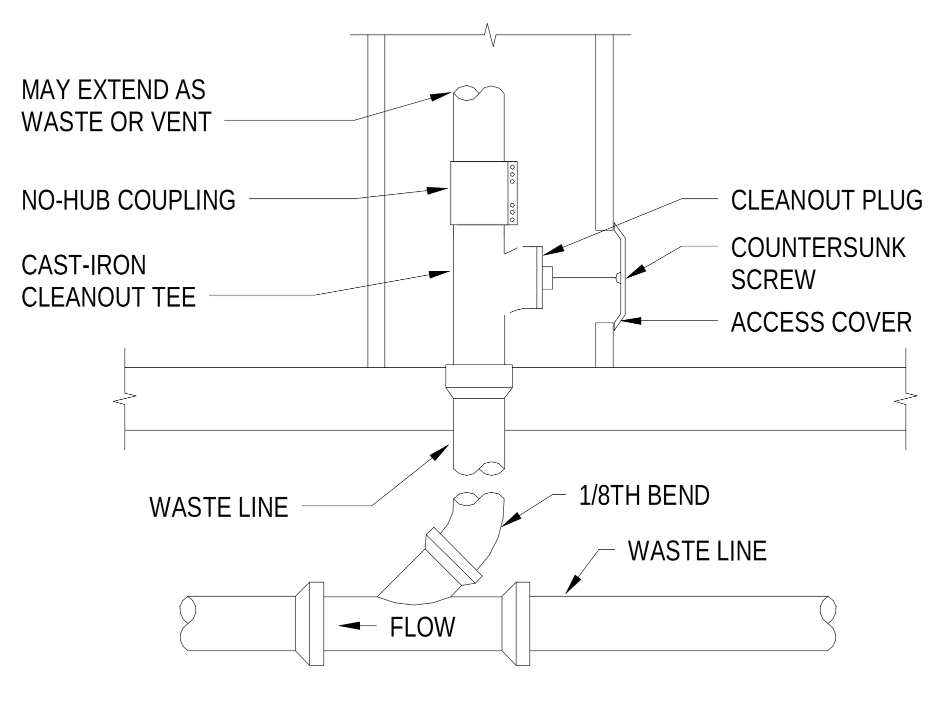
APPROVED	
FOR COMMANDER NAVFAC	8/14/2025
ACTIVITY	APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE	05/27/2025
DES CWH / DRW BPT / CHK MJL	
PM/DM	SMS/CWJ
BRANCH MANAGER	ALG
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING ENLARGED PLANS - DOMESTIC WATER

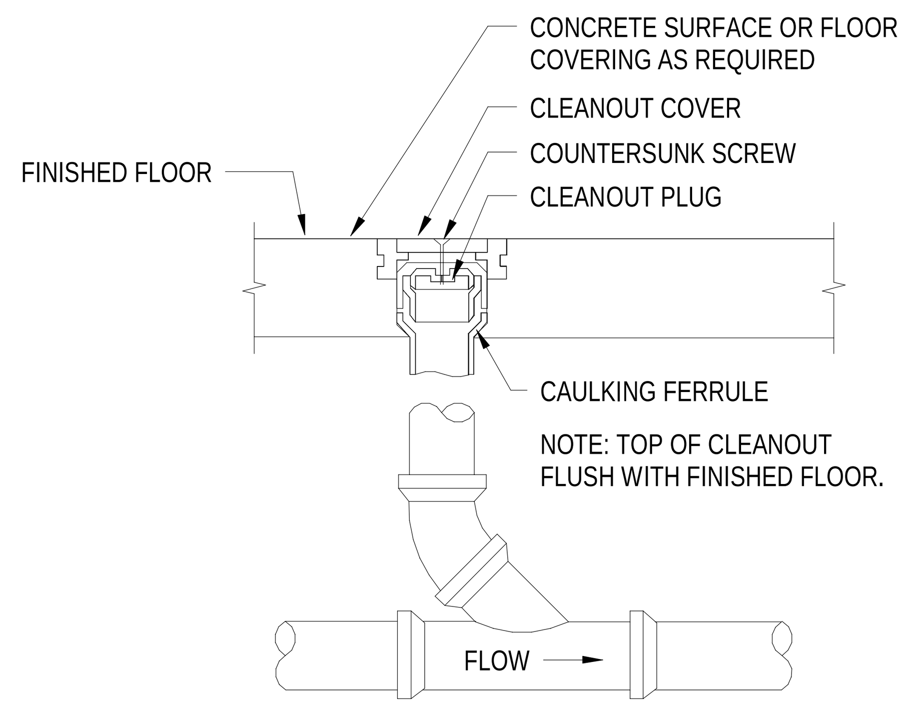
SCALE:	AS NOTED
EPROJECT NO.:	1667770
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12894013
SHEET	63 OF 140

P-402

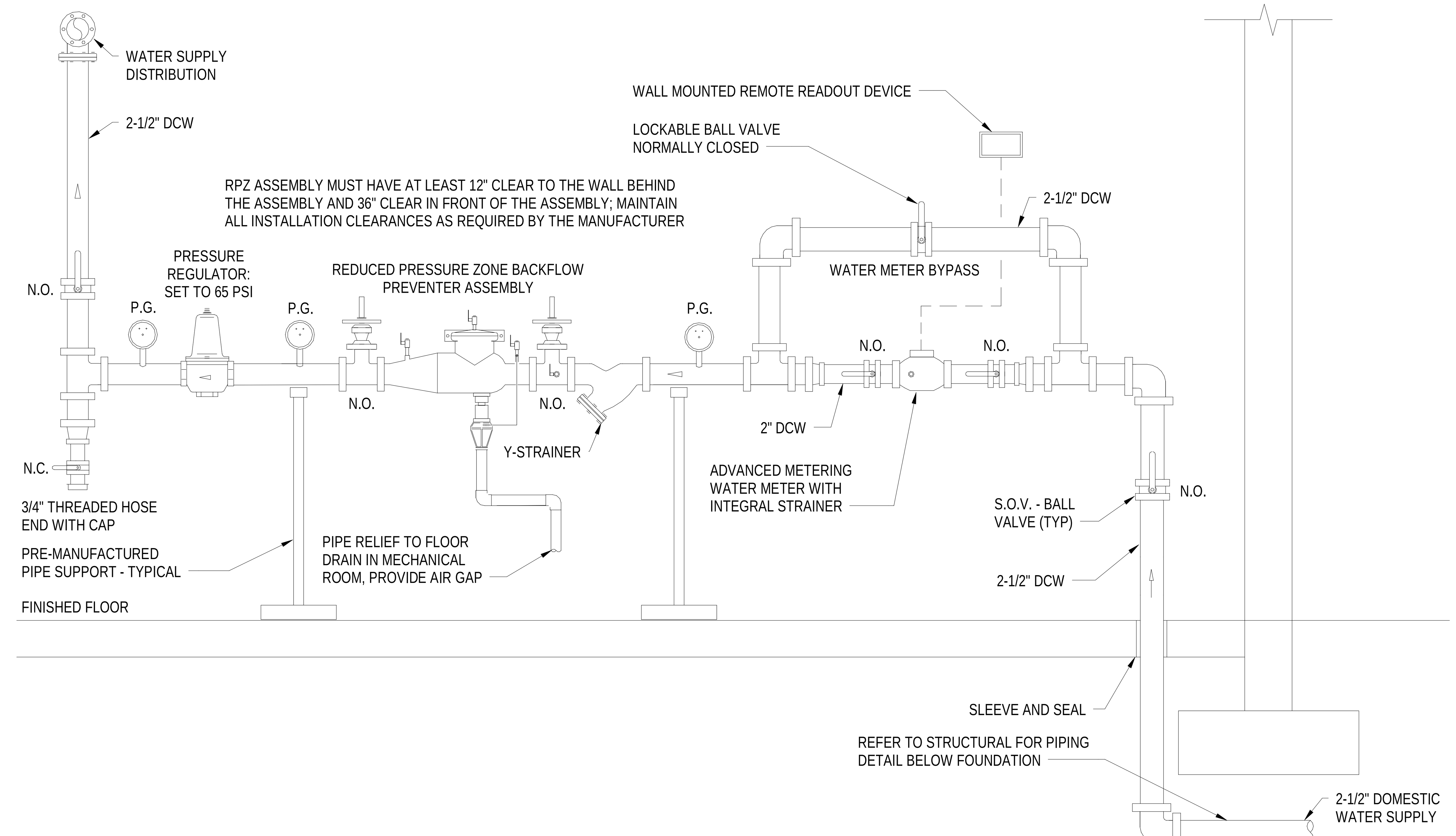
6/20/2025 12:13:24 PM P-402
\\dc1vs01\FEDPRJ\FDZE1628\PLUMBLDG-1667770-P-BIM_v21.rvt



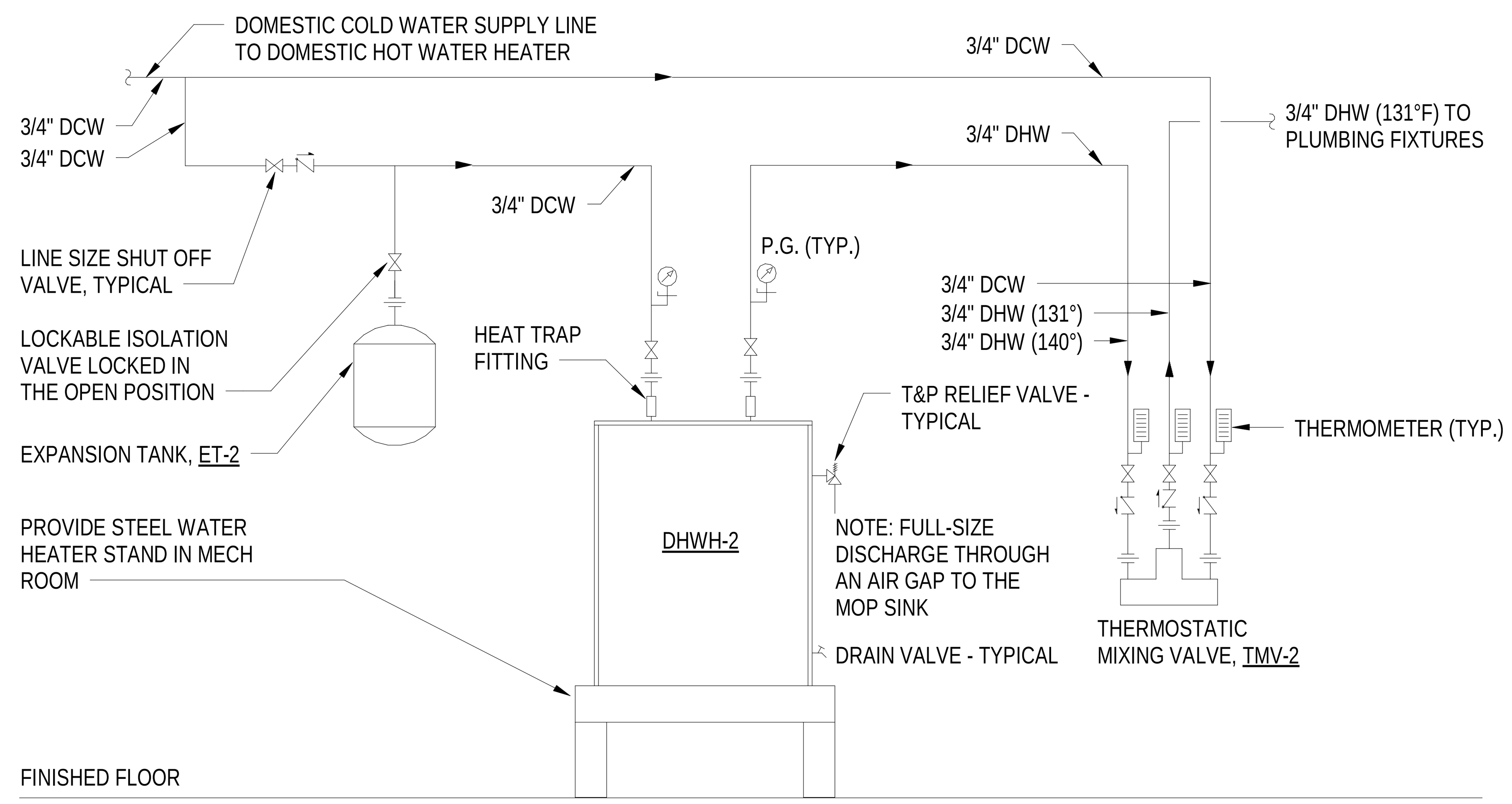
D1 WALL CLEANOUT DETAIL
NTS



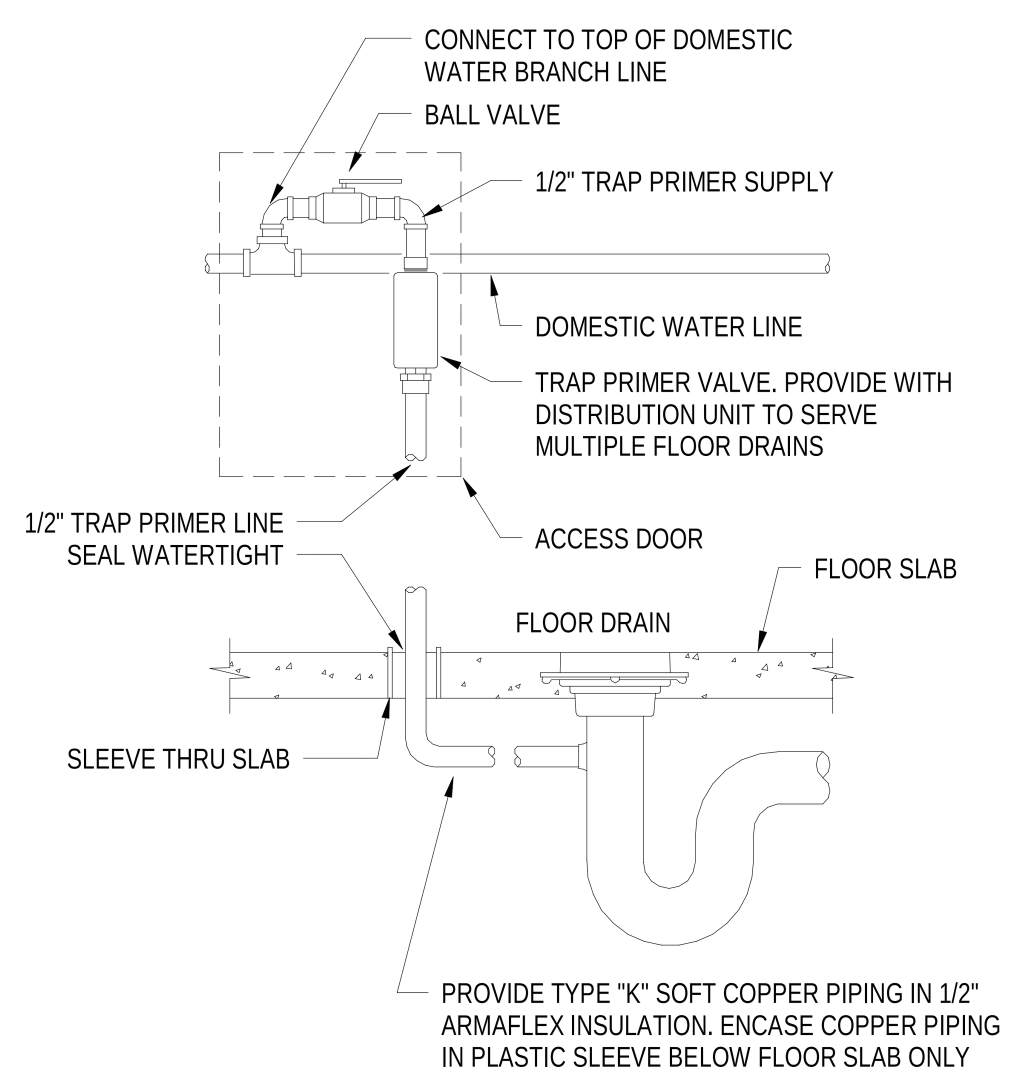
D2 FLOOR CLEANOUT DETAIL
NTS



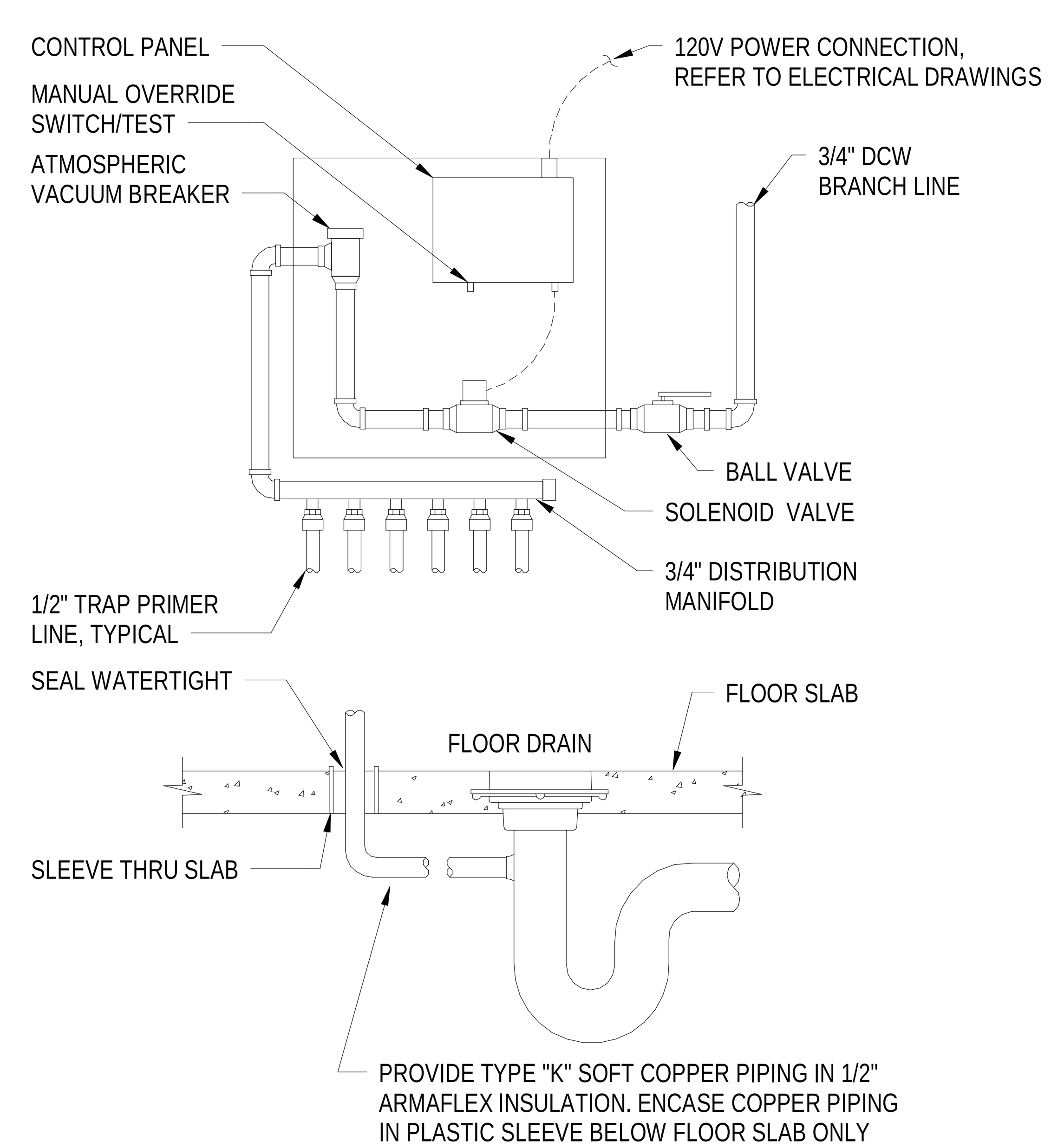
C3 WATER SERVICE ENTRANCE DETAIL
NTS



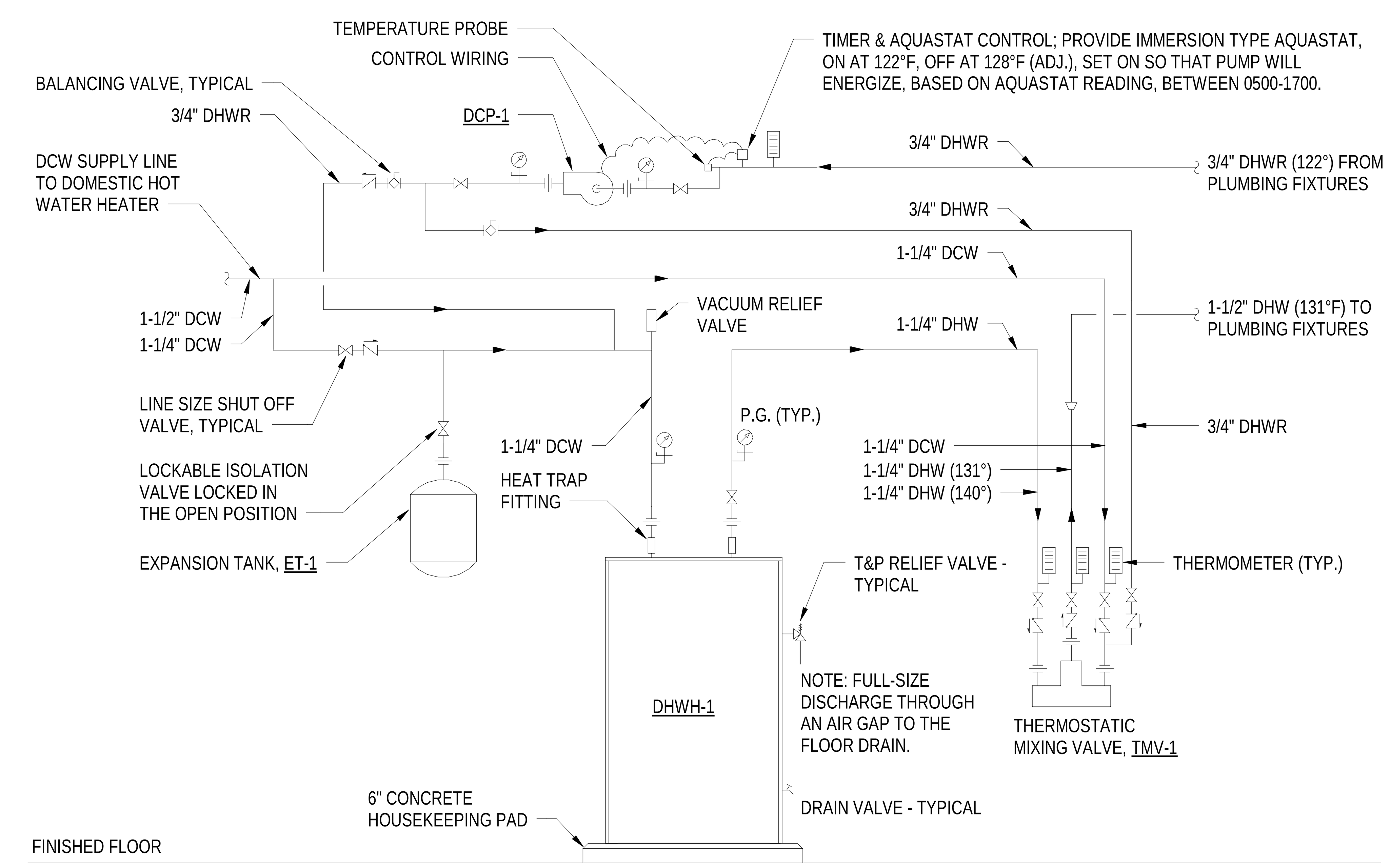
B1 ELECTRIC WATER HEATER DETAIL (MECHANICAL ROOM)
NTS



A1 TRAP PRIMER DETAIL
NTS



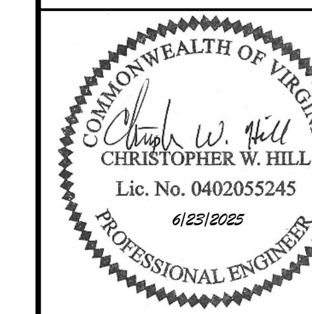
A2 ELECTRONIC TRAP PRIMER DETAIL
NTS



A4 ELECTRIC WATER HEATER DETAIL (JANITOR CLOSET)
NTS

SYM	DESCRIPTION	DATE	APPR

NAVFAC



SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

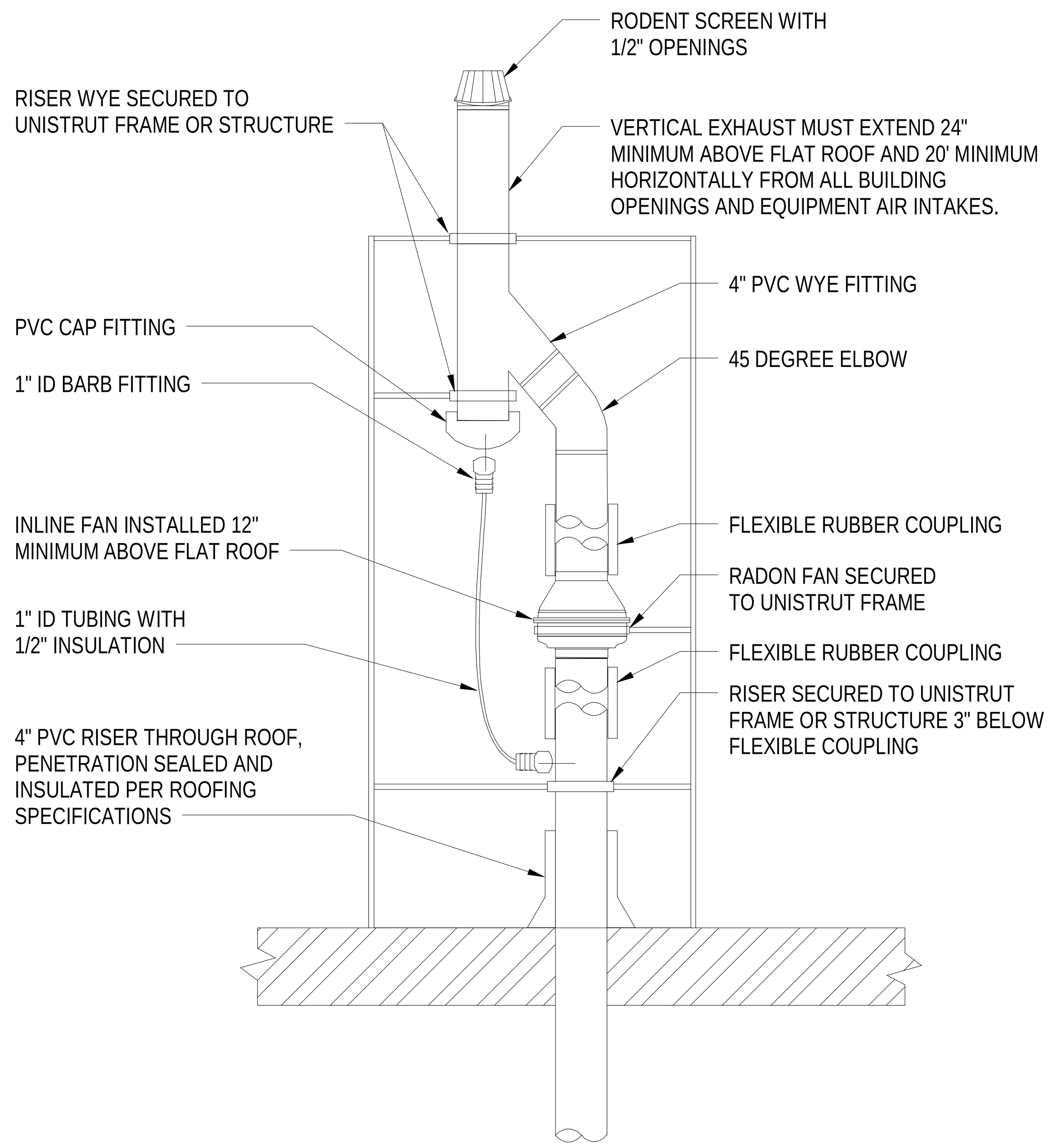
APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	CWH	DRW	BPT	CHK	MJL
PM/DM					SMS/CWJ
BRANCH MANAGER					ALG
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

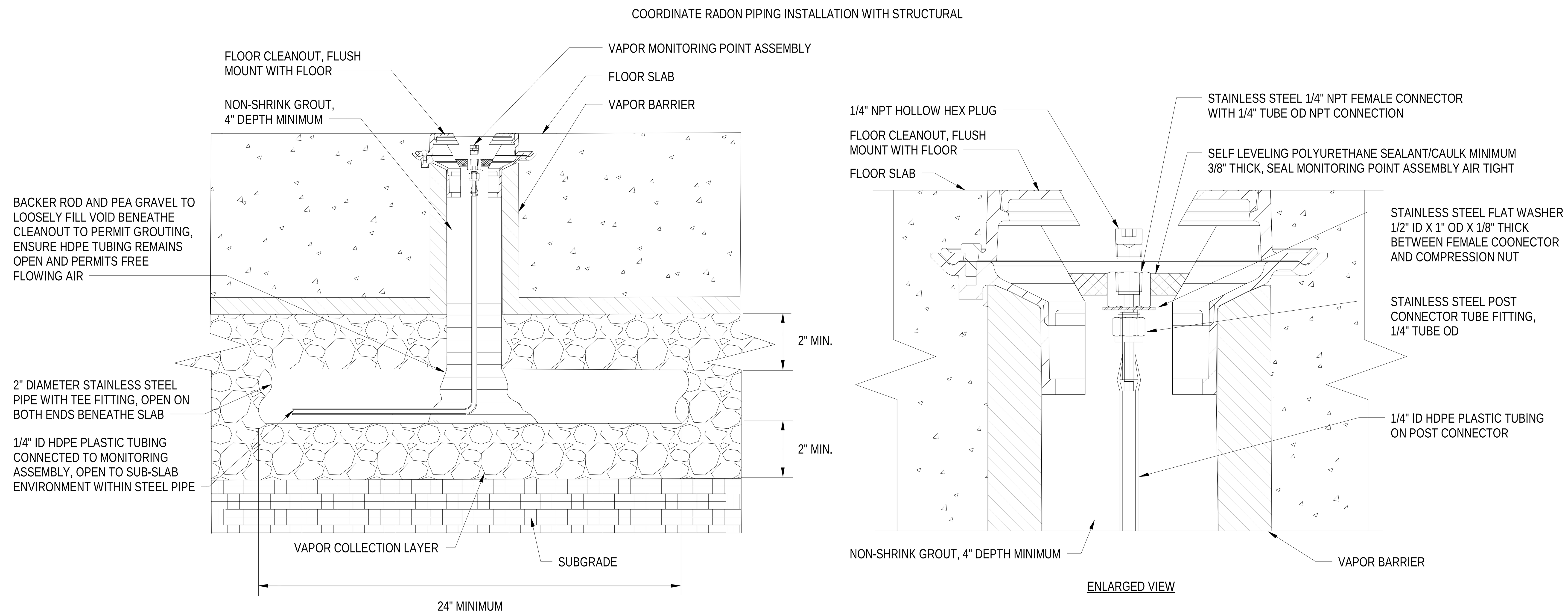
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894014
SHEET 64 OF 140
P-501

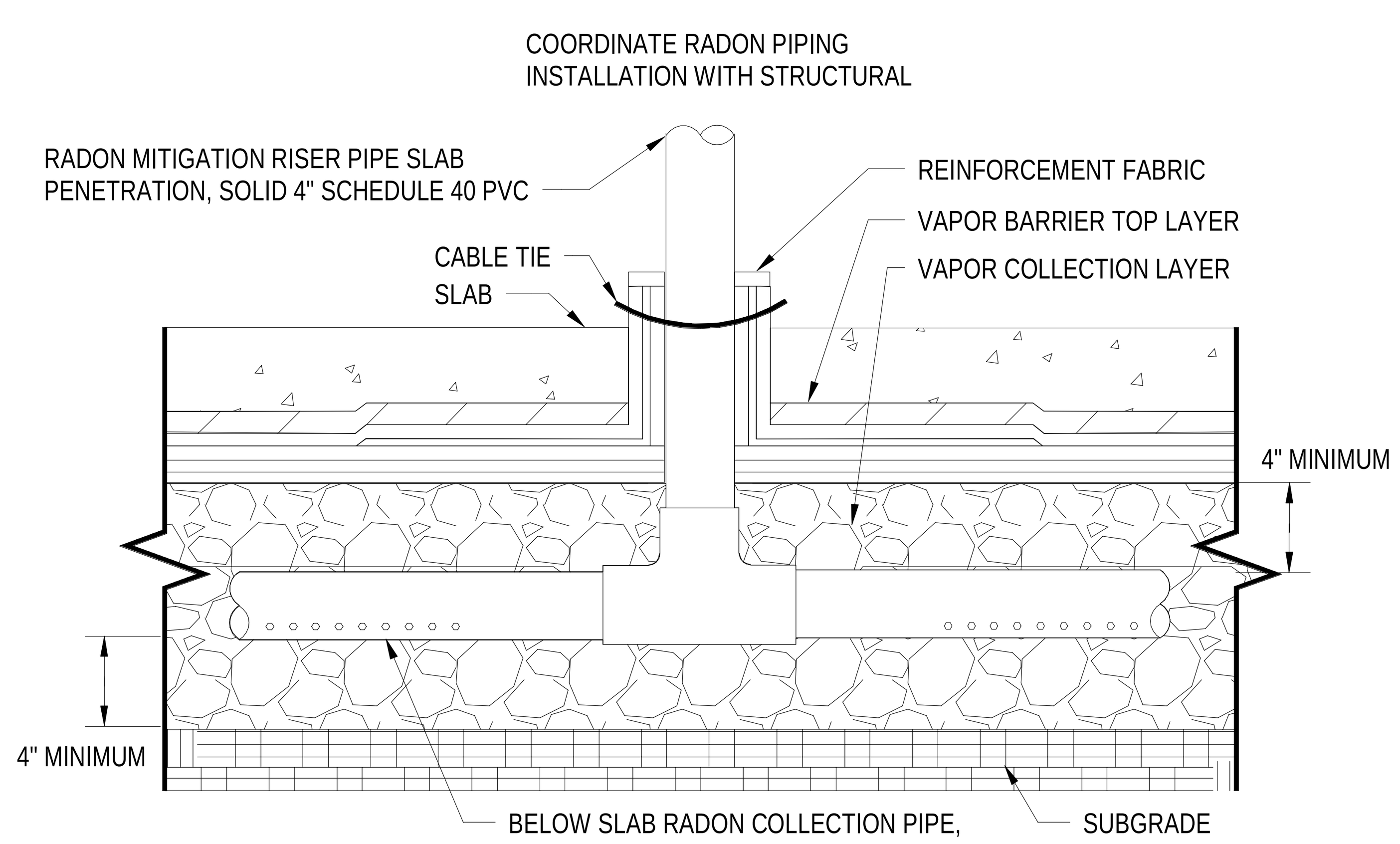
6/20/2025 12:13:27 PM P-501
\\dc1vs01\FEDPRJ\FDZE1628\PLUMB\LDG-1667770-P-BIM_v21.rvt



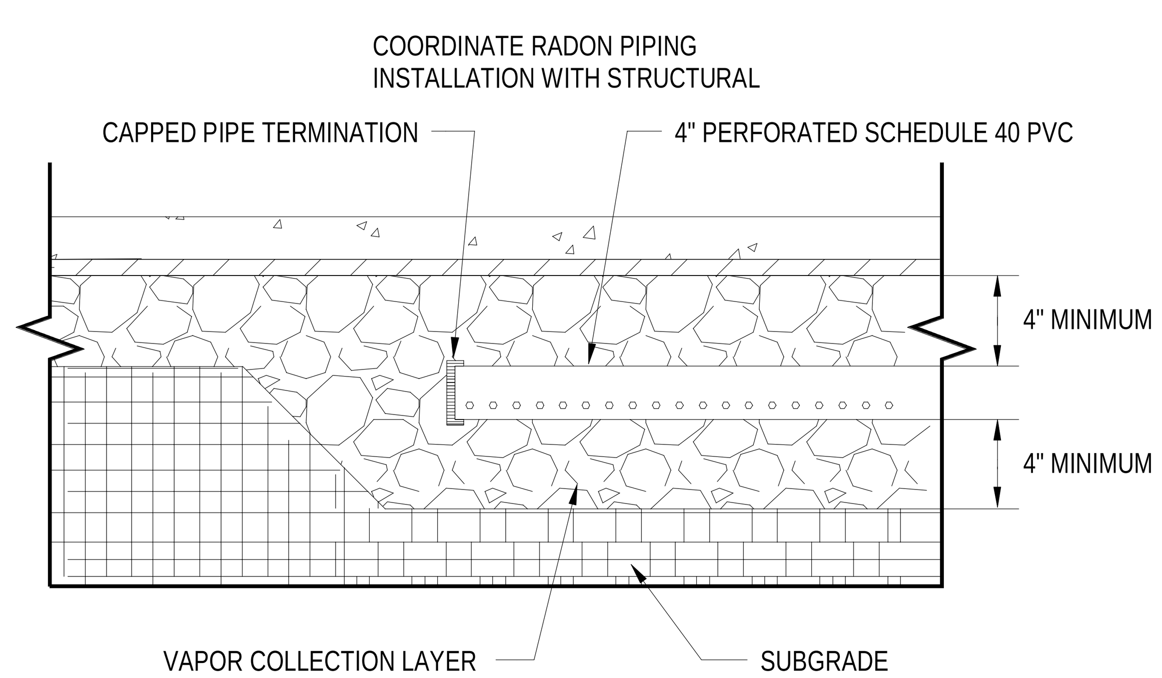
B1 RADON FAN ASSEMBLY DETAIL
NTS



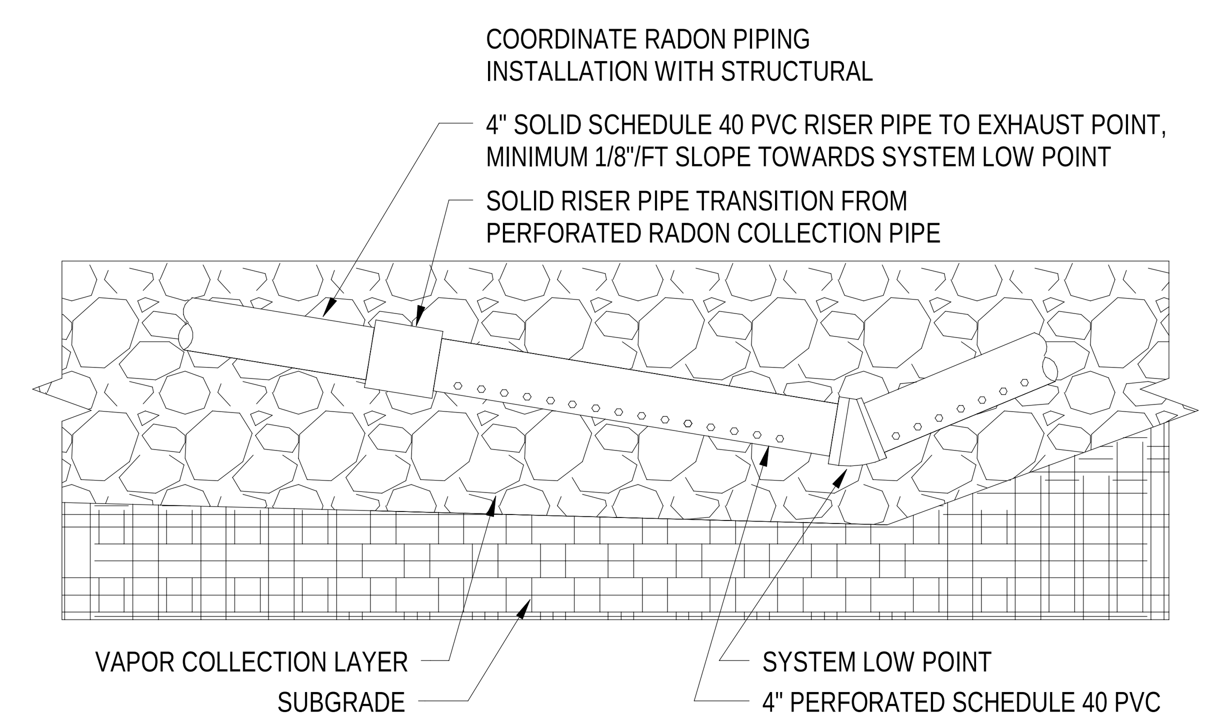
B3 VAPOR MONITORING POINT DETAIL
NTS



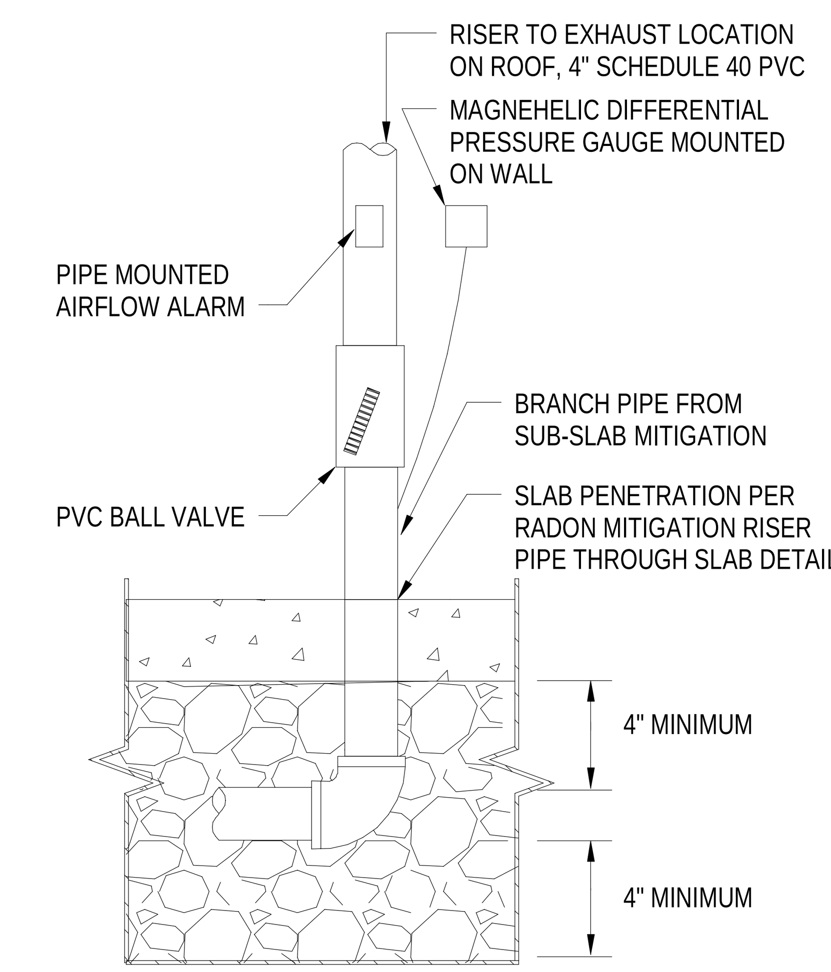
A1 VAPOR BARRIER RISER PENETRATION DETAIL
NTS



A2 AGGREGATE LAYER AND COLLECTION PIPE DETAIL
NTS



A4 VAPOR EXTRACTION POINT DETAIL
NTS



A5 RADON RISER FROM SLAB DETAIL
NTS

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	CWH	DRW	BPT	CHK	MJL
PM/DM					SMS/CWJ
BRANCH MANAGER					ALG
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894015
SHEET 65 OF 140

P-502

6/20/2025 12:13:30 PM
\\dc1vs01\FEDPRJ\FEDE1628\PLUMBLDG-1667770-P-BIM_v21.rvt
P-502

PLUMBING FIXTURE SCHEDULE

TAG	ITEM	CONNECTION SIZE (IN.) WASTE	VENT	DCW	DHW	WSFU	DFU	REMARKS
EWC-1	ACCESSIBLE WATER COOLER W/ BOTTLE FILLER	1-1/2"	1-1/4"	1/2"	-	0.25	0.5	REFRIGERATED ABA COMPLIANT HIGH/LOW WATER COOLER WITH BOTTLE FILLER AND 8 GPH CHILLER, 115V/60HZ POWER
FCO-1	FLOOR CLEANOUT	3"	-	-	-	-	-	EPOXY COATED CAST IRON FLOOR CLEANOUT, NICKEL BRONZE TOP, BRASS PLUG. PROVIDE SQUARE COVER IN AREAS WITH TILE FLOOR AND ROUND COVER FOR OTHER FLOOR TYPES
FCO-2	FLOOR CLEANOUT	4"	-	-	-	-	-	EPOXY COATED CAST IRON FLOOR CLEANOUT, NICKEL BRONZE TOP, BRASS PLUG. PROVIDE SQUARE COVER IN AREAS WITH TILE FLOOR AND ROUND COVER FOR OTHER FLOOR TYPES
FD-1	FLOOR DRAIN	3"	-	-	-	-	2	EPOXY COATED CAST IRON FLOOR DRAIN WITH VANDAL PROOF AND HEEL PROOF NICKEL BRONZE STRAINER. PROVIDE SQUARE STRAINER IN AREAS WITH TILE FLOOR AND ROUND STRAINER FOR OTHER FLOOR TYPES. PROVIDE WITH TRAP PRIMER CONNECTION
FD-2	FLOOR DRAIN	4"	-	-	-	-	2	HEAVY DUTY EPOXY COATED CAST IRON FLOOR DRAIN WITH VANDAL PROOF NICKEL BRONZE STRAINER. PROVIDE SQUARE STRAINER IN AREAS WITH TILE FLOOR AND ROUND STRAINER FOR OTHER FLOOR TYPES. PROVIDE WITH TRAP PRIMER CONNECTION
FD-3	FLOOR DRAIN	6"	-	-	-	-	2	HEAVY DUTY EPOXY COATED CAST IRON FLOOR DRAIN WITH VANDAL PROOF NICKEL BRONZE STRAINER. PROVIDE SQUARE STRAINER IN AREAS WITH TILE FLOOR AND ROUND STRAINER FOR OTHER FLOOR TYPES. PROVIDE WITH TRAP PRIMER CONNECTION
HB-1	HOSE BIBB	-	-	3/4"	-	-	-	MODERATE CLIMATE, INTEGRAL BACKFLOW PREVENTER, EXPOSED WITH LOCKING STEM AND KEY
L-1	LAVATORY, SOLID SURFACE MULTI-LEVEL	1-1/2"	1-1/4"	1/2"	1/2"	2	1	SOLID SURFACE MULTI-LEVEL THEE STATION LAVATORY WITH INTEGRAL BOWLS AND THREE BATTERY-POWERED SENSOR OPERATED FAUCETS WITH LAMINAR 0.5 GPM FLOW RATE AND FLOW CONTROL. PROVIDE WITH ASSE 1070 THERMOSTATIC MIXING VALVE W/ 1/2" MALE NPT CONNECTION. ABA COMPLIANT
LI-1	LINT INTERCEPTOR	2"	-	-	-	-	-	STEEL LINT INTERCEPTOR WITH NON-SKID SECURED COVER WITH REMOVABLE LIFT HANDLE, ALUMINUM LINT INTERCEPTING SECONDARY SCREEN ASSEMBLY, AND PERMANENT STRAINING BAFFLE, MAXIMUM 30 GPM FLOW RATE
MS-1	MOP SINK	3"	1-1/2"	1/2"	1/2"	3	2	PRECAST TERRAZZO, 12" HIGH 2" WIDE WALLS WITH 6" DROP FRONT. STAINLESS STEEL DRAIN BODY, COMBINATION STAINLESS STEEL DOME STRAINER WITH LINT BASKET. PROVIDE DEEP SEAL P-TRAP, DUAL HANDLE 2.2 GPM WALL-MOUNT FAUCET WITH HOSE & INTEGRAL INLET CHECK VALVES. PROVIDE INDIVIDUAL CHECK VALVES ON THE COLD AND HOT WATER SUPPLY PIPING TO THE FAUCET
NFHB-1	NON-FREEZE HOSE BIBB	-	-	3/4"	-	-	-	FREEZE-PROOF WALL HYDRANT, INTEGRAL VACUUM BREAKER, RECESSED BOX, LOCKING COVER AND KEY
OB-1	ICEMAKER OUTLET BOX	-	-	1/2"	-	-	-	RECESSED WHITE POWDER COATED ICE MAKER OUTLET BOX WITH BRASS-PLATED, QUARTER TURN WATER HAMMER ARRESTOR VALVE MEETING LEAD-FREE REQUIREMENTS; PROVIDE 1/2" FLEXIBLE BRAIDED STAINLESS STEEL LINE ON FINAL CONNECTION TO APPLIANCE
OB-2	COFFEE MAKER / WATER DISPENSER OUTLET BOX	-	-	1/2"	-	-	-	RECESSED WHITE POWDER COATED WATER SUPPLY OUTLET BOX WITH BRASS-PLATED, QUARTER TURN WATER HAMMER ARRESTOR VALVE MEETING LEAD-FREE REQUIREMENTS; PROVIDE 3/8" FLEXIBLE BRAIDED STAINLESS STEEL LINE ON FINAL CONNECTION TO APPLIANCE
OB-3	WASHING MACHINE OUTLET BOX	2"	1-1/2"	1/2"	1/2"	3	3	RECESSED BOX WITH 2" DRAIN CONNECTIONS, SINGLE LEVER SHUTOFF WITH 1/2" HOT AND COLD WATER CONNECTIONS AND WITH SHOCK ARRESTORS ON BOTH HOT AND COLD WATER SUPPLIES
SH-1	ACCESSIBLE SHOWER	2"	1-1/2"	1/2"	1/2"	4	2	ACCESSIBLE, 2.0 GPM HAND-HELD SHOWER HEAD, CHROME FINISH, SINGLE HANDLE, PRESSURE BALANCING, TEMPERATURE LIMITING STOPS, VANDAL RESISTANT SCREWS, 5" ROUND CENTER DRAIN, SLIP RESISTANT. PROVIDE WITH SHOWER ROD, CURTAIN, AND HOOKS
SH-2	SHOWER	2"	1-1/2"	1/2"	1/2"	4	2	2.0 GPM WALL MOUNTED SHOWER HEAD, CHROME FINISH, SINGLE HANDLE, PRESSURE BALANCING, TEMPERATURE LIMITING STOPS, VANDAL RESISTANT SCREWS. CENTER DRAIN, NON-SLIP SOLID SURFACE SHOWER PAN WITH WATERPROOFING MEMBRANE AT SHOWER FLOOR
SK-1	BREAK ROOM SINK	1-1/2"	1-1/2"	1/2"	1/2"	1.4	2	COUNTED-MOUNTED STAINLESS STEEL SINGLE BOWL SINK WITH GOOSENECK FAUCET, 1.5 GPM AERATOR, ABA COMPLIANT; PROVIDE WITH ASSE 1070 THERMOSTATIC MIXING VALVE W/ 1/2" MALE NPT CONNECTION
SK-2	LACTATION ROOM SINK	1-1/2"	1-1/2"	1/2"	1/2"	1.4	2	COUNTED-MOUNTED STAINLESS STEEL SINGLE BOWL SINK WITH SINGLE LEVER HANDLE FAUCET, 1.5 GPM AERATOR, ABA COMPLIANT. PROVIDE WITH ASSE 1070 THERMOSTATIC MIXING VALVE W/ 1/2" MALE NPT CONNECTION
TMV-1	THERMOSTATIC MIXING VALVE, MASTER	-	-	1"	1"	-	-	THERMOSTATIC WATER MIXING VALVE, ADJUSTABLE HIGH TEMPERATURE LIMIT STOP, INLET CHECKSTOPS, OUTLET BALL VALVE, 0.5 GPM MINIMUM FLOW CAPACITY
TMV-2	THERMOSTATIC MIXING VALVE, MASTER	-	-	1/2"	1/2"	-	-	THERMOSTATIC WATER MIXING VALVE, ADJUSTABLE HIGH TEMPERATURE LIMIT STOP, INLET CHECKSTOPS, OUTLET BALL VALVE, 0.5 GPM MINIMUM FLOW CAPACITY
TP-1	TRAP PRIMER	-	-	1/2"	-	-	-	PRESSURE DROP ACTIVATED TRAP PRIMER, INTEGRAL VACUUM BREAKER, CAPABLE OF SERVING 4 FLOOR DRAINS WITH DISTRIBUTION UNIT
TP-2	ELECTRONIC TRAP PRIMER MANIFOLD	-	-	1/2"	-	-	-	ELECTRONIC TRAP PRIMER MANIFOLD WITH BOX, INTEGRAL VACUUM BREAKER, 120V POWER, CAPABLE OF SERVING 1-4 FLOOR DRAINS, 115V/60HZ POWER
U-1	URINAL	2"	1-1/2"	3/4"	-	5	2	VITREOUS CHINA, 0.125 GPF BATTERY-POWERED SENSOR OPERATED FLUSH VALVE WITH MANUAL PUSH-BUTTON OVERRIDE
WC-1	ACCESSIBLE WATER CLOSET, FLOOR-MOUNT	4"	2"	1"	-	10	4	ACCESSIBLE, VITREOUS CHINA, FLOOR-MOUNT, ELONGATED BOWL, OPEN FRONT SEAT, LESS COVER, 1.28 GPF BATTERY-POWERED SENSOR OPERATED FLUSH VALVE WITH MANUAL PUSH-BUTTON OVERRIDE
WC-2	WATER CLOSET, FLOOR-MOUNT	4"	2"	1"	-	10	4	VITREOUS CHINA, FLOOR-MOUNT, ELONGATED BOWL, OPEN FRONT SEAT, LESS COVER, 1.28 GPF BATTERY-POWERED SENSOR OPERATED FLUSH VALVE WITH MANUAL PUSH-BUTTON OVERRIDE

DOMESTIC ELECTRIC WATER HEATER SCHEDULE

TAG	LOCATION	SERVICE	TYPE	RECOVERY (GPH)	TEMP RISE (°F)	TEMP SETPOINT (°F)	CAPACITY (GAL)	ELECTRICAL DATA TOTAL KW	NUMBER OF ELEMENTS	VOLTAGE	FLA	PHASE	NOTES
DHWH-1	JANITOR CLOSET	DOMESTIC HOT WATER	ELECTRIC STORAGE	98	100	140	120	24	2	480	28.9	3	1, 2, 3
DHWH-2	MECHANICAL ROOM	DOMESTIC HOT WATER	ELECTRIC STORAGE	6	100	140	6	1.5	1	208	7.2	1	1, 3

NOTES:

1. SET STORAGE TEMPERATURE TO 140°F AND PROVIDE WITH ASSE 1017 THERMOSTATIC MASTER MIXING VALVE SET TO 131°F.
2. KW INDICATED IS THE TOTAL OF TWO 12 KW ELEMENTS OPERATING SIMULTANEOUSLY.
3. PROVIDE THERMAL EXPANSION TANK; REFER TO EXPANSION TANK SCHEDULE.

EXPANSION TANK (DIAPHRAGM TYPE) SCHEDULE

TAG	LOCATION	SERVICE	TYPE	FLUID	ACCEPTANCE VOLUME (GAL)	SIZE VOLUME (GAL)	FILL PRESSURE (PSIG)	RELIEF VALVE SETTING (PSIG)	NOTES
ET-1	JANITOR CLOSET	DHW SYSTEM	DIAPHRAGM	WATER	3.2	6.4	60	150	1, 2, 3
ET-2	MECHANICAL ROOM	DHW SYSTEM	DIAPHRAGM	WATER	0.9	2	60	150	1, 2, 3

NOTES:

1. PROVIDE ASME VERTICAL EXPANSION TANK WITH HEAVY DUTY BUTYL DIAPHRAGM, STEEL CONSTRUCTION.
2. PROVIDE LINE SIZE ISOLATION BALL VALVES ON CONNECTION.
3. EXPANSION TANK CANNOT BE SUSPENDED FROM THE PIPING SYSTEM; IT MUST BE SUPPORTED FROM THE STRUCTURE OF THE BUILDING.

PUMP SCHEDULE

TAG	LOCATION	TYPE	FLUID TYPE	FLOW (GPM)	PUMP HEAD (FT H20)	ELECTRICAL DATA MOTOR HP	VOLTAGE	PHASE	RPM	NOTES
DCP-1	JANITOR CLOSET	DOMESTIC HOT WATER CIRCULATING PUMP	122°F HOT WATER	1	15	0.09	115 V	1	3600	1, 2

NOTES:

1. PROVIDE LEAD FREE STAINLESS STEEL BODY WITH STAINLESS STEEL IMPELLER.
2. PROVIDE IMMERSION AQUASTAT WITH PROGRAMMABLE TIMER.

BACKFLOW PREVENTER SCHEDULE

TAG	LOCATION	CROSS CONNECTION	TYPE	VALVE BODY SIZE (IN.)	DESIGN FLOW RATE (GPM)	PRESSURE DROP AT DESIGN FLOW RATE (PSI)	NOTES
RPZ-1	MECHANICAL ROOM	DOMESTIC WATER	REDUCED PRESSURE ZONE	2 1/2"	84	10	1, 2, 3, 4

NOTES:

1. PROVIDE LEAD FREE ASSE 1013 REDUCED PRESSURE ZONE BACKFLOW PREVENTER WITH INLET STRAINER.
2. ROUTE RELIEF DRAIN THROUGH AIR GAP TO FLOOR DRAIN.
3. PRESSURE DROP ACROSS BACKFLOW PREVENTER MUST BE 10 PSI OR LESS AT THE DESIGN FLOW RATE.
4. MAXIMUM RELIEF VALVE DISCHARGE RATE MUST NOT EXCEED 275 GPM AT 60 PSI.

WATER METER SCHEDULE

TAG	LOCATION	TYPE	REGISTER TYPE	INLET/OUTLET (IN.)	FLOW RANGE MIN/MAX (GPM)	NOTES
WM-1	MECHANICAL ROOM	NUTATING DISC, POSITIVE DISPLACEMENT	MAGNETIC DRIVE	2"	2.5 / 170	1, 2, 3, 4

NOTES:

1. PROVIDE WATER METER ENCODER WITH PULSE OUTPUT TO INTERFACE TO THE BASE AMI METERING SYSTEM.
2. PROVIDE USER-PROGRAMMABLE REMOTE MOUNTED DISPLAY PANEL MONITORING CURRENT AND TOTAL WATER FLOW RATE.
3. PRESSURE DROP ACROSS WATER METER MUST BE 6 PSI OR LESS AT THE DESIGN FLOW RATE.
4. PROVIDE WATER METER WITH INTEGRAL STRAINER.

WATER HAMMER ARRESTOR SCHEDULE

TAG	FIXTURE UNITS	PIPE SIZE (IN.)	NOTES
WHA-A	1 - 11	1/2"	1, 2, 3
WHA-C	33 - 60	1"	1, 2, 3

NOTES:

1. PROVIDE PISTON OPERATED, COPPER BODY, LEAD FREE WATER HAMMER ARRESTOR.
2. WATER HAMMER ARRESTOR SIZING MUST BE BASED ON SPECIFIC MANUFACTURER'S FIXTURE UNIT CAPACITY TABLE.
3. WATER HAMMER ARRESTORS MUST BE CERTIFIED BY THE PLUMBING AND DRAINAGE INSTITUTE.

RADON FAN SCHEDULE

TAG	SERVICE	TYPE	AIRFLOW (CFM)	ESP (IN. WC)	RPM	ELECTRICAL DATA WATTS	VOLTAGE	PHASE	HZ	NOTES
RF-1	RADON MITIGATION	INLINE CENTRIFUGAL	120	3.0	4290	142	120 V	1	60	1, 2, 3
RF-2	RADON MITIGATION	INLINE CENTRIFUGAL	120	3.0	4290	142	120 V	1	60	1, 2, 3
RF-3	RADON MITIGATION	INLINE CENTRIFUGAL	120	3.0	4290	142	120 V	1	60	1, 2, 3

NOTES:

1. PROVIDE HIGH EFFICIENCY INLINE RADON FAN WITH INTEGRAL SPEED CONTROL SYSTEM, ANTI-VIBRATION COUPLERS, THERMAL OVERLOAD PROTECTION, AND UV-RESISTANT HOUSING.
2. RADON FAN MUST BE UL LISTED FOR OUTDOOR USE.
3. FINAL RADON FAN SPEED SETTING MUST BE ADJUSTED DURING TAB TO MEET THE REQUIRED CFM AIRFLOW AT THE ACTUAL MEASURED STATIC PRESSURE.

SYM	DESCRIPTION	DATE	APPR

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. D..., RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES CWH | DRW BPT | CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING SCHEDULES

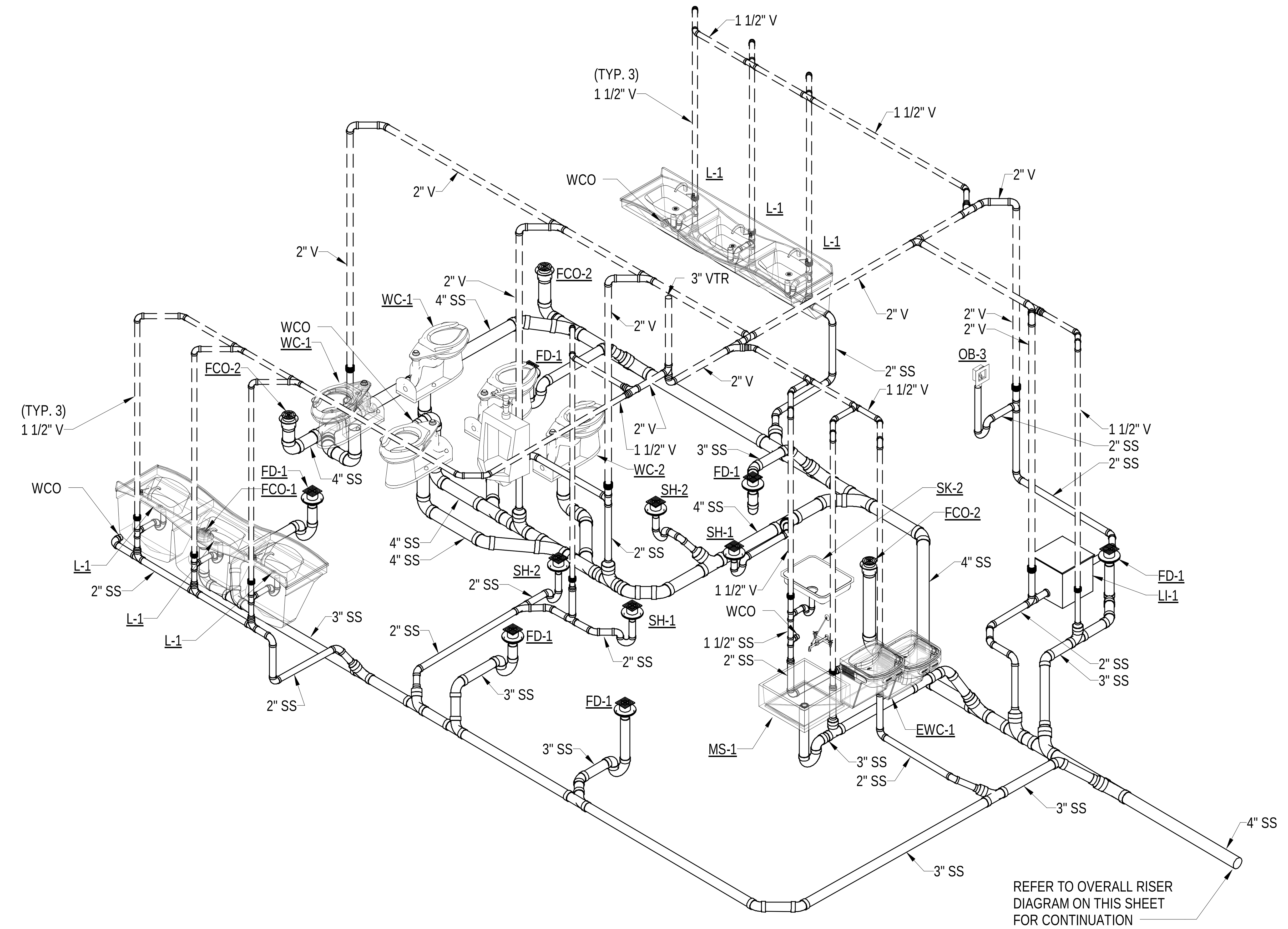
SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894016
SHEET 66 OF 140

P-601

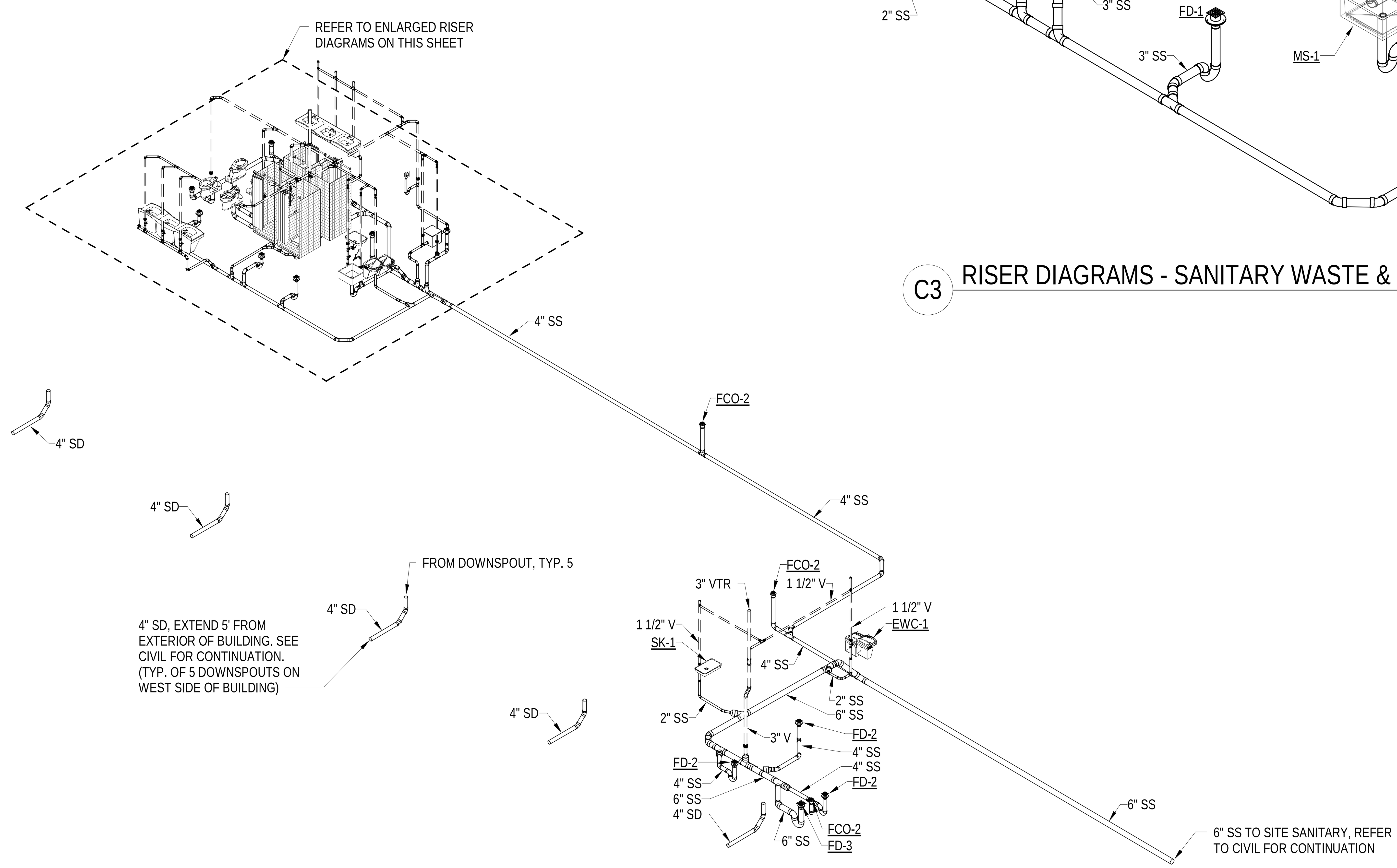
6/20/2025 12:13:32 PM
\\dc1vs01\FEDPRJ\FD2E1628\PLUMB\LDG-1667770-P-BIM_v21.rvt
P-601

GENERAL SHEET NOTES

A. REFER TO SHEET P-001 FOR LEGEND AND GENERAL NOTES.



C3 RISER DIAGRAMS - SANITARY WASTE & VENT - ENLARGED RESTROOMS



A1 RISER DIAGRAMS - SANITARY WASTE & VENT AND STORM DRAINAGE - OVERALL

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES CWH	DRW BPT / CHK MJL
PM/DM	SMS/CWJ
BRANCH MANAGER	ALG
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING RISER DIAGRAMS - SANITARY WASTE & VENT

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894017
SHEET 67 OF 140

P-901

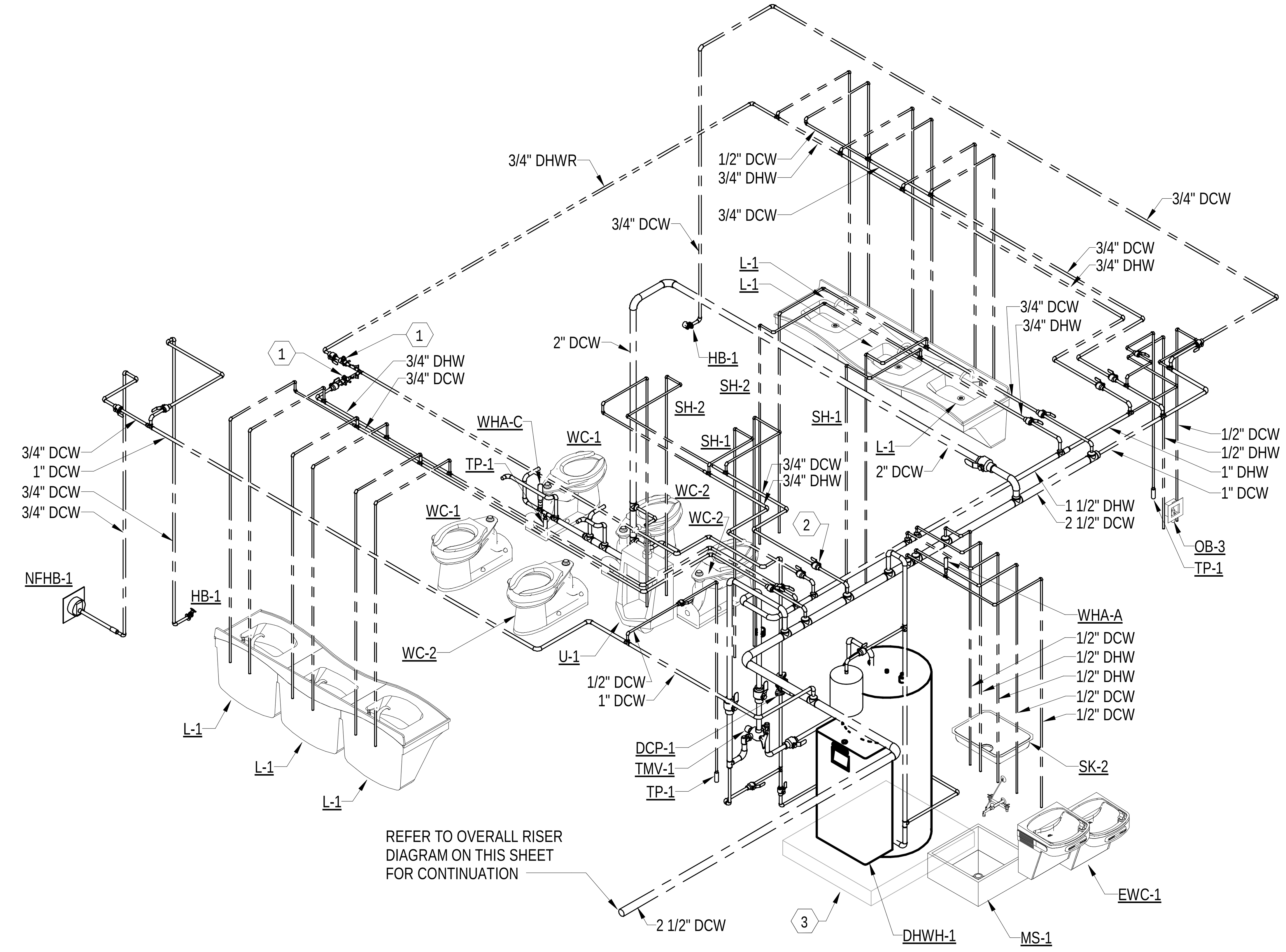
6/20/2025 12:13:37 PM P-901
\\dc1vs01\FEDPRJ\FDZE1628\PLUMB\BLDG-1667770-P-BIM_v21.rvt

GENERAL SHEET NOTES

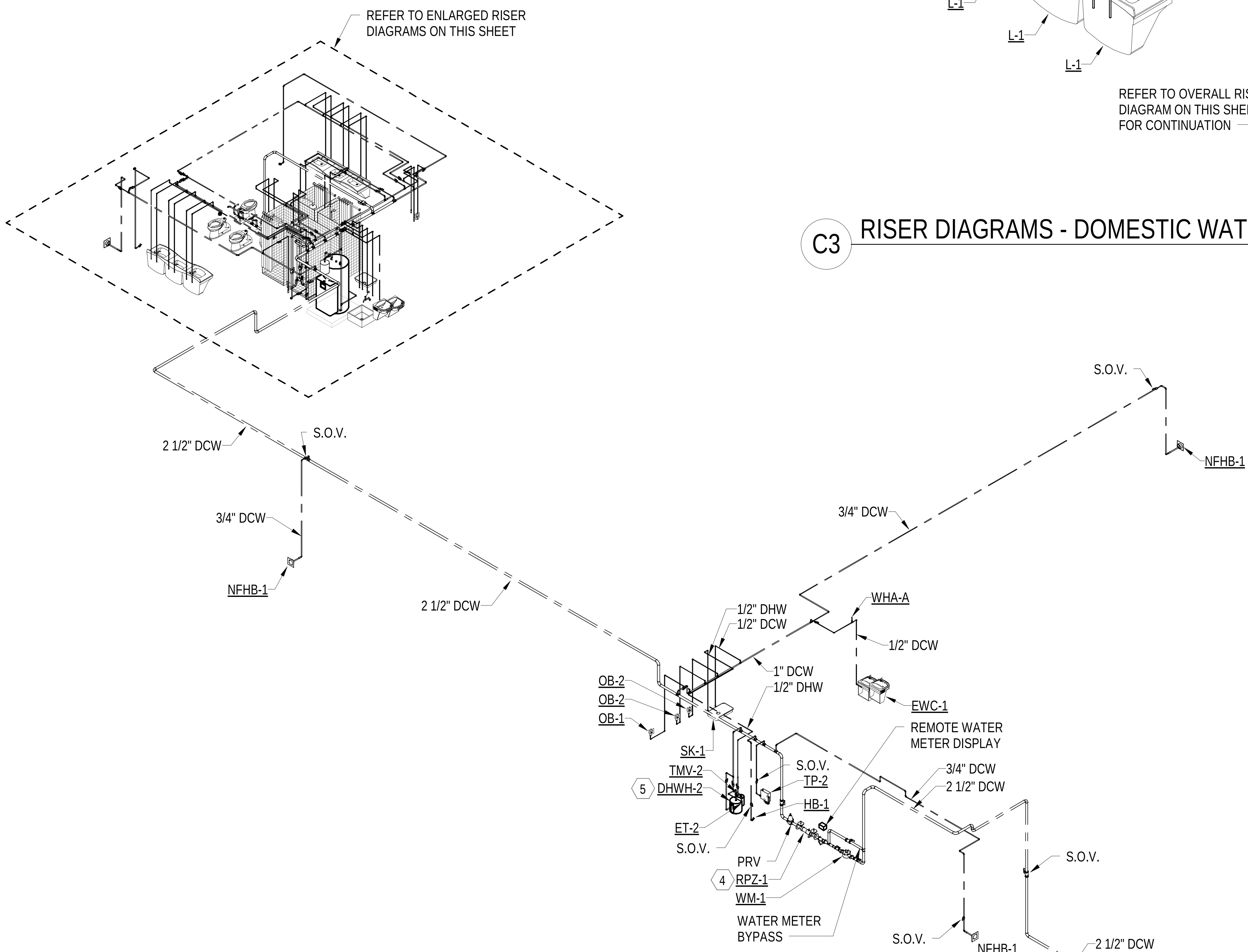
A. REFER TO SHEET P-001 FOR LEGEND AND GENERAL NOTES.

SHEET KEYNOTES

1 PROVIDE BALANCING VALVE AND CHECK VALVE FOR RECIRCULATION BRANCH. SET BALANCING VALVE TO 0.5 GPM. FINALIZE BALANCING VALVE FLOW DURING TAB.
2 PROVIDE LINE SIZE ISOLATION VALVES ON BRANCH LINES (TYPICAL).
3 6" HOUSEKEEPING PAD, EXTEND 6" FROM EQUIPMENT FOOTPRINT. REFER TO STRUCTURAL DRAWINGS FOR DETAILED DIMENSIONS AND SPECIFICATIONS. COORDINATE EQUIPMENT PAD DIMENSIONS WITH FINAL EQUIPMENT SELECTIONS.
4 REFER TO WATER SERVICE ENTRANCE DETAIL ON SHEET P-501 FOR RPZ PIPING SUPPORT LOCATIONS AND DETAILED PIPING ARRANGEMENT.
5 REFER TO ELECTRIC WATER HEATER DETAIL (MECH ROOM) ON SHEET P-501 FOR DETAILED PIPING ARRANGEMENT.



C3 RISER DIAGRAMS - DOMESTIC WATER - ENLARGED RESTROOMS



A1 RISER DIAGRAMS - DOMESTIC WATER - OVERALL

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES CWH	DRW BPT / CHK MJL
PM/DM	SMS/CWJ
BRANCH MANAGER	ALG
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
MID-ATLANTIC
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
PLUMBING RISER DIAGRAMS - DOMESTIC WATER

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894018
SHEET 68 OF 140

P-902

6/20/2025 12:13:43 PM P-902
\\dc1vs01\FEDPRJ\FDZE1628\PLUMB\BLDG-1667770-P-BIM_v21.rvt

MECHANICAL LEGEND

T	TEMPERATURE SENSOR/THERMOSTAT
H	HUMIDITY SENSOR
A 000	AIR TERMINAL TAG WITH AIR FLOW (CFM)
SW	EMERGENCY HVAC SYSTEM SHUTOFF SWITCH
XXX	SPACE CO2 SENSOR / DDC POINT TAG
12"x10"	RECTANGULAR DUCT DIMENSIONS (INCHES)
12"Ø	ROUND DUCT DIMENSION
	FLEXIBLE ROUND DUCT (5'-0" MAX LENGTH)
	DOWNWARD DIRECTION OF SLOPED PIPING
	SUPPLY DUCT, SUPPLY DUCT DOWN/UP
	RETURN DUCT, RETURN DUCT DOWN/UP
	2-WAY VALVE
	3-WAY VALVE
	AUTOMATIC BALANCING VALVE
	BALL VALVE
	BUTTERFLY VALVE
	CHECK VALVE
	DRAIN VALVE
	GATE VALVE
	GLOBE VALVE
	MANUAL FLOW CONTROL BALANCING VALVE
	PRESSURE REDUCING VALVE
	RELIEF VALVE
	SOLENOID VALVE
	SQAURE HEAD BALANCING VALVE
	TEMPERATURE PRESSURE RELIEF VALVE
	BACKFLOW PREVENTER
	AUTOMATIC AIR VENT
	ECCENTRIC REDUCER
	FLANGE
	MANUAL AIR VENT
	PIPE CAP
	PIPE DOWN
	SUCTION REDUCER
	PIPE UP
	PRESSURE GAUGE WITH SHUTOFF COCK
	STEAM TRAP
	STRAINER WITH BLOWDOWN VALVE
	THERMOMETER
	UNION
	VACUUM BREAKER
	WYE STRAINER
DPS	DIFFERENTIAL PRESSURE SENSOR (DUCTWORK)
	NEW WORK NOTE TAG
	DEMOLITION NOTE TAG
	POINT OF CONNECTION TAG
	POINT OF DISCONNECTION TAG
*	POINT SUPPLIED BY UNIT MANUFACTURER
	AVERAGING TEMPERATURE SENSOR

	VERTICAL (WALL) FIRE DAMPER
	HORIZONTAL (CEILING) FIRE DAMPER
AD	ACCESS DOOR (AD), ACCESS PANEL (AP)
	VOLUME DAMPER
	EXHAUST AIR (NEG.) DUCT SECTION
	VRF CEILING CASSETTE
	VRF CEILING CASSETTE DUCTED
	VRF DUCTED FAN COIL UNIT
	VENT THRU FLOOR/ ROOF
	RISE IN DUCT (POINT IS HIGH)

FIRE DAMPER
DUCT THRU WALL
DUCT ACCESS DOOR
HORIZONTAL (FLOOR) FIRE DAMPER
DUCT THRU FLOOR
SECURITY BAR
DUCT THRU WALL
DUCT ACCESS DOOR
AD
DUCT ACCESS DOOR
BD
BALANCE/VOLUME DAMPER (MANUAL)
MD#
MOTOR OPERATED DAMPER (DDC)

DETAIL/SECTION TITLE	DETAIL CALLOUT SYMBOL	SECTION CUT SYMBOL
SECTION, DETAIL OR ELEVATION IDENTIFICATION MARK; SECTION, DETAIL OR ELEVATION TITLE; XX / X-XXX; PLATE NUMBER WHERE SECTION, DETAIL OR ELEVATION IS TAKEN	DETAIL NUMBER; XX / X-XXX; PLATE NUMBER WHERE DETAIL IS DRAWN	SECTION LETTER; PLATE NUMBER WHERE SECTION IS DRAWN; XX / X-XXX; VIEW SECTION

(NAME)	PIPING TO BE REMOVED
CWR	CONDENSER WATER RETURN PIPING
CWS	CONDENSER WATER SUPPLY PIPING
CHWR	CHILLED WATER RETURN PIPING
CHWS	CHILLED WATER SUPPLY PIPING
CND	CONDENSATE DRAIN PIPING
DCW	DOMESTIC COLD WATER PIPING
DHW	DOMESTIC HOT WATER PIPING
DTR	DUAL TEMPERATURE RETURN PIPING
DTS	DUAL TEMPERATURE SUPPLY PIPING
G	NATURAL GAS PIPING
HG	HOT GAS BYPASS PIPING
HPS	HIGH PRESSURE STEAM PIPING
HWR	HOT WATER RETURN PIPING
HWS	HOT WATER SUPPLY PIPING
LPC	LOW PRESSURE CONDENSATE PIPING
LPS	LOW PRESSURE STEAM PIPING
RL	REFRIGERANT LIQUID PIPING
RS	REFRIGERANT SUCTION PIPING
VAC	VACUUM (AIR)

MECHANICAL ABBREVIATIONS

A/C	AIR CONDITIONING
AC	SPLIT SYSTEM AIR CONDITIONER
AD	ACCESS DOOR
AFD	ACCESS FLOOR DIFFUSER
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
AI	ANALOG INPUT
AIP	ABANDON IN PLACE
AO	ANALOG OUTPUT
AP	ACCESS PANEL
APD	AIR PRESSURE DROP
AR	AIRFLOW REGULATOR
AS	AIR SEPARATOR
ATFP	ANTI-TERRORISM FORCE PROTECTION
AV	ANALOG VALUE
B	BOILER
BAS	BUILDING AUTOMATIC SYSTEM (DDC)
BD	BALANCING DAMPER
BDD	BACKDRAFT DAMPER (DUCTWORK)
BG	BELOW GROUND
BHP	BRAKE HORSEPOWER
BI	BINARY INPUT
BO	BINARY OUTPUT
BOD	BOTTOM OF DUCT
BR	BASEBOARD RADIATOR
BSB	BRANCH SELECTOR BOX (VRF)
BTUH	BRITISH THERMAL UNITS PER HOUR
BV	BINARY VALUE
CC	CEILING CASSETTE / COOLING COIL
CD	CONDENSATE DRAIN
CER	CEILING EXHAUST REGISTER
CFM	CUBIC FEET PER MINUTE
CFT	CHEMICAL FEED TANK
CH	CHILLER
CHP	CHILLER PUMP
CHR	CHILLED WATER RETURN
CHS	CHILLED WATER SUPPLY
CLG	CEILING
CND	CONDENSER
COP	COEFFICIENT OF PERFORMANCE
COV	CHANGE OF VALUE
CRAHU	COMPUTER ROOM AIR HANDLING UNIT
CRG	CEILING RETURN GRILLE
CSD	CEILING SUPPLY DIFFUSER
CT	COOLING TOWER
CTG	CEILING TRANSER GRILLE
CU	CONDENSING UNIT
CUH	CABINET UNIT HEATER

MECHANICAL ABBREVIATIONS

Cv	LOSS COEFFICIENT
CW	CHILLED WATER
DB	DRY BULB
DCW	DOMESTIC COLD WATER (POTABLE)
DDC	DIRECT DIGITAL CONTROL
DEG	DEGREE
DIA	DIAMETER
DOAS	DEDICATED OUTSIDE AIR UNIT
DWGS	DRAWINGS
DX	DIRECT EXPANSION
EA	EXHAUST AIR
EAT	ENTERING AIR TEMPERATURE
EC	ELECTRIONICALLY COMMUTATED (MOTOR) / ELECTRIC HEAT COIL
EER	ENERGY EFFICENCY RATIO
EF	EXHAUST FAN
EFF	EFFECTIVENESS / EFFICIENCY
EH	ELECTRIC HEAT
ERV	ENERGY RECOVERY VENTILATOR
ESP	EXTERNAL STATIC PRESSURE
ET	EXPANSION TANK
ETR	EXISTING TO REMAIN
EWH	ELECTRIC WATER HEATER
EWT	ENTERING WATER TEMPERATURE
F	FAHRENHEIT
FCU	FAN COIL UNIT
FD	FIRE DAMPER
FT	FEET
FT WG	FEET WATER GAUGE
G	NATURAL GAS
GP	GLYCOL PUMP
GPM	GALLONS PER MINUTE
GRH	GRAVITY ROOF HOOD
H	HUMIDIFIER (ELECTRIC)
HC	HEAT COIL (DX)
HGRC	HOT GAS REHEAT COIL (DX)
HGRH	HOT GAS REHEAT
HHWS	HOT HEATING WATER SYSTEM
HP	HEAT PUMP, HORSEPOWER
HR	HOUR / HEAT RECOVERY
HT	HEIGHT
HW	HOT WATER
HWP	HOT WATER PUMP
HX	HEAT EXCHANGER
IAW	IN ACCORDANCE WITH
IBS	IN BEAM SPACE
IC	IN CEILING
IDU	INDOOR UNIT (VRF)
IN.	INCH
IWG	INCHES WATER GAUGE
KW	KILOWATT
LBS	POUNDS
LPS	LOW PRESSURE STEAM
LV	LOUVER
LWT	LEAVING WATER TEMPERATURE
MAU	MAKE-UP AIR UNIT
MAX	MAXIMUM
MBH	1,000 BTUH
MCA	MAXIMUM CIRCUIT AMPS
MD	MOTOR OPERATED DAMPER
MIN	MINIMUM
NC	NORMALLY CLOSED, NOISE CRITERIA
NO	NUMBER, NORMALLY OPEN
NP	NOT PROVIDED
OA	OUTSIDE AIR
OAU	OUTSIDE AIR UNIT
OCC	OCCUPANCY
ODU	OUTDOOR UNIT (VRF)
OF	ON FLOOR
OG	ON GRADE
OH	OPPOSITE HAND
OR	ON ROOF
OW	ON WALL
PA	PRIMARY AIR (VAV TERMINAL)
PH	PHASE
PRV	POWER ROOF VENTILATOR
PSIG	POUNDS PER SQUARE INCH GAUGE
PTAC	PACKAGED TERMINAL AIR CONDITIONING UNIT
PTHP	PACKAGED TERMINAL HEAT PUMP
PTP	PRESSURE/TEMPERATURE TEST PORT
QTY	QUANTITY
RA	RETURN AIR
RC-X	REHEAT COIL - NUMBER
RD	RETURN DIFFUSER

MECHANICAL ABBREVIATIONS

RG	RETURN GRILLE
RH	RELATIVE HUMIDITY
RL	REFRIGERANT LIQUID
RS	REFRIGERANT SUCTION
RTU	ROOFTOP UNIT
SA	SUPPLY AIR
SAT	SUPPLY AIR TEMPERATURE
SD	SUPPLY DIFFUSER
SEER	SEASONAL ENERGY EFFICIENCY RATIO
SF	SUPPLY FAN
SIM	SIMILAR
SP	STATIC PRESSURE
SQ	SQUARE
SQFT	SQUARE FEET
TC	THRU CEILING
TF	THRU FLOOR
TR	THRU ROOF
TYP.	TYPICAL
UH	UNIT HEATER
UOI	UNLESS OTHERWISE INDICATED
UON	UNLESS OTHERWISE NOTED
V	VOLTS
VAV	VARIABLE AIR VOLUME
VFD	VARIABLE FREQUENCY DRIVE
VWV	VARIABLE WATER VOLUME
W	WIDTH
WB	WET BULB
WC	WATER COLUMN (PRESSURE)
WH	WALL HOOD
WPD	WATER PRESSURE DROP
°	DEGREES
Ø	DIAMETER
Φ	OVAL

GENERAL MECHANICAL NOTES

1. THESE GENERAL MECHANICAL NOTES APPLY TO EVERY MECHANICAL DRAWING IN THIS SET. SHEET SPECIFIC NOTES ARE INDICATED ON EACH SHEET UNDER GENERAL SHEET NOTES. NOT EVERY SYMBOL AND ABBREVIATION ARE USED.
2. PROVIDE EVERY MATERIAL AND PIECE OF EQUIPMENT AND PERFORM NECESSARY LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED, AND AS REQUIRED BY CODE.
3. INSTALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS, CONTRACT DOCUMENTS, APPLICABLE CODES, AND REGULATIONS.
4. CONCEAL DUCTWORK AND PIPING ABOVE SUSPENDED CEILINGS AND IN WALLS, UNLESS OTHERWISE NOTED.
5. PROVIDE VIBRATION ISOLATORS FOR MECHANICAL EQUIPMENT AND PIPING TO PREVENT TRANSMISSION OF VIBRATION TO BUILDING STRUCTURE.
6. PROVIDE ACCESS PANELS IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVE FIRE DAMPERS, VALVES, SMOKE DETECTORS, AND OTHER CONCEALED MECHANICAL EQUIPMENT THAT REQUIRES ROUTINE MAINTENANCE. ACCESS PANELS MUST BE SIZED AS REQUIRED TO ALLOW WORKING CLEARANCE FOR DUCT ACCESS DOORS.
7. MITERED 90 DEGREE RECTANGULAR DUCT ELBOWS IN SUPPLY DUCTWORK MUST HAVE TURNING VANES. PROVIDE AN ACCESS DOOR UPSTREAM OF ELBOW FOR TURNING VANE INSPECTION AND CLEANING.
8. PROVIDE PIPE SLEEVES FOR PIPING PENETRATIONS THRU WALLS AND FLOORS. EACH PIPE MUST BE SEPARATELY SLEEVED. PROVIDE CONTINUOUS SEALANT JOINT AROUND PIPE SLEEVE, BOTH SIDES OF WALL, WATERTIGHT.
9. OPENINGS IN FIRE WALLS DUE TO DUCTWORK, PIPING, CONDUIT, ETC., MUST BE FIRE STOPPED.
10. PROVIDE P-TRAPS IN CONDENSATE DRAIN PIPING AT EACH AC UNIT UNLESS THE UNIT IS EQUIPPED WITH AN INTERNAL TRAP.
11. CONDENSATE DRAIN PIPING CLEANOUTS MUST BE INSTALLED AT EACH CHANGE IN DIRECTION GREATER THAN 45 DEGREES. WHERE MORE THAN ONE CHANGE OF DIRECTION OCCURS IN A RUN OF PIPING, ONLY ONE CLEANOUT MUST BE REQUIRED FOR EACH 40 FEET OF DEVELOPED LENGTH. COMBINATION WYE AND 1/8 BENDS MUST BE USED IN LIEU OF TEES. UNLESS OTHERWISE INDICATED CONDENSATE LINES MUST SLOPE DOWNWARD AND BE GRAVITY FED.
12. MOUNT THERMOSTATS 48 INCHES ABOVE THE FINISHED FLOOR AND RECESSED, FLUSH WITH WALL SURFACE, WHERE POSSIBLE UNLESS OTHERWISE NOTED. SEE ELECTRICAL DRAWINGS FOR COORDINATION WITH LIGHT SWITCHES.
13. TESTING AND BALANCING (TAB): TAB EQUIPMENT AND DUCTWORK; 2-SEASONS.
14. MECHANICAL EQUIPMENT, WHICH IS PROVIDED WITH MANUFACTURER'S INSTALLED CONTROLS, MUST BE STARTED UP BY THE MANUFACTURER'S REPRESENTATIVE IN CONJUNCTION WITH THE CONTROLS CONTRACTOR (NOT MECHANICAL CONTRACTOR). TRAINING AND MANUALS FOR THE MANUFACTURER'S INSTALLED CONTROLS MUST BE PROVIDED BY THE EQUIPMENT MANUFACTURER'S REPRESENTATIVE. CONTROLS CONTRACTOR IS RESPONSIBLE FOR COORDINATING WITH MANUFACTURER'S REPRESENTATIVE FOR THE COMMUNICATION OF THE DDC SYSTEM TO THE MANUFACTURER'S INSTALLED CONTROLS.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

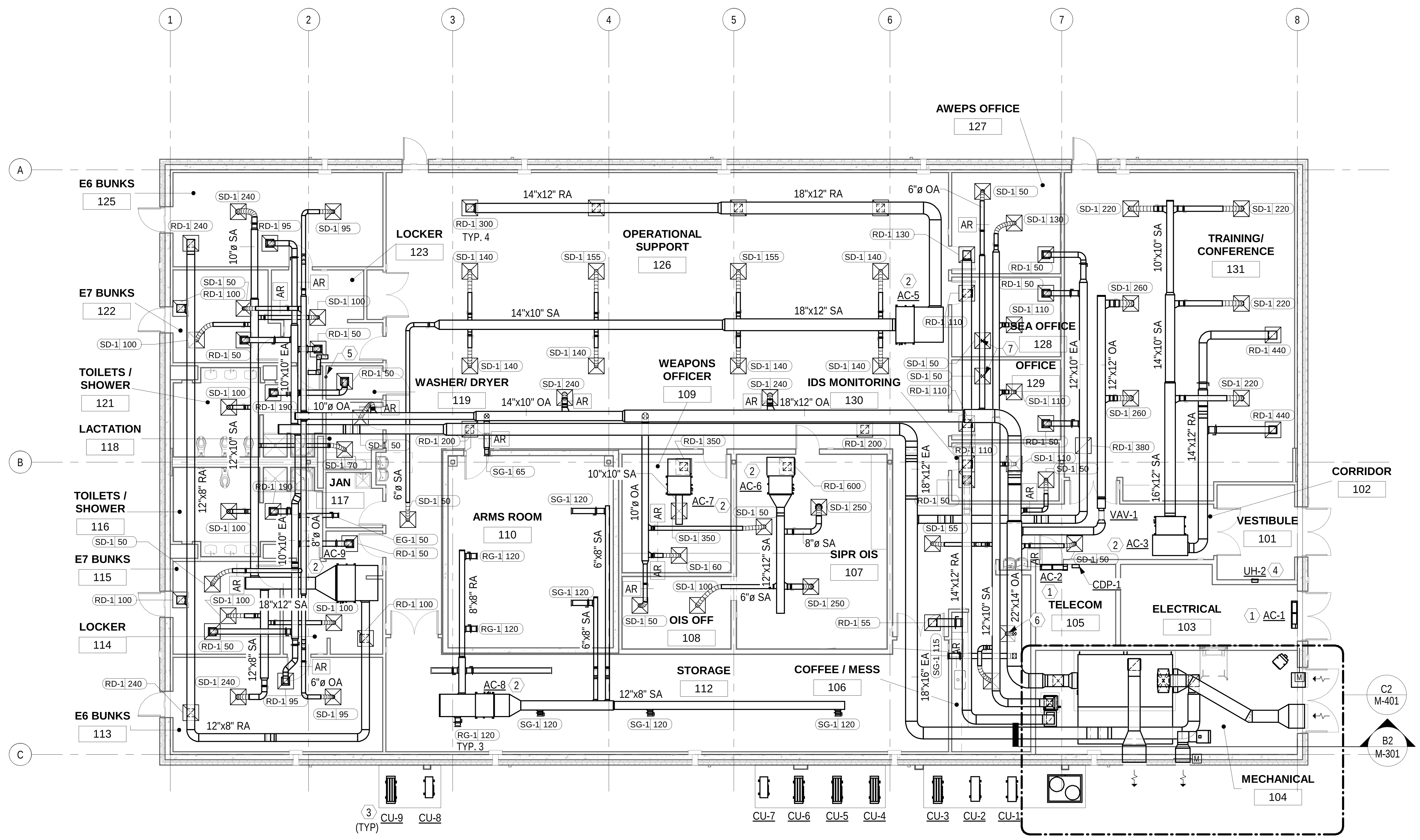
COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH DRW CWH CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL LEGEND, ABBREVIATIONS AND NOTES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894019
SHEET 69 OF 140
M-001



B1 FIRST FLOOR PLAN - DUCTWORK

1/8" = 1'-0"

GENERAL SHEET NOTES

A. DUCTWORK AND EQUIPMENT SHOWN ON THIS DRAWING IS DIAGRAMMATIC. LOCATE DUCTWORK AND EQUIPMENT SO AS TO CLEAR OBSTACLES.
B. WORK TO BE PREFORMED IN ACCORDANCE WITH APPLICABLE FEDERAL, STATE, AND LOCAL CODES.
C. COORDINATE HVAC INSTALLATIONS WITH APPLICABLE TRADES.
D. MECHANICAL AND PLUMBING EQUIPMENT. CONSTRUCTION DISCREPANCIES TO BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE CONTRACTING OFFICER FOR A RESOLUTION.

SHEET KEYNOTES

1. PROVIDE WALL MOUNTED DUCTLESS AIR CONDITIONING UNIT.
2. PROVIDE SUSPENDED AIR CONDITIONING UNIT.
3. PROVIDE CONDENSING UNIT ON 6-INCH CONCRETE PAD.
4. PROVIDE WALL MOUNTED UNIT HEATER.
5. PROVIDE ALUMINUM DRYER EXHAUST AIR DUCT WITH LINT CLEANOUT AT EACH CHANGE IN DIRECTION. ROUTE TO GOOSENECK AT ROOF. PROVIDE VENT CAP WITH STAINLESS STEEL BIRD SCREEN.
6. SD-1 AT 55 CFM. PROVIDE VOLUME DAMPER PRIOR TO AIRFLOW REGULATOR AT DUCT BRANCH.
7. PROVIDE VOLUME DAMPER PRIOR TO AIRFLOW REGULATOR AT DUCT BRANCH.

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

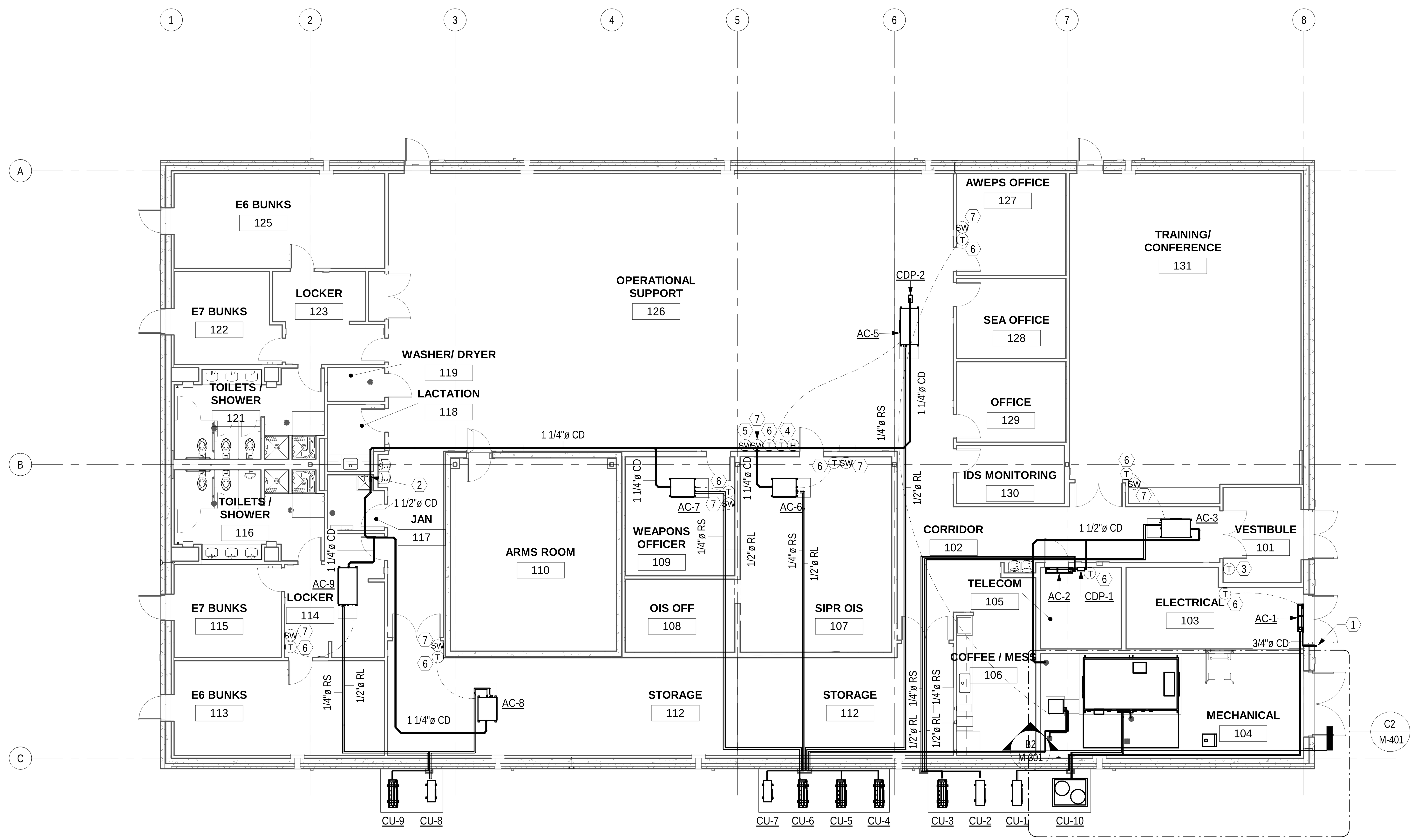
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH DRW SDH CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL FLOOR PLAN - DUCTWORK

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894020
SHEET 70 OF 140
M-101

6/20/2025 11:54:25 AM M-101
\\dc1vs01\fedprj\FDZE1628\MECH\BLDG-1667770-M-BIM_v21.rvt



B1 FIRST FLOOR PLAN - PIPING

1/8" = 1'-0"

PLAN NORTH

0 4' 8' 16' 24'

1/8" = 1'-0"

GENERAL SHEET NOTES

A. REFER TO M-001 FOR ABBREVIATIONS, LEGEND, AND GENERAL NOTES.

B. PROVIDE FIRE STOPPING AT FIRE-RATED PENETRATIONS.

C. COORDINATE WORK WITH OTHER TRADES PRIOR TO FABRICATION AND INSTALLATION TO PREVENT CONFLICTS. NOTIFY CONTRACTING OFFICER WHERE CONFLICTS ARE DISCOVERED PRIOR TO CONSTRUCTION AND BEFORE BEGINNING WORK IN THOSE AREAS. CONFLICTS THAT ARISE DUE TO LACK OF COORDINATION MUST BE RECTIFIED AT NO COST TO THE GOVERNMENT.

D. EQUIPMENT INSTALLATION MUST BE IN STRICT ACCORDANCE TO MANUFACTURER'S INSTRUCTIONS.

E. SLOPES AND INVERT ELEVATIONS OF INTERIOR PIPING MUST BE ESTABLISHED BEFORE ANY PIPING IS INSTALLED, IN ORDER THAT PROPER SLOPES WILL BE MAINTAINED. PIPING MUST BE RUN TO AVOID CONFLICT WITH OTHER TRADES.

F. LAYOUT OF PIPING IS DIAGRAMMATIC. ROUTE PIPING CONCEALED ABOVE CEILING OR IN WALLS OR CHASES UNLESS OTHERWISE NOTED. ROUTE EXPOSED PIPING, AS NOTED, HIGH AS POSSIBLE, UNLESS OTHERWISE NOTED. ALLOW FOR RISES, DROPS, AND OFFSETS AS REQUIRED.

G. CLEANOUTS MUST BE FULL SIZE OF PIPE.

SHEET KEYNOTES

1. ROUTE 3/4" CD DOWN TO FINISHED GRADE.
2. ROUTE 1 1/2" CD DOWN TO MOP SINK.
3. SPACE TEMPERATURE SENSOR ASSOCIATED WITH UNIT HEATER.
4. SPACE TEMPERATURE AND HUMIDITY SENSORS ASSOCIATED WITH DOAS-1.
5. PROVIDE EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH WITH CLEAR HINGED COVER, WARNING SIGN, AND A TAG. SEE DETAIL A4/M501 FOR FACEPLATE DETAILS.
6. THERMOSTAT IS PROVIDED BY UNIT MANUFACTURER FOR INSTALLATION IN FIELD. MOUNT AND CONNECT THERMOSTAT IN ACCORDANCE WITH REQUIREMENTS AND INSTRUCTIONS PROVIDED BY AC UNIT MANUFACTURER.
7. PROVIDE A 2-HOUR IN-WALL TIMER SWITCH TO ALLOW THE OCCUPANT TO PLACE THE UNIT IN OCCUPIED MODE DURING UNOCCUPIED PERIODS. PLACE SIGN ABOVE THE SWITCH TO READ "OCCUPANCY OVERRIDE SWITCH".

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED

FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY: APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE 05/27/2025

DES CWH	DRW SDH / CHK MJL
PM/DM	SMS/CWJ
BRANCH MANAGER	ALG
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL FLOOR PLAN - PIPING

SCALE:	AS NOTED
EPROJECT NO.:	1667770
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12894021
SHEET	71 OF 140

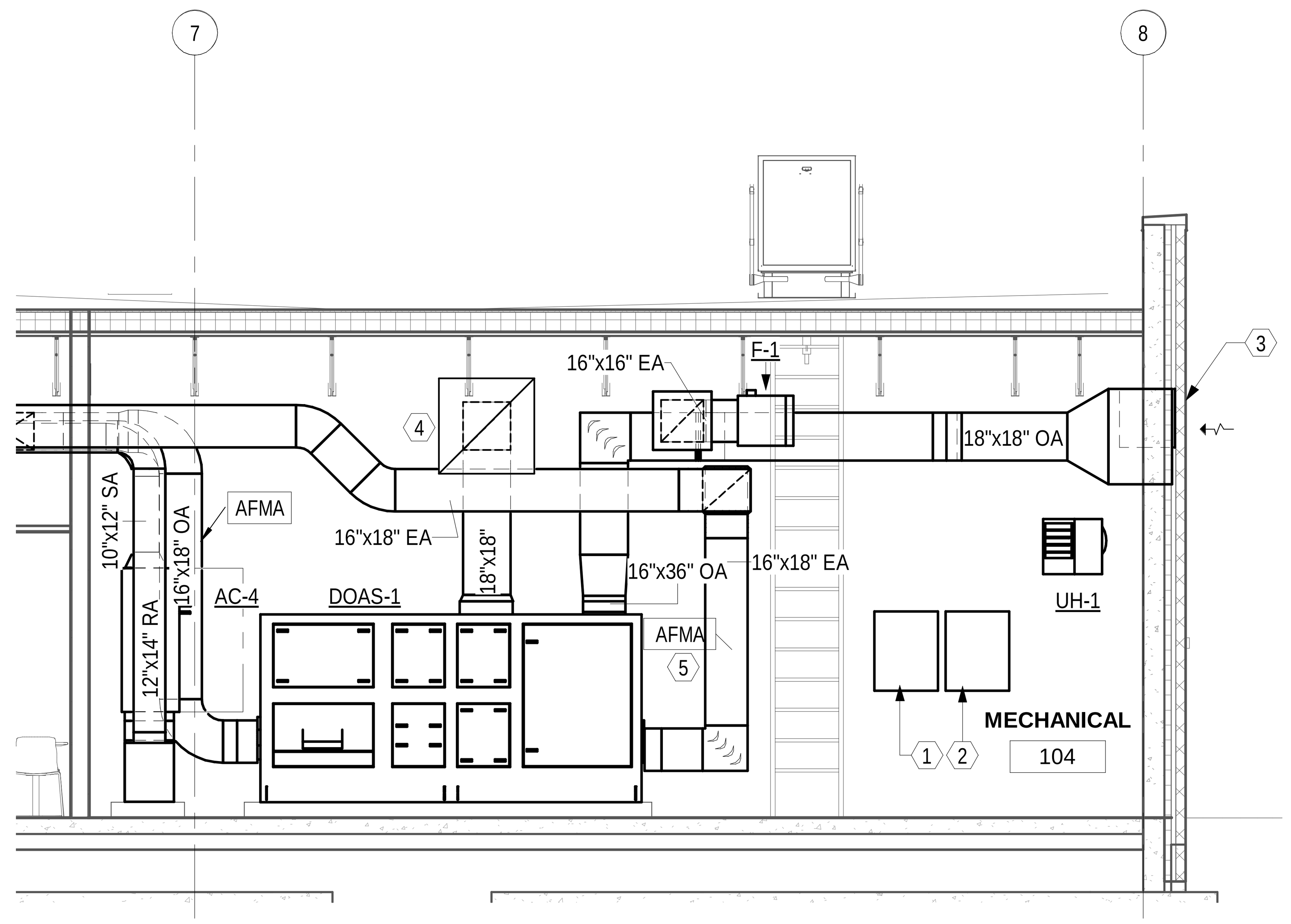
MP101

GENERAL SHEET NOTES

A. DUCTWORK AND EQIUPMENT SHOWN ON THIS DRAWING IS DIAGRAMMATIC. LOCATE DUCTWORK AND EQUIPMENT SO AS TO CLEAR OBSTACLES.
B. WORK TO BE PREFORMED IN ACCORDANCE WITH APPLICABLE FEDERAL, STATE, AND LOCAL CODES.
C. COORDINATE HVAC INSTALLATIONS WITH APPLICABLE TRADES.
D. MECHANICAL AND PLUMBING EQUIPMENT. CONSTRUCTION DISCREPANCIES TO BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE CONTRACTING OFFICER FOR A RESOLUTION.

SHEET KEYNOTES

1 CONTROL PANEL ASSOCIATED WITH DOAS-1.
2 BUILDING NETWORK CONTROL PANEL.
3 36"x36" OUTDOOR AIR INTAKE LOUVER. SEE ARCHITECTURAL PLANS FOR MORE INFORMATION. LOUVER MUST BE MINIMUM 10 FEET ABOVE FINISHED GRADE.
4 36"x36" EXHAUST LOUVER. SEE ARCHITECTURAL PLANS FOR MORE INFORMATION. LOUVER MUST BE MINIMUM 10 FEET ABOVE FINISHED GRADE.
5 PROVIDE AIRFLOW MEASURING ARRAY WITH FILTER BANK.



B2 MECHANICAL ROOM SECTION
1/4" = 1'-0"

0 2' 4' 8' 12'
1/4" = 1'-0"

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

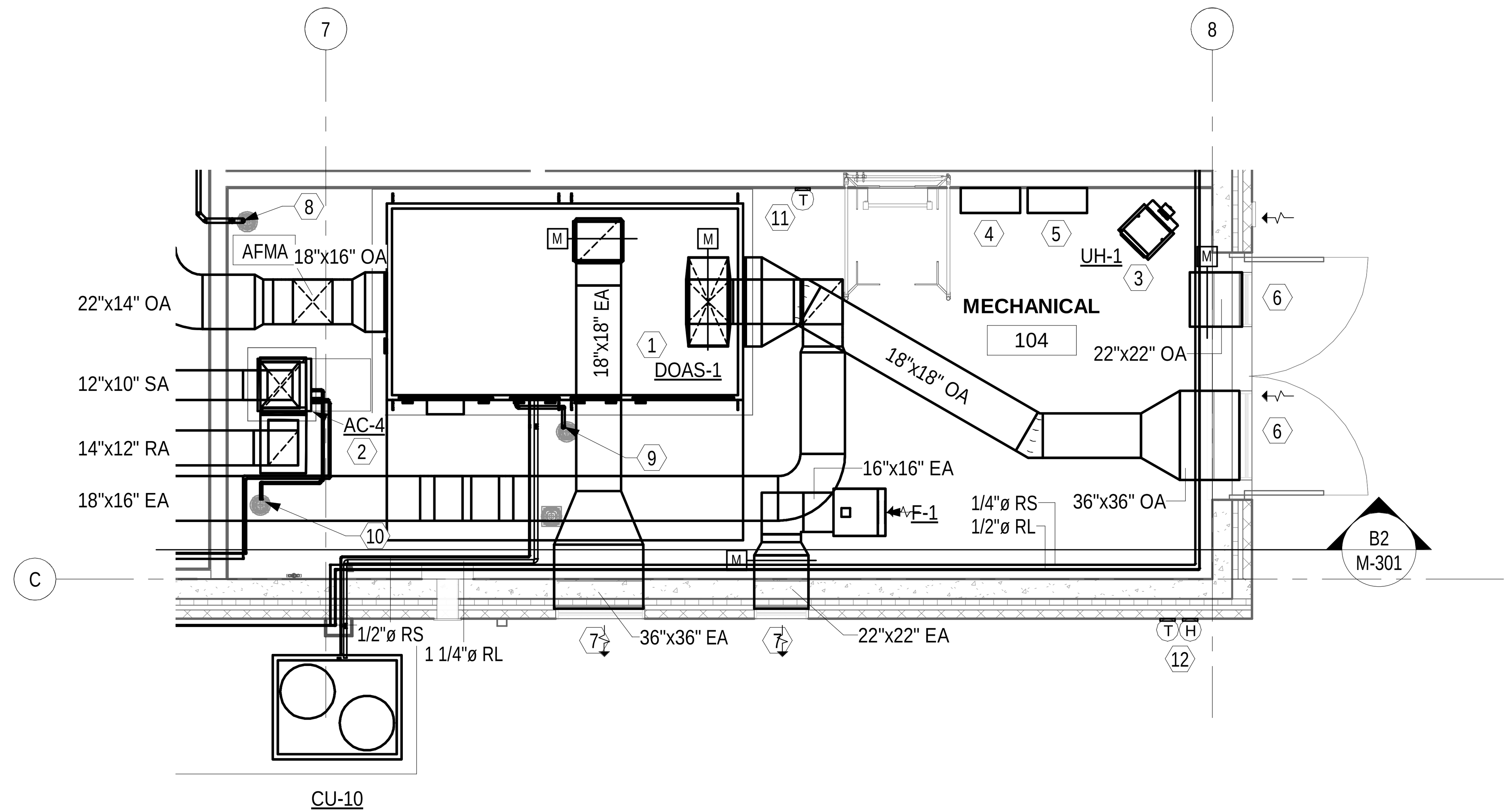
APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH DRW SDH CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL SECTIONS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894022
SHEET 72 OF 140

M-301

M-301
6/20/2025 11:54:28 AM
\\dc1vs01\fedprj\FDZE1628\MECH\BLDG-1667770-M-BIM_v21.rvt



C2 MECHANICAL ENLARGED PLAN - MECHANICAL ROOM

1/4" = 1'-0"

GENERAL SHEET NOTES

A. DUCTWORK AND EQIUPMENT SHOWN ON THIS DRAWING IS DIAGRAMMATIC. LOCATE DUCTWORK AND EQUIPMENT SO AS TO CLEAR OBSTACLES.

B. WORK TO BE PREFORMED IN ACCORDANCE WITH APPLICABLE FEDERAL, STATE, AND LOCAL CODES.

C. COORDINATE HVAC INSTALLATIONS WITH APPLICABLE TRADES.

D. MECHANICAL AND PLUMBING EQUIPMENT. CONSTRUCTION DISCREPANCIES TO BE BROUGHT TO THE IMMEDIATE ATTENTION OF THE CONTRACTING OFFICER FOR A RESOLUTION.

SHEET KEYNOTES

1 PROVIDE DEDICATED OUTDOOR AIR SYSTEM WITH ENERGY RECOVERY.
2 PROVIDE FLOOR MOUNTED VERTICAL AIR CONDITIONING UNIT.
3 PROVIDE SUSPENDED UNIT HEATER.
4 PROVIDE BAS PANEL FOR DOAS-1.
5 PROVIDE BUILDING NETWORK CONTROL PANEL.
6 PROVIDE OUTDOOR AIR LOUVER. LOUVER MUST BE MINIMUM 10 FEET ABOVE FINISHED GRADE.
7 PROVIDE EXHAUST AIR LOUVER. LOUVER MUST BE MINIMUM 10 FEET ABOVE FINISHED GRADE.
8 ROUTE 1 1/2" CD DOWN TO FLOOR DRAIN.
9 ROUTE 1" CD DN TO FLOOR DRAIN.
10 ROUTE 3/4" CD DOWN TO FLOOR DRAIN.
11 SPACE TEMPERATURE SENSOR ASSOCIATED WITH F-1 AND UH-1.
12 PROVIDE OUTSIDE AIR TEMPERATURE AND HUMIDITY SENSORS. MOUNT OUTSIDE AIR SENSORS UNDER SUN SHIELD, AS HIGH ON THE WALL AS POSSIBLE, ON THE NORTH SIDE OF THE WALL. LOCATION MUST BE SELECTED SUCH THAT SOLAR RADIATION, EXHAUST OR COOLING SOURCES WILL NOT AFFECT SENSORS READING. VALUES MUST BE BROADCASTED TO CONTROLLERS IN THE BUILDING THAT REQUIRE OUTSIDE AIR VALUES

0 2' 4' 8' 12'

1/4" = 1'-0"

PLAN NORTH

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED

FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE 05/27/2025

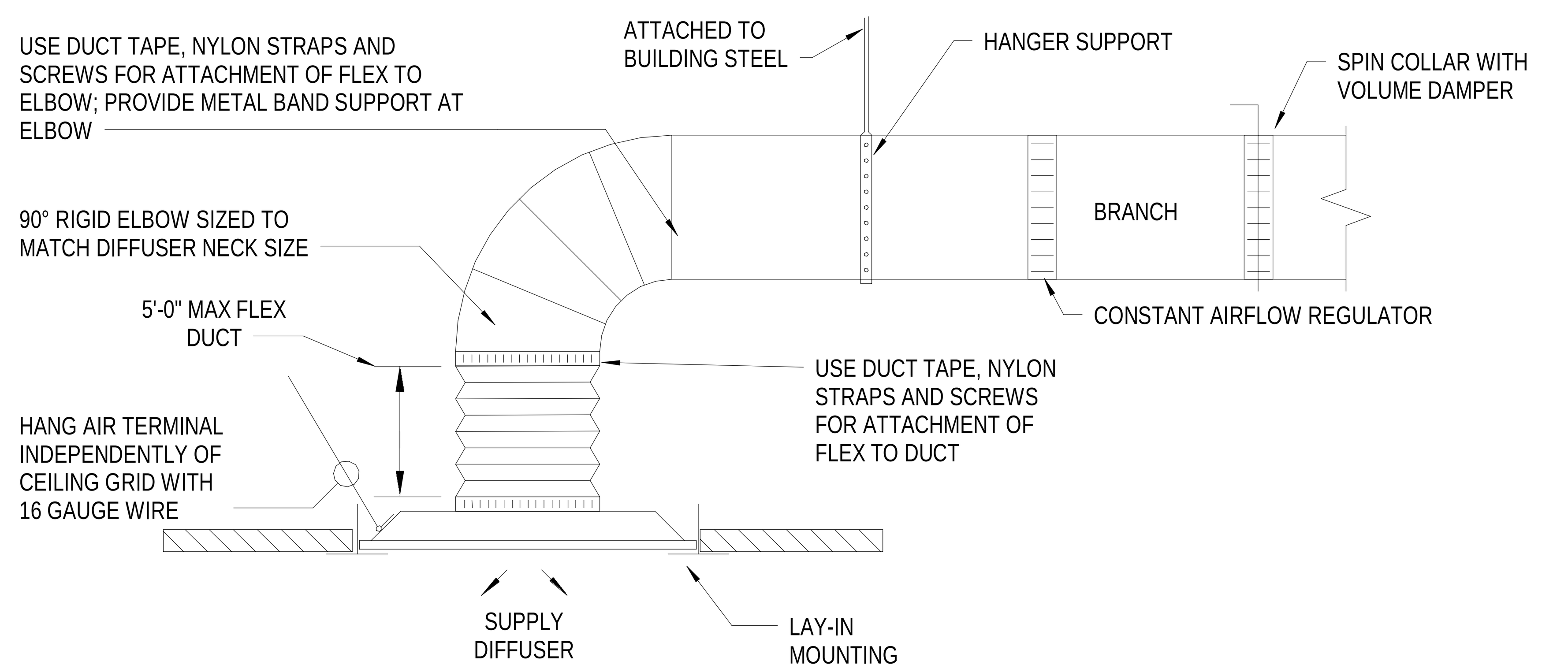
DES	CWH	DRW	SDH	CHK	MJL
PM/DM					SMS/CWJ
BRANCH MANAGER					ALG
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL ENLARGED FLOOR PLAN

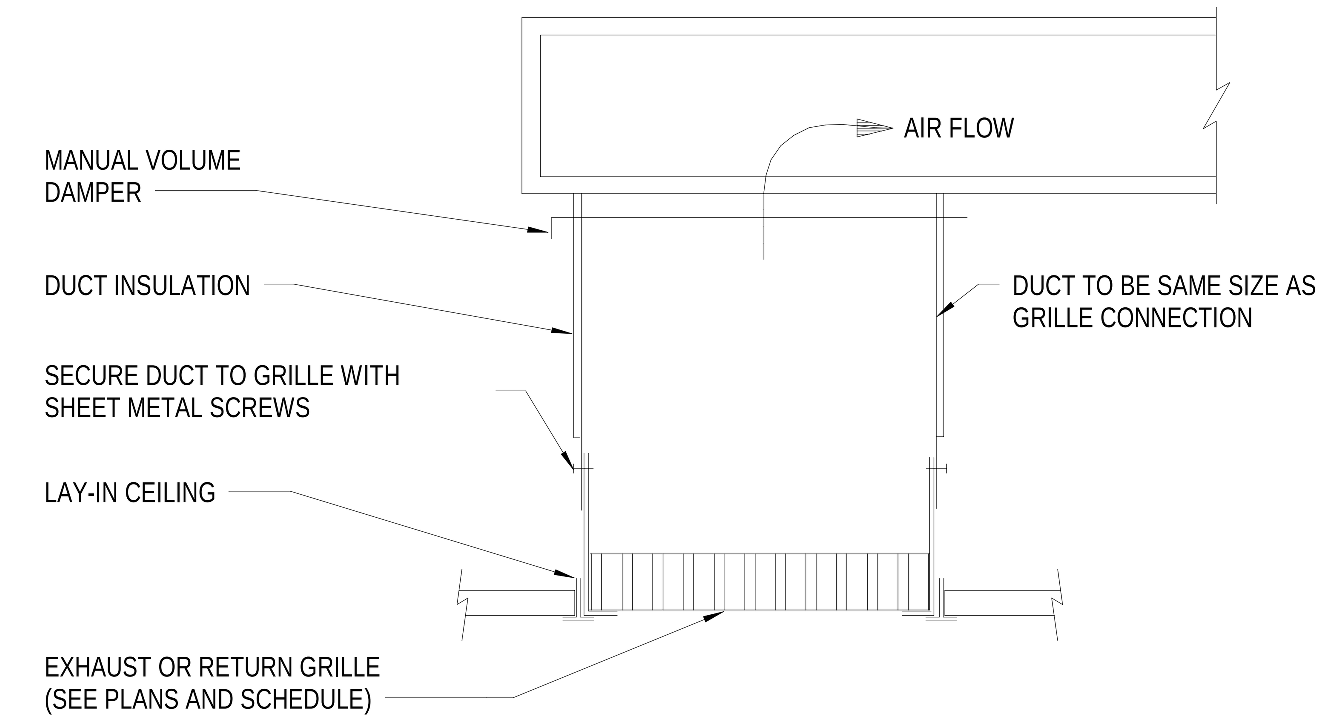
SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894023
SHEET 73 OF 140

M-401

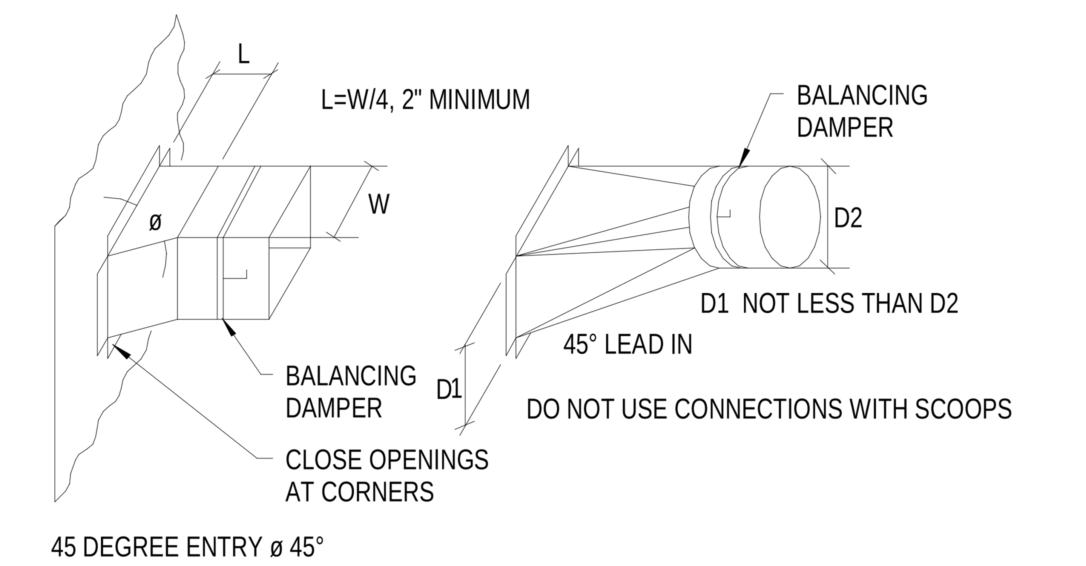
6/20/2025 11:54:31 AM M-401
\\dc1vs01\fedprj\FDZE1628\MECH\BLDG-1667770-M-BIM_v21.rvt



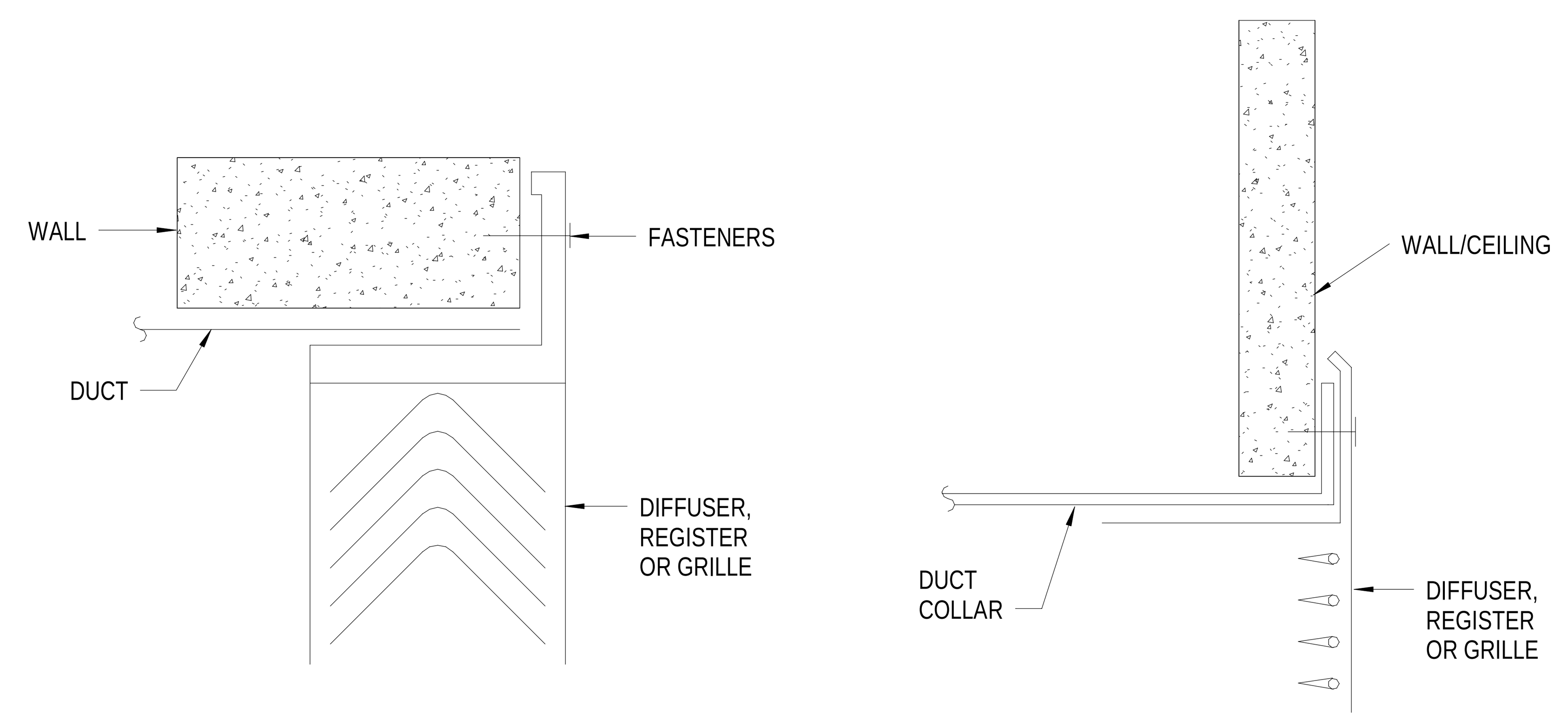
C1 TYPICAL LAY-IN SUPPLY DIFFUSER DETAIL
NTS



C3 TYPICAL LAY-IN RETURN OR EXHAUST GRILLE DETAIL
NTS

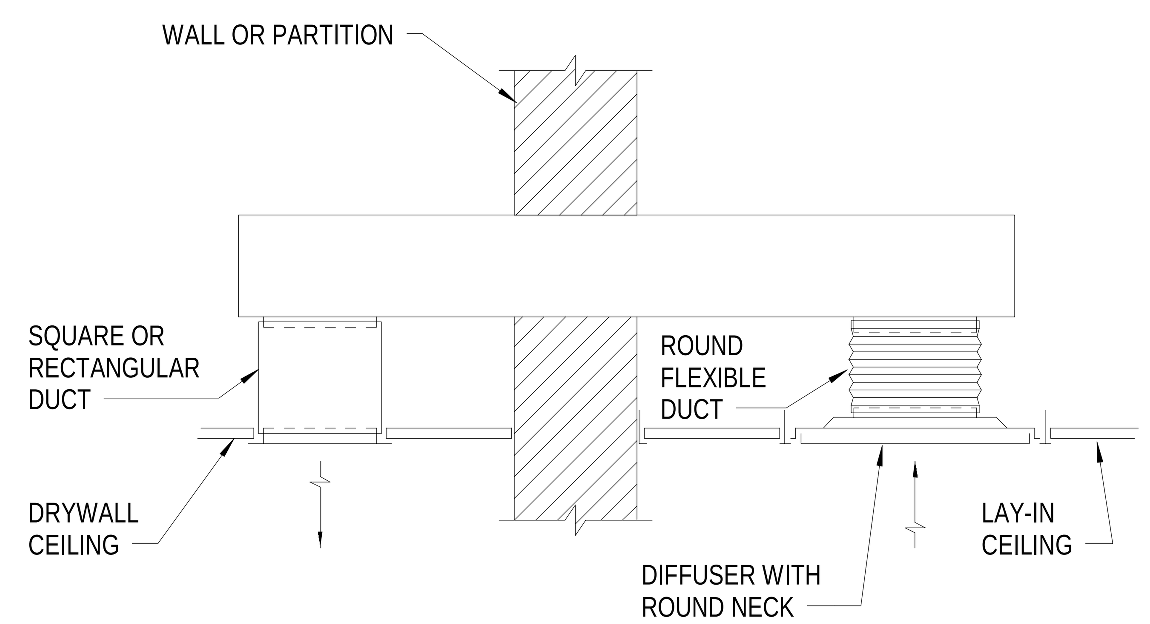


C4 TYPICAL BRANCH DUCT CONNECTION DETAIL
NTS

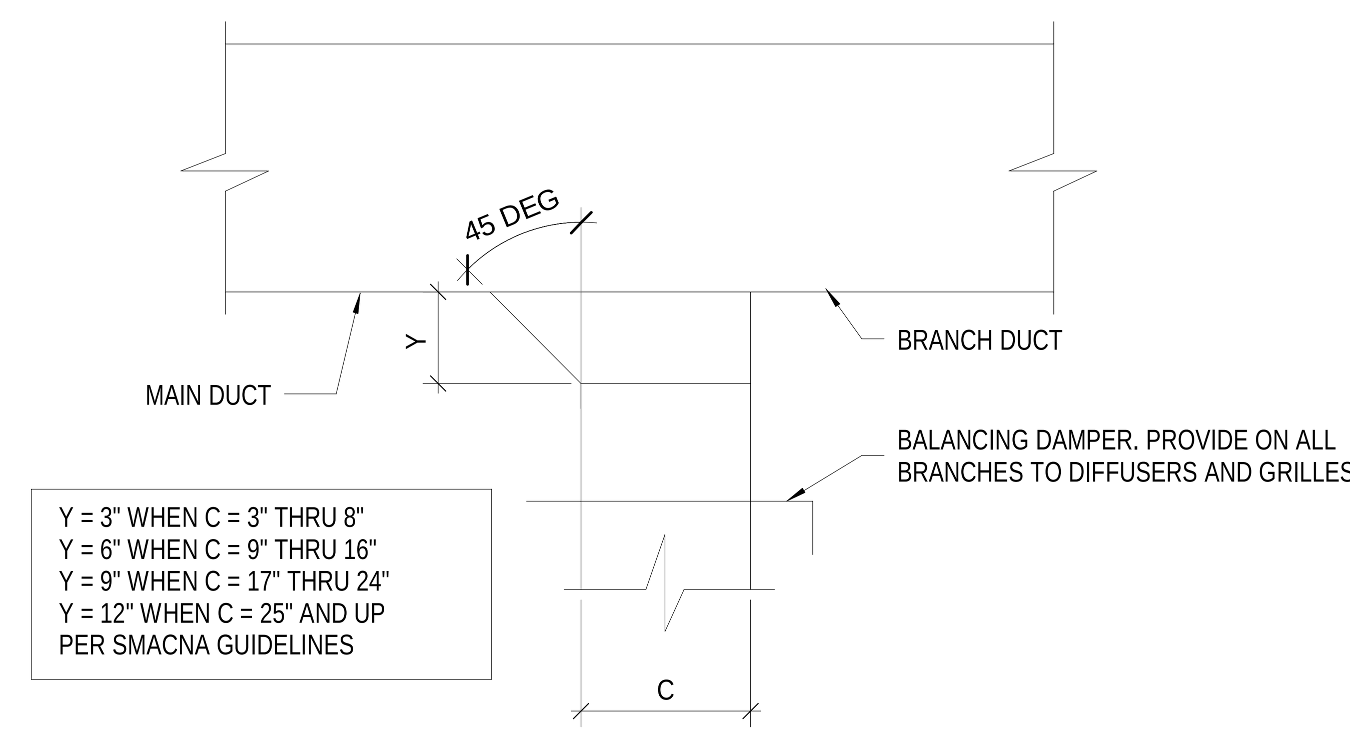


NOTE: DEVICE FLANGES MUST COVER DUCT FLANGES.

B1 TYPICAL DIFFUSER, GRILLE AND REGISTER CONNECTIONS
NTS

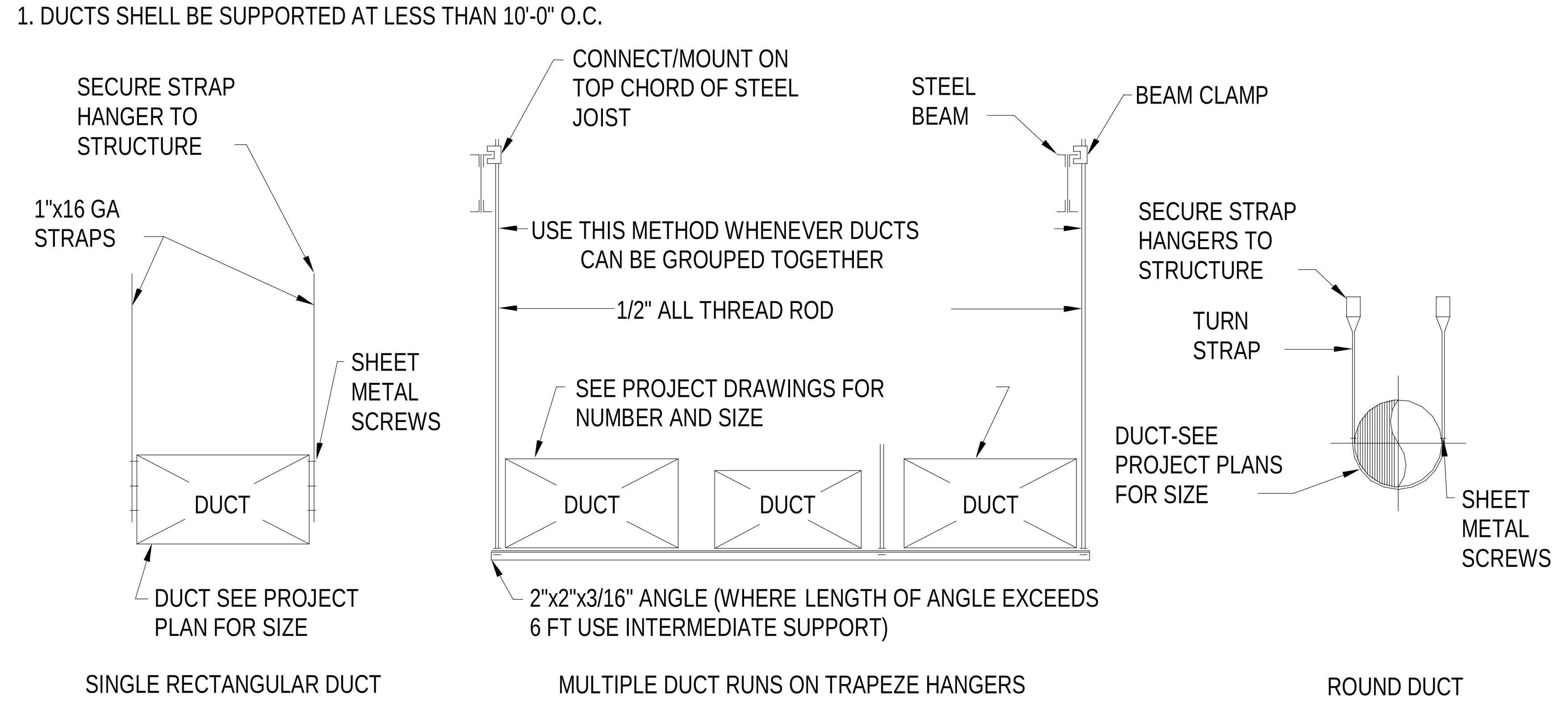


B3 TOILET TRANSFER DUCT DETAIL
NTS

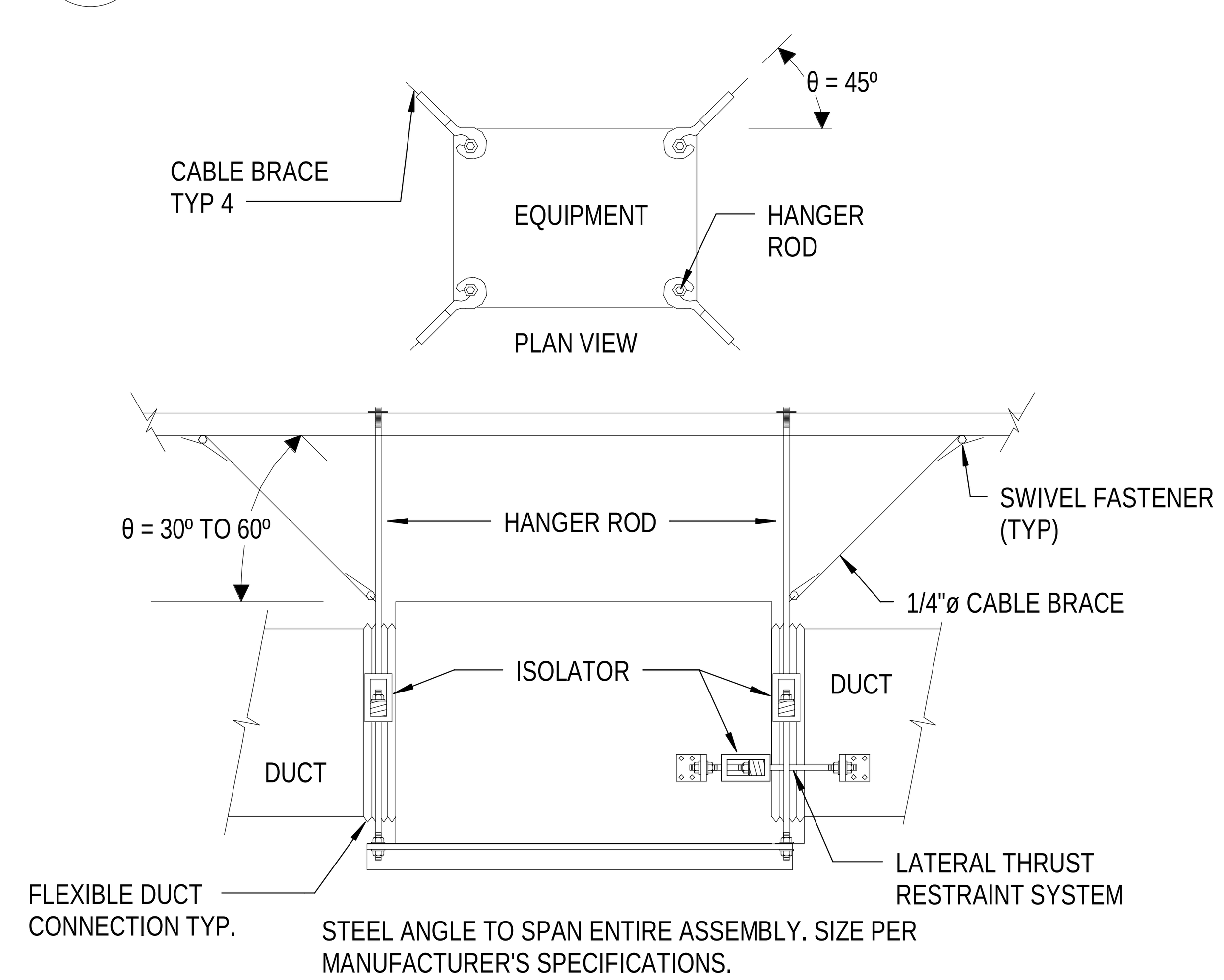


B4 TYPICAL RECTANGULAR DUCT BRANCH CONNECTION DETAIL
NTS

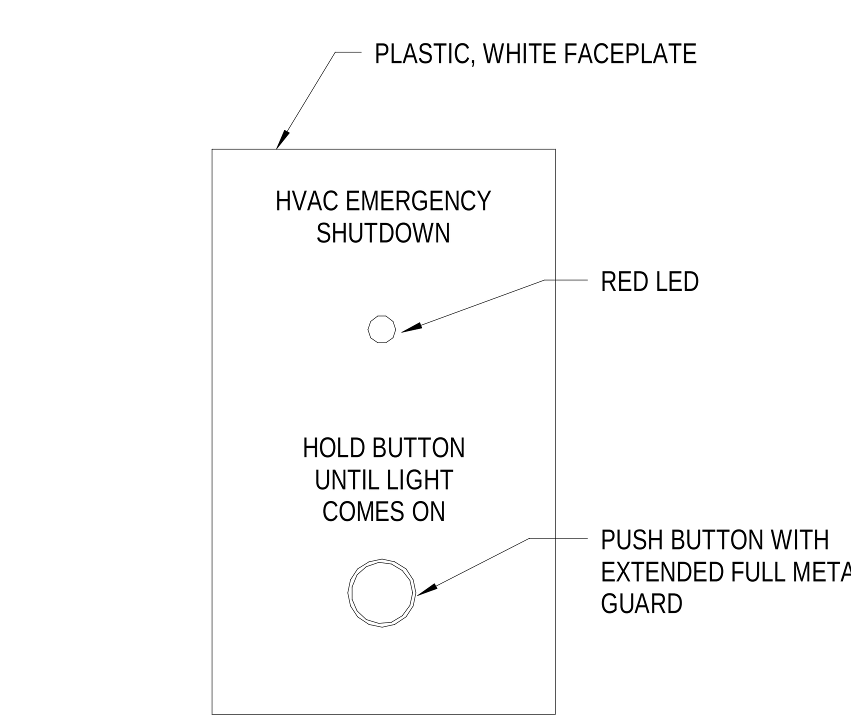
NOTE:
1. DUCTS SHELL BE SUPPORTED AT LESS THAN 10'-0" O.C.



A1 TYPICAL DUCT HANGER AND SUPPORT DETAIL
NTS



A3 HVAC UNIT SUPPORT WITH BRACING DETAIL
NTS



NOTES:
1. TEXT MUST BE ENGRAVED INTO THE FACEPLATE.

A4 AIR DISTRIBUTION EMERGENCY SHUTDOWN SWITCH
NTS

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

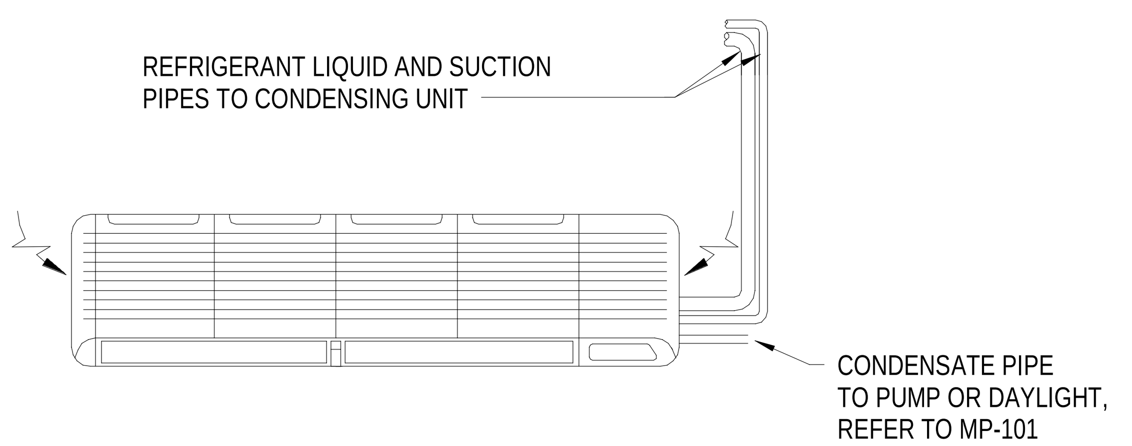
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH DRW CWH CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION DSN

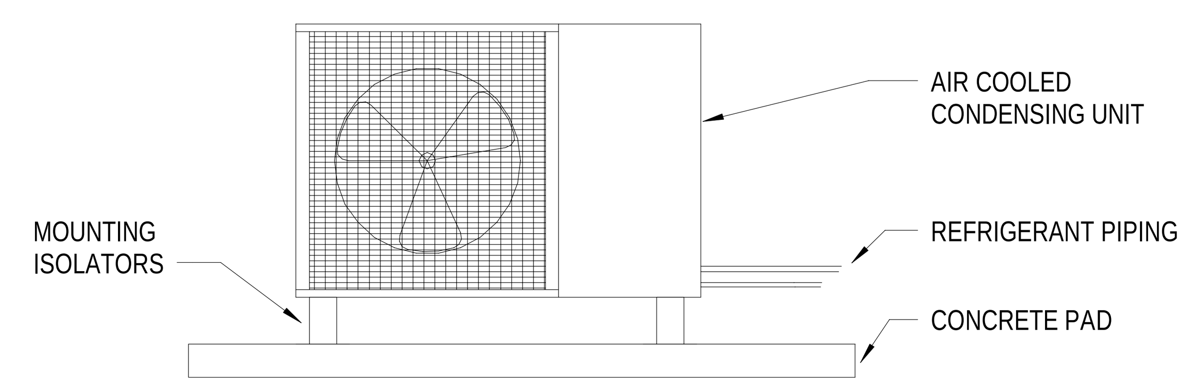
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
MID-ATLANTIC
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894024
SHEET 74 OF 140
M-501

6/20/2025 11:54:33 AM
\\dc1vs01\fedprj\FDZE1628\MECH\BLDG-1667770-M-BIM_v21.rvt
M-501

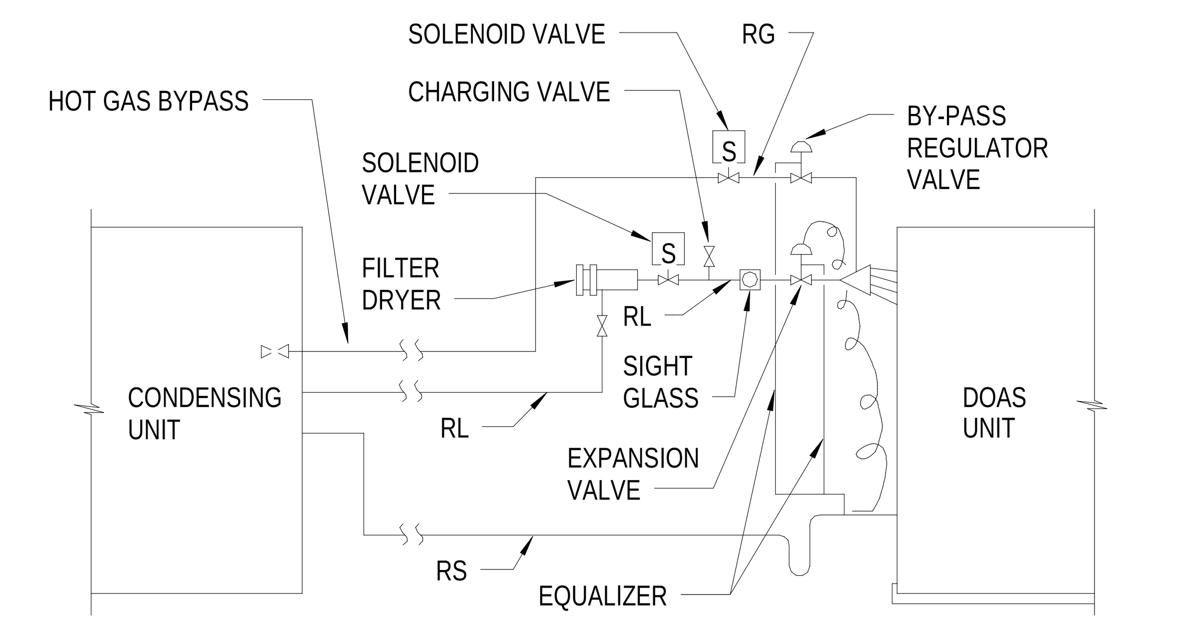


C1 WALL-MOUNTED DUCTLESS SPLIT SYSTEM UNIT DETAIL
NTS

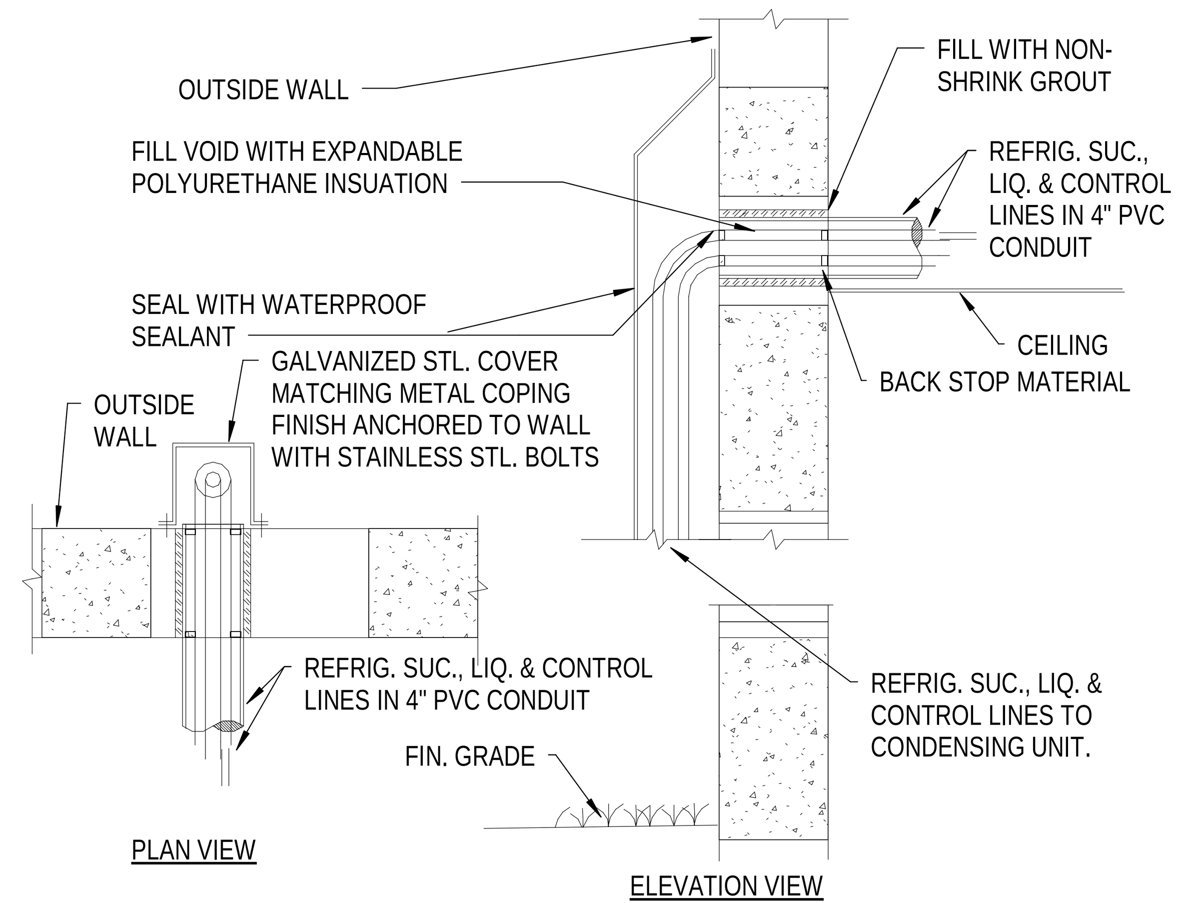


NOTES:
1. MOUNT CONDENSING UNIT TO 6-INCH CONCRETE PAD. SHIM TO LEVEL.
2. ATTACHING AND SECURING OF EQUIPMENT TO PAD MUST BE IN ACCORDANCE WITH MANUFACTURER'S INSTALLATION GUIDELINES.

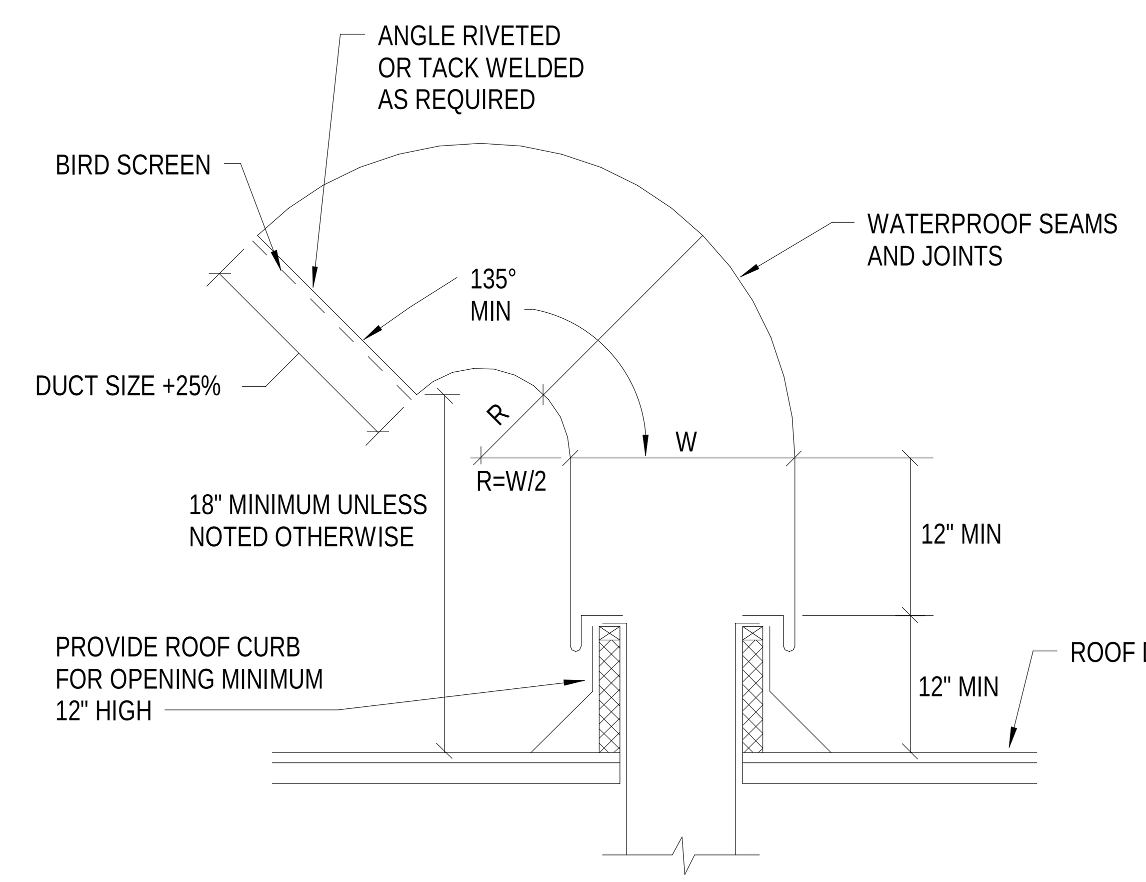
C2 SPLIT SYSTEM CONDENSING UNIT DETAIL
NTS



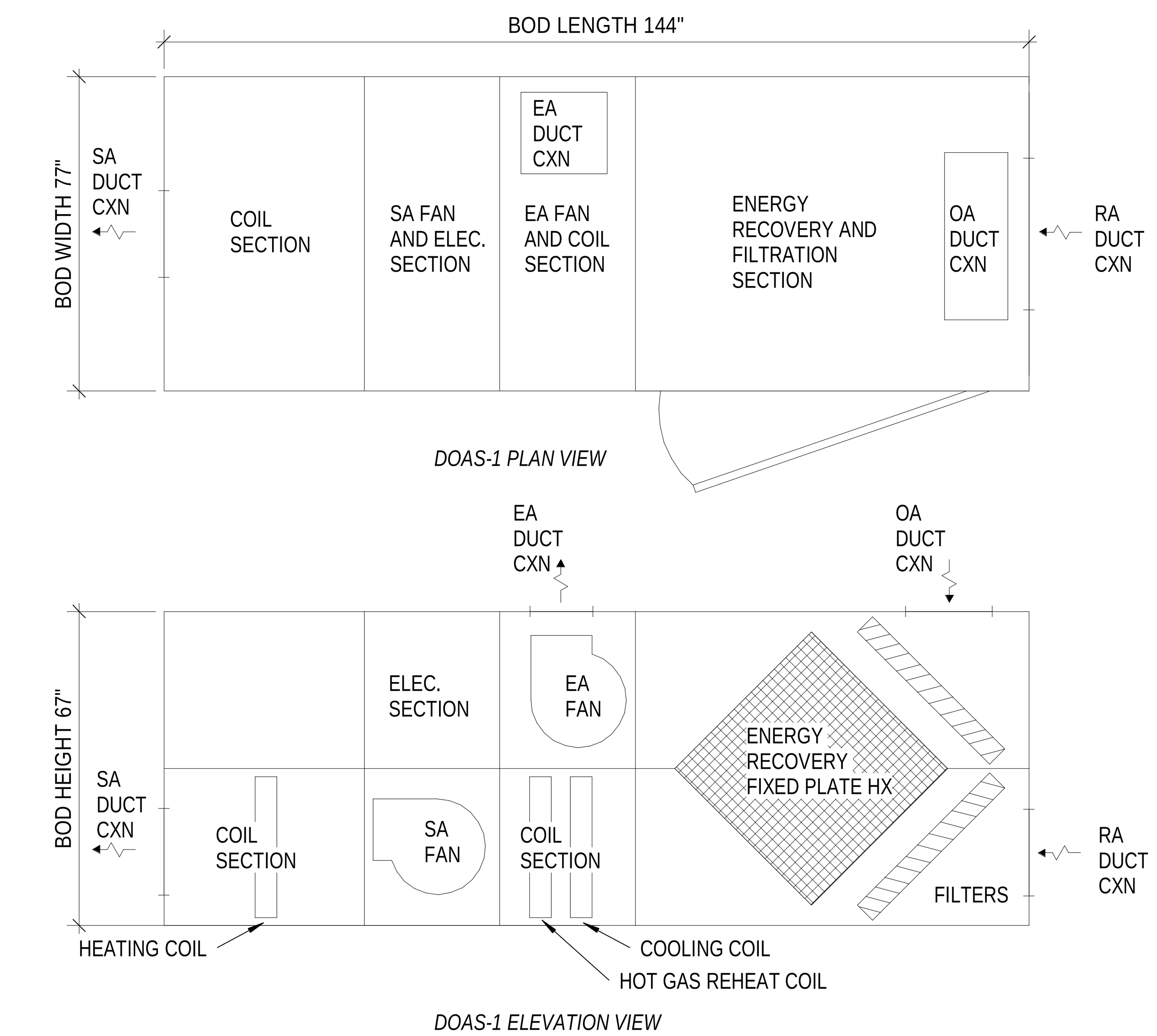
C4 DOAS REFRIGERANT PIPING DETAIL
NTS



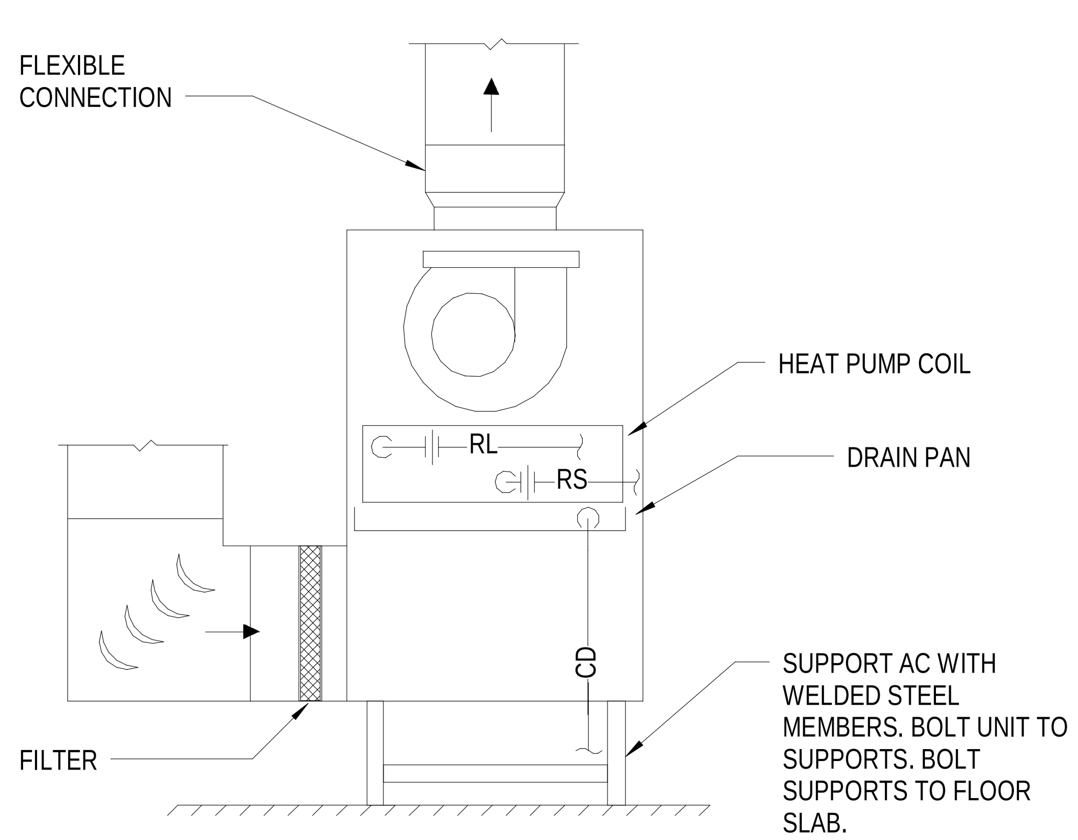
B1 REFRIGERANT PIPE WALL PENETRATIONS
NTS



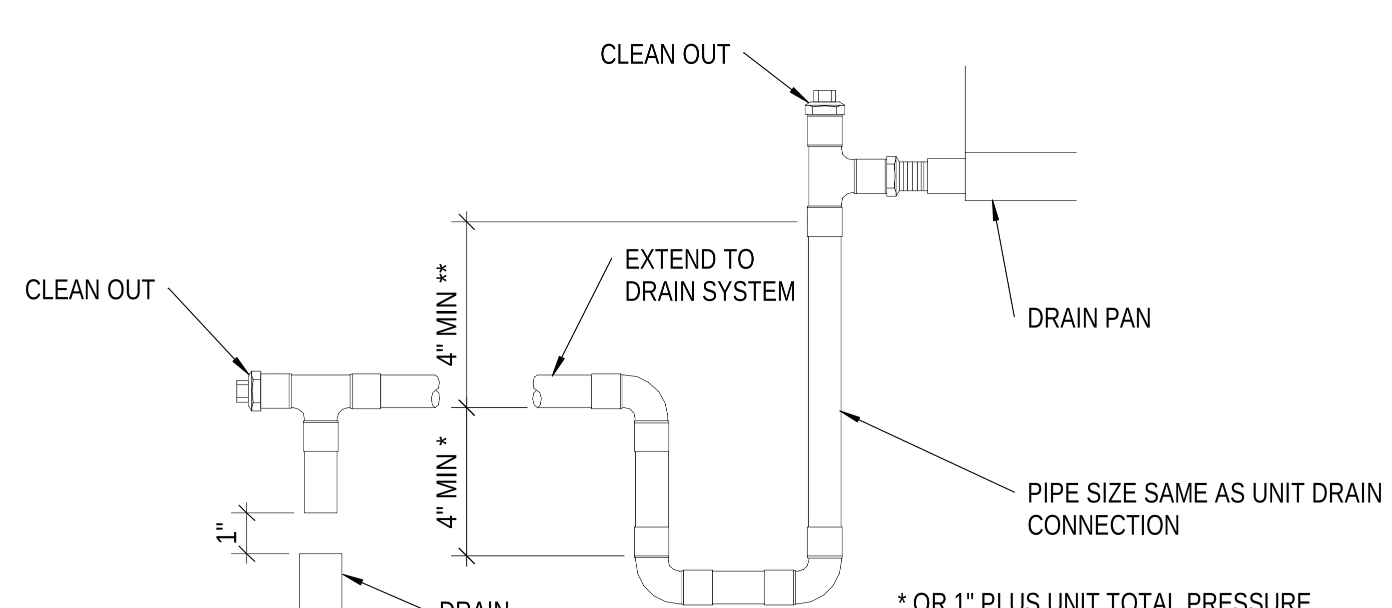
B2 LAUNDRY VENT GOOSENECK DETAIL
NTS



B4 DOAS SECTIONS DETAIL
NTS



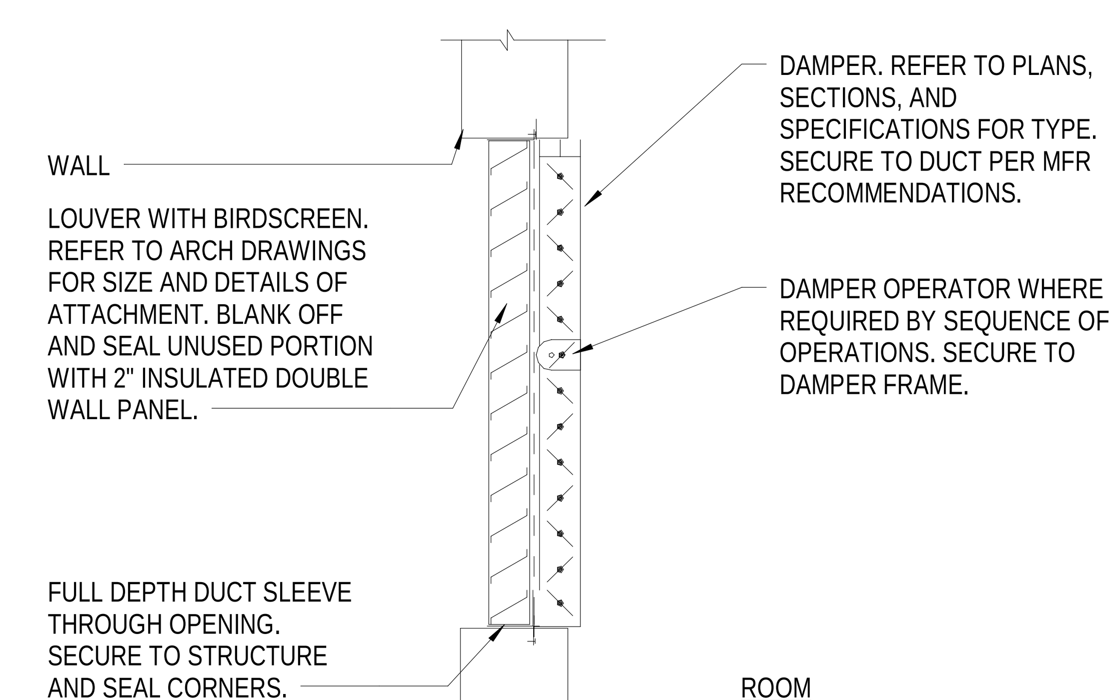
A1 VERTICAL CABINET AC UNIT
NTS



* OR 1" PLUS UNIT TOTAL PRESSURE WHICHEVER IS GREATER FOR BLOW THRU UNIT.

** OR 1" PLUS UNIT TOTAL PRESSURE WHICHEVER IS GREATER FOR DRAW THRU UNIT.

A2 CONDENSATE DRAIN TRAP DETAIL
NTS



A4 LOUVER AND CONTROL DAMPER DETAIL
NTS

SYM	DESCRIPTION	DATE	APPR

NAVFAC

CHRISTOPHER W. HILL
Lic. No. 0402055245
COMMONWEALTH OF VIRGINIA
PROFESSIONAL ENGINEER
6/23/2025
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES CWH	DRW CWH	CHK MJL
PM/DM		SMS/CWJ
BRANCH MANAGER		ALG
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894025
SHEET 75 OF 140

M-502

6/20/2025 11:54:35 AM M-502
\\dc1vs01\fedprj\FDZE1628\MECH\BLDG-1667770-M-BIM_v21.rvt

DEDICATED OUTSIDE AIR UNIT WITH ENERGY RECOVERY SCHEDULE

EQUIPMENT TAG	LOCATION	ENERGY RECOVERY PLATE EXCHANGER								DX COOLING COIL DATA			HOT GAS REHEAT COIL DATA		ELEC HEATING COIL DATA		REFRIGERANT TYPE	FILTER TYPE	FAN DATA				ELECTRICAL DATA			EQUIPMENT WEIGHT (LBS)	NOTES
		MIN. SUPPLY CFM	MAX. SUPPLY CFM	OUTDOOR AIR EAT SUMMER WINTER (°FDB/°FWB)	MIN. EXHAUST CFM	MAX. EXHAUST CFM	EXHAUST EAT SUMMER WINTER (°FDB/°FWB)	EXCHANGER LAT SUMMER WINTER (°FDB/°FWB)	EFFECTIVENESS SUMMER WINTER (%)	LAT DB/WB (°F)	TOTAL CAPACITY (MBH)	SENSIBLE CAPACITY (MBH)	LAT DB/WB (°F)	TOTAL CAPACITY (MBH)	LAT DB/WB (°F)	TOTAL CAPACITY (KW)			SUPPLY MOTOR (HP)	EXHAUST MOTOR (HP)	SUPPLY E.S.P. (IN WG)	EXHAUST E.S.P. (IN WG)	V/PH/HZ	MCA	MOP		
DOAS-1	MECH RM 104	1,565	1,985	S - 82 / 72 W - 14 / 12	1,725	1,800	S - 78.0 / 64.8 W - 68.0 / 56.8	S - 79.2 / 68.5 W - 51.2 / 45.3	S - 58.0 W - 74.1	53.0 / 52.9	98.9	57.3	68.0 / 58.5	32.0	68.0 / 52.8	15.0	R-410A	MERV 8 - EA MERV 13 - OA	5	5	1.0	1.0	460/3/60	39.3	40	2,500	1 THRU 7

NOTES :
1. PROVIDE VARIABLE SPEED DEDICATED OUTDOOR AIR UNIT WITH AIR SOURCE DIRECT EXPANSION COOLING, HOT GAS REHEAT DEHUMIDIFICATION, ELECTRIC RESISTANCE HEATING, ENERGY RECOVERY, AND ECONOMIZER CONTROL.
2. PROVIDE MANUFACTURER'S FACTORY APPLIED COATING SUITABLE FOR A CORROSIVE SALT SPRAY ENVIRONMENT ON EQUIPMENT CASING AND COILS.
3. COORDINATE DUCT ROUTING AND CONNECTIONS WITH FIELD CONDITIONS.
4. PROVIDE SINGLE POINT POWER CONNECTION WITH FACTORY MOUNTED AND WIRED NON-FUSED DISCONNECT SWITCH.
5. PROVIDE FACTORY CONFIGURED VFD FOR FIELD INSTALLATION ON DOAS CASING, DOAS MUST BE FACTORY WIRED.
6. PROVIDE PREMIUM EFFICIENCY, INVERTER DUTY RATED MOTORS SUITABLE FOR USE WITH VFDS. EACH FAN MOTOR MUST HAVE A DEDICATED VFD.
7. PROVIDE FACTORY ISOLATION DAMPERS. OUTDOOR AND EXHAUST AIR DAMPERS MUST BE LOW LEAKAGE DAMPERS WITH A 3 CFM/SF OR LESS LEAKAGE RATE.
8. ENERGY RECOVERY PLATE EXCHANGER TERMPERATURE AND EFFECTIVENESS DATA PROVIDED FOR MAXIMUM SUPPLY AND EXHAUST FLOWS.

SPLIT SYSTEM AIR CONDITIONER SCHEDULE

EQUIPMENT TAG	LOCATION	SUPPLY AIR (CFM)	OUTDOOR AIR (CFM)	ESP (IN WG)	REFRIGERANT TYPE	ELECTRICAL DATA	DX COIL DATA							OPERATING WEIGHT (LBS)	FILTER TYPE	NOTES
							COOLING			HEATING AT 47°F		HEATING AT 17°F				
							TOTAL CAPACITY (MBH)	EAT DB/WB (°F)	LAT DB/WB (°F)	TOTAL CAPACITY (MBH)	RATED CAPACITY (MBH)	TOTAL CAPACITY (MBH)	RATED CAPACITY (MBH)			
AC-1	ELEC RM 103	775	-	-	R-410A	POWER FROM CU	24	85.0 / 70.8	55.4	--	-	-	-	46	MERV8	1 THRU 6, 9, 11
AC-2	TELECOM RM 105	775	-	-	R-410A	POWER FROM CU	30	75.0 / 62.5	52.3	-	-	-	-	46	MERV8	1 THRU 6, 9, 11
AC-3	ENTRY CORRIDOR	880	-	0.60	R-410A	POWER FROM CU	30	78.0 / 65.3	55.3	34	32	32	21	69	MERV13	1 THRU 8, 11
AC-4	MECH RM 104	875	-	0.80	R-410A	POWER FROM CU	30	78.0 / 65.3	53.2	34	32	32	21	141	MERV13	1 THRU 6, 10, 11
AC-5	OPEN OFFICE126	1,200	-	0.60	R-410A	POWER FROM CU	36	78.0 / 65.3	56.2	40	38	38	25.4	86	MERV13	1 THRU 8, 11
AC-6	SIPR 107	600	-	0.60	R-410A	POWER FROM CU	18	78.0 / 65.3	55.6	22	19	13.5	11	62	MERV13	1 THRU 8, 11
AC-7	WEP OFF 109	350	-	0.60	R-410A	POWER FROM CU	12	78.0 / 65.3	59.4	18	14	10.5	8.7	58	MERV13	1 THRU 8, 11
AC-8	STORAGE 112	600	-	0.60	R-410A	POWER FROM CU	18	78.0 / 65.3	56.3	22	19	13.5	11	62	MERV13	1 THRU 8, 11
AC-9	LOCKER RM 114	1,200	-	0.60	R-410A	POWER FROM CU	36	78.0 / 65.3	56.2	40	38	38	25.4	86	MERV13	1 THRU 8, 11

NOTES :
1. PROVIDE SPLIT SYSTEM VARIABLE CAPACITY AIR CONDITIONING UNIT, DX COOLING CONTROL.
2. PROVIDE FACTORY MOUNTED, WIRED, AND CONFIGURED WATER LEVEL SENSOR/SWITCH IN THE CONDENSATE DRAIN PAN INTERLOCKED TO STOP UNIT OPERATION AT DRAIN PAN HIGH LEVEL.
3. WHERE CONDENSATE CANNOT BE GRAVITY DRAINED, PROVIDE MANUFACTURER'S INTEGRAL, FACTORY MOUNTED, WIRED, AND CONFIGURED CONDENSATE PUMP.
4. PROVIDE MANUFACTURER'S DISPOSABLE FILTER MEDIA AT INDOOR UNIT, MINIMUM MERV 8.
5. PROVIDE MANUFACTURER'S MICROPROCESSOR BASED CONTROLLER WITH BUILT IN SOURCE CODE SEQUENCE AND CONTROLS, INCLUDING NECESSARY INTERLOCKS, SAFETIES, AND ALARMS.
6. COORDINATE INTERFACE BETWEEN INDOOR AND OUTDOOR UNITS.
7. SUSPENDED HORIZONTAL DUCTED UNIT. PROVIDE SUPPORT BRACING AND VIBRATION ISOLATION FROM BUILDING STRUCTURE.
8. PROVIDE FACTORY ASSEMBLED FILTER BOX WITH ACCESS PANEL AND 2-INCH THICK MERV13 DISPOSIBLE FILTERS.
9. WALL MOUNTED DUCTLESS UNIT. PROVIDE MANUFACTURER'S FACTORY MOUNTING BRACKETS AND HARDWARE.
10. FLOOR MOUNTED VERTICAL DUCTED UNIT. PROVIDE SUPPORT AND HOUSEKEEPING PAD.
11. WALL-MOUNTED THERMOSTAT PROVIDED BY UNIT MANUFACTURER.

VAV BOX SCHEDULE

EQUIPMENT TAG	LOCATION	AREA SERVED	INLET SIZE (IN)	OUTLET SIZE (IN)	MAXIMUM AIRFLOW (CFM)	MINIMUM AIRFLOW (CFM)	INLET STATIC PRESSURE (IN. WG.)	NOTES
VAV-1	CORRIDOR	TRANING/CONFERENCE ROOM 131	8	12x10	520	100	0.50	1

NOTES :
1. PROVIDE VARIABLE AIR VOLUME TERMINAL UNIT, WITH DEMAND VENTILATION CONTROL.

ELECTRIC UNIT HEATER SCHEDULE

EQUIPMENT TAG	AREA SERVED	TYPE	ELECTRICAL DATA					WEIGHT (LBS)	NOTES
			CAPACITY (KW)	V	PH	HZ	FLA		
UH-1	MECH ROOM 104	PROPELLER	5	208	1	60	24.0	34	1,2
UH-2	VESTIBULE 101	WALL MOUNTED	2	208	1	60	9.6	-	1,2

NOTES :
1. PROVIDE WITH MANUFACTURER'S INTEGRAL UNIT MOUNTED NON-FUSED DISCONNECT SWITCH.
2. PROVIDE WITH MANUFACTURER'S MOUNTING BRACKETS AND HARDWARE.

AIR TERMINAL SCHEDULE

TAG	TYPE	CFM RANGE	NECK SIZE (IN)	NOTES
EG-1	LOUVERED EXHAUST GRILLE	0-75	6x6	1,2,5,7,10
RD-1	SQUARE RETURN/EXHAUST DIFFUSER	0-500	12x12	1,2,3,5,9
RG-1	RETURN GRILLE	0-150	8x6	1, 2, 4
SD-1	SQUARE CEILING DIFFUSER	0-140 141-240 241-300	6" DIA. 8" DIA. 10" DIA.	1,2,3,4,5,8
SG-1	SUPPLY GRILLE	0-120	8x6	1,2,3,4,5,6

NOTES :
1. COORDINATE AIR TERMINAL DEVICE FRAME WITH CEILING TYPE.
2. PROVIDE ALUMINUM GRILLES AND DIFFUSERS.
3. PROVIDE REMOTE DAMPER CONTROLLER FOR DAMPER LOCATIONS ABOVE INACCESIBILE CEILINGS WITH ACCESSORIES FOR PROPER OPERATION.
4. MAXIMUM FLEX DUCT LENGTH MUST BE 5 FT.
5. FLEX DUCT MUST NOT BE USED ABOVE INACCESSIBLE CEILINGS.
6. PROVIDE 1/2" SPACED DOUBLE DEFLECTION, INDIVIDUALLY ADJUSTABLE, LONG DIMENSION BLADES IN FRONT.
7. PROVIDE 1/2" SPACED SINGLE DEFLECTION, INDIVIDUALLY ADJUSTABLE, LONG DIMENSION BLADES IN FRONT.
8. PROVIDE PLAQUE TYPE DIFFUSER WITH 4-WAY THROW PATTERN.
9. PROVIDE PERFORATED FACE RETURN.
10. PROVIDE OPPOSED BLADE DAMPERS FOR GRILLES.

CONDENSING UNIT SCHEDULE

EQUIPMENT TAG	ASSOCIATED INDOOR UNIT	OUTDOOR DESIGN TEMPERATURE		ELECTRICAL DATA					WEIGHT (LBS)	NOTES
		SUMMER DB/WB (F)	WINTER DB (F)	V	PH	HZ	MCA	MOCP		
CU-1	AC-1	87 / 76	14	208	1	60	19	26	151	1 THRU 6
CU-2	AC-2	87 / 76	14	208	1	60	19	26	151	1 THRU 6
CU-3	AC-3	87 / 76	14	208	1	60	24	40	261	1 THRU 6
CU-4	AC-4	87 / 76	14	208	1	60	24	40	261	1 THRU 6
CU-5	AC-5	87 / 76	14	208	1	60	24	40	261	1 THRU 6
CU-6	AC-6	87 / 76	14	208	1	60	11	28	99	1 THRU 6
CU-7	AC-7	87 / 76	14	208	1	60	11	28	93	1 THRU 6
CU-8	AC-8	87 / 76	14	208	1	60	11	28	99	1 THRU 6
CU-9	AC-9	87 / 76	14	208	1	60	24	40	261	1 THRU 6
CU-10	DOAS-1	87 / 76	14	460	3	60	15.1	20	345	1 THRU 3, 6

NOTES :
1. PROVIDE SPLIT SYSTEM VARIABLE CAPACITY CONDENSING UNIT, WITH VARIABLE COMPRESSOR.
2. PROVIDE MANUFACTURER'S FACTORY APPLIED COATING SUITABLE FOR A CORROSIVE SALT SPRAY ENVIRONMENT ON CASING AND COILS.
3. PROVIDE SINGLE POINT POWER CONNECTION WITH FACTORY MOUNTED AND WIRED FUSED DISCONNECT SWITCH, NEMA 4X OUTDOOR USE RATED.
4. OUTDOOR CONDENSER UNIT MUST POWER INDOOR EVAPORATOR UNIT.
5. PROVIDE MANUFACTURER'S FACTORY LOW AMBIENT KIT AND WIND BAFFLE.
6. COORDINATE INTERFACE BETWEEN INDOOR AND OUTDOOR UNITS.

DUCTWORK CONSTRUCTION AND LEAKAGE TESTING SCHEDULE

SYSTEM	DUCT PRESSURE CLASS				DUCTWORK				DUCT TEST PRESSURE: INCHES OF WATER COLUMN	NOTES
	INCHES OF WATER COLUMN				ROUND / OVAL		RECTANGULAR			
	SUPPLY DUCT	RETURN DUCT	EXHAUST DUCT	OUTSIDE AIR DUCT	DUCT SEAL CLASS	DUCT LEAKAGE CLASS	DUCT SEAL CLASS	DUCT LEAKAGE CLASS		
AC-3,4,5,6,7,8,9	+1	-1	-	-	A	8	A	16	1.0	1,2,3
DOAS-1	+2	-2	-2	+2	A	4	A	8	2.0	1,2,3

NOTES:
1. BALANCER MUST NOT SUM THE AIRFLOWS AT EACH AIR TERMINAL DEVICE WITHOUT ACCURATELY MEASURING THE TOTAL FLOW RATE AT THE FAN FOR COMPARISON AND DETERMINATION OF APPROXIMATE LEAKING RATES.
2. TEST IN ACCORDANCE WITH UFGS 23 05 93, TESTING, ADJUSTING AND BALANCING FOR HVAC AND THE PROCEDURES IN SMACNA'S HVAC AIR DUCT LEAKAGE TEST MANUAL.
3. STEEL SHEET AND STRIP USED FOR DUCTWORK AND CONNECTORS MUST BE G-90 COATED GALVANIZED STEEL OF LOCKFORMING GRADE CONFORMING TO ASTM A653 AND A924 STANDARDS. MINIMUM YIELD STRENGTH FOR SHEET STEEL AND REINFORCEMENTS MUST BE 30,000 PSI.

OUTDOOR AIR SCHEDULE

ZONE	OCCUPANCY COUNT (SEE NOTE 1)	OA CFM
3 - CONFERENCE ROOM	51	520
4 - OFFICES (127,128,129,130) & COFFEE	10	305
5 - OPEN OFFICE	32	480
6 - SIPR & OIS	3	100
7 - OFFICE 109	3	60
8 - ARMS ROOM & STORAGE	0	180
9 - BUNKS, RRs, LOCKERS, JAN, & LAC	33	340

NOTES:
1. THE NUMBER OF OCCUPANTS LISTED IS FOR INFORMATION PURPOSES ONLY.

FAN SCHEDULE

TAG	LOCATION	AREA SERVED	FAN TYPE	DRIVE	LOW SPEED AIRFLOW (CFM)	HIGH SPEED AIRFLOW (CFM)	ESP (IN. WC)	MOTOR DATA			ELECTRICAL DATA				NOTES
								BHP	NOM. HP	RPM	FLA	MCA	MOP	V/PH/HZ	
F-1	MECH RM 104	MECH RM 104	INLINE	DIRECT	400.00	700	0.40	0.09	1/2	976	4	5	15	208/1/60	1, 2, 3, 4

NOTES:
1. PROVIDE WITH FACTORY MOUNTED AND WIRED NON-FUSED DISCONNECT.
2. PROVIDE ELECTRONICALLY COMMUTATED MOTOR WITH THERMAL OVERLOAD PROTECTION AND SPEED CONTROL.
3. PROVIDE OSHA APPROVED FAN MOTOR GUARD.
4. PROVIDE MANUFACTURER'S FACTORY APPLIED COATING SUITABLE FOR A CORROSIVE SALT SPRAY ENVIRONMENT.

CONDENSATE PUMP SCHEDULE

TAG	DIMENSIONS (IN)			FLOW (GPH)	HEAD (FT)	FLOW (GPH) @ HEAD (FT)				ELECTRICAL					NOTES
	LENGTH	WIDTH	HEIGHT			1'	5'	10'	20'	HP	V	HZ	P	FLA	
CDP-1	11	5	7	48-80	1-10	80	70	48	0	1/30	115	60	1	1.5	1, 3
CDP-2	11	5	7	48-80	1-10	80	70	48	0	1/30	115	60	1	1.5	1, 2

NOTES:
1. PROVIDE 0.5 GALLON MINIMUM RECEIVER TANK.
2. PROVIDE HARDWIRED POWER CONNECTION.
3. PROVIDE PLUG-IN TYPE CONNECTION.

DAMPER SCHEDULE

DAMPER TAG	SYSTEM EQUIPMENT	DESCRIPTION	DESIGN DUCT SIZE		BLADE ACTION	FAIL POSITION	CONTROL ACTUATOR			NOTES
			WIDTH (INCHES)	HIGHT (INCHES)			CONTROL SIGNAL	CONTROL VOLTAGE	POSITION FEEDBACK	
OA-D	DOAS-1	OUTSIDE AIR DAMPER	16	36	PARALLEL	CLOSE	TWO POSITION	24 VAC	YES	1
EA-D	DOAS-1	EXHAUST AIR DAMPER	16	20	PARALLEL	CLOSE	TWO POSITION	24 VAC	YES	1
FBYP-D	DOAS-1	HEAT EXCHANGER OUTSIDE AIR BYPASS DAMPER	---	---	PARALLEL	----	MODULATING	24 VAC	YES	1, 2
OA-D-1	F-1	OUTSIDE AIR DAMPER	22	22	PARALLEL	CLOSE	TWO POSITION	24 VAC	YES	
OA-D-2	F-1	OUTSIDE AIR DAMPER	16	16	PARALLEL	CLOSE	TWO POSITION	24 VAC	YES	

NOTES:
1. DAMPER PROVIDED WITH THE UNIT.
2. SEE DOAS-1 CONTROL DIAGRAM FOR DAMPER FAIL POSITION

DESIGN CONDITIONS

14th WEATHER SQUADRON STATION ID	ICAO_KGON
LOCATION	GROTON NEW LONDON, CONNECTICUT
CLIMATE ZONE	5A
ELEVATION	9 FEET
OUTDOOR DESIGN CONDITIONS - COMFORT COOLING	
SUMMER	82°F DB / 72°F WB (1% DATA)
SUMMER DEHUMIDIFICATION	121 GR/LB (75°F WB) / 80°F MCDB (0.4% DATA)
CONDENSING UNITS	87°F DB (5°F ADDED TO SUMMER DESIGN)
WINTER	14°F DB / 12 WB (99% DATA)
INDOOR DESIGN SET POINTS - GENERAL OCCUPIED AREAS	
COOLING	78°F
HEATING	68°F
DEHUMIDIFICATION	50% RH / 57.9°F DEWPOINT
INDOOR DESIGN SET POINTS - ELECTRICAL AND TELECOM ROOMS	
COOLING	75°F
INDOOR DESIGN SET POINTS - VENTILATED MECHANICAL ROOM	
COOLNIG	10°F ABOVE AMBIENT
HEATING	55°F WINTER

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH DRW CWH CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL SCHEDULES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894026
SHEET 76 OF 140
M-601

CONTROL SCHEMATIC SYMBOLS

W-X-Y-Z ## — DEVICE SYMBOL, GENERIC
USED TO REPRESENT SENSORS, INSTRUMENTS SFETIES, AND OTHER CONTROL DEVICES
W-X-Y-Z (SEE ABBREVIATIONS AND ACRONYMS)
GENERIC USAGE TO SHOW LOCATION OF DEVICE, PROCESS OR DEVICE BEING MEASURED OR CONTROLLED, MODIFIERS AND OTHER APPLICATION INFORMATION

W-X-Y-Z ## — DEVICE SYMBOL, GENERIC
MULTIPLE OUTPUT DEVICE WHERE AT LEAST 1 OUTPUT IS ALWAYS SAFETY INTERLOCK

W-X-Y-Z DDC ## DIR REV — DEVICE SYMBOL, CONTROLLER
DDC : DIRECT DIGITAL CONTROL
: DEVICE NUMBER
DIR : DIRECT CONTROL ACTION, OR
REV : REVERSE CONTROL ACTION
W-X-Y-Z : (SEE ABBREVATIONS AND ACRONYMS)
GENERIC USAGE TO SHOW LOCATION OF DEVICE, PROCESS OR DEVICE BEING MEASURED OR CONTROLLED, SIGNAL TYPE, MODIFIERS AND OTHER APPLICATION INFORMATION

ACTUATOR, ELECTRIC

AFMA — AIR FLOW MEASUREMENT ARRAY, WITH FLOW TRANSMITTER
SEE "DEVICE SYMBOL, GENERIC"

BOILER

CHILLER

C — COIL, COOLING

DX — COIL, COOLING, DIRECT-EXPANSION

HC — COIL, HOT GAS REHEAT

H — COIL, HEATING

GF — COIL, GAS FURNACE

PH — COIL, PREHEAT

CONTACT, PRESSURE, HIGH LIMIT (HL)
SHOWN IN ACTIVATED POSITION
OPEN WHEN PRESSURE IS ABOVE SETPOINT
CLOSED WHEN PRESSURE IS BELOW SETPOINT

CONTACT, PRESSURE, HIGH LIMIT (HL)
SHOWN IN UNACTIVATED POSITION
OPEN WHEN PRESSURE IS ABOVE SETPOINT
CLOSED WHEN PRESSURE IS BELOW SETPOINT

CONTACT, PRESSURE, LOW LIMIT (LL)
SHOWN IN ACTIVATED POSITION
OPEN WHEN PRESSURE IS BELOW SETPOINT
CLOSED WHEN PRESSURE IS ABOVE SETPOINT

CONTACT, PRESSURE, LOW LIMIT (LL)
SHOWN IN UNACTIVATED POSITION
OPEN WHEN PRESSURE IS BELOW SETPOINT
CLOSED WHEN PRESSURE IS ABOVE SETPOINT

CONTACT, TEMPERATURE, HIGH LIMIT (HL)
SHOWN IN ACTIVATED POSITION
OPEN WHEN TEMPERATURE IS ABOVE SETPOINT
CLOSED WHEN TEMPERATURE IS BELOW SETPOINT

CONTACT, TEMPERATURE, HIGH LIMIT (HL)
SHOWN IN UNACTIVATED POSITION
OPEN WHEN TEMPERATURE IS ABOVE SETPOINT
CLOSED WHEN TEMPERATURE IS BELOW SETPOINT

CONTACT, TEMPERATURE, LOW LIMIT (LL)
SHOWN IN ACTIVATED POSITION
OPEN WHEN TEMPERATURE IS BELOW SETPOINT
CLOSED WHEN TEMPERATURE IS ABOVE SETPOINT

CONTACT, TEMPERATURE, LOW LIMIT (LL)
SHOWN IN UNACTIVATED POSITION
OPEN WHEN TEMPERATURE IS BELOW SETPOINT
CLOSED WHEN TEMPERATURE IS ABOVE SETPOINT

CONTROL SIGNAL LINE, LOW VOLTAGE

CURRENT TRANSFORMER/SWITCH

DAMPER, BALANCING

D-## — DAMPER, PARALLEL BLADE
NORMALLY OPEN OR CLOSED AS SHOWN
= DAMPER NUMBER AS SHOWN IN DAMPER SCHEDULE

D-## — DAMPER, OPPOSED BLADE
NORMALLY OPEN OR CLOSED AS SHOWN
= DAMPER NUMBER AS SHOWN IN DAMPER SCHEDULE

HEAT EXCHANGER

MAIN AIR — MAIN AIR

M # — MOTOR

MS-# — MOTOR STARTER

Rxx — RELAY COIL
XX = SEQUENTIAL RELAY (SOLENOID) NUMBER

Rxx-YY — REAL COIL CONTACT
NORMALLY OPEN
XX = RELAY COIL NUMBER
YY = CONTACT NUMBER

Rxx-YY — REAL COIL CONTACT
NORMALLY OPEN
XX = RELAY COIL NUMBER
YY = CONTACT NUMBER

FAN

FILTER

LDP ## — LOCAL DISPLAY PANEL
= SEQUENTIAL NUMBER (WHEN MORE THAN 1 LDP)

P — PRESSURE GAGE

P — PRESSURE WITCH, DIFFERENTIAL, WITH GAGE
H = HIGH PRESSURE TAP
L = LOW PRESSURE TAP
SEE "DEVICE SYMBOL, GENERIC"

PUMP

RADIATOR

RST-BUT — RESET BUTTON FOR SAFETY RESET

SMOKE DETECTOR, DUCT-MOUNTED
SEE "DEVICE SYMBOL, GENERIC"

SWITCH, MANUAL

T — TEMPERATURE GAGE

TEMPERATUR SENSOR, AVARAGE or TEMPERATURE LOW LIMIT / FREEZESTAT
SEE "DEVICE SYMBOL, GENERIC"

TEMPERATUR SENSOR, POINT
SEE "DEVICE SYMBOL, GENERIC"

STAT — THERMOSTAT WITH SPECIFIED I/O FUNCTIONS

THERMOWELL IN PIPE

VALVE, BALANCING

VRF INTELLIGENT ROUND DAMPER

V-## NC — VALVE, NORMALLY CLOSED
WITH SPRING RETURN FAILSAFE
= VALVE NUMBER A SHOWN IN VALVE SCHEDULE

V-## NO — VALVE, NORMALLY OPEN
WITH SPRING RETURN FAILSAFE
= VALVE NUMBER A SHOWN IN VALVE SCHEDULE

V-## COM NO — VALVE, 3-WAY DIVERTING WITH SPRING RETURN FAILSAFE
NO = NORMALLY OPEN
NC = NALRMALLY CLOSED
COM = COMMON PORT
= VALVE NUMBER AS SHOWN IN VALVE SCHEDULE

V-## NO COM — VALVE, 3-WAY MIXING WITH SPRING RETURN FAILSAFE
NO = NORMALLY OPEN
NC = NALRMALLY CLOSED
COM = COMMON PORT
= VALVE NUMBER AS SHOWN IN VALVE SCHEDULE

VFD XX-C XX-SS XX-S — VARIABLE FREQUENCY DRIVE
XX = PROCESS/DEVICE BEING CONTROLLED
SS = START/STOP COMMAND
S = STATUS FEEDBACK (ON/OFF)
C = 4 - 20 mA, VDC, or SNVT CONTROL SIGNAL

ABBREVIATIONS AND ACRONYMS

2P	TWO - POSITION (CONTROL)
ADJ	ADJUSTABLE
ALM	ALARM
BA	BYPASS AIR
BLDG	BUILDING
BLR	BOILER
BUT	BUTTON
C	COMMAND (MODULATING CONTROL SIGNAL)
CF	CONDESER FAN
CHLR	CHILLER
CLG	COOLING
CO2	CARBON DIOXIDE
COM	COMMON
COMP	COMPRESSOR
CR	CONDENSER/CONDENSATE WATER RETURN
CS	CONDENSER/CONDENSATE WATER SUPPLY
CSR	CURRENT SENSING RELAY
CT	CURRENT TRANSFORMER /SWITCH
CTF	COOLING TOWER FAN
CHWR	CHILLED WATER RETURN
CHWS	CHILLED WATER SUPPLY
CHW	CHILLED WATER
D	DAMPER
DA	DISCHARGE AIR
DB	DEADBAND
DDC	DIRECT DIGITAL CONTROL(LER)
DIFF	DIFFERENCE
DIR	DIRECT (CONTROL ACTION)
DIS	DIABLE
DISP	DISPLAY
DT	DUAL TEMP
DX	DIRECT EXPANSION (UNIT)
EA	EXHAUST AIR
ECO	ECONOMIZER
EF	EXHAUST FAN
EMCS	ENERGY MONITORING AND CONTROL SYSTEM
ENA	ENABLE
F	FLOW
FAP	FIRE ALARM PANEL
FLT	FILTER
HD	HOT DECK
HL	HIGH LIMIT
HTG	HEATING
HTHW	HIGH TEMPERATURE HOT WATER
HUM	HUMIDIFIER
HW	HOT WATER
HWR	HOT WATER RETURN
HWS	HOT WATER SUPPLY
I/O	INPUT/OUTPUT
L1	LOAD
LDR	LOCAL DISPLAY PANEL
LL	LOW LIMIT
M	MOTOR or MAIN
M&C	MONITORING & CONTROL (SOFTWARE)
MA	MIXED AIR
MINOA	MINIMUM OUTSIDE AIR
MS	MOTOR STARTER
N	NEUTRAL
N/A	NOT APPLICABLE
NC	NORMALLY CLOSED
NCI	NETWORK CONFIGURATION INPUT
NO	NORMALLY OPEN
OA	OUTSIDE AIR
OCC	OCCUPIED
ODT	ON DELAY TIMET
OWS	OPERATOR WORKSTATION
P	PRESSURE
PC	PRE-COOLING
PCW	PRIMARY CHILLER WATER
PCWR	PRIMARY CHILLER WATER RETUR
PCWS	PRIMARY CHILLER WATER SUPPLY
PH	PREHEAT
PID	PROPORTIONAL INTEGRAL DERIVATIVE (CONTROL)
PMP	PUMP
R	RELAY
RA	RETURN AIR
REV	REVERSE (CONTROL ACTION)
RF	RETURN FAN
RH	RELATIVE HUMIDITY
RLA	RELIEF AIR
RM	ROOM
RST	RESET
RTN	RETURN
S	STATUS
SA	SUPPLY AIR
SEC	SECONDARY
SF	SUPPLY FAN
SMK	SMOKE
SNVT	STANDARD NETWORK VARIABLE TYPE
SP	SETPOINT
SS	START/STOP COMMAND
STAT	THERMOSTAT
STM	STEAM
SYS	SYSTEM
SCHD	SCHEDULER
T	TEMPERATURE
TAP	TAP PRESSURE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

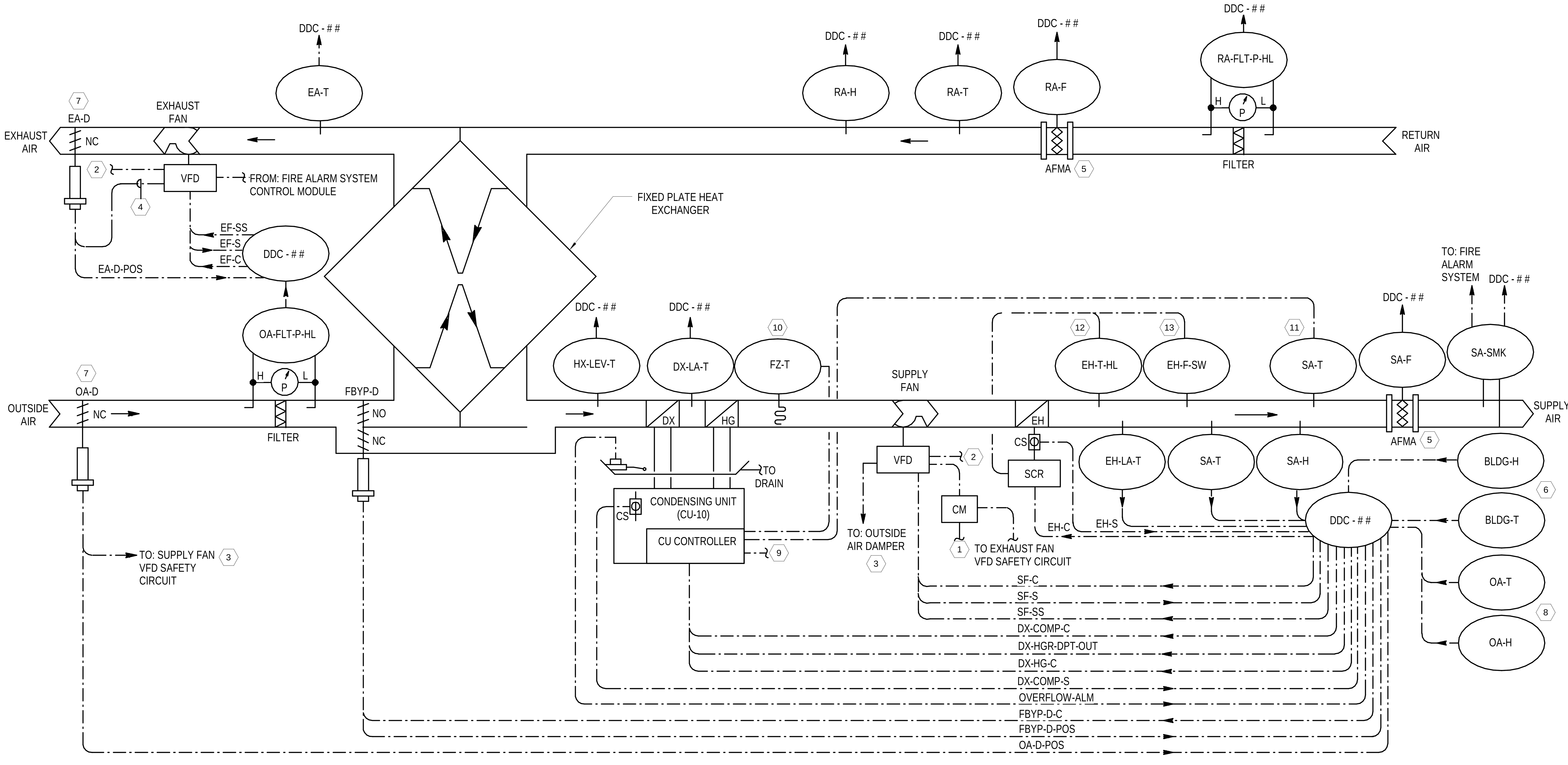
APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES CWH	DRW CWH
CHK	MJL
PM/DM	SMS/CWJ
BRANCH MANAGER	ALG
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	DSN

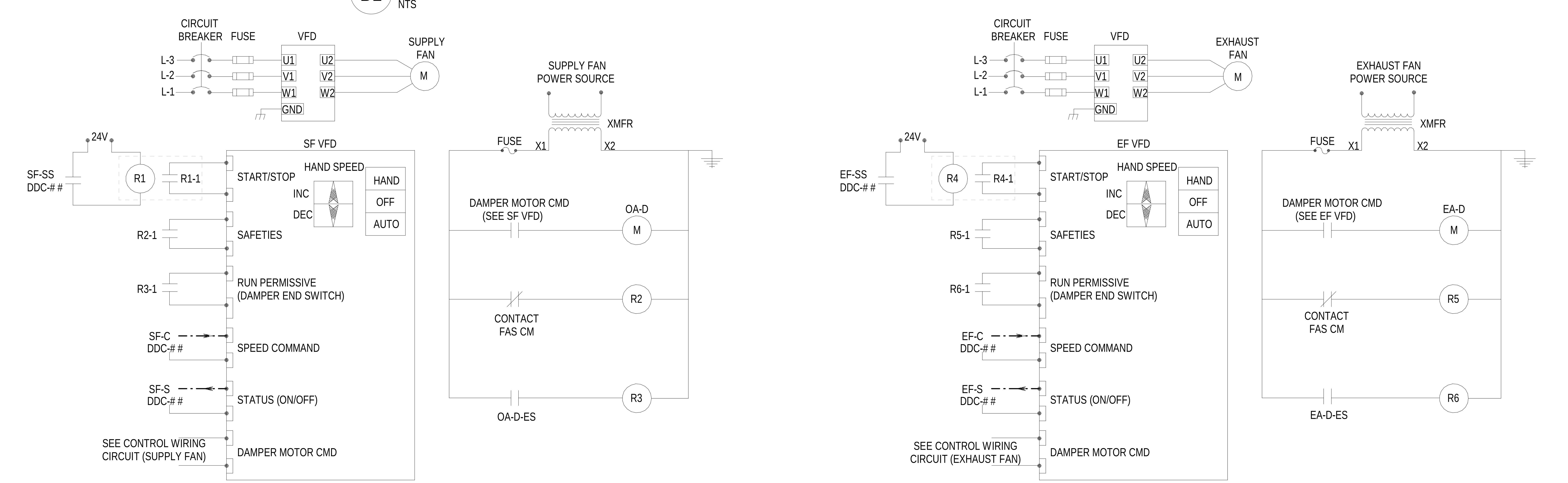
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL CONTROLS LEGEND AND ABBREVIATIONS

SCALE:	AS NOTED
EPROJECT NO.:	1667770
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12894027
SHEET	77 OF 140

M-800



B1 DEDICATED OUTSIDE AIR SYSTEM (DOAS-1) - CONTROL DIAGRAM
NTS



A1 DOAS-1 SUPPLY & EXHAUST FAN WIRING DIAGRAM
NTS

GENERAL SHEET NOTES

A. SEE SHEET M-001 AND M-800 FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS.
B. DUCT SMOKE DETECTOR AND FIRE ALARM CONTROL MODULE PROVIDED AND CONNECTED TO FIRE ALARM SYSTEM BY DIVISION 28. SEE DIVISION 28 "FIRE DETECTION AND ALARM SYSTEM" FOR ADDITIONAL INFORMATION.
C. REFER TO SHEET M-802 FOR DOAS-1 SEQUENCES OF OPERATION.
D. REFER TO SHEET M-805, M-806, AND M-807 FOR DOAS-1 POINTS SCHEDULE.

SHEET KEYNOTES

1. PROVIDE INTERLOCK WIRING BETWEEN SUPPLY FAN AND EXHAUST FAN AND FIRE ALARM SYSTEM THROUGH FIRE ALARM CONTROL MODULE (CM) TO STOP FANS WHEN SUPPLY DUCT SMOKE DETECTOR SENSES PARTICLES OF COMBUSTION. THE INTERLOCK MUST BE HARDWIRED AND NOT PERFORMED THROUGH THE BAS.
2. PROVIDE NETWORK CONNECTION TO VFD AND CONNECT VFD BACNET COMMUNICATION INTERFACE TO BAS BACNET NETWORK. PROVIDE VFD MONITORING POINTS INDICATED IN THE POINTS LIST TO THE BAS THROUGH THE COMMUNICATION INTERFACE.
3. PROVIDE INTERLOCK WIRING BETWEEN SUPPLY FAN AND OUTSIDE AIR DAMPER. WHEN FAN IS COMMANDED TO START IN EITHER HAND OR AUTO MODE, THE DAMPER MUST OPEN. ONCE THE DAMPER IS PROVEN OPEN BY DAMPER LIMIT SWITCH, THE SUPPLY FAN MUST START. FAN MUST STOP IF OUTSIDE AIR DAMPER CLOSES.
4. PROVIDE INTERLOCK WIRING BETWEEN EXHAUST FAN AND ASSOCIATED DAMPER. WHEN FAN IS COMMANDED TO START IN EITHER HAND OR AUTO MODE, THE DAMPER MUST OPEN. ONCE THE DAMPER IS PROVEN OPEN BY DAMPER LIMIT SWITCH, THE EXHAUST FAN MUST START. FAN MUST STOP IF ISOLATION DAMPER CLOSES.
5. PROVIDE AIR FLOW MEASURING STATION (AFMS) AS INDICATED. INSTALL AND CONNECT AFMS IN ACCORDANCE WITH REQUIREMENTS AND INSTRUCTIONS PROVIDED BY THE AFMS MANUFACTURER.
6. PROVIDE SPACE TEMPERATURE AND HUMIDITY SENSORS FOR DOAS-1. SEE MECHANICAL FLOOR PLAN ON MP101 SHEET FOR SENSORS LOCATION.
7. PROVIDE LOW LEAKAGE AIR DAMPERS TO MEET ATFP REQUIREMENTS.
8. PROVIDE OUTDOOR AIR TEMPERATURE AND HUMIDITY SENSORS. SEE SHEET M-401 FOR SENSORS LOCATION.
9. PROVIDE NETWORK CONNECTION TO CONDENSING UNIT CONTROLLER AND CONNECT BACNET CONTROLLER TO BAS BACNET NETWORK. COORDINATE NETWORK POINTS (MINIMUM 20 POINTS) FOR CONDENSING UNIT WITH THE SITE.
10. PROVIDE FROSTAT FOR CONDENSING UNIT. INSTALL AND CONNECT FROSTAT IN ACCORDANCE WITH REQUIREMENTS AND INSTRUCTIONS PROVIDED BY THE CONDENSING UNIT MANUFACTURER.
11. DISCHARGE AIR TEMPERATURE SENSOR PROVIDED BY CONDENSING UNIT MANUFACTURER. INSTALL AND CONNECT SENSOR IN ACCORDANCE WITH REQUIREMENTS AND INSTRUCTIONS PROVIDED BY THE CONDENSING UNIT MANUFACTURER.
12. INTERLOCK ELECTRIC HEATER WITH TEMPERATURE LIMIT SWITCH. WHEN TEMPERATURE LIMIT SWITCH DETECTS TEMPERATURE GREATER THAN CUTOUT POINT SET BY UNIT MANUFACTURER, ELECTRIC HEATER MUST BE DISABLED. THE TEMPERATURE LIMIT SWITCH MUST BE PROVIDED WITH THE ELECTRIC HEATER.
13. INTERLOCK ELECTRIC HEATER WITH AIR FLOW SWITCH. WHEN AIR FLOW SWITCH INDICATES THAT THERE IS NO AIR FLOW, ELECTRIC HEATER MUST BE DISABLED. THE AIRFLOW SWITCH MUST BE PROVIDED WITH THE ELECTRIC HEATER.

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	CWH	DRW	CWH	CHK	MJL
PM/DM					SMS/CWJ
BRANCH MANAGER					ALG
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL CONTROL DIAGRAMS & SEQUENCES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894028
SHEET 78 OF 140

M-801

M-801
6/20/2025 11:54:43 AM
\\dc1vs01\fedgrp\FDZE1628\MECH\BLDG-1667770-M-BIM_v21.rvt

DOAS SEQUENCES OF OPERATION

GENERAL:
1. THE CONTROL SEQUENCE MUST CONTAIN ALGORITHMS THAT WILL MAKE HEATING, COOLING, AND VENTILATION DECISIONS IN RESPONSE TO SIGNALS FROM THE BAS. THE CONTROL ALGORITHM MUST MAINTAIN ACCURATE TEMPERATURE CONTROL AND MINIMIZE DRIFT FROM SET POINT.
2. PROGRAM AN ADJUSTABLE SOFTWARE TIME DELAY INTO THE BAS TO STAGGER THE RESTART OF THE UNIT AFTER A POWER FAILURE TO ASSURE THE GENERATOR IS NOT OVERLOADED WHEN IT RUNS AFTER A POWER FAILURE. SEE "SEQUENTIAL UNIT START-UP SCHEDULE AFTER A POWER FAILURE" FOR THE RESTART OF EACH EQUIPMENT.

DOAS RUN: UNIT MUST BE AUTOMATICALLY OR MANUALLY ENABLED TO RUN IN OCCUPIED OR UNOCCUPIED MODE AS FOLLOWS:
1. AUTOMATIC OPERATION IN OCCUPIED/UNOCCUPIED MODES BASED ON A DAILY PROGRAM (TIME) SCHEDULE WHICH INCLUDES A HOLIDAY FEATURE. COORDINATE THE TIME SCHEDULE WITH THE FACILITY MAINTENANCE PERSONNEL.
   A. OCCUPIED TIME: MON-FRIDAY: 0600 - 1800 / WEEKEND: ADJUSTABLE
   B. UNOCCUPIED TIME: MON-FRIDAY: 1800 - 0600 / WEEKEND: 0000 - 2400
2. OR MANUALLY SELECTED BY USER FROM GRAPHIC INTERFACE.

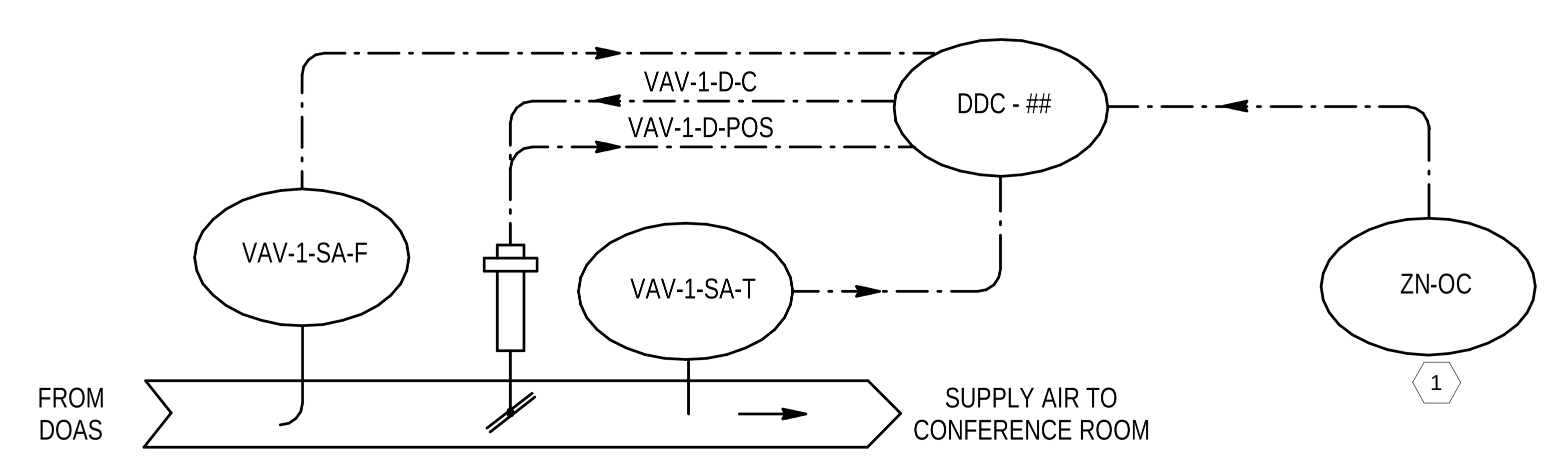
DOAS UNOCCUPIED MODE:
1. THE DOAS MUST BE OFF, UNLESS UNIT IS CALLED FOR DEHUMIDIFICATION MODE.
2. WHEN THE BUILDING AIR DEW POINT TEMPERATURE, AS DETERMINED USING SPACE AIR TEMPERATURE AND HUMIDITY SENSORS, RISES ABOVE 57 °F (ADJ.) FOR MORE THAN 15 MINUTES (ADJ.), THE DOAS MUST START AND THE DEHUMIDIFICATION MODE MUST BE ENABLED.
   A. THE BAS MUST MODULATE THE COMPRESSOR TO MAINTAIN THE COOLING COIL LEAVING AIR TEMPERATURE OF 53 °F (ADJ.). DURING THIS MODE, THE HOT GAS REHEAT COIL MUST REHEAT THE AIR TO 68 °F (ADJ.) TO AVOID SPACE OVERCOOLING.
   B. WHEN THE BUILDING RELATIVE HUMIDITY DROPS BELOW DEHUMIDIFICATION SETPOINT FOR MORE THAN 15 MINUTES (ADJ.), THE BAS MUST DE-ACTIVATE DEHUMIDIFICATION MODE AND STOP THE UNIT.
3. IN THE EVENT ANY OF AC-3 THRU AC-9 IS ENABLED BY ASSOCIATED TIMER SWITCH DURING UNOCCUPIED HOURS, THE DOAS MUST START AND OPERATE ACCORDING TO THE OCCUPIED MODE DESCRIBED BELOW. DOAS MUST RUN FOR A MINIMUM OF 30 MINUTES (ADJUSTABLE) OR UNTIL DEACTIVATION OF ALL AC-3 THRU AC-9.

DOAS OCCUPIED MODE: DURING THIS MODE, THE AIR HANDLING UNIT MUST OPERATE AS DESCRIBED BELOW:
1. SUPPLY FAN HAND-OFF-AUTO OPERATION: HAND-OFF-AUTO SETTINGS MUST BE PROVIDED AS PART OF THE VARIABLE FREQUENCY DRIVE (VFD) THROUGH THE DRIVE'S KEYPAD. IN THE OFF MODE, THE FAN MUST BE STOPPED AND ASSOCIATED ISOLATION SUPPLY AIR DAMPER MUST BE CLOSED. IN THE HAND MODE, OUTSIDE AIR DAMPER MUST OPEN, AND DAMPER POSITION SWITCH MUST ENERGIZE THE SUPPLY FAN THROUGH A HARDWIRED INTERLOCK WITH THE DAMPER POSITION SWITCH. WHEN THE DAMPER IS FULLY OPEN. THE SUPPLY FAN MUST RUN AT THE SPEED SETPOINT MANUALLY SET AT VFD CONTROL PANEL. IN THE AUTO MODE, THE BAS MUST CONTROL SUPPLY FAN AS DESCRIBED BELOW. UPON ACTIVATION, SAFETIES MUST BE HARD WIRED TO THE VFD TO STOP FAN IN THE HAND AND AUTO MODES.
   A. SUPPLY FAN OPERATION AUTOMATIC MODE: TO START THE FAN, THE BAS MUST SEND A COMMAND SIGNAL TO SUPPLY FAN VFD, WHICH, THROUGH A HARDWIRE INTERLOCK WILL OPEN THE OUTSIDE AIR DAMPER. ONCE DAMPER IS FULLY OPEN, AS SENSED BY THE ASSOCIATED DAMPER POSITION SWITCH, THE FAN MUST BE ENERGIZED. TO STOP THE FAN, THE BAS MUST SEND A STOP COMMAND TO THE VFD WHICH MUST DE-ENERGIZE THE SUPPLY FAN AND CLOSE THE OUTSIDE AIR DAMPER.
   B. SUPPLY FAN VFD SPEED CONTROL: WHEN THE SUPPLY FAN IS STARTED, THE BAS MUST CONTROL THE SPEED OF THE VFD TO MAINTAIN SUPPLY AIR FLOW RATE ACCORDING TO THE FOLLOWING RELATIONSHIP:

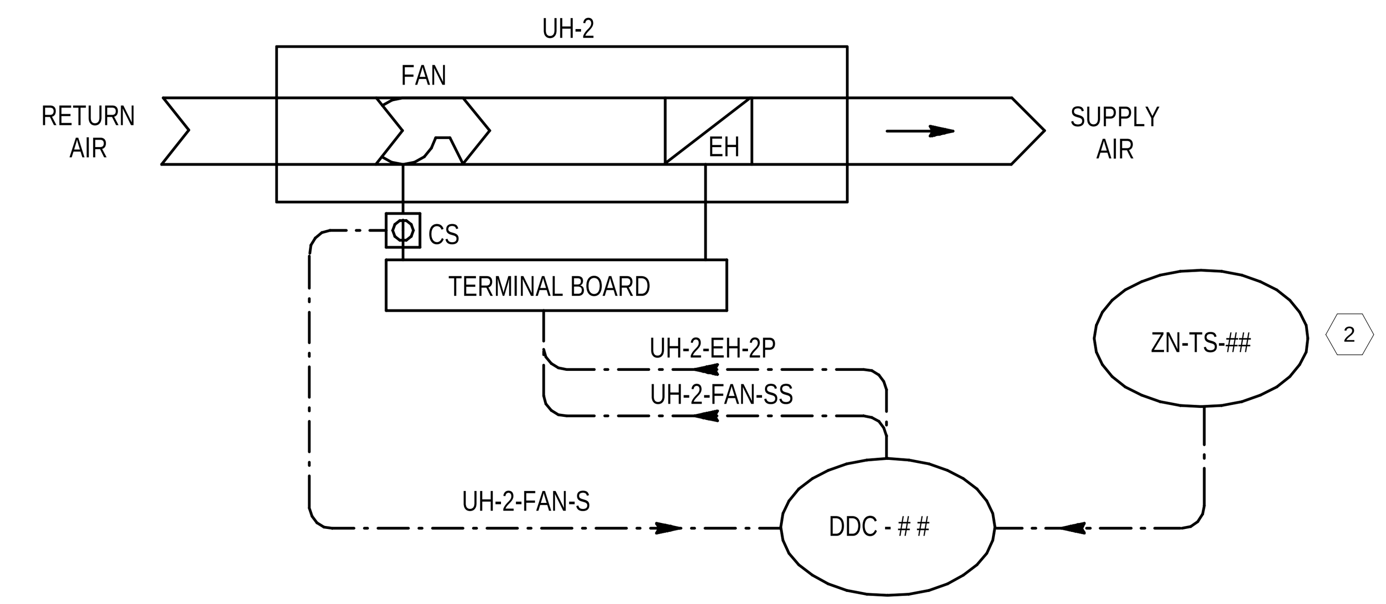
SUPPLY AIRFLOW SETPOINT(CFM) = MINIMUM DESIGN SUPPLY AIR FLOW (CFM) + VAV-1 AIR FLOW (CFM)

2. EXHAUST FAN HAND-OFF-AUTO OPERATION: HAND-OFF-AUTO SETTINGS MUST BE PROVIDED AS PART OF THE VARIABLE FREQUENCY DRIVE (VFD) THROUGH THE DRIVE'S KEYPAD. IN THE OFF MODE, THE FAN MUST BE STOPPED AND ASSOCIATED ISOLATION DAMPER MUST BE CLOSED. IN THE HAND MODE, ISOLATION DAMPER MUST OPEN, AND DAMPER POSITION SWITCH MUST ENERGIZE THE EXHAUST FAN THROUGH A HARDWIRED INTERLOCK WITH THE DAMPER POSITION SWITCH. WHEN THE DAMPER IS FULLY OPEN. THE EXHAUST FAN MUST RUN AT THE SPEED SETPOINT MANUALLY SET AT VFD CONTROL PANEL. IN THE AUTO MODE, THE BAS MUST CONTROL EXHAUST FAN AS DESCRIBED BELOW. UPON ACTIVATION, SAFETIES MUST BE HARD WIRED TO THE VFD TO STOP FAN IN THE HAND AND AUTO MODES.
   A. EXHAUST FAN OPERATION AUTOMATIC MODE: TO START THE EXHAUST FAN, THE BAS MUST SEND A START SIGNAL TO THE EXHAUST FAN VFD. THIS MUST ENABLE THE HARD WIRE INTERLOCK THAT OPENS THE ISOLATION DAMPER. WHEN THE DAMPER IS FULLY OPEN, AS SENSED BY THE DAMPER POSITION SWITCH, THE EXHAUST FAN MUST BE ENERGIZED THROUGH THE HARD WIRE INTERLOCK. TO STOP THE FAN, THE BAS MUST SEND A STOP COMMAND TO THE VFD WHICH MUST DE-ENERGIZE THE EXHAUST FAN AND CLOSE THE ISOLATION DAMPER.
   B. EXHAUST FAN VFD SPEED CONTROL: WHEN THE EXHAUST FAN IS STARTED, THE BAS MUST CONTROL THE SPEED OF THE VFD TO MAINTAIN EXHAUST AIR FLOW RATE ACCORDING TO THE FOLLOWING RELATIONSHIP:
   EXHAUST AIRFLOW SETPOINT (CFM) = SUPPLY AIRFLOW RATE (CFM) - OFFSET (CFM)
   C. THE OFFSET MUST BE DETERMINED BY TAB CONTRACTOR DURING TAB PROCESS.
3. OUTSIDE AIR HEAT EXCHANGER FACE AND BYPASS DAMPER CONTROL:
   A. WHEN THE UNIT IS OFF, THE DAMPER MUST BE NORMALLY OPEN TO THE HEAT EXCHANGER AND CLOSED TO BYPASS.
   B. WHEN THE UNIT IS ON, THE BAS MUST CONTROL THE OUTSIDE AIR HEAT EXCHANGER FACE AND BYPASS DAMPER AS FOLLOWS:
      a. ECONOMIZER MODE: WHEN OUTSIDE AIR DRY BULB TEMPERATURE IS LESS THAN 62 °F (ADJ.), THE BAS MUST ENABLE THE ECONOMIZER MODE. ONCE ENABLED, OA FACE AND BYPASS DAMPER AT HEAT EXCHANGER MUST MODULATE OPEN TO BYPASS THE HEAT EXCHANGER TO MAINTAIN SUPPLY AIR TEMPERATURE SETPOINT. WHEN OUTSIDE AIR DRY BULB TEMPERATURE IS LESS THAN 52 °F (ADJ.), THE BAS MUST DISABLE THE ECONOMIZER MODE.
      b. NON-ECONOMIZER MODE: THE HEAT EXCHANGER FACE AND BYPASS DAMPER MUST BE NORMALLY OPEN TO THE HEAT EXCHANGER AND CLOSED TO BYPASS DURING HEATING OR COOLING MODES.
      c. WINTER DEFROST MODE: THE HEAT EXCHANGER FACE AND BYPASS DAMPER MUST MODULATE TO PREVENT THE EXHAUST AIR TEMPERATURE FROM DROPPING BELOW 33 °F(ADJ.).
4. COOLING CONTROL:
   A. WHEN THE SUPPLY FAN IS OFF, THE DX COOLING COIL MUST BE DISABLED.
   B. WHEN DOAS SUPPLY FAN IS ON AND DISCHARGE AIR TEMPERATURE RISES ABOVE UNIT'S DISCHARGE AIR TEMPERATURE COOLING SETPOINT, INITIALLY SET AT 73 °F (ADJ.), THE BAS MUST MODULATE THE COMPRESSOR TO MAINTAIN THE UNIT'S DISCHARGE AIR TEMPERATURE AT COOLING SETPOINT.
5. DEHUMIDIFICATION CONTROL:
   A. WHEN RETURN AIR DEW POINT TEMPERATURE, AS DETERMINED USING UNIT RETURN AIR TEMPERATURE AND HUMIDITY SENSORS, RISES ABOVE 57 °F (ADJ.), DEHUMIDIFICATION MODE MUST BE ENABLED. DURING THIS MODE, THE COMPRESSOR MUST MODULATE TO MAINTAIN THE COOLING COIL LEAVING AIR TEMPERATURE OF 53 °F (ADJ.) AND THE HOT GAS REHEAT COIL MUST REHEAT THE AIR TO 68 °F (ADJ.) TO AVOID SPACE OVERCOOLING.
   B. WHEN RETURN AIR DEW POINT TEMPERATURE DROPS BELOW SETPOINT, THE BAS MUST DE-ACTIVATE DEHUMIDIFICATION MODE AND CONTROL OF THE DISCHARGE AIR TEMPERATURE MUST RETURN TO NORMAL OPERATION.
6. HEATING CONTROL:
   A. WHEN THE SUPPLY FAN IS OFF, THE ELECTRIC HEATING COIL MUST BE DISABLED.
   B. WHEN DOAS SUPPLY FAN IS ON AND DISCHARGE AIR TEMPERATURE DROPS BELOW UNIT'S DISCHARGE AIR TEMPERATURE SETPOINT, INITIALLY SET AT 68 °F (ADJ.), THE BAS MUST MODULATE THE ELECTRIC HEATING COIL VIA SCR CONTROLS TO MAINTAIN THE UNIT'S DISCHARGE AIR TEMPERATURE AT SETPOINT.

SAFETIES AND ALARMS:
1. SMOKE DETECTION CONTROL: UPON DETECTION OF SMOKE FROM THE SUPPLY SMOKE DETECTOR, THE UNIT FANS MUST STOP, THE OUTDOOR AND EXHAUST AIR DAMPERS MUST CLOSE, AND AN ALARM MUST BE SENT TO THE BAS. WHEN THE FIRE ALARM CONDITION HAS BEEN CLEARED AND THE FIRE ALARM SYSTEM HAS BEEN RESET, THE UNIT MUST RETURN TO NORMAL OPERATION.
2. SUPPLY/EXHAUST FAN FAILURE: THE FAN VFD STATUS CONTACT MUST BE USED TO MONITOR THE STATUS OF THE UNIT FANS. ONCE THE FAN HAS BEEN RUNNING, THE SUPPLY FAN DOES NOT PROVIDE A "FAN RUNNING" STATUS AFTER A 120 SECOND (ADJ.) TIME DELAY, A FAN FAILURE ALARM MUST BE ISSUED AT THE BAS AND THE FAN START COMMAND MUST BE CANCELED.
3. ELECTRIC HEATING COIL
   A. ELECTRIC COIL ALARM: A CURRENT SENSOR MUST BE USED TO MONITOR THE STATUS OF ELECTRIC COIL WHEN THE COIL IS ENABLED. IF NO STATUS IS INDICATED FOR THE ELECTRIC COIL AFTER A 60 SECONDS (ADJUSTABLE) TIME DELAY, A CRITICAL ALARM MUST BE SENT TO THE BAS AND THE ELECTRIC COIL MUST BE DISABLED.
   B. HIGH TEMPERATURE LIMIT: ELECTRIC HEAT COIL MUST BE HARDWIRE INTERLOCKED WITH A HEATER HIGH TEMPERATURE SWITCH WHICH MUST DISABLE THE ELECTRIC HEAT COIL IF TEMPERATURE RISES ABOVE THE CUTOUT POINT SET BY UNIT MANUFACTURER.
   C. AIRFLOW LIMIT: ELECTRIC HEATING COIL MUST BE HARD WIRE INTERLOCKED WITH AN AIRFLOW SWITCH, WHICH MUST DISABLE HEATING IF NO AIRFLOW IS PRESENT. THE AIRFLOW SWITCH MUST BE PROVIDED BY THE ELECTRIC HEATING COIL MANUFACTURER.
4. COMPRESSOR ALARM: A CURRENT SENSOR MUST BE USED TO MONITOR THE STATUS OF COMPRESSOR WHEN THE COMPRESSOR IS ENABLED. IF NO STATUS IS INDICATED FOR THE COMPRESSOR AFTER A 60 SECONDS (ADJUSTABLE) TIME DELAY, A CRITICAL ALARM MUST BE SWENT TO THE BAS AND THE COMPRESSOR MUST BE DISABLED.
5. DISCHARGE AIR TEMPERATURE ALARM: A CRITICAL ALARM MUST BE ANNUNCIATED TO THE BAS IN THE EVENT THAT DISCHARGE AIR TEMPERATURE DROPS 5 °F (ADJUSTABLE) BELOW HEATING SET POINT OR RISES 5 °F (ADJUSTABLE) ABOVE COOLING SET POINT.
6. EXHAUST AIR TEMPERATURE ALARM: A CRITICAL ALARM MUST BE ANNUNCIATED TO THE BAS IN THE EVENT THAT EXHAUST AIR TEMPERATURE DROPS BELOW WINTER DEFROST SET POINT.
7. RETURN AIR DEW POINT TEMPERATURE ALARM: A CRITICAL ALARM MUST BE ANNUNCIATED TO THE BAS IN THE EVENT THAT RETURN AIR DEW POINT TEMPERATURE RISES 5 °F (ADJUSTABLE) ABOVE RETURN AIR DEW POINT TEMPERATURE SET POINT.
8. BUILDING AIR DEW POINT TEMPERATURE ALARM: A CRITICAL ALARM MUST BE ANNUNCIATED TO THE BAS IN THE EVENT THAT BUILDING AIR DEW POINT TEMPERATURE RISES 5 °F (ADJUSTABLE) ABOVE BUILDING AIR DEW POINT TEMPERATURE SET POINT.
9. DRAIN PAN OVERFLOW ALARM: THE UNIT CONTROLLER MUST MONITOR THE DRAIN PAN OVERFLOW SWITCH LOCATED ON THE DRIP PAN. WHEN OVERFLOW SWITCH IS ACTIVATED, A CRITICAL ALARM MUST BE ISSUED AT THE UNIT CONTROLLER AND THE UNIT MUST BE SHUTDOWN.
10. FILTER ALARM:
   A. A DIFFERENTIAL PRESSURE SWITCH MUST MONITOR THE DIFFERENTIAL PRESSURE ACROSS THE OUTDOOR AIR FILTER WHEN THE FAN IS RUNNING. IF DURING NORMAL OPERATION THE DIFFERENTIAL AIR PRESSURE EXCEEDS FILTER CLEAN VALUE PLUS 0.25 INCHES WC (ADJ.) FOR 60 SECONDS (ADJ.), AN ALARM MUST BE SENT TO THE BAS.
   B. A DIFFERENTIAL PRESSURE SWITCH MUST MONITOR THE DIFFERENTIAL PRESSURE ACROSS THE RETURN AIR FILTER WHEN THE FAN IS RUNNING. IF DURING NORMAL OPERATION THE DIFFERENTIAL AIR PRESSURE EXCEEDS FILTER CLEAN VALUE PLUS 0.25 INCHES WC (ADJ.) FOR 60 SECONDS (ADJ.), AN ALARM MUST BE SENT TO THE BAS.
11. AIRFLOW ALARM:
   A. THE BAS MUST MONITOR THE SUPPLY AIR FLOW WHENEVER THE DOAS IS RUNNING. AN ALARM MUST BE ANNUNCIATED AT THE BAS WHENEVER THE SUPPLY AIR FLOW RATE VARIES BY MORE THAN ±10% FROM ITS SET POINT.
   B. THE BAS MUST MONITOR THE RETURN AIR FLOW WHENEVER THE DOAS IS RUNNING. AN ALARM MUST BE ANNUNCIATED AT THE BAS WHENEVER THE RETURN AIR FLOW RATE VARIES BY MORE THAN ±10% FROM ITS SET POINT.



A1 VAV BOX (VAV-1) - CONTROL DIAGRAM
NTS



A3 ELECTRIC UNIT HEATER (UH-2) - CONTROL DIAGRAM
NTS

VAV BOX SEQUENCES OF OPERATION

GENERAL: THE VARIABLE AIR VOLUME TERMINAL MUST BE FULLY CONTROLLED BY THE BAS. SCHEDULE MUST BE THE SAME AS THE PARENT DOAS.

VAV BOX DAMPER CONTROL:
1. UNOCCUPIED MODE: THE VAV BOX MUST BE INDEXED IN UNOCCUPIED MODE AND THE VAV BOX DAMPER MUST BE CLOSED.
2. OCCUPIED MODE:
   A. DURING THIS MODE, THE BAS MUST MODULATE THE VAV BOX DAMPER TO MAINTAIN MINIMUM AIRFLOW SET POINT.
   B. WHEN THE ROOM OCCUPANCY SENSOR DOES DETECT OCCUPANCY, THE VAV BOX DAMPER MUST OPEN TO INCREASE AIRLOW TO MAXIMUM SET POINT. THE VAV BOX DAMPER MUST REMAIN AT MAXIMUM FOR 30 MINUTES (ADJUSTABLE) AFTER LAST DETECTION OF OCCUPANCY BY ROOM OCCUPANCY SENSOR , AND THEN THE VAV BOX DAMPER MUST RETURN TO THE NORMAL MODE.
   C. SEE VAV TERMINAL SCHEDULE ON M-601 FOR VAV BOX MINIMUM AND MAXIMUM AIRFLOW SET POINTS.

VAV AIRFLOW ALARM: THE BAS MUST MONITOR THE VAV SUPPLY AIR FLOW WHENEVER THE VAV BOX IS INDEXED IN OCCUPIED MODE. AN ALARM MUST BE ANNUNCIATED AT THE BAS WHENEVER THE SUPPLY AIR FLOW RATE VARIES BY MORE THAN ±10% FROM ITS SET POINT.

ELECTRIC UNIT HEATER SEQUENCES OF OPERATION

GENERAL: THE ELECTRIC UNIT HEATER (UH-2) IS CONTROLLED BY THE BAS AS DESCRIBED BELOW.

ELECTRIC UNIT HEATER CONTROL:
1. WHEN THE SPACE TEMPERATURE FALLS BELOW THE TEMPERATURE SET POINT, INITIALLY SET AT 55 DEGREES F (ADJUSTABLE), THE BAS MUST START THE UNIT HEATER FAN AND ENABLE THE ELECTRIC HEATING COIL TO MAINTAIN THE SPACE TEMPERATURE SET POINT. THE ELECTRIC HEATING COIL MUST NOT BE ENABLED UNTIL THE FAN OPERATION HAS BEEN PROVED.
2. WHEN THE SPACE TEMPERATURE RISES ABOVE SPACE TEMPERATURE SET POINT, THE BAS MUST DISABLE THE ELECTRIC HEATING COIL AND STOP UNIT HEATER FAN.

FAN PROOF: A CURRENT SWITCH MUST BE USED TO MONITOR THE STATUS OF THE FAN AND SEND A STATUS SIGNAL TO THE BAS WHEN THE FAN IS RUNNING. IF THE STATUS INDICATED DOES NOT MATCH THE COMMANDED OUTPUT FOR THE FAN, AFTER A 30 SECONDS (ADJUSTABLE) TIME DELAY, A FAN FAILURE ALARM MUST BE SENT TO THE BAS, AND THE FAN START COMMAND MUST BE CANCELED.

SPACE TEMPERATURE ALARM: AN ALARM MUST BE ANNUNCIATED TO THE BAS IN THE EVENT THAT SPACE TEMPERATURE DROPS BELOW 50 DEGREES F (ADJUSTABLE).

HEATING SHUT OFF: WHEN THE OUTDOOR AIR TEMPERATURE RISES ABOVE 45 DEG F (ADJUSTABLE), THE BAS MUST DISABLE THE ELECTRIC HEATING COIL AND STOP UNIT HEATER FAN.

GENERAL SHEET NOTES

A. SEE SHEET M-001 AND M-800 FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS.
B. REFER TO SHEET M-801 FOR DOAS-1 CONTROL DIAGRAM.
C. REFER TO SHEET M-809 FOR VAV BOX (VAV-1) POINTS SCHEDULE.
D. REFER TO SHEET M-809 FOR UNIT HEATER (UH-2) POINTS SCHEDULE.

SHEET KEYNOTES

1. PROVIDE CONTROL WIRING AND CONNECT OUTPUT SIGNAL FROM OCCUPANCY SENSOR(S) TO BAS. REFER TO ELECTRICAL FLOOR PLANS FOR OCCUPANCY SENSOR LOCATION. COORDINATE WORK WITH DIVISION 26.
2. PROVIDE ROOM TEMPERATURE SENSOR FOR SPACE TEMPERATURE CONTROL AND MONITORING. SEE MECHANICAL FLOOR PLAN MP-101 FOR SPACE TEMPERATURE SENSOR'S LOCATION.

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

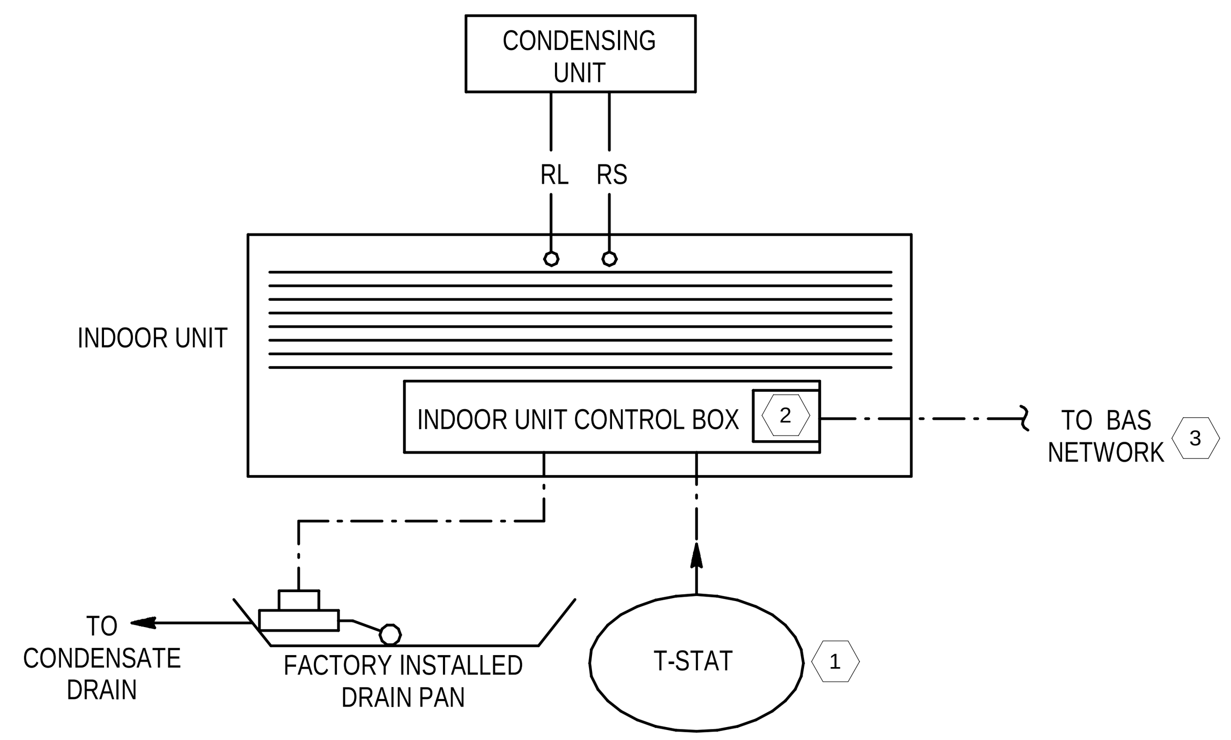
APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH DRW CWH CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER
CHIEF ENG/ARCH JWJ
FIRE PROTECTION

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL CONTROL DIAGRAMS & SEQUENCES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894029
SHEET 79 OF 140
M-802

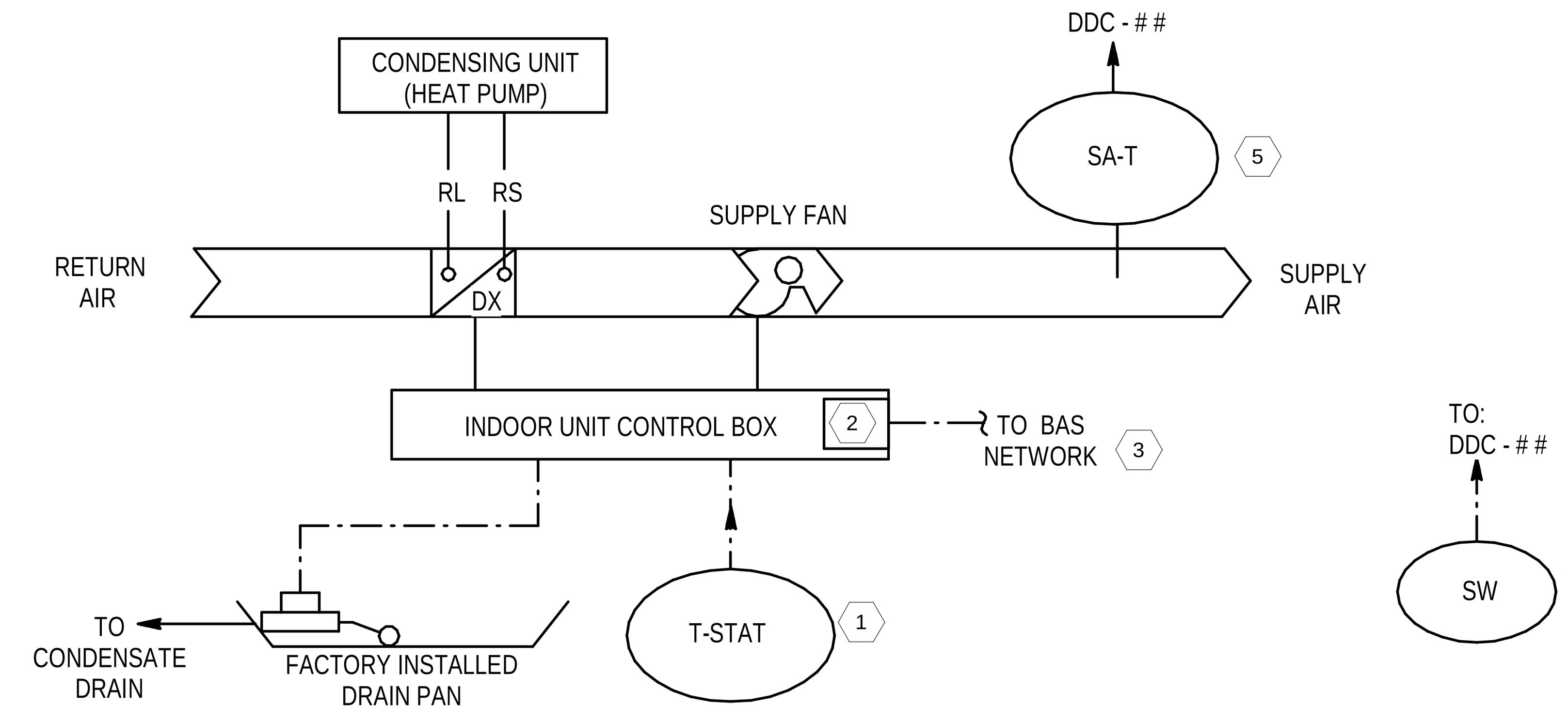
6/20/2025 11:54:50 AM M-802
\\dc1vs01\fedpj\FD2E1628\MECH\BLDG-1667770-M-BIM_v21.rvt

1	2	3	4	5



C1 AIR CONDITIONING UNIT (AC-1 & AC-2) - CONTROL DIAGRAM

NTS



C2 AIR CONDITIONING UNIT (AC-3 THRU 9) - CONTROL DIAGRAM

NTS

AC UNIT SEQUENCES OF OPERATION

GENERAL:
1. THE SPLIT SYSTEM AIR CONDITIONING UNIT MUST BE CONTROLLED BY A PACKAGED CONTROLS WITH BACNET INTERFACE PROVIDED BY THE UNIT MANUFACTURER. THE UNIT MUST BE DIRECTLY CONNECTED TO THE BAS NETWORK USING BACNET MS/TP AND COMMUNICATE WITH THE BAS TO ALLOW MONITORING, CONTROL, AND ALARM NOTIFICATION.
2. PROGRAM AN ADJUSTABLE SOFTWARE TIME DELAY INTO THE BAS TO STAGGER THE RESTART OF UNIT AFTER A POWER FAILURE TO ASSURE THE GENERATOR IS NOT OVERLOADED WHEN IT RUNS AFTER A POWER FAILURE. SEE "SEQUENTIAL UNIT START-UP SCHEDULE AFTER A POWER FAILURE" FOR THE RESTART OF EACH EQUIPMENT.

AC ON/OFF CONTROL: THE SYSTEM MUST BE CONTROLLED EITHER BY THE BAS OR BY THE ROOM THERMOSTAT AS DESCRIBED BELOW.

SPACE TEMPERATURE CONTROL: WHEN THE SPACE TEMPERATURE RISES ABOVE THE SPACE TEMPERATURE SET POINT, INITIALLY SET AT 75 DEGREES F (ADJ.), THE AC FAN MUST START AND THE COMPRESSOR MUST CYCLE ON TO MAINTAIN SPACE TEMPERATURE AT SET POINT. WHEN THE SPACE TEMPERATURE FALLS BELOW THE SET POINT, THE AC FAN AND COMPRESSOR MUST STOP. THE SET POINT MUST BE ADJUSTABLE AT THE ROOM THERMOSTAT, OR THROUGH BAS INTERFACE.

FAN CONTROL: THE AC FAN MUST START OR STOP AS DESCRIBED UNDER "SPACE TEMPERATURE CONTROL". FAN SPEED MUST BE ADJUSTABLE BY THE USER AT THE ROOM THERMOSTAT, OR THE BAS.

CONDENSATE OVERFLOW SWITCH: WHEN THE WATER LEVEL IN THE DRAIN PAN REACHES AN UNSAFE LEVEL, THE UNIT MUST BE DE-ENERGIZED. THE UNIT AUTOMATICALLY RESETS W HEN THE DRAIN PAN IS CLEARED OF WATER.

TEMPERATURE ALARM: AN ALARM MUST BE ANNUNCIATED TO THE BAS IN THE EVENT THAT SPACE TEMPERATURE RISES 5 DEGREES F (ADJ.) ABOVE SPACE TEMPERATURE SETPOINT.

LOSS OF COMMUNICATION ALARM: IF COMMUNICATION IS LOST, A CRITICAL ALARM MUST BE GENERATED AT THE BAS OPERATOR'S WORKSTATION AND THE UNIT CONTROLLER MUST OPERATE USING DEFAULT MODE AND SETPOINT.

AC UNIT SEQUENCES OF OPERATION

GENERAL:
1. THE AIR-CONDITIONING UNIT MUST BE CONTROLLED BY A PACKAGED CONTROLS WITH BACNET INTERFACE PROVIDED BY THE UNIT MANUFACTURER. THE UNIT MUST BE DIRECTLY CONNECTED TO THE BAS NETWORK USING BACNET MS/TP AND COMMUNICATE WITH THE BAS.
2. PROGRAM AN ADJUSTABLE SOFTWARE TIME DELAY INTO THE BAS TO STAGGER THE RESTART OF UNIT AFTER A POWER FAILURE TO ASSURE THE GENERATOR IS NOT OVERLOADED WHEN IT RUNS AFTER A POWER FAILURE. SEE "SEQUENTIAL UNIT START-UP SCHEDULE AFTER A POWER FAILURE" FOR THE RESTART OF EACH EQUIPMENT.

AC UNIT RUN: UNIT MUST BE AUTOMATICALLY OR MANUALLY ENABLED TO RUN IN OCCUPIED OR UNOCCUPIED MODE AS FOLLOWS:
1. AUTOMATIC OPERATION IN OCCUPIED/UNOCCUPIED MODES BASED ON A DAILY PROGRAM (TIME) SCHEDULE WHICH INCLUDES A HOLIDAY FEATURES. COORDINATE THE TIME SCHEDULE WITH THE FACILITY MAINTENANCE PERSONNEL.
   A. OCCUPIED TIME: MON-FRIDAY: 0600 - 1800 / WEEKEND: ADJUSTABLE
   B. UNOCCUPIED TIME: MON-FRIDAY: 1800 - 0600 / WEEKEND: 0000 - 2400
2. OR MANUALLY SELECTED BY USER FROM GRAPHIC INTERFACE.

UNOCCUPIED MODE: DURING THIS MODE, THE AIR-CONDITIONING UNIT MUST BE OFF, UNLESS UNIT IS CALLED FOR COOLING OR HEATING, OR AN AFTER-HOUR OVEERIDE IS INITIATED AS DESCRIBED BELOW:
1. WHEN THE SPACE TEMPERATURE RISES ABOVE 83°F (ADJ.) THE BAS MUST START THE UNIT. THE UNIT MUST OPERATE UNTIL THE SPACE TEMPERATURE DROPS BELOW 80°F (ADJ.).
2. WHEN THE SPACE TEMPERATURE DROPS BELOW 60°F (ADJ.) THE BAS MUST START THE UNIT. THE UNIT MUST OPERATE UNTIL THE SPACE TEMPERATURE RISES ABOVE 63°F (ADJ.).
3. OR, WHEN THE TIMER SWITCH IS MANUALLY SET BY THE OCCUPANT TO OPERATE THE SYSTEM FOR UP TO TWO HOURS, THE AC UNIT MUST BE INDEXED TO THE OCCUPIED MODE. WHEN THE PRESET AMOUNT OF TIME HAS EXPIRED, THE AC UNIT MUST RETURN TO THE NORMAL MODE.

OPTIMUM START MODE:
THE BAS MUST USE AN OPTIMIZED START-UP ROUTINE, INCORPORATING OA TEMPERATURE, SPACE TEMPERATURE, AND PAST HISTORY OF BUILDING RESPONSE TO DETERMINE THE OPTIMUM TIME TO START THE AIR-CONDITIONING UNIT. COOL-DOWN MODE MUST BE USED IF THE SPACE TEMPERATURE, AS SENSED BY UNIT THERMOSTAT, IS ABOVE THE OCCUPIED COOLING SET POINT. WARM-UP MODE MUST BE USED IF THE SPACE TEMPERATURE, AS SENSED BY UNIT THERMOSTAT, IS BELOW THE OCCUPIED HEATING SET POINT. COOL-DOWN MODE OR WARM-UP MODE MUST OCCUR ONLY ONCE IN A DAY. AT THE SCHEDULED OCCUPANCY TIME, THE AC UNIT MUST START IF NOT ALREADY STARTED BY THE OPTIMAL START ROUTINE, AND THE UNIT MUST BE SWITCHED TO THE OCCUPIED MODE.

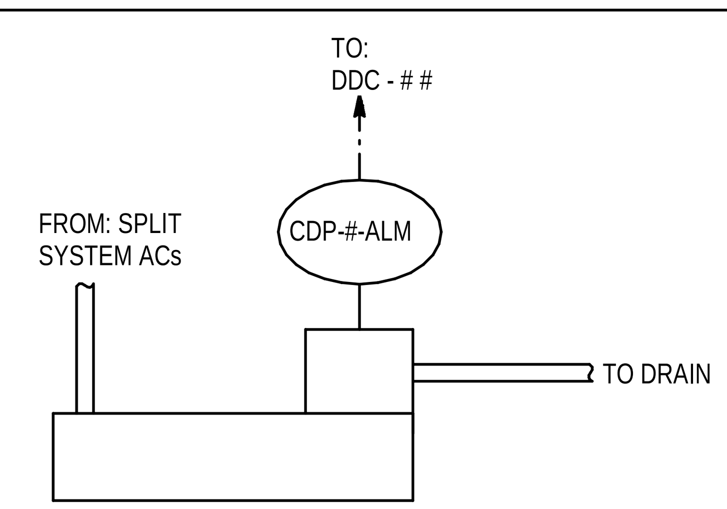
OCCUPIED MODE: DURING THIS MODE, THE AIR-CONDITIONING UNIT MUST OPERATE AS DESCRIBED BELOW:
1. FAN CONTROL: THE BAS MUST SEND AN ENABLE SIGNAL TO THE AIR-CONDITIONING UNIT CONTROLLER AND THE UNIT CONTROLLER START THE FAN. FAN SPEED MUST BE ADJUSTABLE BY THE USER AT THE BAS AND/OR THE WALL-MOUNTED THERMOSTAT.
2. COOLING CONTROL: WHEN THE SPACE TEMPERATURE RISES ABOVE THE COOLING SET POINT, INITIALLY SET AT 78°F (ADJUSTABLE), THE UNIT COOLING CAPACITY MUST BE MODULATED AS REQUIRED TO MAINTAIN THE SPACE AIR TEMPERATURE AT SETPOINT. ON A DECREASE IN DEMAND FOR COOLING THE UNIT COOLING CAPACITY MUST BE DE-ENERGIZED/MODULATED AS REQUIRED.
3. HEATING CONTROL: WHEN THE SPACE TEMPERATURE DROPS BELOW THE HEATING SET POINT, INITIALLY SET AT 68°F (ADJUSTABLE), THE UNIT HEATING CAPACITY MUST BE MODULATED AS REQUIRED TO MAINTAIN THE SPACE AIR TEMPERATURE AT SETPOINT. ON A DECREASE IN DEMAND FOR HEATING THE UNIT HEATING CAPACITY MUST BE DE-ENERGIZED/MODULATED AS REQUIRED.

CONDENSATE OVERFLOW SWITCH: WHEN THE WATER LEVEL IN THE DRAIN PAN REACHES AN UNSAFE LEVEL, THE UNIT MUST BE DE-ENERGIZED. THE UNIT AUTOMATICALLY RESETS W HEN THE DRAIN PAN IS CLEARED OF WATER.

SPACE TEMPERATURE ALARM: A CRITICAL ALARM MUST BE ANNUNCIATED TO THE BAS IN THE EVENT THAT SPACE TEMPERATURE DROPS 5 °F (ADJUSTABLE) BELOW HEATING SET POINT OR RISES 5 °F (ADJUSTABLE) ABOVE COOLING SET POINT.

LOSS OF COMMUNICATION ALARM: IF COMMUNICATION IS LOST, A CRITICAL ALARM MUST BE GENERATED AT THE BAS OPERATOR'S WORKSTATION AND THE UNIT CONTROLLER MUST OPERATE USING DEFAULT MODE AND SETPOINT.

ADDITIONAL POINTS MONITORED BY THE BAS: IN ADDITION TO THE POINTS BEING MONITORED BY THE BAS THROUGH THE BACNET COMMUNICATION NETWORK, A POINT HAS BEEN INCLUDED FOR DIRECT MONITORING OF SUPPLY AIR TEMPERATURE SENSOR AS INDICATED ON CONTROL DIAGRAM. THE POINT MUST BE DIRECTLY MONITORED BY THE BAS AND INSTALLED AS PART OF THE BAS WORK.



B1 CONDENSATE PUMP (CDP-1 & CDP-2) - CONTROL DIAGRAM

NTS

SEQUENCES OF OPERATION

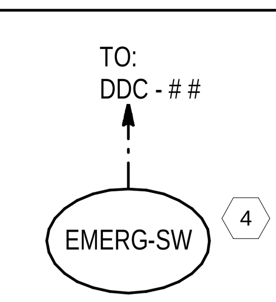
GENERAL: THE BAS MUST MONITOR THE CONDENSATE PUMP HIGH LEVEL. UPON RECEIVING A CONDENSATE PUMP HIGH LEVEL SIGNAL, THE BAS MUST DISABLE UNITS DISCHARGING TO ASSOCIATED CONDENSATE PUMP AND GENERATE AN ALARM AT THE OPERATOR'S WORKSTATION.

SEQUENTIAL UNIT START-UP SCHEDULE AFTER A POWER FAILURE

GENERAL: FOR THE FOLLOWING EQUIPMENT, AN ADJUSTABLE SOFTWARE TIME DELAY MUST BE PROVIDED TO STAGGER THE RESTART OF THE UNITS AFTER A POWER FAILURE. IT MUST BE INDIVIDUALLY SET SO THE UNITS START IN THE FOLLOWING SEQUENTIAL ORDER TO ASSURE THE GENERATOR IS NOT OVERLOADED WHEN IT RUNS AFTER A POWER FAILURE:
1. AC-1, AC-2, AND CDP-1
2. UH-1 AND F-1
3. AC-3 THRU 9 AND CDP-2
4. UH-2
5. DOAS-1
6. VAV-1

A1 SEQUENTIAL UNIT START-UP SCHEDULE AFTER A POWER FAILURE

NTS



A3 AIR DISTRIBUTION EMERGENCY SHUT OFF SWITCH - CONTROL DIAGRAM

NTS

SEQUENCES OF OPERATION

GENERAL: WHEN THE EMERGENCY AIR DISTRIBUTION SHUTOFF SWITCH IS ACTIVATED, DOAS-1 MUST SHUTDOWN AND ASSOCIATED OUTDOOR AND EXHAUST AIR DAMPERS MUST CLOSE; F-1 FAN MUST STOP AND ASSOCIATED OUTSIDE INTAKE LOUVER DAMPERS MUST CLOSE. WHEN THE EMERGENCY CONDITION HAS BEEN CLEARED, AND THE MANUAL SWITCH HAS BEEN RESET, THE MECHANICAL EQUIPMENT MUST BE RETURNED TO NORMAL OPERATION.

GENERAL SHEET NOTES

A. SEE SHEET M-001 AND M-800 FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS.
B. REFER TO SHEET M-807 FOR AIR CONDITIONING UNIT (AC-1 & AC-2) POINTS SCHEDULE.
C. REFER TO SHEET M-808 FOR AIR CONDITIONING UNIT (AC-3 THRU 9) POINTS SCHEDULE.
D. REFER TO SHEET M-807 FOR MISCELLANEOUS POINTS SCHEDULE FOR POINTS ASSOCIATED WITH AC UNIT, DUCT TEMPERATURE SENSOR, AC UNIT OCCUPANCY SWITCH, CONDENSATE PUMP, AND AIR DISTRIBUTION EMERGENCY SHUT-OFF SWITCH.

SHEET KEYNOTES

1. WALL-MOUNTED THERMOSTAT PROVIDED BY SPLIT SYSTEM AIR CONDITIONING UNIT MANUFACTURER. REFER TO MECHANICAL FLOOR PLAN SHEET MP101 FOR THERMOSTAT'S LOCATION.
2. UNIT MOUNTED BACNET INTERFACE CARD PROVIDED BY UNIT MANUFACTURER.
3. PROVIDE COMMUNICATION WIRING AND CONNECT THE SPLIT SYSTEM AIR CONDITIONING UNIT TO THE BAS NETWORK. PROVIDE MONITORING AND CONTROL POINTS INDICATED IN THE POINTS LIST TO THE BAS.
4. PROVIDE AIR DISTRIBUTION EMERGENCY SHUT-OFF SWITCH WHERE INDICATED ON MECHANICAL FLOOR PLAN SHEET MP101 SHEET. EMERGENCY SWITCH MUST BE DIRECTLY CONNECTED TO THE BAS FOR MONITORING.
5. PROVIDE DUCT TEMPERATURE SENSOR ON THE SUPPLY AIR DUCT. DUCT TEMPERATURE SENSOR MUST BE DIRECTLY CONNECTED TO THE BAS FOR MONITORING.
6. PROVIDE A TWO-HOUR IN-WALL TIMER SWITCH TO ALLOW OCCUPANT TO PLACE THE UNIT IN OCCUPIED MODE DURING PERIODS. TIMER SWITCH MUST BE DIRECTLY CONNECTED TO THE BAS FOR MONITORING. SEE MECHANICAL FLOOR PLAN SHEET MP101 FOR TIMER SWITCH LOCATION.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/25/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

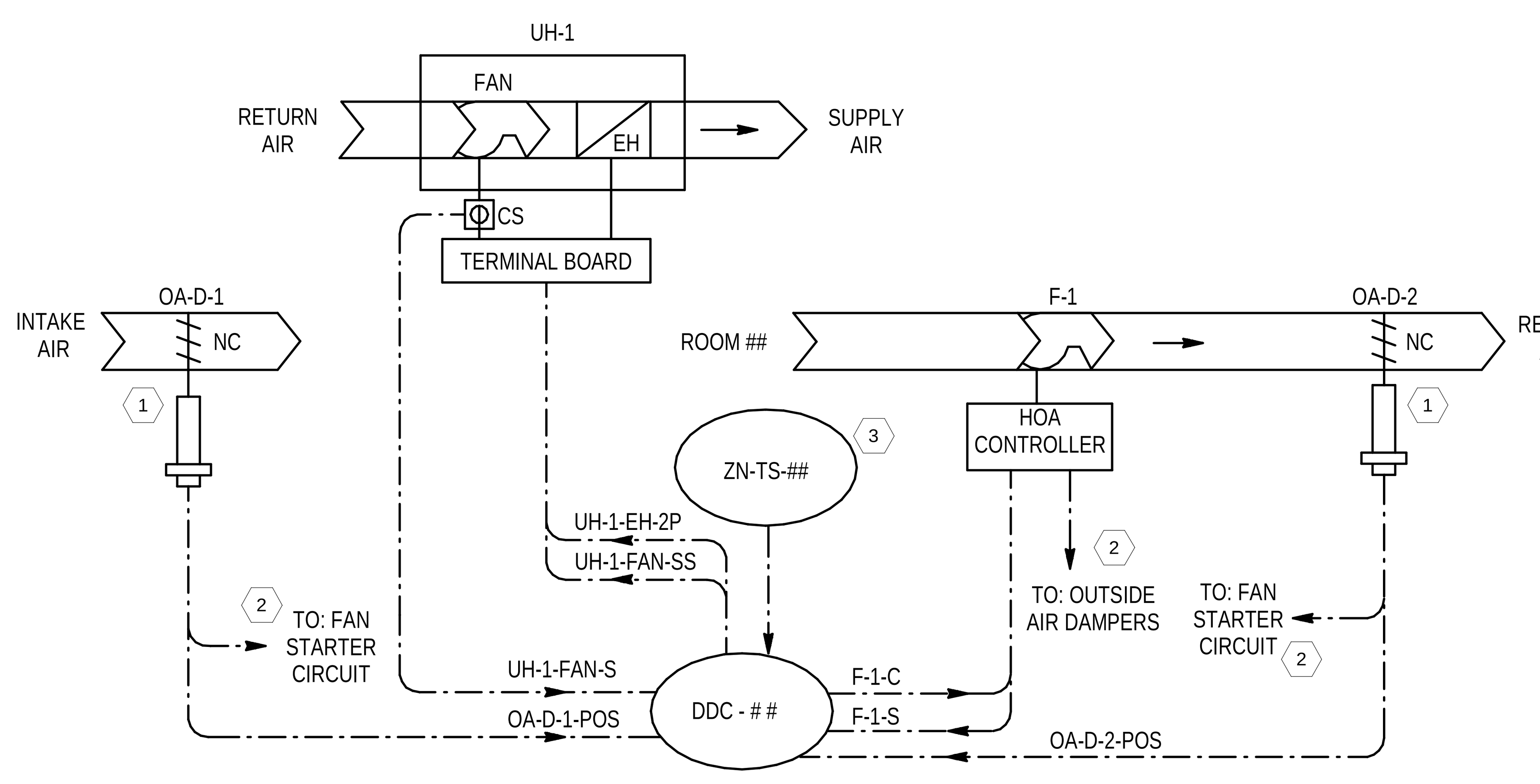
DES	CWH	DRW	CWH	CHK	MJL
PM/DM					SMS/CWJ
BRANCH MANAGER					ALG
CHIEF ENG/ARCH					JWJ
FIRE PROTECTION					DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL CONTROL DIAGRAMS & SEQUENCES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894030
SHEET 80 OF 140

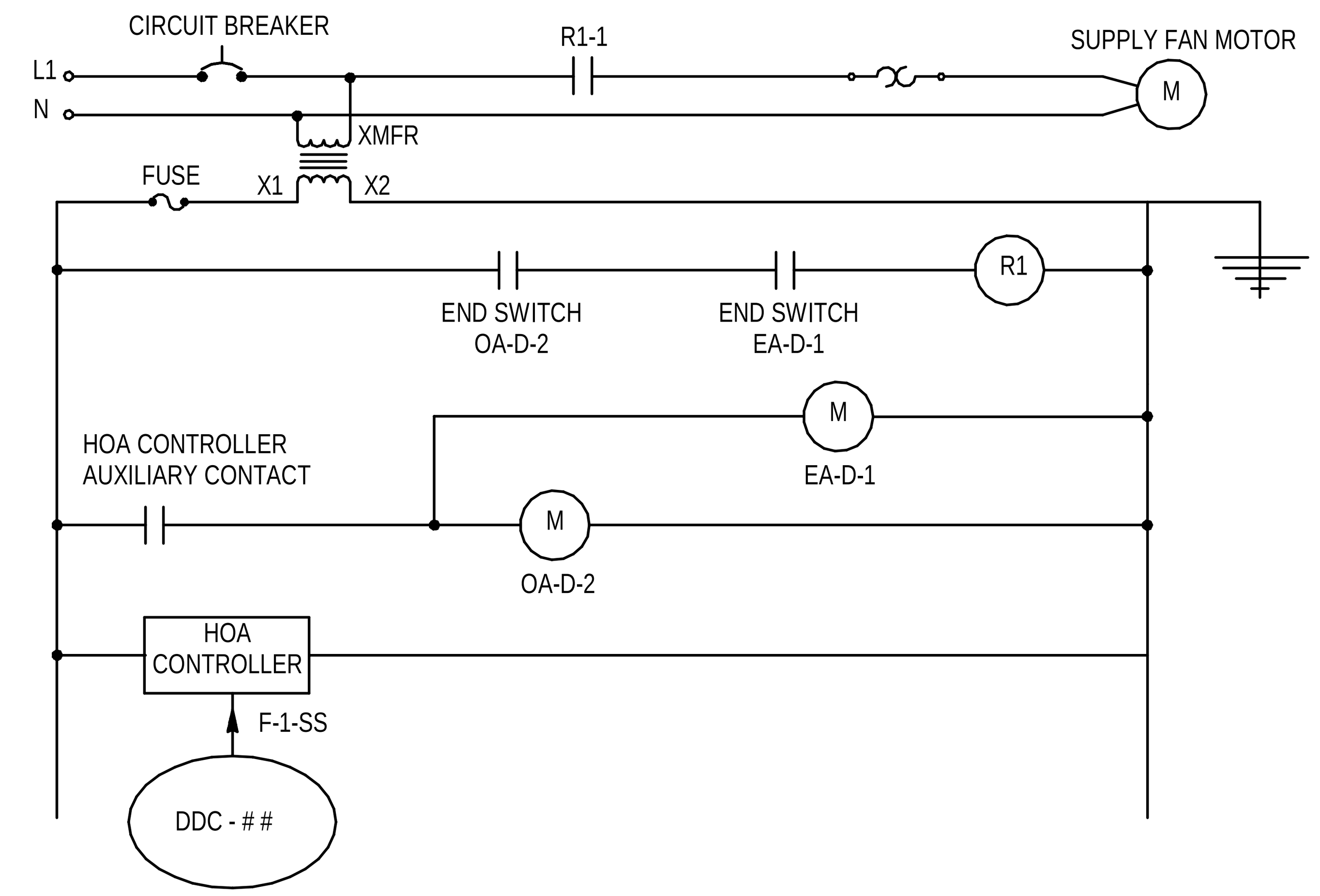
M-803

1	2	3	4	5



C1 FAN (F-1) AND UNIT HEATER (UH-1) - CONTROL DIAGRAM 8

NTS



FAN (F-1) AND UNIT HEATER (UH-1) SEQUENCE OF OPERATION

GENERAL:
1. THE F-1 FAN IS CONTROLLED BY THE BAS AS DESCRIBED BELOW.
2. PROGRAM AN ADJUSTABLE SOFTWARE TIME DELAY INTO THE BAS TO STAGGER THE RESTART OF UNIT AFTER A POWER FAILURE TO ASSURE THE GENERATOR IS NOT OVERLOADED WHEN IT RUNS AFTER A POWER FAILURE. SEE "SEQUENTIAL UNIT START-UP SCHEDULE AFTER A POWER FAILURE" FOR THE RESTART OF EACH EQUIPMENT.

FAN OPERATION: FAN HAND-OFF-AUTO OPERATION: A HAND-OFF-AUTO (HOA) CONTROLLER IS PROVIDED WITH ELECTRONICALLY COMMUTATED MOTOR (ECM). IN THE OFF MODE, THE OUTSIDE AIR INTAKE DAMPERS MUST BE CLOSED AND THE FAN MUST BE STOPPED. IN THE HAND MODE, THE OUTSIDE AIR INTAKE DAMPERS MUST OPEN. WHEN OUTSIDE AIR INTAKE DAMPERS ARE FULLY OPEN, THE ASSOCIATED DAMPER POSITION SWITCHES MUST START THE FAN. THE FAN MUST RUN AT THE SPEED SETPOINT MANUALLY SET AT HOA CONTROLLER. IN THE AUTO MODE, THE FAN MUST BE CONTROLLED BY THE BAS AS DESCRIBED BELOW.

AUTOMATIC MODE START/STOP CONTROL:
1. FAN START/STOP: THE BAS MUST START AND STOP THE FAN BASED ON SPACE TEMPERATURE. WHEN THE SPACE TEMPERATURE EXCEEDS THE TEMPERATURE SETPOINT, THE BAS MUST SEND A START SIGNAL TO THE FAN HOA CONTROLLER. THIS MUST ENABLE THE HARD WIRE INTERLOCK THAT OPENS THE OUTSIDE AIR INTAKE DAMPERS. WHEN THE DAMPERS ARE FULLY OPENED, AS SENSED BY THE DAMPER POSITION SWITCHES, THE FAN MUST BE ENERGIZED THROUGH THE HARD WIRE INTERLOCK AND RUN AS DESCRIBED BELOW.
2. FAN LOW SPEED CONTROL: WHEN THE SPACE TEMPERATURE RISES ABOVE 80 °F (ADJUSTABLE), THE BAS MUST SEND A SIGNAL TO THE ECM FAN WHICH MUST START THE FAN AND IT MUST OPERATE CONTINUOUSLY AT LOW SPEED. WHEN THE SPACE TEMPERATURE DROPS BELOW SPACE TEMPERATURE SET POINT, THE BAS MUST SEND A STOP SIGNAL TO THE FAN HOA CONTROLLER, WHICH DE-ENERGIZES THE FAN AND CLOSE THE DAMPERS.
3. FAN HIGH SPEED CONTROL: WHEN THE SPACE TEMPERATURE RISES ABOVE 100 °F (ADJUSTABLE), WHILE THE FAN IS RUNNING AT LOW SPEED, THE BAS MUST RUN THE FAN AT HIGH SPEED. WHEN THE SPACE TEMPERATURE DROPS BELOW THE SET POINT, THE REVERSE OCCURS.

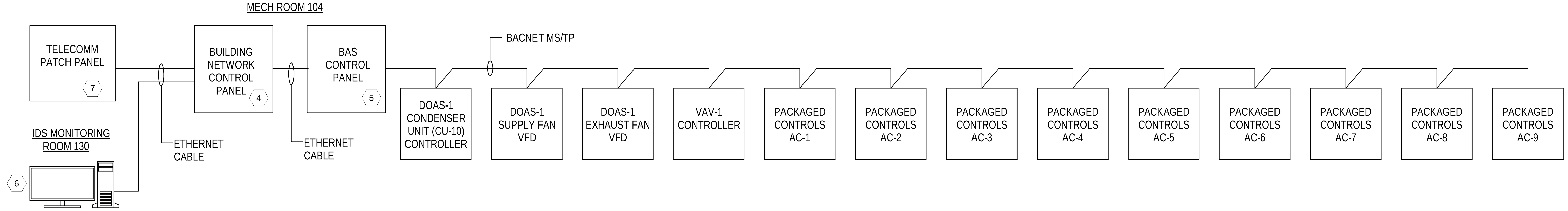
FAN PROOF: THE RUN STATUS FROM FAN HOA CONTROLLER MUST BE USED TO MONITOR THE STATUS OF THE FAN. IF THE STATUS INDICATED DOES NOT MATCH THE COMMANDED OUTPUT FOR THE FAN AFTER A 120 SECONDS (ADJUSTABLE) TIME DELAY, AN ALARM MUST BE GENERATED AND SENT TO THE BAS, AND THE FAN START COMMAND MUST BE CANCELED.

GENERAL:
1. THE UH-1 UNIT HEATER (UH) IS CONTROLLED BY THE BAS AS DESCRIBED BELOW.
2. PROGRAM AN ADJUSTABLE SOFTWARE TIME DELAY INTO THE BAS TO STAGGER THE RESTART OF UNIT AFTER A POWER FAILURE TO ASSURE THE GENERATOR IS NOT OVERLOADED WHEN IT RUNS AFTER A POWER FAILURE. SEE "SEQUENTIAL UNIT START-UP SCHEDULE AFTER A POWER FAILURE" FOR THE RESTART OF EACH EQUIPMENT.

UNIT HEATER CONTROL:
1. WHEN THE SPACE TEMPERATURE FALLS BELOW THE TEMPERATURE SET POINT, INITIALLY SET AT 55 DEGREES F (ADJUSTABLE), THE BAS MUST START THE UNIT HEATER FAN AND ENABLE THE ELECTRIC HEATING COIL TO MAINTAIN THE SPACE TEMPERATURE SET POINT. THE ELECTRIC HEATING COIL MUST NOT BE ENABLED UNTIL THE FAN OPERATION HAS BEEN PROVED.
2. WHEN THE SPACE TEMPERATURE RISES ABOVE SPACE TEMPERATURE SET POINT, THE BAS MUST DISABLE THE ELECTRIC HEATING COIL AND STOP UNIT HEATER FAN.

FAN PROOF: A CURRENT SWITCH MUST BE USED TO MONITOR THE STATUS OF THE FAN AND SEND A STATUS SIGNAL TO THE BAS WHEN THE FAN IS RUNNING. IF THE STATUS INDICATED DOES NOT MATCH THE COMMANDED OUTPUT FOR THE FAN, AFTER A 30 SECONDS (ADJUSTABLE) TIME DELAY, A FAN FAILURE ALARM MUST BE SENT TO THE BAS, AND THE FAN START COMMAND MUST BE CANCELED.

SPACE TEMPERATURE ALARM: A CRITICAL ALARM MUST BE ANNUNCIATED TO THE BAS IN THE EVENT THAT SPACE TEMPERATURE DROPS BELOW 50°F (ADJUSTABLE) OR RISES ABOVE 105°F (ADJUSTABLE).



A1 BAS NETWORK RISER DIAGRAM

NOT TO SCALE

GENERAL SHEET NOTES

A. SEE SHEET M-001 AND M-800 FOR GENERAL NOTES, SYMBOLS, AND ABBREVIATIONS.
B. REFER TO SHEET M-808 FOR FAN (F-1) AND UNIT HEATER (UH-1) POINTS SCHEDULE.

SHEET KEYNOTES

1. PROVIDE LOW LEAKAGE AIR DAMPER FOR OUTSIDE AIR INTAKE DAMPERS.
2. PROVIDE INTERLOCK WIRING BETWEEN FAN AND OUTSIDE AIR INTAKE DAMPERS. WHEN FAN IS COMMANDED TO START IN EITHER HAND OR AUTO MODE, THE DAMPERS MUST OPEN. ONCE THE DAMPERS ARE PROVEN OPEN BY DAMPER LIMIT SWITCHES, THE FAN MUST START. THE FAN MUST STOP IF ANY DAMPER CLOSES.
3. PROVIDE SPACE TEMPERATURE SENSOR FOR ROOM TEMPERATURE CONTROL AND ALARM. REFER TO MECHANICAL FLOOR PLAN SHEET M-401 FOR SPACE TEMPERATURE SENSOR'S LOCATION.
4. PROVIDE BUILDING NETWORK CONTROL PANEL TO SERVE AS BUILDING POINT OF CONTACT TO CONNECT LOCAL BAS TO BASE-WIDE UMCS. REFER TO MECHANICAL FLOOR ON M-401 SHEET FOR PANEL'S LOCATION.
5. PROVIDE BAS CONTROL PANEL FOR DOAS-1 AND MISCELLANEOUS HVAC EQUIPMENT. REFER TO MECHANICAL FLOOR PLAN SHEET M-401 FOR PANEL'S LOCATION.
6. PROVIDE OPERATOR WORKSTATION (OWS) LOCATED IN IDS MONITORING ROOM. OWS MUST ACT AS PRIMARY ACCESS POINT FOR LOCAL BAS INCLUDING SCHEDULE AND ALARM HANDLING, SET POINT ADJUSTMENT AND OTHER SUPERVISORY FUNCTIONS. ALL DDC CONTROLLERS, SOFTWARE AND OTHER COMPONENTS THAT ARE NOT PART OF EQUIPMENT MANUFACTURER'S PACKAGED CONTROLS MUST BE PROVIDED BY KMC NIAGARA CONTROLS. COORDINATE WORK AND ADDITIONAL REQUIREMENTS WITH THE GOVERNMENT.
7. REFER TO TELECOMMUNICATIONS DRAWINGS FOR THE PATCH PANEL'S LOCATION.
8. THE WIRING DIAGRAM IS SHOWN/PROVIDED AS A GUIDE TO THE CONTROLS CONTRACTOR. THE CONTROLS CONTRACTOR MUST DEVELOP THE WIRING DIAGRAM TO BE PROJECT SPECIFIC AND CONSISTENT WITH THE PROJECT SUBMITTALS.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES MP DRW AH CHK CH
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL CONTROL DIAGRAMS & SEQUENCES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894031
SHEET 81 OF 140
M-804

DEDICATED OUTSIDE AIR SYSTEM (DOAS) POINTS SCHEDULE												ALARMING			PRIMARY POINT INFORMATION		NETWORK DATA EXCHANGE		OVERRIDES	ALARM INFORMATION		
FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARM CONDITION	ALARM PRIORITY	M&C ROUTING	OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	NIAGARA ID	DATA FROM	DATA TO	OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION
PROOFS & SAFETIES	SF-S	SUPPLY FAN STATUS	<__>		ON/OFF	BI			V	V	Y	SUPPLY FAN PROOF FAILED	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EF-S	EXHAUST FAN STATUS	<__>		ON/OFF	BI			V	V	Y	EXHAUST FAN PROOF FAILED	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SA-SMK	SUPPLY AIR SMOKE	<__>		ALM/NORMAL	BI		H	V	V	Y	ALM	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	DX-COMP-S	COMPRESSOR STATUS	<__>		ON/OFF	BI			V	V	Y	COMPRESSOR PROOF FAILED	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	OVERFLOW-ALM	CONDENSATE DRAIN PAN OVERFLOW ALARM	<__>		ALM/NORMAL	BI			V	V	Y	ALM	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EH-S	ELECTRIC HEAT COIL STATUS	<__>		ON/OFF	BI			V	V	Y	ELECTRIC HEAT COIL PROOF FAILED	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	FBYP-D-POS	FACE AND BYPASS AIR DAMPER POSITION	<__>		OPEN/CLOSED	BI			V	V				[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	OA-D-POS	OUTDOOR AIR DAMPER POSITION	<__>		OPEN/CLOSED	BI			V	V				[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EA-D-POS	EXHAUST AIR DAMPER POSITION	<__>		OPEN/CLOSED	BI			V	V				[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
START/STOP	SYS-OCC	OCCUPANCY INPUT (FROM SYSTEM SCHEDULER)			OCC/UNOCC	NET-IN			V	V				[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SYS-OCC-STATUS	OCCUPANCY INPUT (ACTUAL)			OCC/UNOCC	NET-OUT			V	V				[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	AC-#-OCC-STATUS	AC-#-OCCUPANCY STATUS (AC-3 THRU AC-9)			OCC/UNOCC	NET-IN			V	V				[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EMERG-SHDN	EMERGENCY SHUTDOWN			NORMAL/ SHUTDOWN	NET-IN				VO		ALM	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SF-SS	SUPPLY FAN START/STOP	<__>		ON/OFF	BO	Y		VO	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EF-SS	EXHAUST FAN START/STOP	<__>		ON/OFF	BO	Y		VO	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	UNIT STATUS	UNIT STATUS			HEAT/COOL/ DEH/OFF	NET-OUT			V	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
SUPPLY FAN CONTROL	SA-F	SUPPLY AIR FLOW	<__>		<__>CFM	AI, NET-IN			V	V	Y	SYSTEM HAS BEEN IN OCCUPIED MODE FOR MORE THAN 15 MIN (ADJ.) AND SA-F IS MORE THAN 10% (ADJ.) ABOVE OR BELOW SA-F-SP		[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VAV-1-SA-F	VAV-1 SUPPLY AIR FLOW			<__>CFM	NET-IN			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SA-F-SP	SUPPLY AIR FLOW SETPOINT		SEE SEQUENCE				C	VO	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	SF-C	SUPPLY FAN COMMAND	<__>		0-100%	AO			VO	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
EXHAUST FAN CONTROL	RA-F	RETURN AIR FLOW	<__>		<__>CFM	AI			V	V	Y	SYSTEM HAS BEEN IN OCCUPIED MODE FOR MORE THAN 15 MIN (ADJ.) AND RA-F IS MORE THAN 10% (ADJ.) ABOVE OR BELOW EA-F-SP	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SA-F	SUPPLY AIR FLOW			<__>CFM	NET-IN			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	OFFSET (CFM)	OFFSET (CFM)		<__>CFM		NET-IN			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EA-F-SP	EXHAUST AIR FLOW SETPOINT		SEE SEQUENCE				C	VO	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	EF-C	EXHAUST FAN COMMAND	<__>		0-100%	AO			VO	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
OA HEAT EXCHANGER FACE AND BYPASS DAMPER CONTROL	OA-T	OUTSIDE AIR TEMPERATURE			<__>	NET-IN			V	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	ECO-HL-SP	ECONOMIZER HIGH LIMIT SETPOINT		62 DEG F				C	VO	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	ECO-LL-SP	ECONOMIZER LOW LIMIT SETPOINT		52 DEG F				C	VO	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	HX-LEV-T	HEAT EXCHANGER LEAVING AIR TEMPERATURE	<__>		<__>DEG F	AI			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EA-T	EXHAUST AIR TEMPERATURE	<__>		<__>DEG F	AI			V	V	Y	SYSTEM HAS BEEN IN OCCUPIED MODE FOR MORE THAN 15 MINUTES AND EA-T DROPS BELOW EA-T-SP	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EA-T-SP	EXHAUST AIR TEMPERATURE SETPOINT		33 DEG F				C	VO	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	FBYP-D-C	FACE AND BYPASS AIR DAMPER COMMAND	<__>		0-100%	AO			VO	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>

POINTS SCHEDULE LEGEND

[__] INDICATES INFORMATION WILL BE PROVIDED AS PART OF INTEGRATION	C CONFIGURABLE	INFO INFORMATIONAL ALARM	NET-OUT NETWORK OUTPUT POINT	V VIEWABLE
<__> INDICATES ENTRY MADE BY BUILDING CONTRACTOR	CRIT CRITICAL ALARM	N/A INDICATES NO VALID ENTRY AVAILABLE	O OVERRIDABLE	VO VIEWABLE AND OVERRIDABLE
~ INDICATES NO ENTRY REQUIRED	H HARDWARE	NET-IN NETWORK INPUT POINT	OC OPERATOR CONFIGURABLE	Y REQUIRED
INDICATES NO VALID ENTRY AVAILABLE				

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE 05/27/2025

DES	CWH
DRW	CWH
CHK	MJL
PM/DM	SMS/CWJ
BRANCH MANAGER	ALG
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA

NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS

MECHANICAL CONTROL POINTS SCHEDULE

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894032
SHEET 82 OF 140

M-805

DEDICATED OUTSIDE AIR SYSTEM (DOAS) POINTS SCHEDULE (CONTINUED)

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING: ALARM CONDITION	ALARMING: ALARM PRIORITY	ALARMING: M&C ROUTING	PRIMARY POINT INFORMATION: OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	PRIMARY POINT INFORMATION: NIAGARA ID	NETWORK DATA EXCHANGE: DATA FROM	NETWORK DATA EXCHANGE: DATA TO	OVERRIDES: OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM INFORMATION: ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM INFORMATION: NOTIFICATION CLASS OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION
OA HEAT EXCHANGER FACE AND BYPASS DAMPER CONTROL (CONTINUED)	OA HX FACE & BYPASS DAMPER PID LOOP SETTINGS	PROPORTIONAL CONSTANT		<__>				C						[__]								<__>
		INTEGRAL CONSTANT		<__>				C						[__]								<__>
		BIAS		<__>				C						[__]								<__>
		MAXIMUM OUTPUT		<__>				C						[__]								<__>
		MINIMUM OUTPUT		<__>				C						[__]								<__>
SUPPLY AIR TEMPERATURE CONTROL: HEATING COIL & COOLING COIL	SA-T	SUPPLY AIR TEMPERATURE	<__>		<__>DEG F	AI			V	V	Y	SYSTEM HAS BEEN IN OCCUPIED MODE FOR MORE THAN 15 MINUTES AND SA-T IS MORE THAN 5 DEG F ABOVE OR BELOW SA-T-SP	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SA-T-SP	SUPPLY AIR TEMPERATURE SETPOINT		HTG = 68 DEG F; CLG = 73 DEG F				C	VO	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	DX-COMP-MODE	COMPRESSOR COOLING/HEATING MODE			CLG/HTG	NET-IN			VO					[__]	<__>	<__>	<__>	<__>	<__>			<__>
	DX-COMP-C	COMPRESSOR COMMAND	<__>		ON/OFF	BO			VO					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EH-LA-T	ELECTRIC HEAT COIL LEAVING AIR TEMPERATURE	<__>		<__>DEG F	AI			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EH-C	ELECTRIC HEATING COIL COMMAND	<__>		0-100%	AO			VO	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	HEATING COIL PID LOOP SETTINGS	PROPORTIONAL CONSTANT		<__>				C														<__>
		INTEGRAL CONSTANT		<__>				C														<__>
		BIAS		<__>				C														<__>
		MAXIMUM OUTPUT		<__>				C														<__>
		MINIMUM OUTPUT		<__>				C														<__>
DEHUMIDIFICATION CONTROL	BLDG-T	BUILDING AIR TEMPERATURE	<__>		<__>DEG F	AI			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	[__]	[__]	<__>
	BLDG-H	BUILDING AIR HUMIDITY	<__>		<__>% RH	AI			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	[__]	[__]	<__>
	BLDG-DWP-T	BUILDING AIR DEW POINT TEMPERATURE (CALCULATED)			<__>DEG F	NET-IN			V	V	Y	SYSTEM IS IN UNOCCUPIED MODE AND BLDG-DWP-T IS MORE THAN 5 DEG F ABOVE BLDG-DWP-T-SP	CRIT	[__]	<__>	<__>	<__>	<__>	<__>			<__>
	BLDG-DWP-T-SP	BUILDING AIR DEW POINT TEMPERATURE SETPOINT		57 DEG F		NET-IN			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>			<__>
	RA-T	RETURN AIR TEMPERATURE	<__>		<__>DEG F	AI			V				CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	RA-H	RETURN AIR HUMIDITY	<__>		<0 - __>% RH	AI			V					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	RA-DWP-T	BUILDING AIR DEW POINT TEMPERATURE (CALCULATED)			<__>DEG F	NET-IN			V	V	Y	SYSTEM IS IN OCCUPIED MODE AND RA-DWP-T IS MORE THAN 5 DEG F ABOVE RA-DWP-T-SP	CRIT	[__]	<__>	<__>	<__>	<__>	<__>			<__>
	RA-DWP-T-SP	BUILDING AIR DEW POINT TEMPERATURE SETPOINT		57 DEG F		NET-IN			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>			<__>
	RA-H-SP	RETURN AIR HUMIDITY SETPOINT		50 % RH				C	VO	VO				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	DX-LA-T	COOLING COIL LEAVING AIR TEMPERATURE	<__>		<__>DEG F	AI			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	DX-LA-T-SP	COOLING COIL LEAVING AIR TEMPERATURE SETPOINT		53 DEG F		NET-IN			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>			<__>
	SA-T	SUPPLY AIR TEMPERATURE			<__>DEG F	NET-IN			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SA-T-SP	SUPPLY AIR TEMPERATURE SETPOINT		68 DEG F				C	VO	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	DX-COMP-MODE	COMPRESSOR COOLING/HEATING MODE			CLG/HTG	NET-IN			VO					[__]	<__>	<__>	<__>	<__>	<__>	[__]	[__]	<__>
	DX-COMP-SS	COMPRESSOR START/STOP			ON/OFF	BO			VO					[__]	<__>	<__>	<__>	<__>	<__>			<__>
	DX-HGR-DPT-OUT	HOT GAS DEW POINT TEMPERATURE OUTPUT	<__>		0-100%	AO			VO	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>
	DX-HGR-C	HOT GAS REHEAT COMMAND	<__>		0-100%	AO			VO	V				[__]	<__>	<__>	<__>	<__>	<__>			<__>

POINTS SCHEDULE LEGEND

[__] INDICATES INFORMATION WILL BE PROVIDED AS PART OF INTEGRATION	C CONFIGURABLE	INFO INFORMATIONAL ALARM	NET-OUT NETWORK OUTPUT POINT	V VIEWABLE
<__> INDICATES ENTRY MADE BY BUILDING CONTRACTOR	CRIT CRITICAL ALARM	N/A INDICATES NO VALID ENTRY AVAILABLE	O OVERRIDABLE	VO VIEWABLE AND OVERRIDABLE
~ INDICATES NO ENTRY REQUIRED	H HARDWARE	NET-IN NETWORK INPUT POINT	OC OPERATOR CONFIGURABLE	Y REQUIRED
INDICATES NO VALID ENTRY AVAILABLE				

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/25/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. Dobratz, RA
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH DRW CWH CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL CONTROL POINTS SCHEDULE

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894033
SHEET 83 OF 140

M-806

DEDICATED OUTSIDE AIR SYSTEM (DOAS) POINTS SCHEDULE (CONTINUED)

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING: ALARM CONDITION	ALARMING: ALARM PRIORITY	ALARMING: M&C ROUTING	PRIMARY POINT INFORMATION: OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	PRIMARY POINT INFORMATION: NIAGARA ID	NETWORK DATA EXCHANGE: DATA FROM	NETWORK DATA EXCHANGE: DATA TO	OVERRIDES: OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM INFORMATION: ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM INFORMATION: NOTIFICATION CLASS OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION
MINIMUM UNIT FAN VFD NETWORK POINTS (TYPICAL)	VFD MODE	VFD RUN MODE			HAND/AUTO	NET-OUT			V					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VFD ALM	VFD ALARM			ALARM/NORMAL	NET-OUT			V			ALM	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VFD OUT-SPEED	VFD OUTPUT SPEED			<__RPM>	NET-OUT			V					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VFD OUT-FREQ	VFD OUTPUT FREQUENCY			<__HZ>	NET-OUT			V					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VFD OUT CURRENT	VFD OUTPUT CURRENT			<__AMPS>	NET-OUT			V					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VFD OUT-PWR	VFD OUTPUT POWER			<__KW>	NET-OUT			V					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VFD KW-HOURS	VFD ENERGY USAGE			<__KWH>	NET-OUT			V					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VFD COMM ERROR	VFD COMMUNICATION LOSS			ALARM/NORMAL	NET-OUT			V			ALM	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
OTHER POINTS	OA-T	OUTSIDE AIR TEMPERATURE	<__>		<__>DEG F	AI			V					[__]	<__>	<__>	<__>	<__>		[__]	[__]	
	OA-H	OUTSIDE AIR HUMIDITY	<__>		<__>% RH	AI			V					[__]	<__>	<__>	<__>	<__>		[__]	[__]	
	SA-H	SUPPLY AIR HUMIDITY	<__>		<0 - __>% RH	AI			V					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	OA-FLT-P-HL	OUTSIDE AIR FILTER PRESSURE HIGH LIMIT SWITCH	<__>	<__>	ALM/NORMAL	BI		H	V			ALM	INFO	[__]	<__>	<__>	<__>	<__>		[__]	[__]	
	RA-FLT-P-HL	RETURN AIR FILTER PRESSURE HIGH LIMIT SWITCH	<__>	<__>	ALM/NORMAL	BI		H	V			ALM	INFO	[__]	<__>	<__>	<__>	<__>		[__]	[__]	

SPLIT SYSTEM AIR CONDITIONER (AC-1 & AC-2) POINTS SCHEDULE

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING: ALARM CONDITION	ALARMING: ALARM PRIORITY	ALARMING: M&C ROUTING	PRIMARY POINT INFORMATION: OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	PRIMARY POINT INFORMATION: NIAGARA ID	NETWORK DATA EXCHANGE: DATA FROM	NETWORK DATA EXCHANGE: DATA TO	OVERRIDES: OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM INFORMATION: ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM INFORMATION: NOTIFICATION CLASS OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION
PROOFS & SAFETIES	UNIT-SAF	UNIT MANUFACTURER'S SAFETIES			ALM/NORMAL	NET-OUT		C				UNIT ALM	CRIT	[__]	<__>	<__>	<__>	<__>		<__>	<__>	<__>
	UNIT-COMM	UNIT COMMUNICATION STATUS			ALM/NORMAL	NET-OUT						ALM	INFO	[__]	<__>	<__>	<__>	<__>		<__>	<__>	<__>
START/STOP	MODE-SEL	OFF-COOL MODE SELECTION SWITCH INPUT			OFF/COOL	NET-IN									<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SF-SEL	SUPPLY FAN ON/AUTO SELECT SWITCH			ON/AUTO	NET-IN									<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	UNIT STATUS	UNIT STATUS			HVAC_COOL/ HVAC_OFF	NET-OUT			V						<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
ZONE TEMPERATURE CONTROL	ZN-T	ZONE TEMPERATURE			< - __DEG>	NET-OUT			V	V	Y	ZN-T IS MORE THAN 5 DEG F ABOVE ZN-T-SP	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	ZN-T-SP	ZONE TEMPERATURE SETPOINT		75 DEG F	60 - 82 DEG F	NET-IN		C	VO		Y			[__]	<__>	<__>	<__>	<__>	<__>			<__>
	SF-SS	SUPPLY FAN START/STOP COMMAND			ON/OFF	NET-IN			VO					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	COMP-MODE	COMPRESSOR COOLING/ HEATING MODE			CLG/HTG	NET-IN			VO					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	COMP-SS	COMPRESSOR START/STOP			ON/OFF	NET-IN			VO					[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>

MISCELLANEOUS POINTS SCHEDULE

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING: ALARM CONDITION	ALARMING: ALARM PRIORITY	ALARMING: M&C ROUTING	PRIMARY POINT INFORMATION: OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	PRIMARY POINT INFORMATION: NIAGARA ID	NETWORK DATA EXCHANGE: DATA FROM	NETWORK DATA EXCHANGE: DATA TO	OVERRIDES: OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM INFORMATION: ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM INFORMATION: NOTIFICATION CLASS OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION
MISCELLANEOUS POINTS	AC-#-SA-T	AC-# SUPPLY AIR TEMPERATURE (TYPICAL OF 7)	<__>		<__>DEG F	AI			V	V	Y				<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	AC-#-OCC-SW	AC-# OCCUPANCY SWITCH (TYPICAL OF 7)	<__>		OCC/UNOCC	BI			V	V	Y				<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	CDP-#-ALM	CONDENSATE PUMP ALARM (TYPICAL OF 2)	<__>		ALM/NORMAL	BI			V	V	Y	ALM	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	EMERG-SW	AIR DISTRIBUTION EMERGENCY SHUT-OFF SWITCH	<__>		NORMAL/ SHUTDOWN	BI		H	V	V		ALM	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>

POINTS SCHEDULE LEGEND

[__] INDICATES INFORMATION WILL BE PROVIDED AS PART OF INTEGRATION	C CONFIGURABLE	INFO INFORMATIONAL ALARM	NET-OUT NETWORK OUTPUT POINT	V VIEWABLE
<__> INDICATES ENTRY MADE BY BUILDING CONTRACTOR	CRIT CRITICAL ALARM	N/A INDICATES NO VALID ENTRY AVAILABLE	O OVERRIDABLE	VO VIEWABLE AND OVERRIDABLE
~ INDICATES NO ENTRY REQUIRED	H HARDWARE	NET-IN NETWORK INPUT POINT	OC OPERATOR CONFIGURABLE	Y REQUIRED
■ INDICATES NO VALID ENTRY AVAILABLE				

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES CWH | DRW CWH | CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL CONTROL POINTS SCHEDULE

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894034
SHEET 84 OF 140

M-807

SPLIT SYSTEM AIR CONDITIONER (AC-3 THRU AC-9) POINTS SCHEDULE

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING: ALARM CONDITION	ALARMING: ALARM PRIORITY	ALARMING: M&C ROUTING	PRIMARY POINT INFORMATION: OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	PRIMARY POINT INFORMATION: NIAGARA ID	NETWORK DATA EXCHANGE: DATA FROM	NETWORK DATA EXCHANGE: DATA TO	OVERRIDES: OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM INFORMATION: ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM INFORMATION: NOTIFICATION CLASS OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION
PROOFS & SAFETIES	UNIT-SAF	UNIT MANUFACTURER'S SAFETIES			ALM/NORMAL	NET-OUT		C				UNIT ALM	CRIT	[__]	<__>	<__>	<__>	<__>		<__>	<__>	<__>
	UNIT-COMM	UNIT COMMUNICATION STATUS			ALM/NORMAL	NET-OUT						ALM	INFO	[__]	<__>	<__>	<__>	<__>		<__>	<__>	<__>
START/STOP	OPT-SS-MODE	OPTIMUM START MODE			WARM-UP/ COOL-DOWN/OFF	NET-IN									<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SYS-OCC	OCCUPANCY INPUT (FROM SCHEDULER)			OCC/UNOCC	NET-IN			VO	V					<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	AC-OCC-SW	AC OCCUPANCY SWITCH			OCC/UNOCC	NET-IN									<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SF-SEL	SUPPLY FAN ON/AUTO SELECT SWITCH			ON/AUTO	NET-IN									<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	UNIT STATUS	UNIT STATUS			HVAC_COOL/ HVAC_HEAT/ HVAC_OFF	NET-OUT			V						<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
ZONE TEMPERATURE CONTROL	ZN-T	ZONE TEMPERATURE				NET-OUT			V		Y	ZN-T IS MORE THAN 5 DEG F ABOVE COOLING SETPOINT OR BELOW HEATING SETPOINT DURING OCCUPIED OR UNOCCUPIED MODE	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	ZN-T-SP	ZONE TEMPERATURE SETPOINT		68 DEG F FOR HEATING, 78 DEG F FOR COOLING	60 - 82 DEG F	NET-IN		C	VO		Y				<__>	<__>	<__>	<__>	<__>			<__>
	ZN-T-SP-UNOCC	ZONE TEMPERATURE SETPOINT FOR UNOCC MODE		60 DEG F FOR HEATING, 83 DEG F FOR COOLING		NET-IN		C	VO						<__>	<__>	<__>	<__>	<__>			<__>
	SF-SS	SUPPLY FAN START/STOP COMMAND			ON/OFF	NET-IN			VO						<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	COMP-MODE	COMPRESSOR COOLING/ HEATING MODE			CLG/HTG	NET-IN			VO						<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	COMP-SS	COMPRESSOR START/STOP			ON/OFF	BO			VO						<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>

F-1 FAN & UH-1 UNIT HEATER POINTS SCHEDULE

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING: ALARM CONDITION	ALARMING: ALARM PRIORITY	ALARMING: M&C ROUTING	PRIMARY POINT INFORMATION: OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	PRIMARY POINT INFORMATION: NIAGARA ID	NETWORK DATA EXCHANGE: DATA FROM	NETWORK DATA EXCHANGE: DATA TO	OVERRIDES: OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM INFORMATION: ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	ALARM INFORMATION: NOTIFICATION CLASS OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION
PROOFS & SAFETIES	F-1-S	F-1 FAN STATUS	<__>		ON/OFF	BI			V	V	Y	F-1 FAN PROOF FAILED	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	UH-1-FAN-S	UH-1 FAN STATUS	<__>		ON/OFF	BI			V	V	Y	UNIT HEATER FAN PROOF FAILED	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	OA-D-1-POS	OUTSIDE AIR DAMPER POSTION	<__>		OPEN/CLOSED	BI			V	V					<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	OA-D-2-POS	OUTSIDE AIR DAMPER POSTION	<__>		OPEN/CLOSED	BI			V	V					<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
FAN SHUTDOWN	EMERG-SHDN	EMERGENCY SHUTDOWN			NORMAL/ SHUTDOWN	NET-IN				VO		ALM	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
ZONE TEMPERATURE CONTROL	ZN-TS-##	ZONE TEMPERATURE	<__>		45-120 DEG F	AI			V	V	Y	ZN-T IS ABOVE 105 DEG F (ADJ.) OR BELOW 50 DEG F (ADJ.)	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	ZN-T-SP	ZONE TEMPERATURE SETPOINT		SEE SEQUENCE		NET-IN		C	VO	V					<__>	<__>	<__>	<__>				<__>
	F-1-C	SUPPLY FAN COMMAND	<__>		0-100%	AO			VO	V	Y				<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	UH-1-FAN-SS	SUPPLY FAN START/STOP	<__>		ON/OFF	BO			V	V	Y				<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	UH-1-EH-2P	ELECTRIC COIL COMMAND	<__>		ON/OFF	BO			VO	V	Y				<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>

POINTS SCHEDULE LEGEND

- [__] INDICATES INFORMATION WILL BE PROVIDED AS PART OF INTEGRATION
- <__> INDICATES ENTRY MADE BY BUILDING CONTRACTOR
- ~ INDICATES NO ENTRY REQUIRED
- (solid black) INDICATES NO VALID ENTRY AVAILABLE
- C CONFIGURABLE
- CRIT CRITICAL ALARM
- H HARDWARE
- INFO INFORMATIONAL ALARM
- N/A INDICATES NO VALID ENTRY AVAILABLE
- NET-IN NETWORK INPUT POINT
- NET-OUT NETWORK OUTPUT POINT
- O OVERRIDABLE
- OC OPERATOR CONFIGURABLE
- V VIEWABLE
- VO VIEWABLE AND OVERRIDABLE
- Y REQUIRED

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/23/2025
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES CWH | DRW CWH | CHK MJL
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION DSN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL CONTROL POINTS SCHEDULE

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894035
SHEET 85 OF 140
M-808

ELECTRIC UNIT HEATER (UH-2) POINTS SCHEDULE

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING: ALARM CONDITION	ALARM PRIORITY	M&C ROUTING	PRIMARY POINT INFORMATION: OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	NIAGARA ID	NETWORK DATA EXCHANGE: DATA FROM	DATA TO	OVERRIDES: OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM INFORMATION: ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION
PROOFS & SAFETIES	UH-2-FAN-S	UH-2 FAN STATUS	<__>		ON/OFF	BI			V	V	Y	UNIT HEATER FAN PROOF FAILED	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
SYSTEM SHUT OFF	OA-T	OUTSIDE AIR TEMPERATURE			<0 - ___>DEG F	NET-IN			V						<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	OA-T-SP	OUTSIDE AIR TEMPERATURE SETPOINT		45 DEG F		NET-IN			V						<__>	<__>	<__>	<__>				<__>
ZONE TEMPERATURE CONTROL	ZN-TS-##	ZONE TEMPERATURE	<__>		45-120 DEG F	AI			V	V	Y	ZN-T IS BELOW 50 DEG F (ADJ.)	CRIT	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	ZN-T-SP	ZONE TEMPERATURE SETPOINT		55 DEG F		NET-IN		C	VO	V					<__>	<__>	<__>	<__>				<__>
	UH-2-FAN-SS	UH-2 SUPPLY FAN START/STOP	<__>		ON/OFF	BO			V	V	Y				<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	UH-2-EH-2P	UH-2 ELECTRIC COIL COMMAND	<__>		ON/OFF	BO			VO	V	Y				<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>

VAV-1 BOX POINTS SCHEDULE

FUNCTION	NAME	DESCRIPTION	DDC HARDWARE ID	SETTING (WITH UNITS)	RANGE (WITH UNITS)	IO TYPE	HOA REQ'D	CONFIG. TYPE	M&C VIEW AND OVRD	LDP VIEW AND OVRD	TREND REQ'D	ALARMING: ALARM CONDITION	ALARM PRIORITY	M&C ROUTING	PRIMARY POINT INFORMATION: OBJECT TYPE AND INSTANCE NUMBER (AND PROPERTY ID)	NIAGARA ID	NETWORK DATA EXCHANGE: DATA FROM	DATA TO	OVERRIDES: OBJECT TYPE & INSTANCE NUMBER OR COMMANDABLE	ALARM INFORMATION: ALARM TYPE (PROVIDE DDC ID FOR REMOTE IN NOTES)	NOTIFICATION CLASS OBJECT INSTANCE NUMBER	CONFIGURATION INFORMATION
PROOFS & SAFETIES	VAV-1-COMM	VAV COMMUNICATION STATUS			ALM/NORMAL	NET-OUT						ALM	INFO	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VAV-1-D-POS	VAV DAMPER POSTION	<__>		0-100% OPEN	AI			V	V				[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
SCHEDULING	SYS-OCC	OCCUPANCY INPUT (FROM SYSTEM SCHEDULER)			OCC/UNOCC	NET-IN			VO						<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	ZN-OCC	ZONE OCCUPANCY INPUT (ROOM OCCUPANCY SENSOR)	<__>		OCC/UNOCC	BI									<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	SYS-OCC-STATUS	OCCUPANCY INPUT (ACTUAL)			OCC/UNOCC	NET-OUT			V						<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
VAV BOX CONTROL	VAV-1-SA-F	VAV SUPPLY AIR FLOW	<__>		[0 - ___ CFM]	AI			V	V	Y	VAV HAS BEEN IN OCCUPIED MODE FOR MORE THAN 15 MIN (ADJ.) AND SA-F IS MORE THAN 10% (ADJ.) ABOVE OR BELOW SA-F-SP	INFO	[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VAV-1-SA-F-SP	VAV SUPPLY AIR FLOW SETPOINT		SEE SCHEDULE				C						[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VAV-1-D-C	VAV DAMPER COMMAND	<__>		0-100% OPEN	AO			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>
	VAV DAMPER PID LOOP SETTINGS	PROPORTIONAL CONSTANT		<__>				C														<__>
		INTEGRAL CONSTANT		<__>				C														<__>
		BIAS		<__>				C														<__>
		MAXIMUM OUTPUT		<__>				C														<__>
		MINIMUM OUTPUT		<__>				C														<__>
OTHER POINTS	VAV-1-SA-T	VAV SUPPLY AIR TEMPERATURE	<__>		<__>DEG F	AI			V	V	Y			[__]	<__>	<__>	<__>	<__>	<__>	<__>	<__>	<__>

THERMOSTAT SCHEDULE / OCCUPANCY SCHEDULE

SYSTEM/UNIT IDENTIFIER	SPACES SERVED	STAT LOCATION	ROOM-T	ROOM-T-SP ADJUST	UNOCC OVERRIDE PUSHBUTTON	UNOCC OVERRIDE TIME	OCC SENSOR	OCCUPIED	UN-OCCUPIED	WARM-UP/ COOL-DOWN	WARM-UP/COOL-DOWN TIME (BAS & LOCAL DEFAULT): MON-FRI	WEEKEND/ HOLIDAY	OCCUPIED TIME (BAS & LOCAL DEFAULT): MON-FRI	WEEKEND/ HOLIDAY	UNOCCUPIED TIME (BAS & LOCAL DEFAULT): MON-FRI	WEEKEND/ HOLIDAY
AC-3 THRU AC-9 (TYPICAL)	OFFICE SPACES	OFFICE SPACES	X	X	X	2 HOUR	~	X	X	X	0500-0600	~	0600-1800	~	1800-0600	0000-2400
AC-1/AC-2 (TYPICAL)	TELECOM/ELEC ROOMS	TELECOM/ELEC ROOMS	X	X	~	~	~	~	~	~	~	~	~	~	~	~
F-1/UH-1	MECH RM 104	MECH RM 104	X	~	~	~	~	~	~	~	~	~	~	~	~	~
UH-2	VESTIBULE	VESTIBULE	X	~	~	~	~	~	~	~	~	~	~	~	~	~

NOTES:

"~" INDICATES NO ENTRY REQUIRED.

POINTS SCHEDULE LEGEND

[__] INDICATES INFORMATION WILL BE PROVIDED AS PART OF INTEGRATION
<__> INDICATES ENTRY MADE BY BUILDING CONTRACTOR
~ INDICATES NO ENTRY REQUIRED
INDICATES NO VALID ENTRY AVAILABLE

C CONFIGURABLE
CRIT CRITICAL ALARM
H HARDWARE

INFO INFORMATIONAL ALARM
N/A INDICATES NO VALID ENTRY AVAILABLE
NET-IN NETWORK INPUT POINT

NET-OUT NETWORK OUTPUT POINT
O OVERRIDABLE
OC OPERATOR CONFIGURABLE

V VIEWABLE
VO VIEWABLE AND OVERRIDABLE
Y REQUIRED

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
CHRISTOPHER W. HILL
Lic. No. 0402055245
6/25/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES MP | DRW AH | CHK CH
PM/DM SMS/CWJ
BRANCH MANAGER ALG
CHIEF ENG/ARCH JWJ
FIRE PROTECTION

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
MECHANICAL CONTROL POINTS SCHEDULE

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894036
SHEET 86 OF 140
M-809

GENERAL NOTES:

1. WHERE "SPLIT CIRCUIT" IS INDICATED, EXTEND SPLIT CIRCUIT HOME RUNS TO A COMMON JUNCTION BOX ABOVE THE PANEL INDICATED. TIE HOME RUNS TOGETHER USING SPLICE KIT AND EXTEND A SINGLE SET OF CONDUCTORS TO CIRCUIT BREAKER NUMBER INDICATED. WIRING USED IN SPLIT CIRCUITS MUST BE THE SAME SIZE.
2. THE ELECTRICAL DRAWINGS ARE DIAGRAMMATICAL IN NATURE AND INDICATE THE GENERAL LOCATION OF OUTLETS, EQUIPMENT, AND THE CIRCUIT ARRANGEMENT OF THE REQUIRED WIRING. ALTHOUGH THE DRAWINGS DO NOT NECESSARILY INDICATE THE ACTUAL ROUTES OF SURFACE MOUNTED CONDUITS, WHERE INDICATED, THEY MUST BE FOLLOWED AS CLOSELY AS COORDINATION WITH THE WORK OF OTHER TRADES AND SPACE WILL PERMIT. WHERE CONDUIT RUNS ARE NOT SHOWN ON THE DRAWINGS, SUCH AS FOR MECHANICAL EQUIPMENT, COORDINATE CONDUIT RUNS WITH THE WORK OF OTHER TRADES AND STRUCTURE. SIMPLIFY INSTALLATION WHEREVER POSSIBLE, BUT SUBJECT TO APPROVAL BY THE COR FOR VISUAL AND STRUCTURAL REASONS. IT IS NOT WITHIN THE SCOPE OF THE DRAWINGS TO SHOW EVERY NECESSARY OFFSETS, BENDS, PULL BOXES, AND OBSTRUCTIONS. THE DRAWINGS ARE NOT INTENDED TO BE SCALED, REFER TO THE ARCHITECTURAL DRAWINGS FOR DIMENSIONS.
3. NOT EVERY SYMBOL SHOWN IN LEGENDS ARE USED - SEE DRAWINGS FOR APPLICABLE DEVICES.
4. MOUNTING HEIGHTS INDICATED AS '+##' ARE MEASURED FROM FINISHED FLOOR UNLESS OTHERWISE NOTED.
5. PROVIDE LABELS DENOTING EQUIPMENT SERVED AND CORRESPONDING CIRCUIT NUMBER ON DEDICATED RECEPTACLES.
6. SEAL PENETRATIONS OF FIRE- AND/OR SMOKE-RATED CONSTRUCTION WITH U.L. APPROVED STOPPING MATERIAL TO RESTORE THE ORIGINAL RATING OF THE PENETRATED CONSTRUCTION.
7. SEAL PENETRATIONS OF NON-RATED PARTITIONS WITH APPROVED MATERIAL TO PREVENT AIRFLOW AND DEBRIS.
8. MOUNTING HEIGHTS SHOWN ON ELECTRICAL, COMMUNICATIONS, AND SYSTEMS DRAWINGS ARE FOR BIDDING PURPOSES ONLY. REFER TO INTERIOR ELEVATION DRAWINGS FOR MOUNTING HEIGHTS OF DEVICES AND EQUIPMENT. WHERE DISCREPANCIES OCCUR, INTERIOR ELEVATIONS TAKE PRECEDENCE.
9. SUPPORT BOXES AND CONDUIT ABOVE CEILING INDEPENDENTLY. DO NOT SUPPORT FROM SUSPENDED CEILING SYSTEM.
10. WHERE CONCEALED BY INACCESSIBLE FINISHES, PROVIDE ACCESS DOORS FOR JUNCTION & PULL BOXES, CONTROL DEVICES & EQUIPMENT, DISCONNECT SWITCHES, AND OTHER ITEMS REQUIRING SERVICE, MAINTENANCE, OPERATION, OR ADJUSTMENT. SIZE AND LOCATE ACCESS DOORS FOR EASE OF PERFORMANCE OF THE FUNCTION INTENDED. DOORS MUST NOT BE SMALLER THAN 18" x 18". COORDINATE LOCATION OF ACCESS DOORS WITH AFFECTED TRADES. SUBMIT PANEL LOCATIONS FOR APPROVAL BY THE ARCHITECT PRIOR TO INSTALLATION.
11. PROVIDE ADJACENT DEVICES UNDER A COMMON FACEPLATE, WITH NO MORE THAN FOUR DEVICES IN A SINGLE FACEPLATE. WHERE MORE THAN FOUR ADJACENT DEVICES ARE PRESENT, EVENLY DIVIDE DEVICES INTO SEPARATE GROUPS OF NO LESS THAN TWO DEVICES PER GROUP. PROVIDE BARRIERS WITHIN BOXES TO SEPARATE DEVICES WHERE DIFFERENT CIRCUITS AND DIFFERENT VOLTAGES ARE PRESENT.
12. PROVIDE LUGS AS NECESSARY FOR TERMINATING OVERSIZED CONDUCTORS OR OTHERWISE MAKE PROVISIONS FOR TERMINATING OVERSIZED CONDUCTORS IN DEVICES WHOSE LUGS DO NOT ACCEPT THE OVERSIZED CONDUCTOR SIZE.
13. SHARED NEUTRAL CONDUCTORS ARE PROHIBITED.
14. OUTAGES MUST BE SCHEDULED WITH A MINIMUM OF 21 CALENDAR DAYS IN ADVANCE. NOTIFY THE BASE PERSONNEL (CONTRACTING OFFICER). MAKE AN APPLICATION FOR OUTAGES TO GOVERNMENT HIGH VOLTAGE FACILITY DEPARTMENT WITH THE PROVIDED REQUEST FORMS AND SUBMIT FOR APPROVAL FOR THE OUTAGES. NO OUTAGES WILL TAKE PLACE WITHOUT WRITTEN AUTHORIZATION TO THE GOVERNMENT PERSONNEL BY THE CONTRACTOR. OUTAGES CAN BE CANCELED BY THE GOVERNMENT UP UNTIL THE SCHEDULED TIME OF THE OUTAGE. OUTAGES MUST BE COORDINATED WITH THE GOVERNMENT PERSONNEL (CONTRACTING OFFICER) TO MINIMIZE TOTAL OUTAGES TO THE FACILITY.
15. ADJUST WIRE SIZES TO COMPENSATE FOR VOLTAGE DROP. THE LENGTH OF A SINGLE PHASE 20A CIRCUIT (ONE-WAY) MUST NOT EXCEED THE FOLLOWING VALUES (BASED ON 16A LOAD):

CONDUCTOR SIZE (AWG)	MAXIMUM 120V CIRCUIT LENGTH	MAXIMUM 277V CIRCUIT LENGTH
#12	60'-0"	135'-0"
#10	90'-0"	210'-0"
#8	140'-0"	330'-0"
#6	225'-0"	525'-0"

MOUNTING HEIGHT SCHEDULE

DEVICES/EQUIPMENT	MOUNTING HEIGHT (AFF)	MEASURED TO:
RECEPTACLES	AS INDICATED ON LEGEND/PLANS	CENTER
TOGGLE SWITCHES	4'-0"	CENTER
WALL DIMMERS	4'-0"	CENTER
MANUAL MOTOR STARTERS	4'-0"	CENTER
OCCUPANCY SENSORS - WALL MOUNTED	4'-0"	CENTER
LIGHTING CONTROL PANEL	6'-6"	TOP
LIGHTING CONTROL STATION	4'-0"	CENTER
JUNCTION BOXES	AS INDICATED ON LEGEND/PLANS	CENTER
MONITORING/CONTROL PANEL	5'-0"	TOP
SURFACE METAL RACEWAYS	AS INDICATED ON LEGEND/PLANS	TOP
PANELBOARDS	6'-6"	TOP
DISCONNECT SWITCHES	5'-0"	TOP
AUTOMATIC TRANSFER SWITCHES	6'-6"	TOP
MANUAL TRANSFER SWITCHES	6'-6"	TOP
CONTROL STATIONS	6'-6"	TOP
SPD (NON-INTEGRAL)	6'-0" (NTE)	TOP
MAGNETIC MOTOR CONTROLLERS	6'-6"	TOP
LIGHTING CONTACTORS ENCLOSURES	6'-6"	TOP
INDIVIDUAL CKT BREAKERS ENCLOSURES	6'-6"	TOP
EMERGENCY POWER OFF SWITCHES	5'-6"	CENTER
TIME SWITCHES	6'-6"	TOP
TELECOMMUNICATIONS OUTLETS	1'-6"	CENTER
TELECOMMUNICATIONS OUTLETS - OVER COUNTER	6"	CENTER (OVER TOP OF COUNTER)
GROUND FAULT CIRCUIT INTERRUPTER	4'-0"	CENTER
GROUND BUS BARS	2'-0"	CENTER
INTEGRATED COMMUNICATIONS PANELS	6'-0"	TOP
GENERATOR REMOTE ALARM ANNUNCIATOR	5'-0"	TOP
GENERATOR TERMINAL BLOCKS	5'-0"	TOP
BAS TERMINAL BLOCKS	5'-0"	TOP
GENERATOR STOP SWITCH	5'-6"	CENTER
GENERATOR TANK FUEL LEVEL INDICATORS	5'-0"	TOP
UPS REMOTE MONITORING PANEL	5'-0"	TOP
UPS BATTERY MONITORING / CONTROL PANEL	5'-0"	TOP
MAIN SWITCHGEAR BATTERY CHARGER	5'-0"	TOP
RELAY CONTROL PANELS	6'-6"	TOP

MOUNTING HEIGHT SCHEDULE NOTES:

1. USE ABOVE INDICATED MOUNTING HEIGHTS UNLESS INDICATED OTHER WISE ON DRAWINGS OR SPECIFICATIONS.
2. COORDINATE MOUNTING HEIGHTS WITH FIELD CONDITIONS, OTHER TRADES, AND RELATED EQUIPMENT.
3. REDUCE MOUNTING HEIGHTS OF FIRE ALARM VISUAL AND AUDIO/VISUAL ALARMS WHERE REQUIRED TO MAINTAIN 6" MINIMUM SPACING FROM CEILING.

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED James E. DeMarle, RA
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES JHP	DRW JHP	CHK KB

PM/DM	SMS/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL GENERAL NOTES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894037
SHEET 87 OF 140

E-001

A

A, AMP	AMPERE
AA	AMBIENT AIR
ABBREV	ABBREVIATION
ABV	ABOVE
AC	ALTERNATING CURRENT
ACC	ACCESS CONTROL
ACT	ABOVE COUNTER TOP
AF	AMPERE FRAME, AMPERE FUSE
AFC	ABOVE FINISHED CEILING
AFF	ABOVE FINISHED FLOOR
AFG	ABOVE FINISHED GRADE
AHU	AIR HANDLING UNIT
AI	ANALOG INPUT
AIC	AMPERE INTERRUPTING CURRENT
AL	ALUMINUM
AM	AMPERE METER
ANN	ANNUNCIATOR
ANSI	AMERICAN NATIONAL STANDARDS INSTITUTE
AO	ANALOG OUTPUT
AP	ANNUNCIATOR POINT
AS	AMETER SWITCH
ASYM	ASYMMETRICAL
AT	AMPERE TRIP
ATS	AUTOMATIC TRANSFER SWITCH
ATX	AUTO TRANSFORMER
AV	AUDIO VISUAL
AVG	AVERAGE
AWG	AMERICAN WIRE GAUGE

B

BAS	BUILDING AUTOMATION SYSTEM
BATT	BATTERY, BATTERIES
BB	BACKBONE
BC	BARE COPPER
BKR	BREAKER
BLDG	BUILDING
BMS	BALANCED MAGNETIC SWITCH
BOC	BOTTOM OF CONDUIT
BOT	BOTTOM OF TRAY

C

°C	CELSIUS
C	CONDUIT, COIL
CAT. NO, CAT	CATALOG
CBM	CERTIFIED BALLAST MANUFACTURER
CCTV	CLOSED CIRCUIT TELEVISION CAMERA
CEG	COMMON EQUIPMENT GROUND
CFM	CUBIC FEET PER MINUTE
CKT	CIRCUIT
CLF	CURRENT LIMITING FUSE
CLO	CLOSET
cm	CENTIMETER
CMS	COMBINATION MOTOR STARTER
COMPT	COMPARTMENT
COND	CONDUIT
CONT	CONTINUOUS, CONTINUED
CPT	CONTROL POWER TRANSFORMER
CPU	CENTRAL PROCESSING UNIT
CRAC	COMPUTER ROOM AIR CONDITIONING UNIT
CRI	COLOR RENDERING INDEX
CSCCN	CENTRAL SCIENTIFIC COMMUNICATIONS NETWORK
CT	CURRENT TRANSFORMER
CU	COPPER COEFFICIENT OF UTILIZATION
CY	CYCLES

D

D	DEPTH, DEEP
DB	DUCT BANK
DC	DIRECT CURRENT
DEPT	DEPARTMENT
DESIG	DESIGNATION
DIA	DIAMETER
DISC	DISCONNECT
DIST, DISTRIB	DISTRIBUTION
DIV	DIVISION
DPDT	DOUBLE POLE - DOUBLE THROW
DPST	DOUBLE POLE - SINGLE THROW
DS, DSW	DISCONNECT SWITCH
DWG	DRAWING

E

(E), EX	EXISTING
E	EMPTY
EGC	EQUIPMENT GROUNDING CONDUCTOR
EL, ELEV	ELEVATION
ELEC	ELECTRICAL / ELECTRIC
EM, EMER	EMERGENCY
EMH	EXISTING MANHOLE, ELECTRICAL MANHOLE
EMO	EMERGENCY MANUAL OFF
EMS	EMERGENCY MANAGEMENT SYSTEM
EMT	ELECTRICAL METALLIC TUBING
ENT	ELECTRICAL NON-METALLIC TUBING
EO	ELECTRICALLY OPERATED
EOL	END OF LIFE
EPO	EMERGENCY POWER OFF
EPP	EXISTING POWER POLE
EPR	ETHYLENE-PROPYLENE RUBBER
EQPT	EQUIPMENT
ES	ENERGY SAVING
ET	ELECTRONIC TRIP
ETC	ETCETERA
ETL	ELECTRICAL TESTING LABORATORIES
EW	EACH WAY
EWC	ELECTRIC WATER COOLER
EWH	ELECTRIC WATER HEATER
EXST	EXISTING, EXHAUST

F

(F)	FUTURE
F	FAHRENHEIT
FA	FORCED AIR, FIRE ALARM
FBO	FURNISHED BY OTHERS, FURNISHED BY OWNER
FC	FLEXIBLE CONDUIT
FDR	FEEDER
FFE	FINISHED FLOOR ELEVATION
FIXT	FIXTURE
FL	FOOT LAMBERTS
FLR	FLOOR
FLUOR	FLUORESCENT
FREQ	FREQUENCY
FRP	FIBERGLASS-REINFORCED POLYESTER
FT	FOOT
FU	FUSE
FUT	FUTURE
FVNR	FULL VOLTAGE NON-REVERSING
FVR	FULL VOLTAGE REVERSING
FWE	FURNISHED WITH EQUIPMENT

G

g	GRAM
G, GND	GROUND
GA	GAUGE
GALV	GALVANIZED
GEC	GROUNDING ELECTRODE CONDUCTOR
GEN	GENERATOR
GFCI	GROUND FAULT EQUIPMENT PROTECTIVE DEVICE
GFE	GOVERNMENT FURNISHED EQUIPMENT, CONTRACTOR INSTALLED
GFEPD	GROUND FAULT EQUIPMENT PROTECTIVE DEVICE
GFP	GROUND FAULT PROTECTION
GFR	GROUND FAULT RELAY
GP	GENERAL PURPOSE

H

H	HEIGHT, HORN
HALOG	HALOGEN
HE	HIGH EXPOSURE
HH	HANDHOLE
HID	HIGH INTENSITY DISCHARGE
HORIZ	HORIZONTAL
HP	HORSEPOWER
HPF	HIGH POWER FACTOR
HPS	HIGH PRESSURE SODIUM
HRG	HIGH RESISTANCE GROUND
HVAC	HEATING, VENTILATING AND AIR CONDITIONING
HZ	HERTZ

I

IAW	IN ACCORDANCE WITH
ICEA	INSULATED CABLE ENGINEERS ASSOCIATION
IDS	INTRUSION DETECTION SYSTEM
IEC	INTERNATIONAL ELECTROTECHNICAL COMMISSION
IEEE	INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS
IESCR	INTEGRATED EQUIPMENT SHORT CIRCUIT RATING
IESNA, IES	ILLUMINATING ENGINEERING SOCIETY OF NORTH AMERICA
IG	ISOLATED GROUND
IMC	INTERMEDIATE METAL CONDUIT
INC, INCAND	INCANDESCENT
INST	INSTRUCTIONS, INSTUMENT / INSTRUMENTATION
INTER	INTERMEDIATE

J

J, JB	JUNCTION BOX

K

°K	DEGREES KELVIN
K	KEY INTERLOCK
KA	KILO-AMPERES
KCM, KCMIL	1000 CIRCULAR MILS
KG	KILO-GRAM
KM	KILO-METER
KV	KILO-VOLTS
KVA	KILO-VOLT-AMPERES
KVAC	KILO-VOLT-AMPERES CONNECTED
KVAD	KILO-VOLT-AMPERES DEMAND
KW	KILOWATT
KWD	KILOWATT DEMAND

L

L	LUMEN, LENGTH
LA	LIGHTNING ARRESTER
LAN	LOCAL AREA NETWORK
LB	POUND
LED	LIGHT EMITTING DIODE
LPF	LOW POWER FACTOR
LRG	LOW RESISTANCE GROUND
LSM	LONG-TIME/SHORT TIME TRIP
LSI	LONG-TIME/SHORT TIME/INSTANTANEOUS TRIP
LSIG	LONG-TIME/SHORT TIME/INSTANTANEOUS/GROUND FAULT TRIP
LSIGM	LONG-TIME/SHORT TIME/INSTANTANEOUS/GROUND FAULT TRIP AND METERING
LSIM	LONG-TIME/SHORT TIME/INSTANTANEOUS TRIP AND METERING
LT	LIQUID TIGHT
LTG	LIGHTING
LTS	LIGHTS
LV	LOW VOLTAGE
LVR	LOW VOLTAGE RELAY

M

m	METER
M/G	MOTOR / GENERATOR
MAX	MAXIMUM
MC	METAL-CLAD CABLE
MCB	MAIN CIRCUIT BREAKER
MCC	MOTOR CONTROL CENTER
MCCB	MOLDED CASE CIRCUIT BREAKER
MECH	MECHANICAL
MEZZ	MEZZANINE
MFR	MANUFACTURER
MH	MOUNTING HEIGHT, MAN HOLE
MHL	METAL HALIDE LAMP
MI	MINERAL INSULATED
MIN	MINIMUM
MIP	MEDICAL ISOLATION PANEL
MLO	MAIN LUGS ONLY
mm	MILLIMETER
MO	MECHANICALLY OPERATED, MOTOR OPERATOR
MP	MOTOR PROTECTOR
MS	MASTER SUBSTATION, MOTOR STARTER
MSP	MOTOR STARTER PANEL
MT	MOUNT
MTD	MOUNTED
MTG	MOUNTING
MTTOE	MOUNTED TO TOP OF EQUIPMENT
MV	MEDIUM VOLTAGE

N

(N)	NEW
N, NEUT	NEUTRAL
NC	NORMALLY CLOSED
NO	NORMALLY OPEN
NOL	NON-OVERLOAD
NA	NOT APPLICABLE
NEC	NATIONAL ELECTRICAL CODE
NEMA	NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION
NIC	NOT IN CONTRACT
NO.	NUMBER
NPF	NORMAL POWER FACTOR
NPT	NATIONAL PIPE THREAD
NTE	NOT TO EXCEED
NTS	NOT TO SCALE

O

OC	ON CENTER
OL	OVERLOAD
OFCI	OWNER FURNISHED, CONTRACTOR INSTALLED
OFE	OWNER FURNISHED EQUIPMENT, CONTRACTOR INSTALLED
OFOI	OWNER FURNISHED, OWNER INSTALLED
OPT	OPTIONAL

P

P	POLE, POWER
PB	PRIVATE BRANCH EXCHANGE
PBRD	PANELBOARD
PCC	POINT OF COMMON COUPLING
PDP	POWER DISTRIBUTION PANEL
PDU	POWER DISTRIBUTION UNIT
PF	POWER FACTOR
PH	PHASE
PLC	PROGRAMMABLE LOGIC CONTROLLER
PMCS	POWER MONITORING AND CONTROL SYSTEM
PNL	PANEL
PT	POTENTIAL TRANSFORMER
PVC	POLYVINYL CHLORIDE

Q

QTY	QUANTITY

R

%R	PERCENT RESISTANCE
(R)	RELOCATE
R	RESISTANCE, RATE, RISE
R/I	RECTIFIER / INVERTER
RCPT	RECEPTACLE
RCR	ROOM CAVITY RATIO
REQD	REQUIRED
RFI	RADIO FREQUENCY INTERFERENCE
RGS	RIGID GALVANIZED STEEL
RM	ROOM
RMS	ROOT MEAN SQUARE
RQD	REQUIRED
RVAT	REDUCED VOLTAGE AUTOTRANSFORMER
RVNR	REDUCED VOLTAGE NON-REVERSING
RVR	REDUCED VOLTAGE REVERSING

S

S	SPEAKER
S/MH	SPACING TO MOUNTING HEIGHT RATIO
SC	SHORT CIRCUIT
SCHED	SCHEDULE
SE	SERVICE ENTRANCE
SEC	SECONDS
SECT	SECTION
SES	SERVICE ENTRANCE SUBSTATION
SHLD	SHIELDED, SHIELD
SKVA	STARTING KILO-VOLT AMPERES
SP	SPARE
SPD	SURGE PROTECTIVE DEVICE
SPDT	SINGLE POLE-DOUBLE THROW
SPST	SINGLE POLE-SINGLE THROW
SQ	SQUARE
SS	STAINLESS STEEL
SSBJ	SUPPLY SIDE BONDING JUMPER
SSL	SOLID STATE LIGHT
SSM	SOLID STATE METERING
SST	SOLID STATE TRIP
ST	SHUNT TRIP
STA	STATION
STD	STANDARD
STP	SHIELDED TWISTED PAIR
SW	SWITCH
SWBD	SWITCHBOARD
SWGR	SWITCHGEAR
SYM	SYMMETRICAL
SYNCH	SYNCHRONIZER

T

T&B	TOP AND BOTTOM
T, TX	TRANSFORMER
T.O.C	TOP OF CONDUIT
TB	TERMINAL BLOCK
TBD	TO BE DETERMINED
TDR	TIME DELAY RELAY
TEL	TELEPHONE
TEL CAB	TELEPHONE CABINET
TJB	TERMINAL JUNCTION BOX
TP	TRANSFORMER PROTECTION
TSTAT	THERMOSTAT
TYP	TYPICAL

U

UDS	UNITIZED DISTRIBUTION SUBSTATION
UG, UGND	UNDERGROUND
UH	UNIT HEATER
UL	UNDERWRITERS' LABORATORIES
UNO, UON	UNLESS NOTED OTHERWISE, UNLESS OTHERWISE NOTED
UPS	UNINTERRUPTIBLE POWER SUPPLY
UTP	UNSHIELDED TWISTED PAIR
UV	ULTRAVIOLET

V

V	VOLT, VOLTS, VOLTMETER
VA	VOLT-AMPERE
VAC	VOLTS ALTERNATING CURRENT
VAR	VOLT AMPERE REACTIVE
VDC	VOLTS DIRECT CURRENT
VERT	VERTICAL
VESDA	VERY EARLY SMOKE DETECTION
VFD	VARIABLE FREQUENCY DRIVE
VM	VOLT METER
VPE	VACUUM PRESSURE ENCAPSULATED
VPI	VACUUM PRESSURE IMPREGNATED
VS	VOLTMETER SWITCH
VSD	VARIABLE SPEED DRIVE
VT	VOLTAGE TRANSFORMER

W

W	WATTS, WIRE, WIDTH
W/	WITH
W/O	WITHOUT
WG	WATER GAGE
WHD	WATTHOUR DEMAND METER
WP	WEATHERPROOF

X

%X	PERCENT REACTANCE
X	REACTANCE
XFMR	TRANSFORMER
XLP	CROSS-LINK POLYETHYLENE INSULATION

Y

Y	WYE-CONNECTED

Z

%Z	PERCENT IMPEDANCE
Z	IMPEDANCE

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE 05/27/2025

DES JHP | DRW JHP | CHK KB

PM/DM SMS/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL ABBREVIATIONS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894038
SHEET 88 OF 140

E-002

6/20/2025 12:26:33 PM E-002
\\dc1vs01\fedprj\FDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt

WIRING DEVICE LEGEND

13	DENOTES ONE OR TWO NUMERIC DIGIT BRANCH CIRCUIT NUMBER
+48"	DENOTES THE MOUNTING HEIGHT IN INCHES AFF
A	ARC FAULT CIRCUIT INTERRUPTER
EWC	ELECTRIC WATER COOLER
GFI	GFCI TYPE, WIRED FOR GFCI TRIP FOR DESIGNATED RECEPTACLE ONLY (NON-FEED-THRU INSTALLATION)
IG	ISOLATED GROUNDING TYPE RECEPTACLE
L	PAD LOCKABLE COVER
PR	PROJECTOR RECEPTACLE LOCATED WITHIN PROJECTOR OR CEILING MOUNT
R	WHEN USED ON DEMOLITION PLANS: DEMOLISH DEVICE AND FACEPLATE, OUTLET BOX TO REMAIN. WHEN USE ON WORK PLANS: PROVIDE DEVICE IN EX. OUTLET BOX.
SPD	SURGE PROTECTIVE DEVICE TYPE RECEPTACLE
TR	TAMPER RESISTANT TYPE RECEPTACLE
TV	TELEVISION RECEPTACLE TYPE RECEPTACLE; COORDINATE MOUNTING HEIGHT AND EXACT LOCATION WITH OWNER'S REPRESENTATIVE PRIOR TO INSTALLATION
U	USB CHARGING
WP	WEATHER-RESISTANT, WEATHERPROOF WITH HINGED COVER
WP-WIN	WEATHER-RESISTANT TYPE, WEATHERPROOF WITH WHILE-IN-USE COVER
XP	EXPLOSION PROOF TYPE, RATED FOR AREA CLASSIFICATION IN WHICH IT IS INSTALLED.
	DUPLEX RECEPTACLE. UNLESS INDICATED OTHERWISE, RECEPTACLES ARE FLUSH MOUNTED, 20A,125V, 1PH, 2P, 3W, NEMA 5-20R.
	DUPLEX RECEPTACLE. UNLESS INDICATED OTHERWISE, RECEPTACLES ARE FLUSH MOUNTED, 20A,125V, 1PH, 2P, 3W, NEMA 5-20R, MOUNTED 6" ABOVE COUNTERTOP OR BACKSPLASH UNLESS OTHERWISE NOTED.
	DUPLEX RECEPTACLE. UNLESS INDICATED OTHERWISE, RECEPTACLES ARE FLUSH MOUNTED, 20A,125V, 1PH, 2P, 3W, NEMA 5-20R ON AN EMERGENCY CIRCUIT.
	SPLIT YOKE DUPLEX RECEPTACLE WITH TOP BEING CONTROLLED, UNLESS OTHERWISE INDICATED. RECEPTACLES ARE FLUSH MOUNTED, 20A,125V, 1PH, 2P, 3W, NEMA 5-20R. UTILIZE VACANCY / OCCUPANCY SENSOR SERVING THE ROOM UNLESS OTHERWISE NOTED. PROVIDE PLUG LOAD CONTROLLER FOR EACH ROOM SERVED. PROVIDE LABEL CLEARLY DIFFERENTIATING WHICH OUTLET IS CONTROLLED/NONCONTROLLED.
	TWO DUPLEX (QUAD) RECEPTACLES. UNLESS INDICATED OTHERWISE, RECEPTACES ARE FLUSH MOUNTED, 20A, 125V, 1PH, 2P, 3W, NEMA 5-20R.
	SWITCHED TWO DUPLEX (QUAD) RECEPTACLES. QUAD RECEPTACLES HAVE CONTROLLED AND NON-CONTROLLED OUTLETS FED FROM THE SAME CIRCUIT, BUT A SPLIT YOKE IS NOT REQUIRED. UNLESS INDICATED OTHERWISE, RECEPTACES ARE FLUSH MOUNTED, 20A, 125V, 1PH, 2P, 3W, NEMA 5-20R. UTILIZE VACANCY / OCCUPANCY SENSOR SERVING THE ROOM UNLESS OTHERWISE NOTED. PROVIDE PLUG LOAD CONTROLLER FOR EACH ROOM SERVED. PROVIDE MARKING CLEARLY DIFFERENTIATING WHICH OUTLET IS CONTROLLED/NONCONTROLLED.
	SIMPLEX RECEPTACLE. UNLESS INDICATED OTHERWISE, RECEPTACLES ARE FLUSH MOUNTED, 20A,125V,1PH, 2P, 3W, NEMA 5-20R.
	SPECIAL PURPOSE RECEPTACLE, UNLESS INDICATED OTHERWISE, RECEPTACLES ARE FLUSH MOUNTED. RECEPTACLE NEMA TYPES INDICATED ON THE SPECIAL RECEPTACLE SCHEDULE.
	CEILING MOUNTED DUPLEX RCPT. UNLESS INDICATED OTHERWISE, 20A,125V, 1PH, 2P, 3W, NEMA 5-20R
A5	CEILING MOUNTED SPECIAL PURPOSE RECEPTACLE. TYPES ARE AS NOTED IN THE SPECIAL RECEPTACLE SCHEDULE.
G	POWER CORD REEL. "G" SUBSCRIPT INDICATES GROUND REEL.
	SURFACE MOUNTED RECEPTACLE
	FLUSH FLOOR TYPE FLOOR BOX. PROVIDE PER MANUFACTURER INSTRUCTION.
	FLUSH FLOOR TYPE FLOOR BOX. PROVIDE PER MANUFACTURER INSTRUCTION. RECEPTACLE NEMA TYPES INDICATED ON THE SPECIAL RECEPTACLE SCHEDULE.
JB	JUNCTION BOX FOR ELECTRICAL EQUIPMENT CONNECTION
F	IN WALL FURNITURE CONNECTION. PROVIDE PER MFR INSTRUCTION.
P	POWER POLE, LENGTH NEEDED TO EXTEND ABOVE CEILING; SHADED AREA DENOTES TELECOMMUNICATION RACEWAY. REFER TO FF&E PACKAGE FOR ADDITIONAL INFORMATION.
P	POWER POLE WITH DUPLEX RECEPTACLES, LENGTH NEEDED TO EXTEND ABOVE CEILING; SHADED AREA DENOTES TELECOMMUNICATION RACEWAY. RECEPTACLE NEMA TYPES INDICATED ON THE SPECIAL RECEPTACLE SCHEDULE.
	WALL MOUNTED TWO-COMPARTMENT SURFACE MOUNTED RACEWAY WITH DUPLEX 120V, 20A, 3-WIRE, NEMA 5-20R RECEPTACLES AT 18" ON CENTER. MOUNT 6" ABOVE COUNTER TOP, UNLESS INDICATED OTHERWISE.

1 +36" WP GFI
- NUMERAL = CIRCUIT NUMBER
- +(INCHES) = HEIGHT IN INCHES ABOVE FINISHED FLOOR
- CAPITAL LETTER(S) = SPECIAL DESIGNATOR SUBSCRIPT DEFINED ABOVE

ELECTRICAL DISTRIBUTION EQUIPMENT SYMBOLS

	FLOOR OR SURFACE MOUNTED 208V PANELBOARD
	FLOOR OR SURFACE MOUNTED 480V PANELBOARD
	RECESSED 208V PANELBOARD
	RECESSED 480V PANELBOARD
T75	GENERAL PURPOSE, DRY TYPE TRANSFORMER. 'T75' IS THE KVA RATING FOR THE TRANSFORMER. IN THIS EXAMPLE, 'T75' REPRESENTS A TRANSFORMER RATED 75KVA.
ATS	AUTOMATIC TRANSFER SWITCH.
SPD	SURGE PROTECTIVE DEVICE.
	MOTOR CONNECTION
	SINGLE POINT CONNECTION FOR PACKAGED EQUIPMENT
	COMBINATION MOTOR STARTER / DISCONNECT
	MOTOR STARTER
	FUSED DISCONNECT
	NON FUSED DISCONNECT
VFD	VARIABLE FREQUENCY DEVICE
	ELECTRICAL MANHOLE

WIRING SYSTEM/RACEWAY/CABLE TRAY LEGEND

#10, 3/4" LP-1,3	BRANCH CIRCUIT "HOMERUN", TIC MARKS INDICATE NUMBER OF CONDUCTORS (INCLUDING NEUTRAL). #10 INDICATES AWG WIRE SIZE, AND 3/4" INDICATES CONDUIT SIZE. WIRE AND CONDUIT SIZE IS APPLICABLE TO EACH CIRCUIT INDICATED FOR ENTIRE LENGTH OF CIRCUIT. EVERY BRANCH CIRCUIT WIRING MUST INCLUDE A #12 AWG GROUND IN 3/4" CONDUIT. ARROW HEADS DENOTE A HOMERUN TO PANELBOARD. NUMBER OF ARROWHEADS INDICATE NUMBER OF CIRCUITS IN CONDUIT. "LP-1,3" INDICATES PANEL AND CIRCUIT NUMBERS.
A	SWITCH LEG CONNECTED TO SWITCH "A"
R1	SWITCH LEG CONNECTED TO RELAY #1 IN LIGHTING CONTROL PANEL INDICATED.
	CONDUIT OR CABLE TURNING UP OR TOWARD THE OBSERVER
	CONDUIT OR CABLE TURNING DOWN OR AWAY FROM THE OBSERVER
	CONDUIT OR CABLE CHANGE IN ELEVATION
	SINGLE CONDUIT CAPPED
3	MULTIPLE CONDUITS, INDIVIDUALLY CAPPED. WHERE NUMERIC DESIGNATION INDICATED, DENOTES QUANTITY OF CAPPED CONDUITS. IN THIS EXAMPLE, THREE CAPPED CONDUITS
	CONDUIT WITH BUSHING
S S	RACEWAY CONTINUATION
	CABLE TRAY

LUMINAIRE LEGEND

- 1'X4' RECESSED LUMINAIRE
- 1'X4' RECESSED CENTER BASKET LUMINAIRE
- 1'X4' SURFACE LUMINAIRE
- 1'X4' SURFACE CENTER BASKET LUMINAIRE
- 1'X4' RECESSED LUMINAIRE CONNECTED TO EMERGENCY SOURCE
- 2'X2' RECESSED LUMINAIRE
- 2'X2' RECESSED CENTER BASKET LUMINAIRE
- 2'X2' SURFACE LUMINAIRE
- 2'X2' SURFACE CENTER BASKET LUMINAIRE
- 2'X2' RECESSED LUMINAIRE CONNECTED TO EMERGENCY SOURCE
- 2'X4' RECESSED LUMINAIRE
- 2'X4' RECESSED CENTER BASKET LUMINAIRE
- 2'X4' SURFACE LUMINAIRE
- 2'X4' SURFACE CENTER BASKET LUMINAIRE
- 2'X4' RECESSED LUMINAIRE CONNECTED TO EMERGENCY SOURCE
- 1'X4' PENDANT LUMINAIRE
- 1'X4' PENDANT LUMINAIRE CONNECTED TO EMERGENCY SOURCE
- 1'X8' LINEAR PENDANT LUMINAIRE
- 1'X8' LINEAR PENDANT LUMINAIRE CONNECTED TO EM. SOURCE
- ROUND PENDANT LUMINAIRE. SEE LUMINAIRE ANCILLARIES BELOW FOR ADDITIONAL REQUIREMENTS.
- DOWNLIGHT LUMINAIRE
- DOWNLIGHT LUMINAIRE CONNECTED TO EMERGENCY SOURCE
- DOWNLIGHT WALL WASHER LUMINAIRE (CHEVRON(S) POINT TO WASHED WALL(S))
- SURFACE OR PENDANT MOUNTED TRACK LUMINAIRE. NUMBERS OF LAMP HEADS AS INDICATED IN LIGHTING FIXTURE SCHEDULE. SEE LUMINAIRE ANCILLARIES BELOW FOR ADDITIONAL REQUIREMENTS.
- SITE POLE LUMINAIRE WITH CONCRETE FOOTING. NUMBERS OF LAMP HEADS AS INDICATED IN LIGHTING FIXTURE SCHEDULE. SEE LUMINAIRE ANCILLARIES BELOW FOR ADDITIONAL REQUIREMENTS.
- WALL MOUNTED EGRESS LIGHT. MOUNT 8'-0" ABOVE FINISHED FLOOR, UNLESS INDICATED OTHERWISE. NUMBERS OF LAMP HEADS AS INDICATED IN LIGHTING FIXTURE SCHEDULE.
- WALL MOUNTED REMOTE LAMP HEAD EMERGENCY LIGHT. MOUNT 8'-0" ABOVE FINISHED FLOOR, UNLESS INDICATED OTHERWISE. NUMBERS OF LAMP HEADS AS INDICATED IN LIGHTING FIXTURE SCHEDULE.
- PENDANT OR CEILING MOUNTED ILLUMINATED EGRESS EXIT SIGN. WHERE USED, DIRECTIONAL ARROWS AS INDICATED. MOUNT 8'-0" ABOVE FINISHED FLOOR, UNLESS INDICATED OTHERWISE.
- WALL MOUNTED ILLUMINATED EGRESS EXIT SIGN. WHERE USED, DIRECTIONAL ARROWS AS INDICATED. MOUNTED 8'-0" AFF UNLESS INDICATED OTHERWISE .
- DENOTES POLE MOUNTED SINGLE ARM LUMINAIRE. POLE HEIGHT AS INDICATED.
- DENOTES POLE MOUNTED TWIN CONFIGURATION SINGLE ARM LUMINAIRE. INDIVIDUAL LUMINAIRE TYPE AS INDICATED; POLE HEIGHT AS INDICATED.
- DENOTES POLE MOUNTED TRIPLEX CONFIGURED SINGLE ARM LUMINAIRE. INDIVIDUAL TYPE AS INDICATED; POLE HEIGHT AS INDICATED.
- DENOTES POLE MOUNTED QUADRAPLEX CONFIGURED SINGLE ARM LUMINAIRE. INDIVIDUAL LUMINAIRE TYPE AS INDICATED; POLE HEIGHT AS INDICATED.
- WALL MOUNTED EXTERIOR LUMINAIRE WITH EMERGENCY POWER

3 A1 a
- NUMERAL = CIRCUIT NUMBER
- LOWERCASE ALPHANUMERIC CHARACTER = LIGHTING SWITCH OR CONTROL ZONE
- ALPHANUMERIC TAG = LUMINAIRE TAG; COORDINATE WITH LIGHTING FIXTURE SCHEDULE

- 4' LINEAR INDUSTRIAL STRIP FIXTURE
- 4' LINEAR INDUSTRIAL STRIP FIXTURE WITH EMERGENCY POWER

LIGHTING CONTROL DEVICE LEGEND

3	THREE WAY.
4	FOUR WAY.
D	WALL DIMMER.
MD	MULT-LOCATION DIMMER, NUMERALS AS INDICATED FOR SINGLE LOCATION DIMMER.
DP	DOUBLE POLE.
XP	EXPLOSION PROOF
F	WIRELESS CEILING FAN CONTROL STATION
OS	OCCUPANCY SENSOR [WITH MANUAL OVERRIDE].
V	VACANCY SENSOR/SWITCH
VD	VACANCY SENSOR/DIMMER
O	OCCUPANCY SENSOR/SWITCH
OD	OCCUPANCY SENSOR/DIMMER
LV	LOW VOLTAGE SWITCH
WP	WEATHERPROOF.
M	SURFACE MOUNTED MOTOR RATED WITH OVERLOAD ELEMENTS, CONTINOUS CURRENT RATING AND QUANTITY OF POLES AS NEEDED FOR EQUIPMENT SERVED.
T	ADJUSTABLE TIMER SWITCH
PL	ILLUMINATED SWITCH (WHEN OFF)
$	RECESSED WALL MOUNTED SWITCH. PROVIDE ADJACENT DEVICES UNDER COMMON FACEPLATE, FOUR DEVICES MAXIMUM PER FACEPLATE. EVENLY DIVIDE DEVICES INTO SEPARATE GROUPS WHERE ADJCENT DEVICES EXCEEDS FOUR.
$	SURFACE WALL MOUNTED SWITCH. PROVIDE ADJACENT DEVICES UNDER COMMON FACEPLATE, FOUR DEVICES MAXIMUM PER FACEPLATE. EVENLY DIVIDE DEVICES INTO SEPARATE GROUPS WHERE ADJCENT DEVICES EXCEEDS FOUR.
	CEILING MOUNTED DUAL-TECHNOLOGY OCCUPANCY SENSOR
	CEILING MOUNTED DUAL-TECHNOLOGY CORRIDOR OCCUPANCY SENSOR
	CEILING MOUNTED PASSIVE INFRARED OCCUPANCY SENSOR
	CEILING MOUNTED PASSIVE INFRARED CORRIDOR OCCUPANCY SENSOR
	CEILING MOUNTED ULTRASONIC OCCUPANCY SENSOR
	CEILING MOUNTED ULTRASONIC CORRIDOR OCCUPANCY SENSOR
V	CEILING MOUNTED DUAL-TECHNOLOGY VACANCY SENSOR
V	CEILING MOUNTED PASSIVE INFRARED VACANCY SENSOR
V	CEILING MOUNTED ULTRASONIC VACANCY SENSOR
P	CEILING MOUNTED PHOTO SENSOR
D	CEILING MOUNTED DAYLIGHT SENSOR
	WALL MOUNTED DUAL-TECHNOLOGY OCCUPANCY SENSOR
	WALL MOUNTED PASSIVE INFRARED OCCUPANCY SENSOR
	WALL MOUNTED ULTRASONIC OCCUPANCY SENSOR
V	WALL MOUNTED DUAL-TECHNOLOGY VACANCY SENSOR
V	WALL MOUNTED PASSIVE INFRARED VACANCY SENSOR
V	WALL MOUNTED ULTRASONIC VACANCY SENSOR

$3a
- LIGHTING CONTROL DEVICE IDENTIFICATION (TYPICAL FOR ALL LIGHTING CONTROL DEVICE SYMBOLS WHERE USED)
- LOWERCASE LETTER(S) = CONTROL ZONE
- CAPITAL LETTER(S) AND/OR NUMERAL = SPECIAL DESIGNATOR SUBSCRIPT DEFINED ABOVE

GROUNDING AND LIGHTNING PROTECTION LEGEND

–G–G–G–G–	GROUNDING CONDUCTOR.
	GROUND ROD.
	GROUND ROD IN TEST WELL.
	GROUND CONNECTION TO BUILDING STEEL OR EQUIPMENT / VESSEL.
	GROUND BAR.
BGA	BARREL GROUNDING BAR ASSEMBLY; MOUNTED 5'-0" AFF.
–LPS—LPS–	MAIN LIGHTNING PROTECTION CONDUCTOR
	LIGHTNING PROTECTION AIR TERMINAL.
	LIGHTNING PROTECTION AIR TERMINAL WITH DOWN CONDUCTOR.

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES JHP DRW JHP CHK KB
PM/DM SMS/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
MID-ATLANTIC
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL SYMBOLS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894039
SHEET 89 OF 140
E-003

6/20/2025 12:26:36 PM E-003

SINGLE LINE DIAGRAM LEGEND

SYMBOL	DESCRIPTION
1200A 52	MEDIUM VOLTAGE DRAWOUT CIRCUIT BREAKER.
1600 AF 800 AS 600 AT LSIG	LOW VOLTAGE DRAWOUT CIRCUIT BREAKER.
1600 AF 800 AS 600 AT LSIG F-800 A C.L.	LOW VOLTAGE DRAWOUT CURRENT LIMITING CIRCUIT BREAKER.
250 AF 225 AT LSIG	LOW VOLTAGE MOLDED CASE CIRCUIT BREAKER.
1200/5 3 LSIG	SOLID STATE TRIP UNIT WITH CT INPUT OR MODULE CAPABLE OF MONITORING AMPS, INSTANTANEOUS KW, PEAK KW AND KWH.
MR 1200/5 3	PHASE CURRENT TRANSFORMER.
50/5 1	ZERO SEQUENCE CURRENT TRANSFORMER.
PT 100 VA	POTENTIAL TRANSFORMER.
480V 10KVA, 1PH 120/240V CPT	CONTROL POWER TRANSFORMER.
T-PD2A 30KVA 480 V- 208/120 V	TRANSFORMER.
KT30Y 480 V- 208/120 V	ISOLATION TRANSFORMER.
	CABLE LUG TERMINATION
LA'S 3	LIGHTNING ARRESTER. NUMERIC VALUE DENOTES QUANTITY.
P S L ATS	TRANSFER SWITCH. 'ATS' DENOTES AUTOMATIC TRANSFER TYPE, 'MTS' DENOTES MANUAL TRANSFER TYPE. 'P' DENOTES PRIMARY POWER SOURCE; 'S' DENOTES SECONDARY POWER SOURCE; 'L' DENOTES TRANSFER SWITCH LOAD SIDE CONNECTION POINT.
PNL L1A	PANELBOARD WITH A VOLTAGE OF 480V OR HIGHER.
PNL L1A	PANELBOARD WITH VOLTAGE UNDER 480V.
	BATTERY.
DM	DIGITAL METER.
	GROUNDING CONNECTION.
13Y	FEEDER / BRANCH CIRCUIT DESIGNATION. IN THIS EXAMPLE, '13Y' REFERS TO THE SYMBOL COLUMN OF THE FEEDER/BRANCH CIRCUIT SCHEDULE. THE ASSOCIATED SYMBOL DESIGNATION DENOTES THE FEEDER / BRANCH CIRCUIT REQUIREMENTS FOR THE ITEM TAGGED.

SINGLE LINE DIAGRAM LEGEND

SYMBOL	DESCRIPTION
SPD	SURGE PROTECTIVE DEVICE.
200 KVAR	VARIABLE POWER FACTOR CORRECTION CAPACITOR BANK.
600 A N.C.	LOAD INTERRUPTER SWITCH.
30A/3P N.C. F-MFR	NON-FUSED DISCONNECT / SAFETY SWITCH.
30A/3P N.C. F-30A	FUSED DISCONNECT / SAFETY SWITCH.
E	ENGINE.
G	GENERATOR.
GFP	GROUND FAULT PROTECTION.
	LOADBANK.
	FUSE
	FUSED DRAWOUT POTENTIAL TRANSFORMER.
	FUSED POTENTIAL TRANSFORMER.
	POWER FILTER.
	SURGE PROTECTIVE DEVICE.
R1	RELAY.
	UPS AND BATTERY.
60A/3P	IN THIS EXAMPLE. DENOTES 60 AMPERES / 3 POLE.
F-30A	IN THIS EXAMPLE, DENOTES FUSED WITH 30 AMPERE FUSES.
F-MFR	DENOTES FUSE RATING IN ACCORDANCE WITH MANUFACTURER'S INSTRUCTIONS FOR THE UTILIZATION EQUIPMENT PROTECTED BY THE FUSES.
N3R	IN THIS EXAMPLE, DENOTES NEMA 3R ENCLOSURE.
SN	DENOTES SOLID NEUTRAL.
N.C.	DENOTES NORMALLY CLOSED.
N.O.	DENOTES NORMALLY OPEN.

1600 AF 800 AT LSIG

DEVICE ADJUSTABLE PROTECTION OPTIONS. LONG, SHORT, INSTANTANEOUS, AND GROUND PICKUPS IN THIS EXAMPLE.

"AF" DENOTES FRAME SIZE, "AT" DENOTES TRIP SETTING. 1600A FRAME WITH 800A TRIP IN THIS EXAMPLE.

30A/3P N.C. (3) F-30A

DENOTES SWITCH SIZE AND POLES. 30A, 3-POLE SWITCH IN THIS EXAMPLE.

DENOTES ORIENTATION OF SWITCH. NORMALLY CLOSED IN THIS EXAMPLE.

DENOTES QUANTITY AND RATING OF FUSES. THREE 30A IN THIS EXAMPLE.

SINGLE LINE DIAGRAM LEGEND

SYMBOL	DESCRIPTION
E 1 RVPW	MOTOR CONTROL CENTER MOTOR STARTER OR LOCAL CONTROLLER. IN THIS EXAMPLE, '1' DENOTES NEMA SIZE, 'RVPW' DENOTES REDUCED VOLTAGE PART-WINDING, WHERE INDICATED, 'E' DENOTES ELECTRONIC OR SOLID STATE OVERLOAD RELAY. UNLESS INDICATED OTHERWISE, LOCAL CONTROLLER ENCLOSURES ARE RATED NEMA 1.
E 1 RVSS	MOTOR CONTROL CENTER MOTOR STARTER OR LOCAL CONTROLLER. IN THIS EXAMPLE, '1' DENOTES NEMA SIZE, 'RVSS' DENOTES REDUCED VOLTAGE SOLID STATE SOFT STARTER, WHERE INDICATED, 'E' DENOTES ELECTRONIC OR SOLID STATE OVERLOAD RELAY. UNLESS INDICATED OTHERWISE, LOCAL CONTROLLER ENCLOSURES ARE RATED NEMA 1.
1 WD	MOTOR CONTROL CENTER MOTOR STARTER OR LOCAL CONTROLLER. IN THIS EXAMPLE, '1' DENOTES NEMA SIZE, 'WD' DENOTES WYE-DELTA STARTER, WHERE INDICATED, 'E' DENOTES ELECTRONIC OR SOLID STATE OVERLOAD RELAY. UNLESS INDICATED OTHERWISE, LOCAL CONTROLLER ENCLOSURES ARE RATED NEMA 1.
1 MSCT	MOTOR CONTROL CENTER MOTOR STARTER OR LOCAL CONTROLLER. IN THIS EXAMPLE, '1' DENOTES NEMA SIZE, 'MSCT' DENOTES MULTI-SPEED CONSTANT TORQUE STARTER, WHERE INDICATED, 'E' DENOTES ELECTRONIC OR SOLID STATE OVERLOAD RELAY. UNLESS INDICATED OTHERWISE, LOCAL CONTROLLER ENCLOSURES ARE RATED NEMA 1.
1 MSVT	MOTOR CONTROL CENTER MOTOR STARTER OR LOCAL CONTROLLER. IN THIS EXAMPLE, '1' DENOTES NEMA SIZE, 'MSVT' DENOTES MULTI-SPEED VARIABLE TORQUE STARTER, WHERE INDICATED, 'E' DENOTES ELECTRONIC OR SOLID STATE OVERLOAD RELAY. UNLESS INDICATED OTHERWISE, LOCAL CONTROLLER ENCLOSURES ARE RATED NEMA 1.
1 MSCH	MOTOR CONTROL CENTER MOTOR STARTER OR LOCAL CONTROLLER. IN THIS EXAMPLE, '1' DENOTES NEMA SIZE, 'MSCH' DENOTES MULTI-SPEED CONSTANT HORSEPOWER STARTER, WHERE INDICATED, 'E' DENOTES ELECTRONIC OR SOLID STATE OVERLOAD RELAY. UNLESS INDICATED OTHERWISE, LOCAL CONTROLLER ENCLOSURES ARE RATED NEMA 1.
50 EF-01	MOTOR; EQUIPMENT IDENTIFICATION AND HORSEPOWER AS INDICATED.
M	MULTIFUNCTION METER
EMM	ELECTRONIC METERING MODULE: MICROPROCESSOR BASED FOR METERING, ANALYZING, WAVEFORM CAPTURE AND CAPABLE OF PROVIDING MONITORED DATA PARAMETERS
LMU	LOCAL MONITORING UNIT FOR SWITCHGEAR ASSEMBLIES CAPABLE OF UP TO 40 SEPARATE DATA INPUTS FROM BREAKER LSIG'S, EMM'S AND MFR'S; CONFIGURABLE DISPLAY SCREEN; DATA OUTPUT PORTS
	DENOTES FUTURE CIRCUIT / EQUIPMENT
	DENOTES BRANCH CIRCUIT / FEEDER IN CABLE TRAY.
	DENOTES BRANCH CIRCUIT / FEEDER IN CONDUIT.
	DENOTES GROUNDING SYSTEM.
	DENOTES ELECTRICAL EQUIPMENT BUS.
	DENOTES DATA CIRCUITRY
	DENOTES EQUIPMENT ENCLOSURE.

ADJUSTABLE PROTECTION DESIGNATIONS

DESIGNATION	PROTECTION
L	LONG PICKUP AND TIME DELAY PROTECTION
S	SHORT PICKUP AND TIME DELAY PROTECTION
I	INSTANTANEOUS PICKUP PROTECTION
G	GROUND (EARTH) PICKUP AND TIME DELAY PROTECTION
N	NEUTRAL PROTECTION

DESIGNATION	PROTECTION DEVICE FEATURES / ACCESSORIES
	NOTE: MULTIPLE DESIGNATIONS MAY BE ADJACENT TO DEVICE.
K	KEY INTERLOCK
ST	SHUNT TRIP
ET	ELECTRONIC TRIP

TELECOMMUNICATIONS LEGEND

SYMBOL	DESCRIPTION
	WIRELESS ACCESS POINT
	DATA/VOICE WALL MOUNTED RECEPTACLE
	DATA WALL MOUNTED RECEPTACLE
	DATA CEILING MOUNTED RECEPTACLE
	DATA WIRELESS ACCESS POINT
	DATA FLOOR BOX
	DATA/VOICE CEILING MOUNTED RECEPTACLE
	DOORBELL PUSHBUTTON MOUNTED 48" AFF TO CENTER OF BOX UNLESS OTHERWISE NOTED.
JB TV	WALL MOUNTED CALBE TV OUTLET MOUNTED 18" AFF TO CENTER OF BOX UNLESS OTHERWISE NOTED.
JB	CEILING MOUNTED JUNCTION BOX.

ELECTRICAL EQUIPMENT NOMENCLATURE

T- L P 2

ALPHANUMERIC CHARACTER DENOTING SEQUENCE NUMBER (A,B,C,ETC).

SINGLE OR DOUBLE ALPHANUMERIC CHARACTER DENOTING EQUIPMENT TYPE DESIGNATION, AS FOLLOWS:

L - 208Y/120V POWER
H - 480Y/277V POWER
T - TELECOM.
F - FIRE PROTECTION
MD - MAIN DISTRIBUTION
P - POWER PANEL

WHERE APPLICABLE, SINGLE TO TRIPLE ALPHANUMERIC CHARACTER DENOTING EQUIPMENT TYPE MODIFIER, AS FOLLOWS:

A - AUTOMATIC TRANSFER SWITCH.
T - TRANSFORMER

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. Dunanue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE 05/27/2025

DES	JHP	DRW	JHP	CHK	KB

PM/DM	SMS/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL SYMBOLS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894040
SHEET 90 OF 140

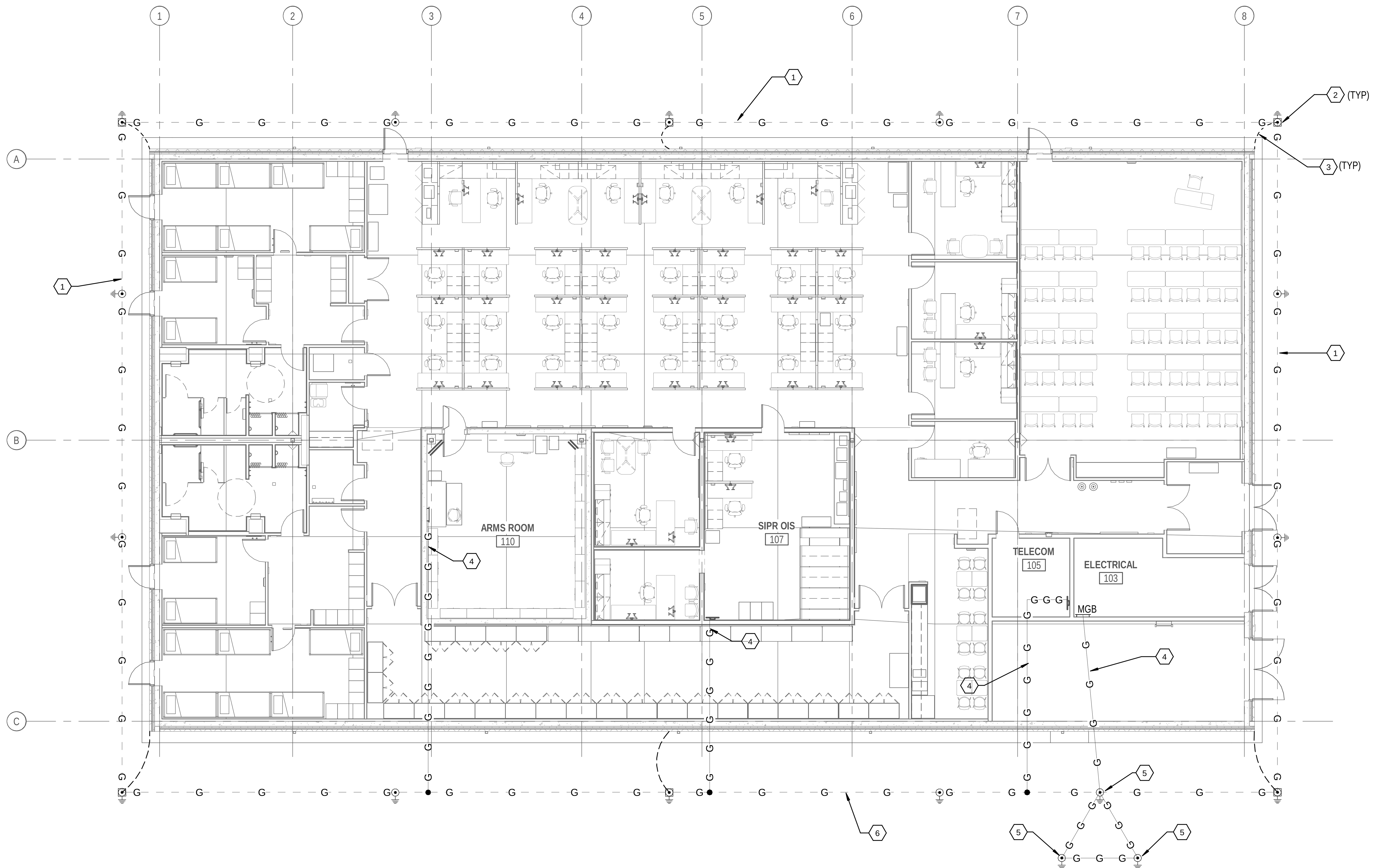
E-004

GENERAL SHEET NOTES

A. REFER TO SHEETS E-001 THROUGH E-004 FOR ADDITIONAL ELECTRICAL INFORMATION.

SHEET KEYNOTES

1. PROVIDE A #2/0 AWG BARE COPPER GROUND LOOP CONDUCTOR AT APPROXIMATELY 3'-0" OUT FROM THE EXTERIOR WALL AT 36" BELOW GRADE.
2. PROVIDE GROUND ROD WITH TEST PIT ENCLOSURE WITH MECHANICAL HIGH STRENGTH CABLE CONNECTORS FOR SYSTEM TESTING.
3. FROM EACH GROUND TEST WELL, BOND THE STRUCTURE WITH A #2/0 AWG BARE COPPER BONDING CONDUCTOR.
4. PROVIDE #2/0 AWG BARE COPPER BONDING CONDUCTOR FROM GROUND BAR TO SECONDARY GROUND RING.
5. NEC REQUIRED SERVICE GROUND ROD.
6. PROVIDE A #2/0 AWG BARE COPPER GROUND LOOP CONDUCTOR AT APPROXIMATELY 8'-0" OUT FROM THE EXTERIOR WALL AT 36" BELOW GRADE TO AVOID CONFLICTS WITH MECHANICAL EQUIPMENT PADS.



B1 GROUNDING PLAN
1/8" = 1'-0"

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

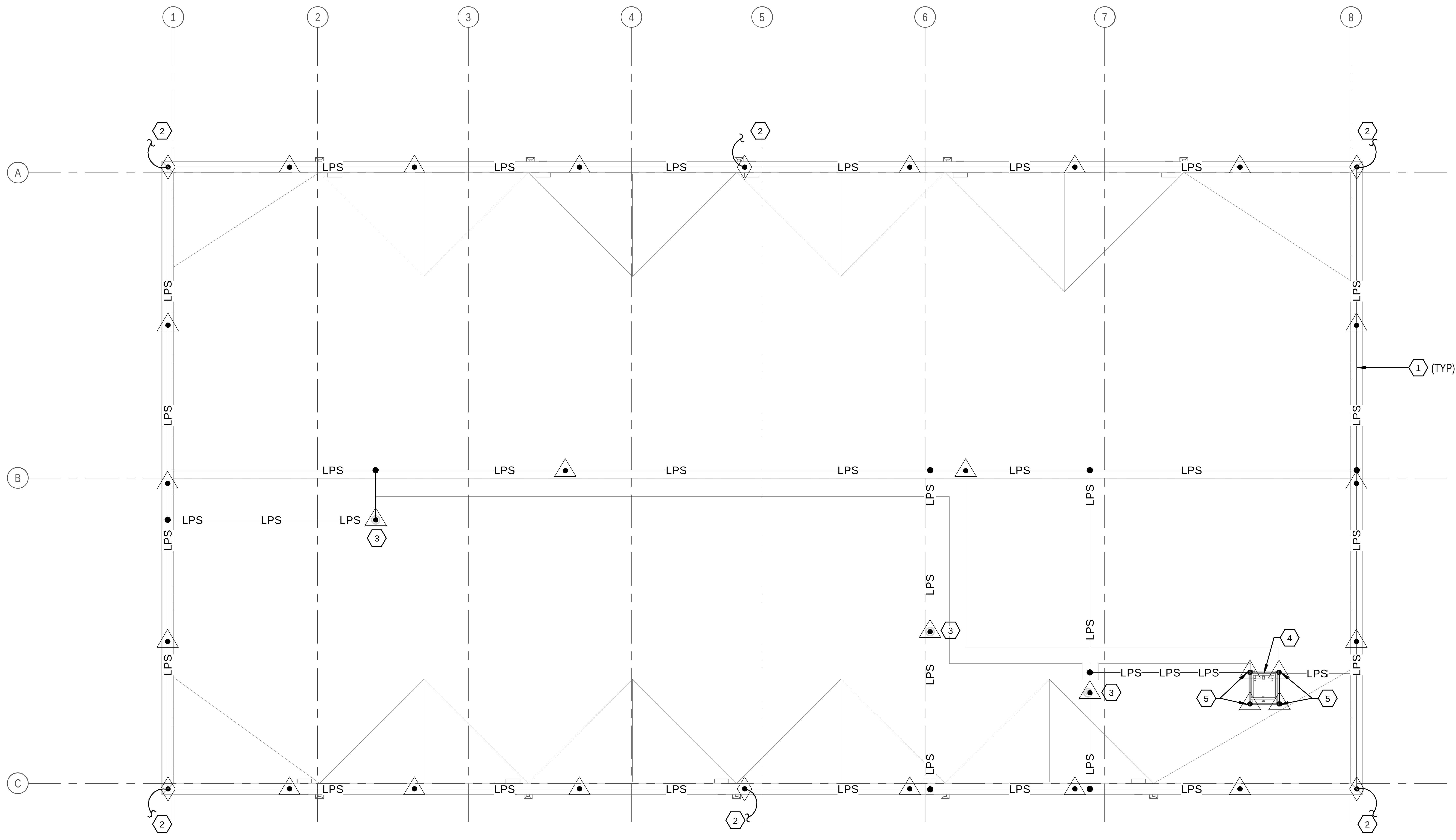
APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	JHP	DRW	JHP	CHK	KB
PM/DM	SMS/CWJ				
BRANCH MANAGER	WRB				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL GROUNDING PLAN

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894041
SHEET 91 OF 140
EG101

6/20/2025 12:27:08 PM
\\dc1vs01\fedprj\FDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt
EG101



B1 ROOF PLAN - LIGHTNING PROTECTION PLAN
1/8" = 1'-0"

GENERAL SHEET NOTES

A. REFER TO SHEETS E-001 THROUGH E-004 FOR ADDITIONAL ELECTRICAL INFORMATION.

B. PROVIDE CLASS 1 LIGHTNING PROTECTION SYSTEM. PROVIDE #2/0AWG BARE COPPER CONDUCTOR WITH STRANDS NOT LESS THAN #17AWG FOR MAIN ROOF LPS AND DOWN CONDUCTORS.

C. EXTERIOR CONDUCTORS WITHIN 8 FEET OF PAVEMENT MUST BE PROTECTED IN 1" SCHEDULE 80 PVC CONDUIT.

SHEET KEYNOTES

1. PROVIDE MAIN ROOF CONDUCTOR INTERCONNECTING THE AIR TERMINALS.
2. PROVIDE DOWN CONDUCTOR FROM AIR TERMINAL TO GROUNDING TEST WELL SHOWN ON SHEET EG101. DOWN CONDUCTOR MUST BE IN A 1" PVC SCHEDULE 80 CONDUIT UNTIL 18" BELOW GRADE.
3. PROVIDE AIR TERMINAL ON RADON FAN, AND BOND TO LPS.
4. BOND ROOF HATCH FRAME, HATCH, AND GATE TO LPS.
5. PROVIDE AIR TERMINAL ON ROOF HATCH SAFETY RAILING.

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

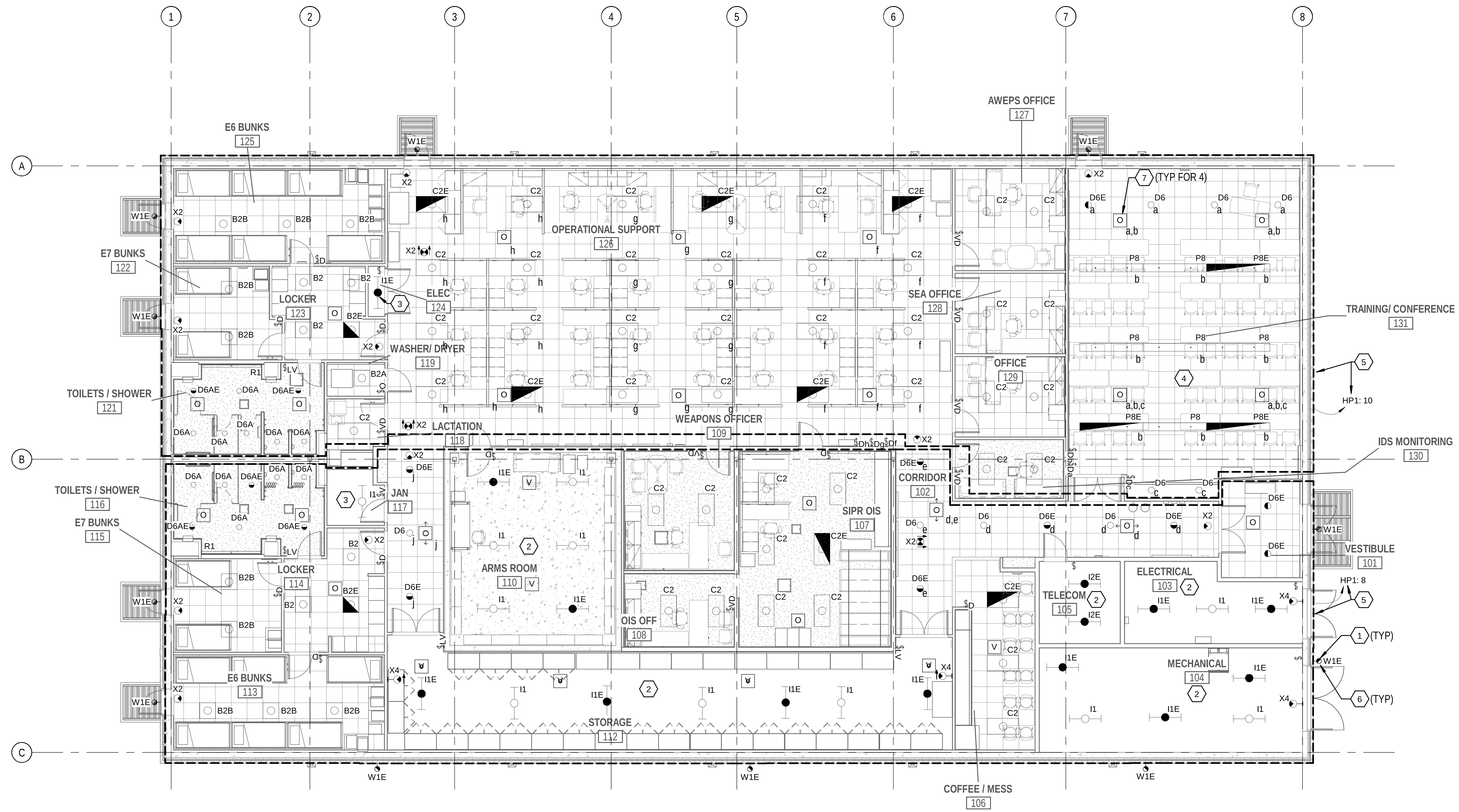
APPROVED	James E. D..., RA
FOR COMMANDER NAVFAC	8/14/2025
ACTIVITY	APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE	05/27/2025
DES JHP / DRW JHP / CHK KB	
PM/DM	SMS/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL LIGHTNING PROTECTION PLAN

SCALE:	AS NOTED
EPROJECT NO.:	1667770
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12894042
SHEET 92	OF 140

EG102

6/20/2025 12:27:11 PM
\\dc1vs01\fedprj\FDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt
EG102



B1 LIGHTING PLAN
1/8" = 1'-0"

GENERAL SHEET NOTES

A. REFER TO SHEETS E-001 THROUGH E-004 FOR ADDITIONAL ELECTRICAL INFORMATION.

B. REFER TO LIGHTING FIXTURE AND CONTROLS SCHEDULE ON SHEET E-701 FOR ADDITIONAL INFORMATION.

C. PROVIDE UNSWITCHED HOT LEG TO EXIT SIGNS.

D. ABSENCE OF CONTROL ZONES INDICATES THAT LIGHTS IN A SPACE ARE CONTROLLED BY THE SWITCH/SENSORS LOCATED WITHIN THAT SPACE.

SHEET KEYNOTES

1. PROVIDE PHOTOCELL MOUNTED ON ROOF FACING NORTHERN SKY FOR CONTROL OF EXTERIOR WALLPACKS.
2. MOUNT PENDANT MOUNTED FIXTURES IN ROOMS NOTED AT 10'-0" AFF TO BOTTOM OF FIXTURE.
3. MOUNT PENDANT MOUNTED FIXTURES IN ROOMS NOTED AT 9'-0" AFF TO BOTTOM OF FIXTURE.
4. MOUNT PENDANT MOUNTED FIXTURES IN ROOMS NOTED AT 8'-6" AFF TO BOTTOM OF FIXTURE.
5. LIGHTING POWER CIRCUIT ZONE. UNLESS OTHERWISE NOTED, LIGHTING FIXTURES LOCATED WITHIN THE DENOTED BOUNDARY MUST BE CIRCUITED TO PANEL AND CIRCUIT NUMBER SHOWN.
6. CIRCUIT EXTERIOR WALLPACK FIXTURES TO PANEL HP1:14.
7. TIE OCCUPANCY SENSORS SERVING TRAINING/CONFERENCE ROOM 131 TO VAV UNIT SERVING THE ROOM. REFER TO MECHANICAL DRAWINGS FOR ADDITIONAL INFORMATION.

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE 05/27/2025

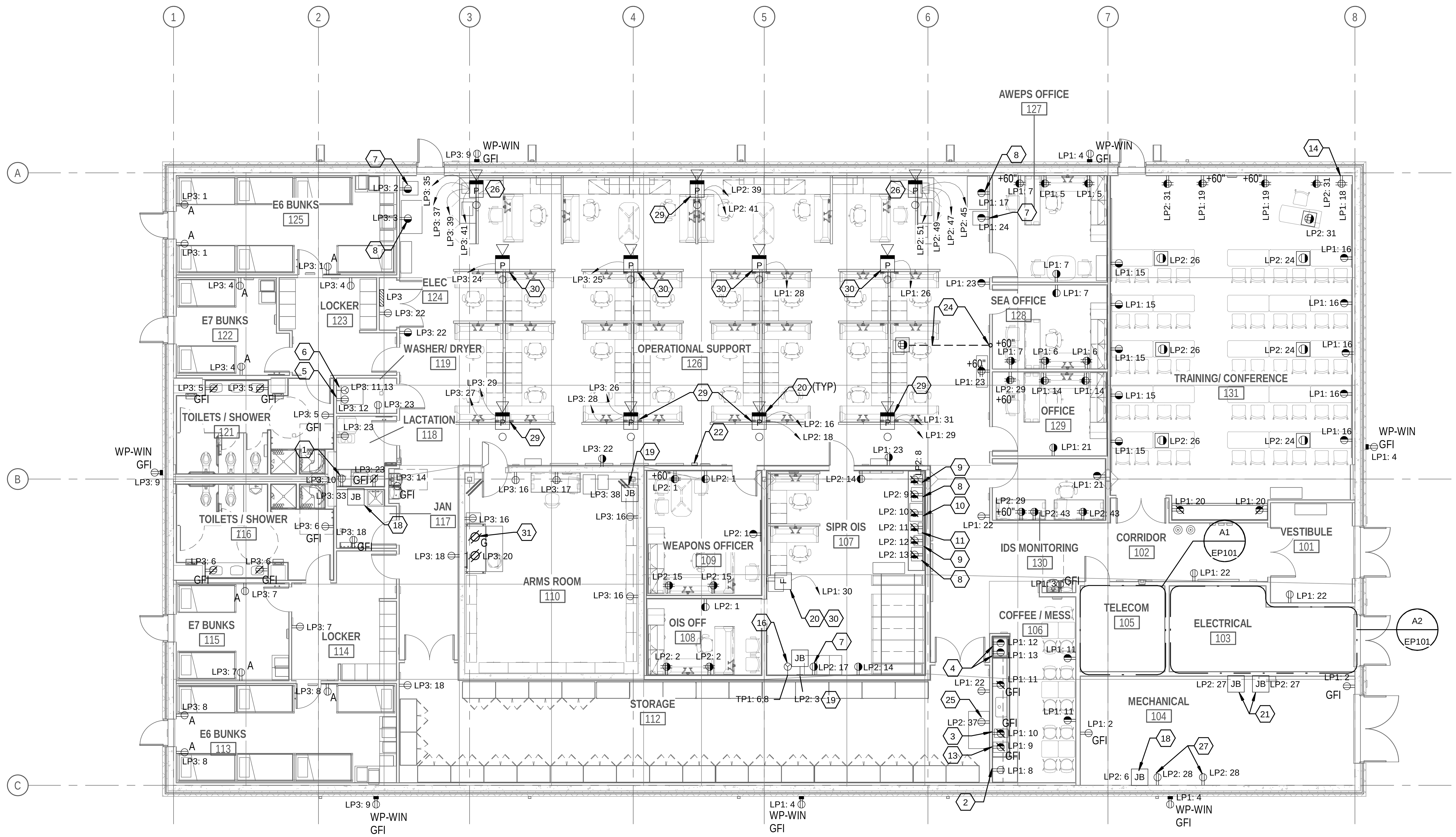
DES JHP	DRW JHP	CHK KB
PM/DM		SMS/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL LIGHTING PLAN

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894043
SHEET 93 OF 140

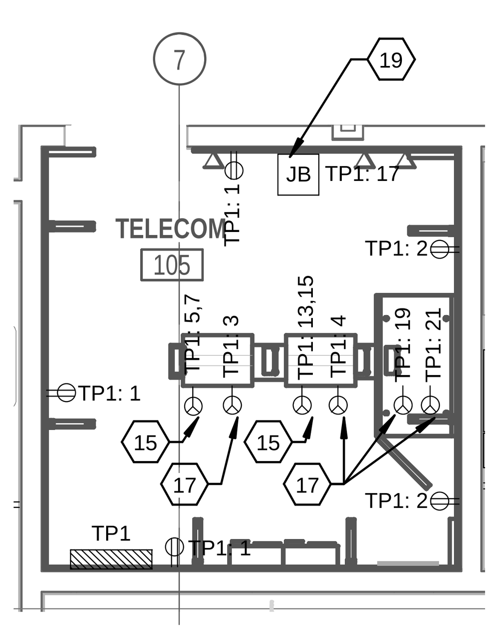
EL101

6/20/2025 12:27:17 PM \\dc1vs01\fedprj\FDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt EL101



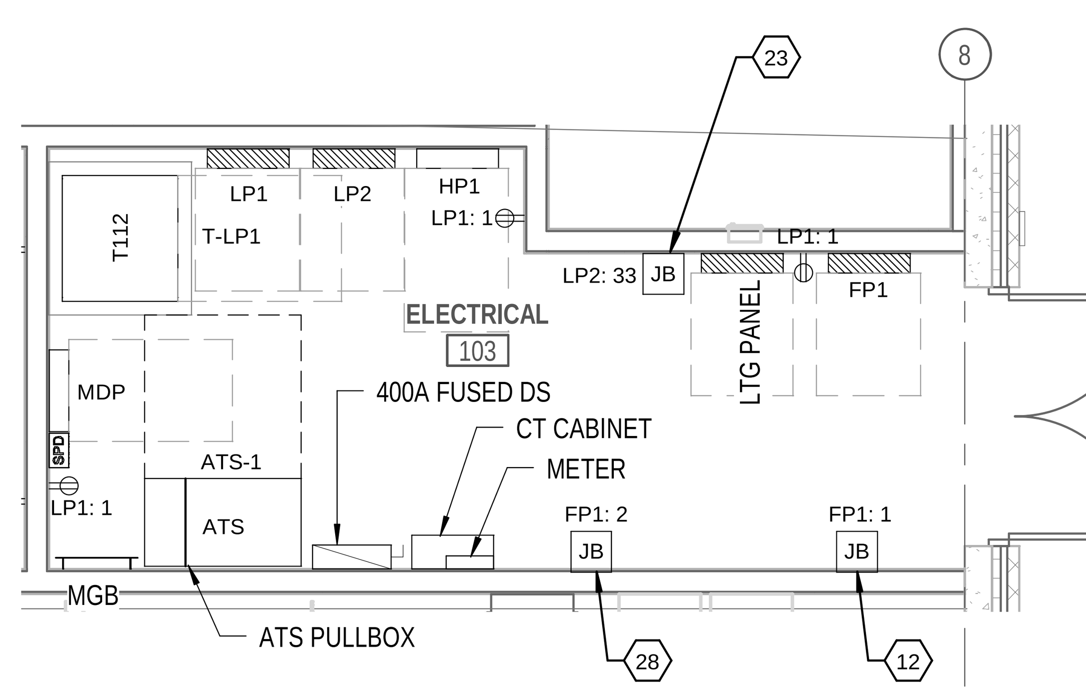
B1 POWER PLAN

1/8" = 1'-0"



A1 TELECOM ROOM 105 - ENLARGED PLAN

1/4" = 1'-0"



A2 ELECTRICAL ROOM - ENLARGED PLAN

1/4" = 1'-0"

GENERAL SHEET NOTES

A. REFER TO SHEETS E-001 THROUGH E-004 FOR ADDITIONAL ELECTRICAL INFORMATION.

SHEET KEYNOTES

1. DEDICATED RECEPTACLE FOR UNDERCOUNTER REFRIGERATOR.
2. DEDICATED RECEPTACLE FOR REFRIGERATOR.
3. DEDICATED RECEPTACLE FOR COFFEE MAKER.
4. DEDICATED RECEPTACLE FOR MICROWAVE.
5. DEDICATED RECEPTACLE FOR WASHING MACHINE.
6. PROVIDE NEMA 14-30R RECEPTACLE FOR DRYER. PROVIDE 3#10 + 1#10G IN 3/4"C.
7. DEDICATED RECEPTACLE FOR SHREDDER.
8. DEDICATED RECEPTACLE FOR PRINTER/COPIER.
9. DEDICATED RECEPTACLE FOR FAX MACHINE.
10. DEDICATED RECEPTACLE FOR SCANNER.
11. DEDICATED RECEPTACLE FOR RADIO CHARGING STATION.
12. PROVIDE DEDICATED 120V HARDWIRED CONNECTION TO FIRE ALARM PANEL.
13. DEDICATED RECEPTACLE FOR WATER/ICE DISPENSER.
14. DEDICATED RECEPTACLE IN CREDENZA FOR A/V RACK. COORDINATE FINAL LOCATION WITH INTERIORS.
15. PROVIDE NEMA L6-30R MOUNTED TO CABLE TRAY (NOT SHOWN) FOR TELECOM RACK POWER. PROVIDE 2#10 + 1#10G IN 3/4"C.
16. PROVIDE NEMA L6-30R FOR WALL-MOUNTED SIPR CABINET POWER. PROVIDE 2#10 + 1#10G IN 3/4"C.
17. PROVIDE NEMA L5-20R MOUNTED TO CABLE TRAY (NOT SHOWN) FOR TELECOM RACK POWER.
18. PROVIDE 120V CONNECTION FOR ELECTRONIC TRAP PRIMER.
19. PROVIDE 120V CONNECTION FOR SECURITY PANELS.
20. PROVIDE PLUG LOAD CONTROL RELAY FOR AT LEAST 25% OF MODULAR SYSTEMS FURNITURE FEEDER CIRCUITS. CIRCUITS MUST RESPOND TO THE OCCUPANCY SENSORS SERVING THE LIGHTING IN A GIVEN AREA.
21. PROVIDE 120V HARDWIRED CONNECTION FOR MECHANICAL CONTROL PANEL.
22. PROVIDE REMOTE GENERATOR ANNUNCIATOR PANEL.
23. PROVIDE 120V CONNECTION FOR LIGHTING CONTROL PANEL.
24. PROVIDE EMPTY 1" CONDUIT FROM FLOORBOX UP TO ROOF FOR ELECTRONIC FIELD MILL (GFGI) AND PROVIDE JUNCTION BOX AT EACH END. VERIFY EXACT LOCATION ON ROOF FOR FIELD MILL. ROOF MOUNTED FIELD MILL, FIELD MILL MONITORING PANEL AND WIRING TO DEVICES ARE PROVIDED BY OTHERS.
25. DEDICATED RECEPTACLE FOR MRE REFRIGERATOR.
26. FAX MACHINE, PRINTER, AND RADIO CHARGING STATION LOCATED AT END TABLE AND SINGLE ADJACENT WORKSTATION ARE POWERED THROUGH POWERPOLE. REFER TO DETAILS A1 AND B1 ON SHEET E-504.
27. PROVIDE RECEPTACLE FOR RADON MONITOR AND REMOTE WATER METER DISPLAY.

SHEET KEYNOTES CONT.

28. PROVIDE 120V CONNECTION FOR FIRE ALARM RADIO ALARM TRANSMITTER.
29. (4) WORKSTATIONS SERVED. REFER TO DETAIL A1 ON SHEET E-502.
30. (2) WORKSTATIONS SERVED. REFER TO DETAIL C1 ON SHEET E-504.
31. PROVIDE 25 FOOT RETRACTABLE GROUND REEL.

0 2' 4' 8' 12'
1/4" = 1'-0"

0 4' 8' 16' 24'
1/8" = 1'-0"

PLAN NORTH

SYM	DESCRIPTION	DATE	APPR

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

KELLEY BOURASSA
Lic. No. 0402061759
PROFESSIONAL ENGINEER
COMMONWEALTH OF VIRGINIA
SEAL

APPROVED
James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

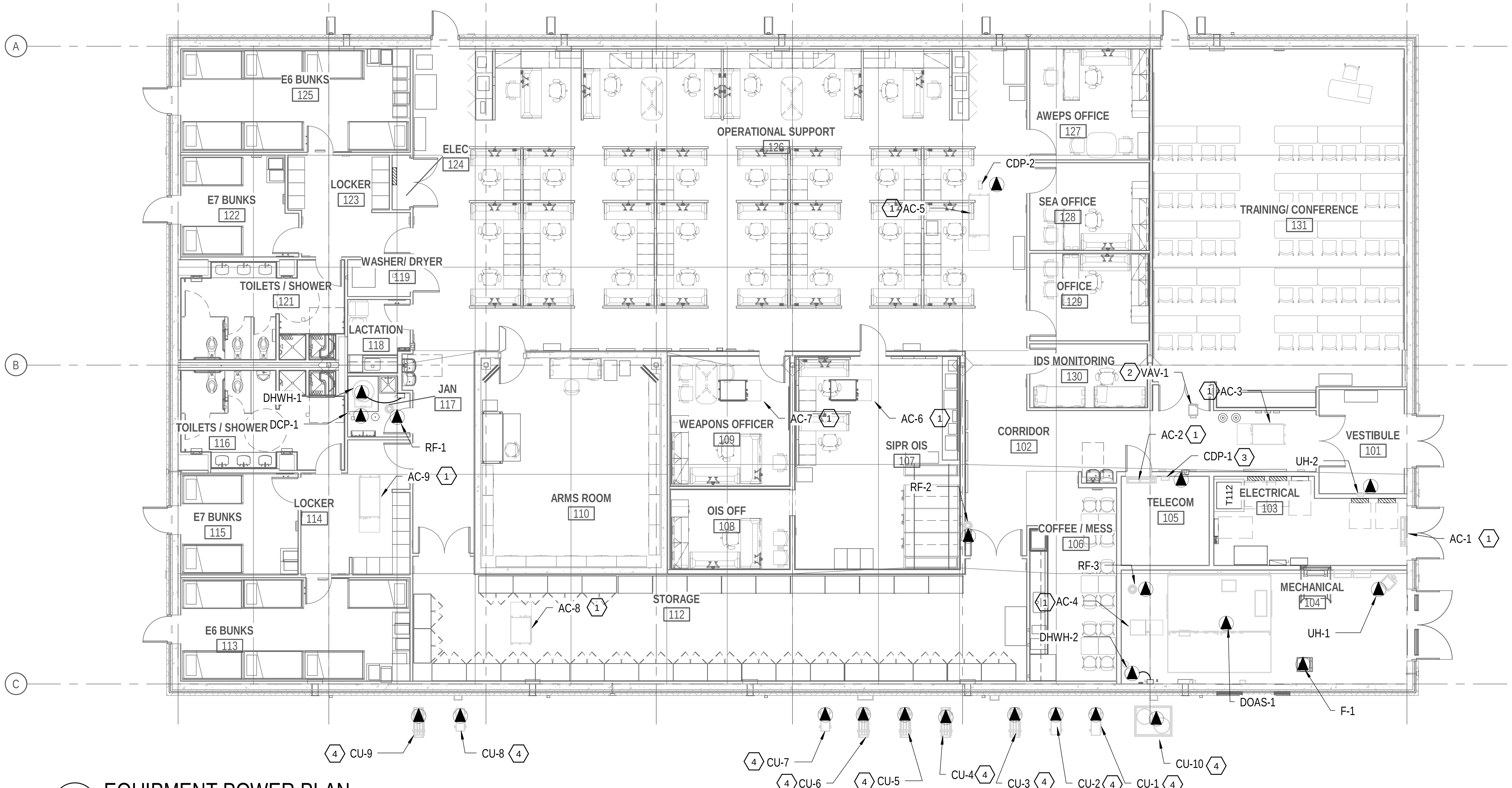
SATISFACTORY TO DATE	05/27/2025
DES JHP / DRW JHP / CHK KB	
PM/DM	SMS/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL POWER PLAN

SCALE:	AS NOTED
EPROJECT NO.:	1667770
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12894044
SHEET 94	OF 140

EP101

6/20/2025 12:27:23 PM
EP101
\\dc1vs01\fedprj\FDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt



B1 EQUIPMENT POWER PLAN

1/8" = 1'-0"

MECHANICAL EQUIPMENT CONNECTION SCHEDULE

TAG	DESCRIPTION	LOCATION	VOLTAGE	PHASE	HP	kVA	FLA	SETS	WIRE AND CONDUIT	CIRCUIT	LOCAL DISCONNECTING MEANS TYPE	DISCONNECT RATING	DISCONNECT TRIP	DISCONNECT ENCLOSURE
CDP-1	CONDENSATE PUMP	TELECOM 105	120	1	NA	0.2	1.5	1	2 - #12+ #12G IN 3/4"C	LP2: 35	RECEPTACLE	NA	NA	NA
CDP-2	CONDENSATE PUMP	OP. SUPPORT 126	120	1	NA	0.2	1.5	1	2 - #12+ #12G IN 3/4"C	LP2: 35	UNIT MOUNTED SWITCH	NA	NA	NA
CU-1	CONDENSING UNIT	EXTERIOR	208	1	NA	3.2	15.2	1	2 - #10 + #10N + #10G IN 3/4"C	LP1: 40,42	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
CU-2	CONDENSING UNIT	EXTERIOR	208	1	NA	3.2	15.2	1	2 - #10 + #10N + #10G IN 3/4"C	TP1: 9,11	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
CU-3	CONDENSING UNIT	EXTERIOR	208	1	NA	4.0	19.2	1	2 - #10 + #10N + #10G IN 3/4"C	LP1: 25,27	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
CU-4	CONDENSING UNIT	EXTERIOR	208	1	NA	4.0	19.2	1	2 - #10 + #10N + #10G IN 3/4"C	LP1: 36,38	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
CU-5	CONDENSING UNIT	EXTERIOR	208	1	NA	4.0	19.2	1	2 - #10 + #10N + #10G IN 3/4"C	LP1: 32,34	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
CU-6	CONDENSING UNIT	EXTERIOR	208	1	NA	1.8	8.8	1	2 - #12 + #12N + #12G IN 3/4"C	LP2: 19,21	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
CU-7	CONDENSING UNIT	EXTERIOR	208	1	NA	1.8	8.8	1	2 - #12 + #12N + #12G IN 3/4"C	LP2: 20,22	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
CU-8	CONDENSING UNIT	EXTERIOR	208	1	NA	1.8	8.8	1	2 - #12 + #12N + #12G IN 3/4"C	LP3: 19,21	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
CU-9	CONDENSING UNIT	EXTERIOR	208	1	NA	4.0	19.2	1	2 - #10 + #10N + #10G IN 3/4"C	LP3: 34,36	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
CU-10	CONDENSING UNIT	EXTERIOR	480	3	NA	10.6	12.8	1	3 - #12 + #12N + #12G IN 3/4"C	HP1: 7,9,11	FACTORY MOUNTED AND WIRED FUSED DISCONNECT SW.	NA	NA	NA
DCP-1	HOT WATER CIRC. PUMP	JAN 118	120	1	0.09	0.2	2.0	1	1 - #12 + #12N + #12G IN 3/4"C	LP3: 15	UNIT MOUNTED SWITCH	15A	NA	NEMA 1
DHWH-1	HOT WATER HEATER	JAN 118	480	3	NA	24.0	28.9	1	3 - #8 + #10G IN 3/4"C	HP1: 1,3,5	NON-FUSIBLE DISCONNECT SWITCH	60A	NA	NEMA 1
DHWH-2	HOT WATER HEATER	MECHANICAL 104	208	1	NA	1.5	7.2	1	1 - #12 + #12N + #12G IN 3/4"C	LP2: 5,7	NON-FUSIBLE DISCONNECT SWITCH	30A	NA	NEMA 1
DOAS-1	OUTSIDE AIR UNIT	MECHANICAL 104	480	3	5	26.1	31.4	1	3 - #8 + #8N + #10G IN 3/4"C	HP1: 2,4,6	FACTORY MOUNTED AND WIRED NONFUSED DISCONNECT SW.	60A	NA	NA
F-1	EXHAUST FAN	MECHANICAL 104	208	1	1/2	0.8	4.0	1	2 - #12+ #12G IN 3/4"C	LP2: 30,32	FACTORY MOUNTED AND WIRED NONFUSED DISCONNECT SW.	NA	NA	NA
RF-1	CENTRIFUGAL FAN	ROOF	120	1	NA	0.1	1.0	1	1 - #12 + #12N + #12G IN 3/4"C	LP3: 31	UNIT MOUNTED SWITCH	NA	NA	NEMA 4X
RF-2	CENTRIFUGAL FAN	ROOF	120	1	NA	0.1	1.0	1	1 - #12 + #12N + #12G IN 3/4"C	LP3: 32	UNIT MOUNTED SWITCH	NA	NA	NEMA 4X
RF-3	CENTRIFUGAL FAN	ROOF	120	1	NA	0.1	1.0	1	1 - #12 + #12N + #12G IN 3/4"C	LP2: 4	UNIT MOUNTED SWITCH	NA	NA	NEMA 4X
UH-1	UNIT HEATER	MECHANICAL 104	208	1	NA	5.0	24.0	1	2 - #10+ #10G IN 3/4"C	LP1: 33,35	FACTORY MOUNTED AND WIRED NONFUSED DISCONNECT SW.	NA	NA	NA
UH-2	UNIT HEATER	VESTIBULE 101	208	1	NA	2.0	9.6	1	2 - #12+ #12G IN 3/4"C	LP2: 23,25	FACTORY MOUNTED AND WIRED NONFUSED DISCONNECT SW.	NA	NA	NA

GENERAL SHEET NOTES

A. REFER TO SHEETS E-001 THROUGH E-004 FOR ADDITIONAL ELECTRICAL INFORMATION.

B. REFER TO EQUIPMENT CONNECTION SCHEDULE ON THIS SHEET FOR ADDITIONAL CIRCUITING AND DISCONNECTING MEANS INFORMATION.

C. COORDINATE FINAL LOCAL DISCONNECTING MEANS REQUIREMENTS FOR ALL EQUIPMENT BASED ON APPROVED EQUIPMENT TO BE PROVIDED.

SHEET KEYNOTES

1. UNIT IS POWERED FROM CORRESPONDING OUTDOOR UNIT. PROVIDE WIRING FROM OUTDOOR UNIT TO INDOOR UNIT AS RECOMMENDED BY EQUIPMENT MANUFACTURER. PROVIDE 30A NON-FUSED DISCONNECT SWITCH IN NEMA 1 ENCLOSURE.
2. MECHANICAL UNIT DOES NOT REQUIRE AC POWER.
3. PROVIDE NEMA 5-20R DUPLEX RECEPTACLE FOR CDP-1. COORDINATE FINAL LOCATION OF RECEPTACLE WITH INSTALLATION OF PUMP.
4. COORDINATE FINAL LOCATION FOR FACTORY PROVIDED AND WIRED DISCONNECT SWITCH FOR CONDENSING UNITS. MAINTAIN NEC REQUIRED CLEARANCES.

PLAN NORTH

0 4' 8' 16' 24'
1/8" = 1'-0"

SYM DESCRIPTION DATE APPR

NAVFAC

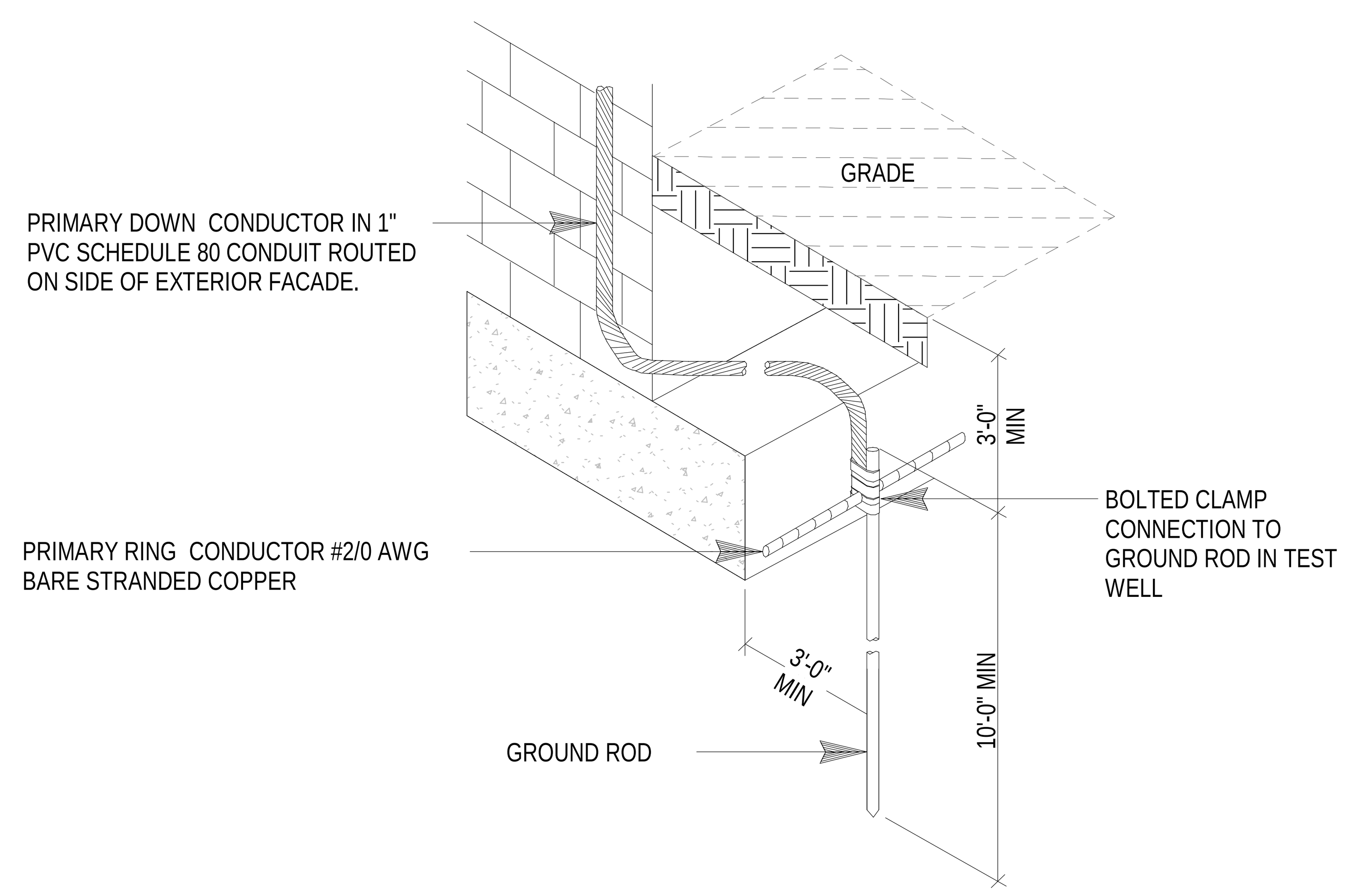
KELLEY BOURASSA
Lic. No. 0402061759
COMMONWEALTH OF VIRGINIA
PROFESSIONAL ENGINEER
6/23/2025

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES JHP DRW JHP CHK KB
PM/DM SMS/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

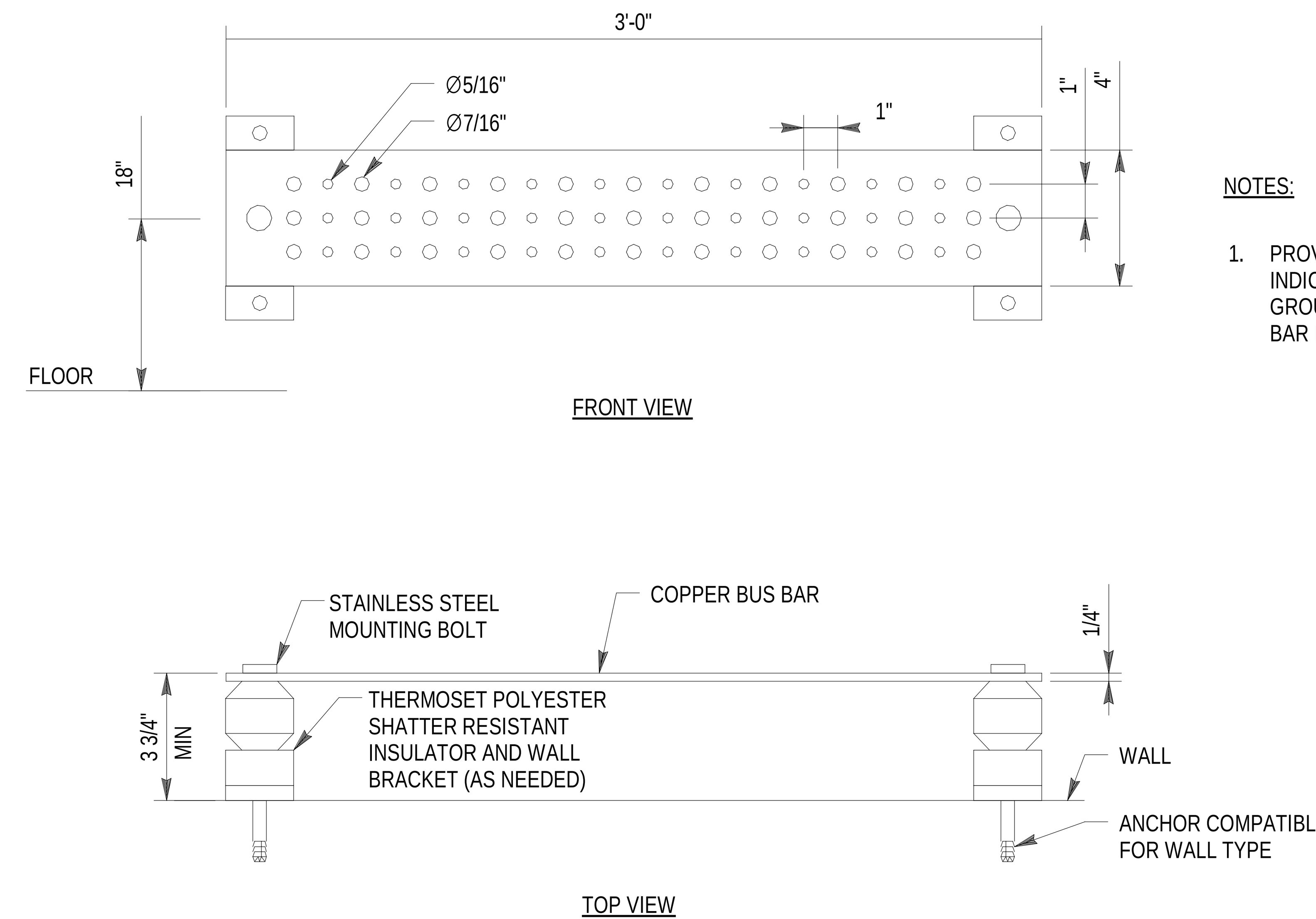
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
MID-ATLANTIC
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL EQUIPMENT POWER PLAN

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894045
SHEET 95 OF 140
EP102



B1 TYPICAL ELECTRICAL DOWN CONDUCTOR DETAIL

NTS

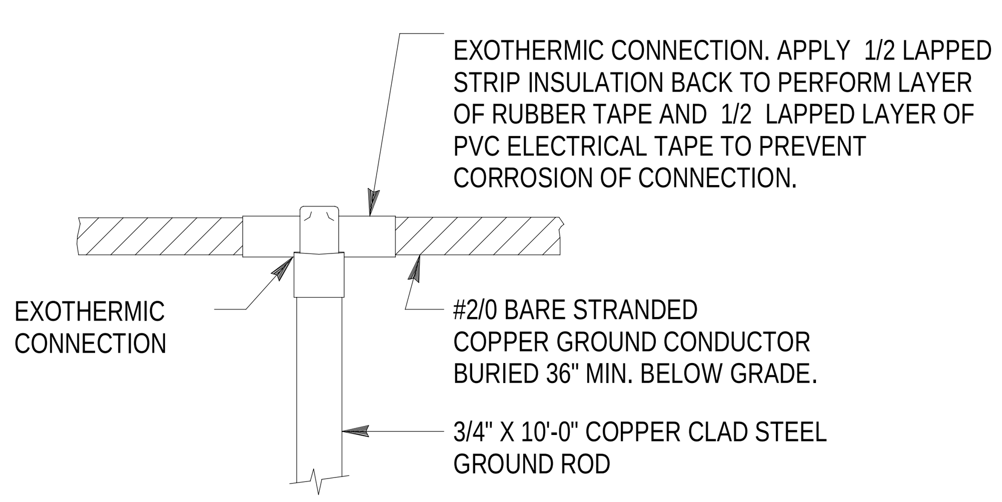


NOTES:

1. PROVIDE WALL MOUNTED NAMEPLATE INDICATING GROUND BAR IDENTITY 12 INCHES ABOVE GROUND BAR. SEE GROUNDING PLAN FOR GROUND BAR IDENTITIES.

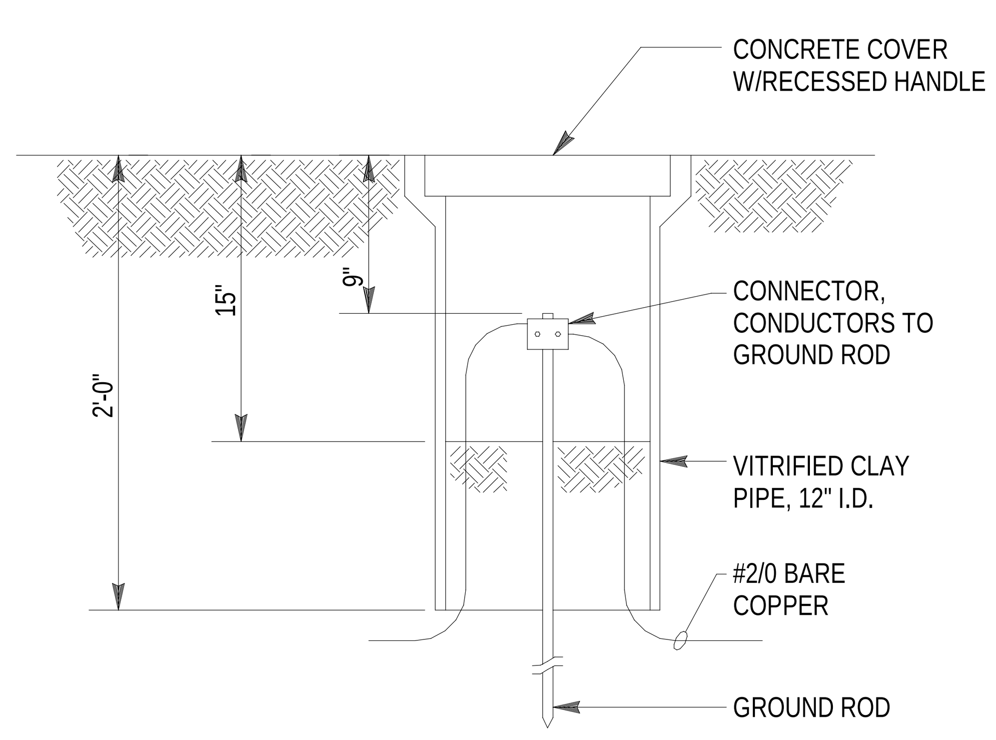
B3 MAIN GROUND BAR (MGB) DETAIL

NTS



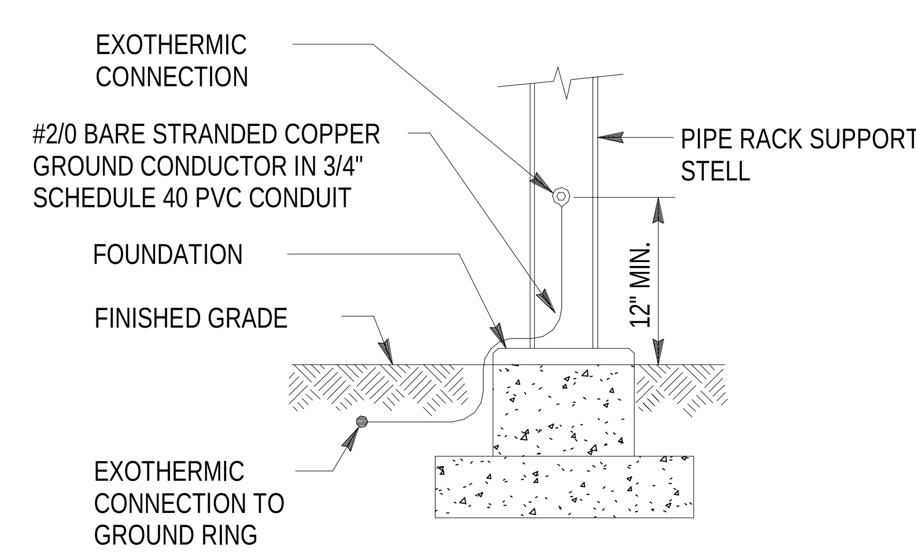
A1 GROUND ROD DETAIL

NTS



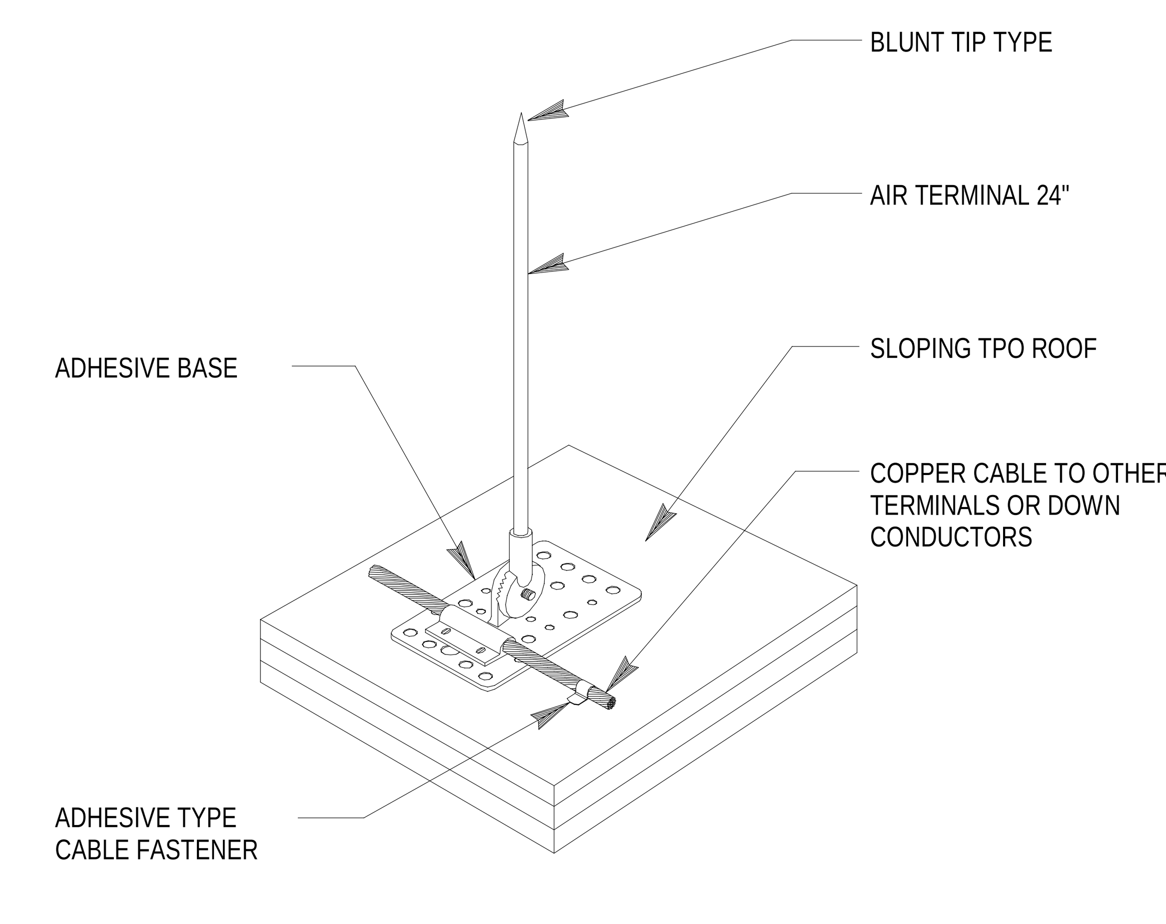
A2 GROUND TEST WELL

NTS



A3 STRUCTURAL STEEL GROUNDING DETAIL

NTS



A4 LIGHTNING PROTECTION AIR TERMINAL

NTS

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER

SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED

FOR COMMANDER NAVFAC 8/14/2025

ACTIVITY

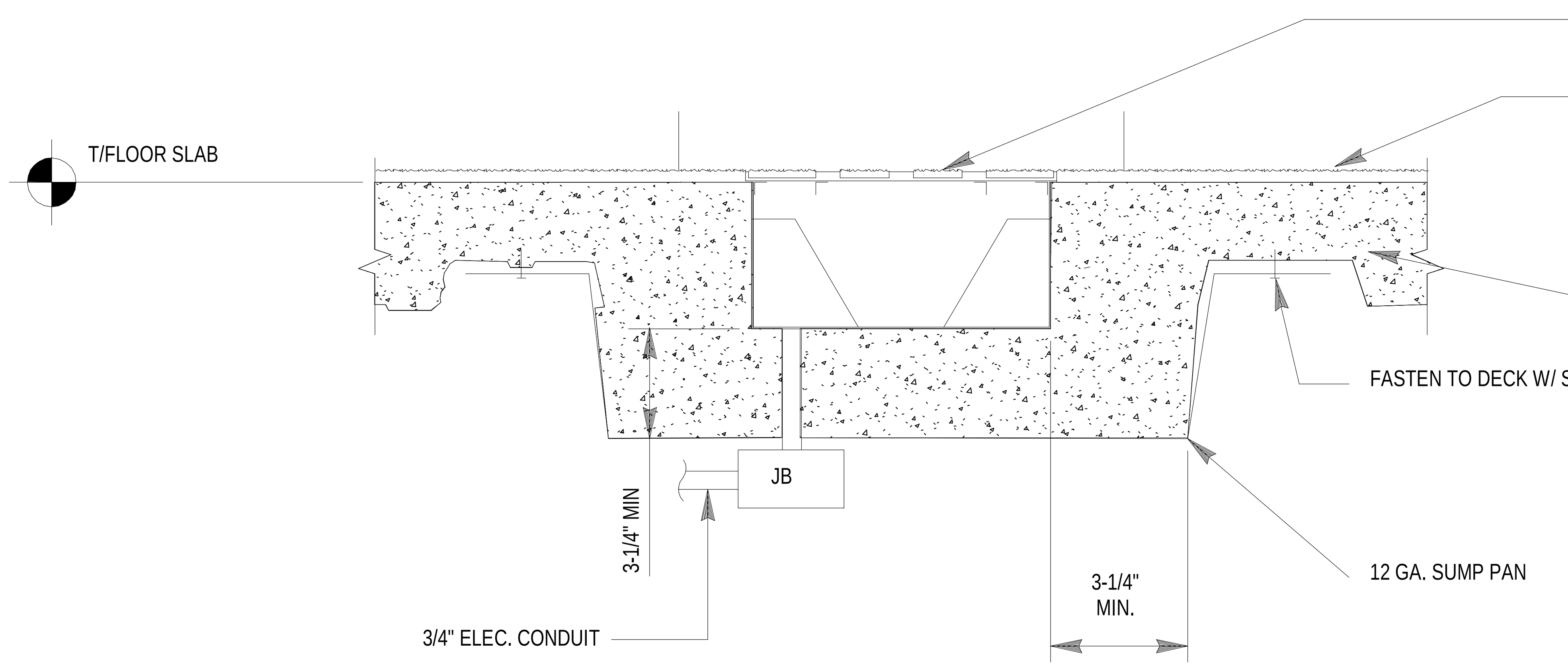
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT

SATISFACTORY TO DATE	05/27/2025
DES JHP DRW JHP	CHK KB
PM/DM	SMS/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL DETAILS

SCALE:	AS NOTED
EPROJECT NO.:	1667770
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	12894046
SHEET 96	OF 140

E-501



FLOOR BOX INSTALLATION DETAIL FOR SLAB ABOVE GRADE

NOTES:

1. SEE TELECOM AND AUDIOVISUAL DRAWINGS FOR FLOOR BOX SIZE AS WELL AS QUANTITY, SIZE AND LOCATION OF CONDUITS.
2. DETAIL BASED UPON FIRE CLASSIFIED LEGRAND EFB8S-FC AV/PWR/DATA FLOOR BOX IN NEW CONSTRUCTION ON INTERMEDIATE FLOOR.

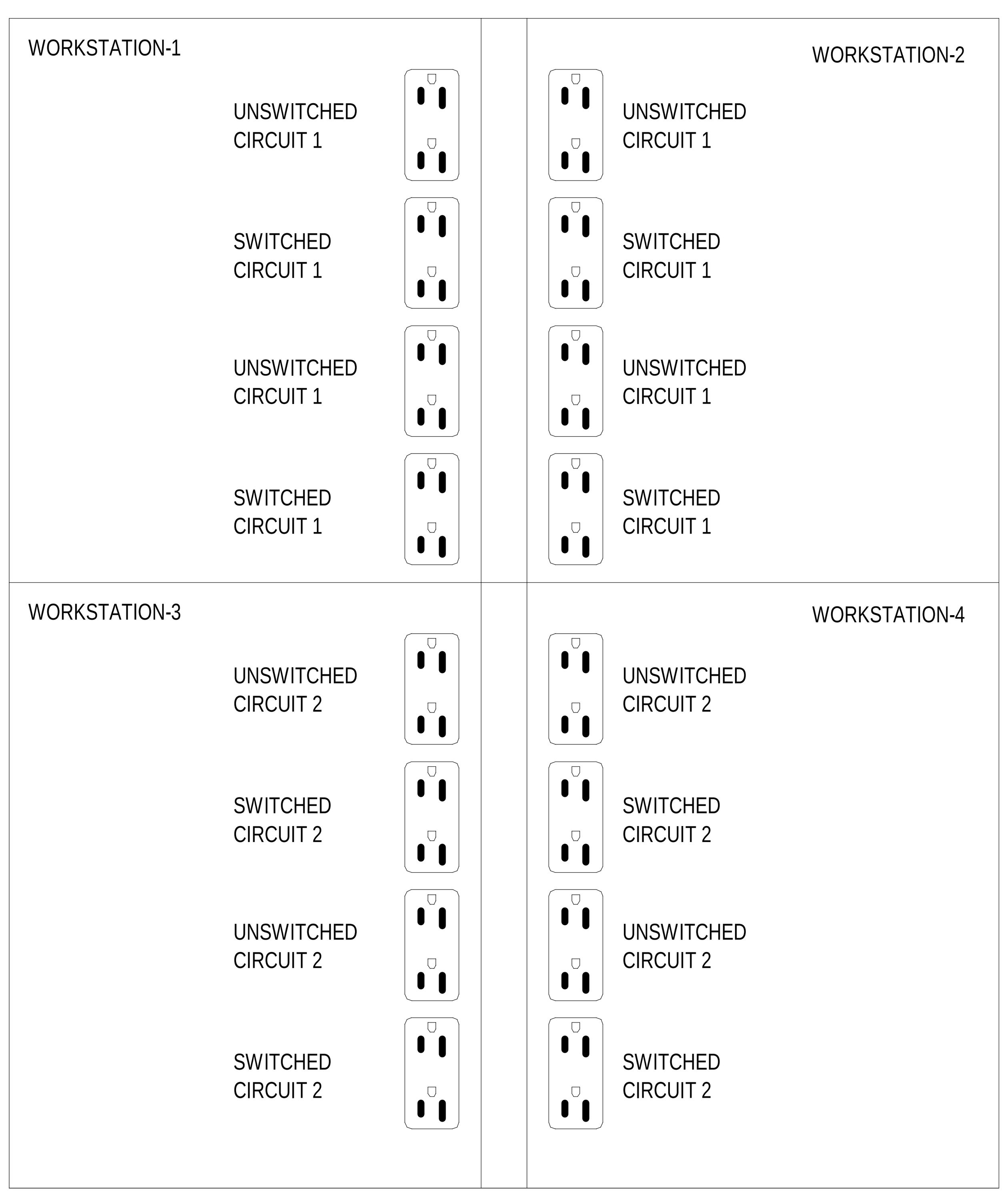
LEGRAND EFB8S AV/PWR/DATA FLOOR BOX HAS FOUR (4) 1-1/4" CONDUIT FEEDS ON BOTTOM

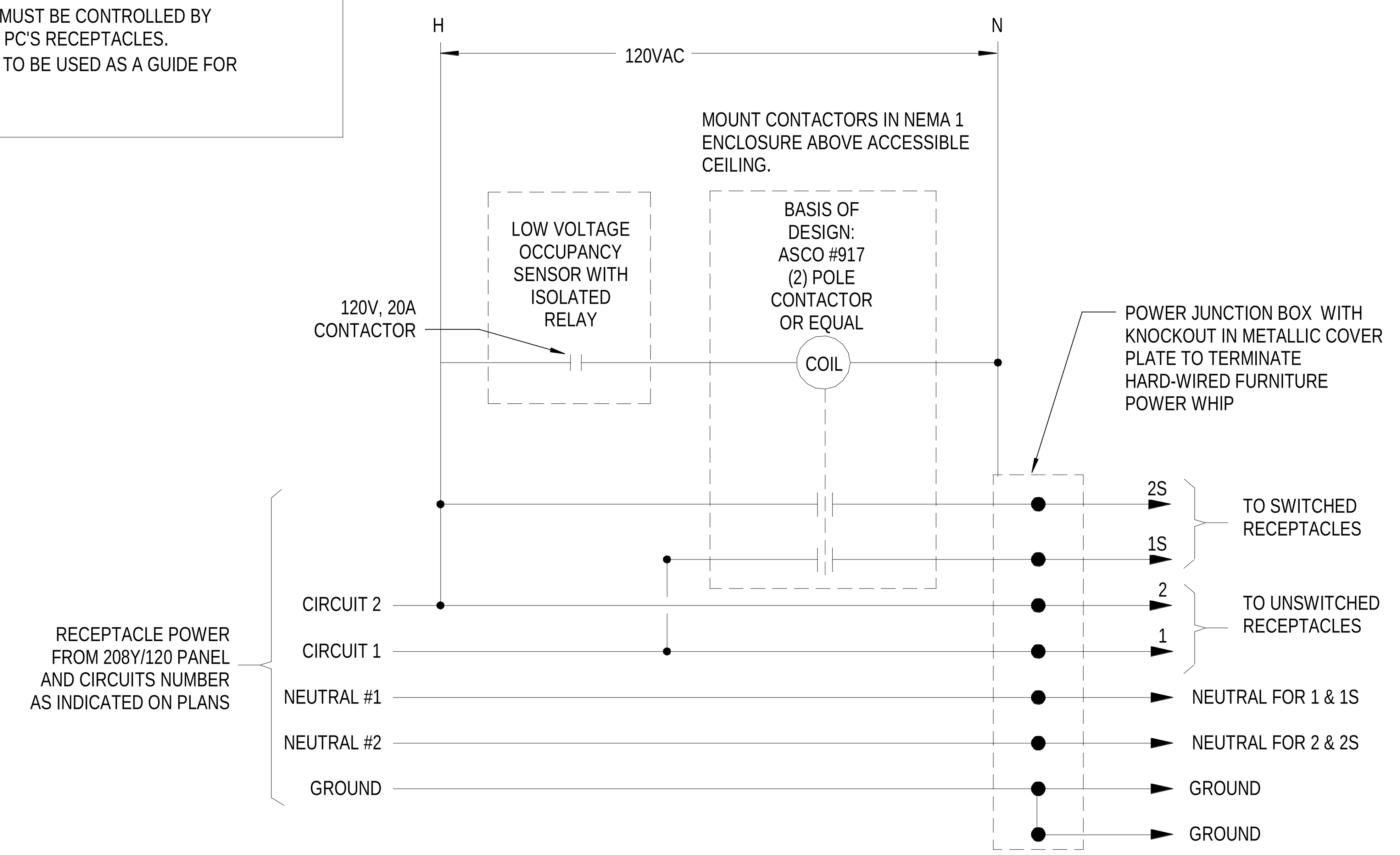
C1 FLOOR BOX INSTALLATION DETAIL FOR SLAB ABOVE GRADE

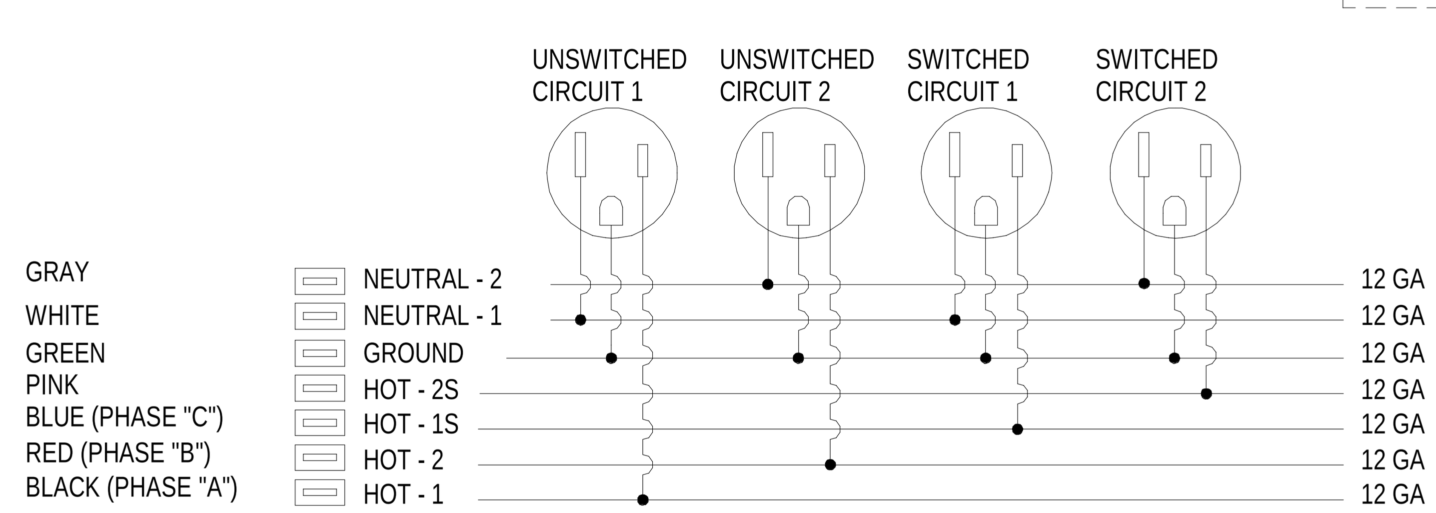
NOT TO SCALE

DUPLEX RECEPTACLE INTEGRAL TO FURNITURE SYSTEM

NOTE:
- AT LEAST 50% OF TOTAL RECEPTACLES INSTALLED IN MODULAR FURNITURE MUST BE CONTROLLED BY OCCUPANCY SENSOR. SWITCH ONLY UTILITY RECEPTACLES . DO NOT SWITCH PC'S RECEPTACLES.
- LAYOUT WITH SUGGESTED CIRCUITING SHOWN BELOW IS CONCEPTUAL AND TO BE USED AS A GUIDE FOR THE ACTUAL FURNITURE LAYOUT.
- MAX FOUR WORKSTATIONS PER ONE PC'S CIRCUIT.



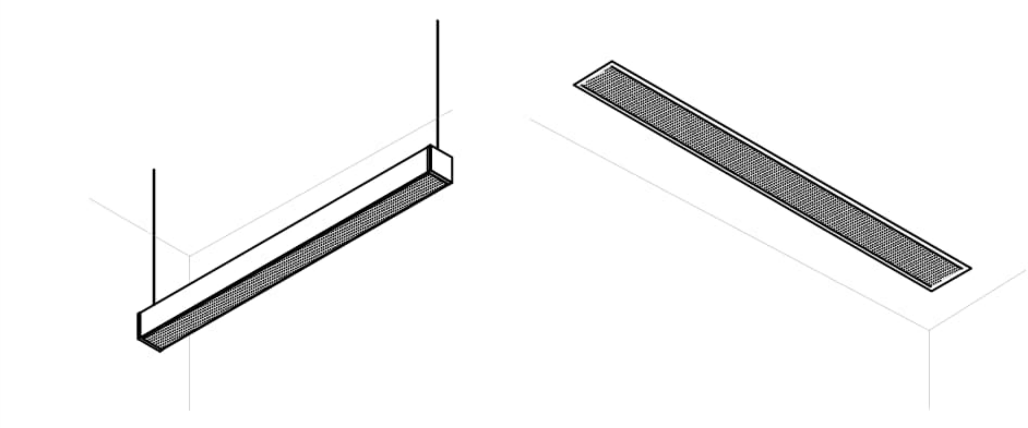




BASED ON USING 4 CIRCUIT, 2+2 WORKSTATION WIRING CONFIGURATION

A1 TYPICAL ELECTRIFIED FURNITURE WIRING - 2+2 - 4 WORKSTATIONS

NTS



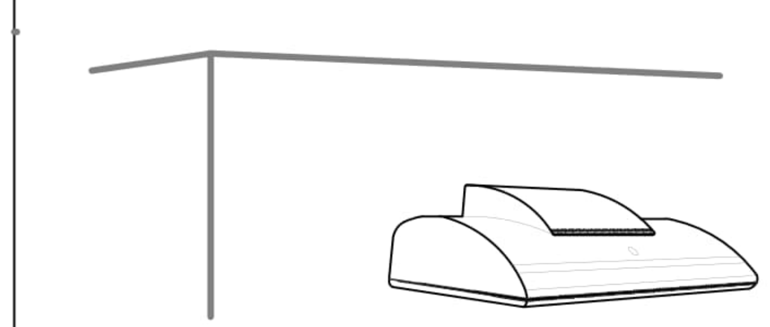
NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING – HEAVY GAUGE COLD ROLLED STEEL, SIZE AS INDICATED IN LUMINAIRE SCHEDULE.
2. OPTICS – SATIN ACRYLIC FLUSH LENS. LAMBERTIAN LIGHT DISTRIBUTION.
3. LIGHT SOURCE – SOLID STATE LEDS, 3500K CCT UON, MINIMUM 80 CRI UON, AND MINIMUM EFFICACY OF 90 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER – REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON-OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE.
5. CERTIFICATION – UL LISTED FOR WET LOCATION, ROHS COMPLIANT, DLC QUALIFIED. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
6. MOUNTING – RECESSED IN HARD CEILING.
7. OPTIONS – VARIOUS PROFILE DIMENSIONS AND RUN LENGTHS, AND VARIOUS CLEAR OR FROSTED POLYCARBONATE LENSES, BAFFLES, OR LOUVERS.

LINEAR LED SLOT

REVISED: NOVEMBER 2020 | LIGHTING PLATE: NL-4



NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING – DIE-CAST ALUMINUM WITH INTEGRAL PASSIVE COOLING MECHANISM. HEAT SINK INCORPORATED DIRECTLY INTO HOUSING COMPARTMENT.
2. OPTICS – PRECISION MOLDED ACRYLIC LENS WITH TYPE IV DISTRIBUTION. BUG UPLIGHT RATING OF U0, WITH GLARE RATING AS DETERMINED BY LIGHTING ZONE INSTALLED.
3. LIGHT SOURCE – SOLID STATE LEDS, 4000K CCT UON, MINIMUM 70 CRI UON, AND MINIMUM EFFICACY OF 80 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER – REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DIMMABLE DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON-OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE.
5. CERTIFICATION – UL LISTED FOR WET LOCATION, ROHS COMPLIANT. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
6. MOUNTING – SURFACE MOUNTED WITH STAINLESS STEEL MOUNTING HARDWARE.
7. OPTIONS – VARIOUS LIGHT DISTRIBUTIONS. BATTERY BACK-UP.

LED WALL PACK

REVISED: NOVEMBER 2020 | LUMINAIRE PLATE: XL-10

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER

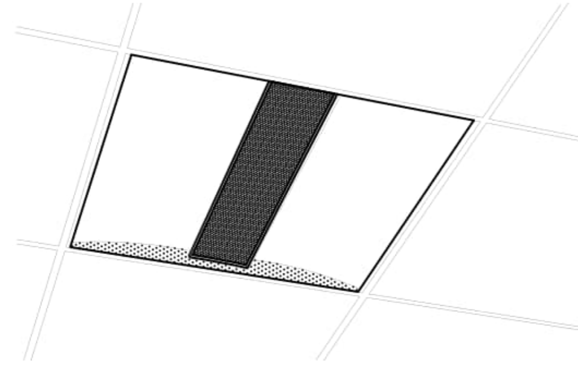
Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES JHP | DRW JHP | CHK KB
PM/DM SMS/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894047
SHEET 97 OF 140
E-502

E-502
6/20/2025 12:26:46 PM



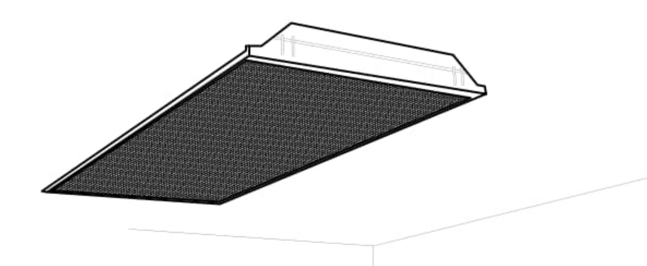
NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING – HEAVY GAUGE COLD ROLLED STEEL. SIZE SHOWN AS INDICATED IN LUMINAIRE SCHEDULE.
2. OPTICS – FROSTED ACRYLIC LENS WITH DIE FORMED COLD ROLLED SHEET STEEL REFLECTORS.
3. LIGHT SOURCE – SOLID STATE LEDS, 3500K CCT UON, MINIMUM 80 CRI UON, AND MINIMUM EFFICACY OF 100 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER – REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DIMMABLE DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON/OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE.
5. CERTIFICATION – UL LISTED FOR DRY LOCATION, ROHS COMPLIANT. DLC QUALIFIED. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
6. MOUNTING – RECESSED IN HARD AND ACOUSTICAL TILE CEILING.
7. OPTIONS – EMERGENCY BATTERY BACK-UP, VARIOUS SIZE AND OUTPUT OPTIONS, SURFACE-MOUNTING KIT.

DIRECT/INDIRECT LED LUMINAIRE

REVISED: NOVEMBER 2020	LIGHTING PLATE: NL-1



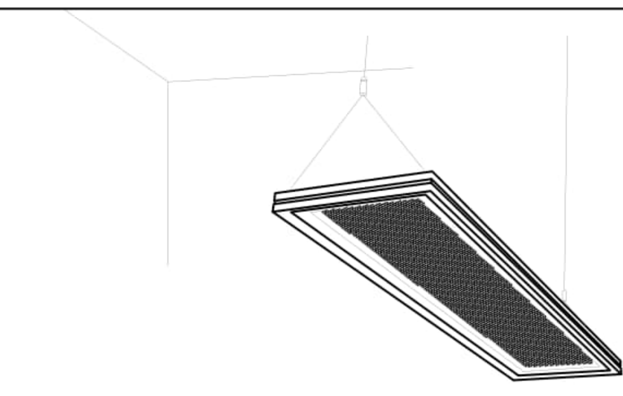
NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING – DIE-FORMED, COLD ROLLED STEEL. SIZE AS INDICATED IN LUMINAIRE SCHEDULE.
2. OPTICS – ACRYLIC LENS.
3. LIGHT SOURCE – SOLID STATE LEDS, 3500K CCT UON, MINIMUM 80 CRI UON, AND MINIMUM EFFICACY OF 100 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER – REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON/OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE.
5. CERTIFICATION – UL LISTED FOR DRY LOCATION, ROHS COMPLIANT, DLC QUALIFIED. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
6. MOUNTING – RECESSED IN HARD AND ACOUSTICAL TILE CEILING.
7. OPTIONS – PRISMATIC LENS. VARIOUS SIZE AND OUTPUT OPTIONS.

FLAT LENS LED LUMINAIRE

REVISED: NOVEMBER 2020	LIGHTING PLATE: NL-2



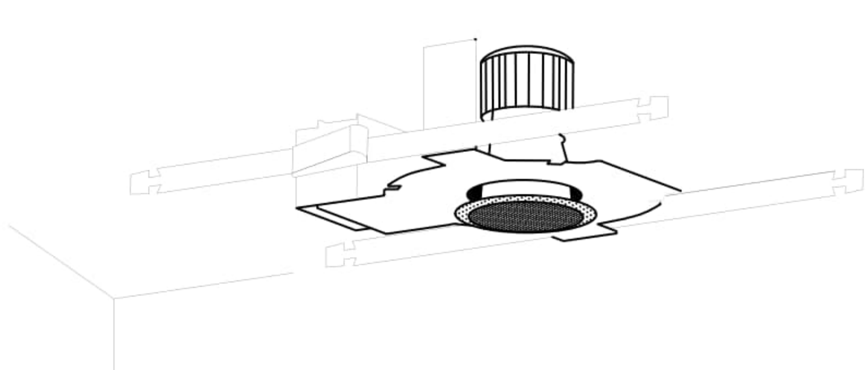
NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING – DIE CAST ALUMINUM FRAME AND HEAT SINK WITH ENAMELED FINISH. SIZE AS INDICATED IN LUMINAIRE SCHEDULE.
2. OPTICS – ACRYLIC LENS. LAMBERTIAN DISTRIBUTION.
3. LIGHT SOURCE – SOLID STATE LEDS, 3500K CCT UON, MINIMUM 80 CRI UON, AND MINIMUM EFFICACY OF 75 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER – REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DIMMABLE DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON-OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE. INTEGRAL TO LUMINAIRE.
5. CERTIFICATION – UL LISTED FOR DRY LOCATION, ROHS COMPLIANT, DLC QUALIFIED. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
6. MOUNTING – PENDANT MOUNTED WITH STAINLESS STEEL MOUNTING HARDWARE.
7. OPTIONS – EMERGENCY BATTERY BACK-UP, VARIOUS PROFILE DIMENSIONS AND RUN LENGTHS, AND VARIOUS CLEAR OR FROSTED POLYCARBONATE LENSES, BAFFLES, OR LOUVERS.

DIRECT/INDIRECT LED PENDANT

REVISED: NOVEMBER 2020	LIGHTING PLATE: NL-6



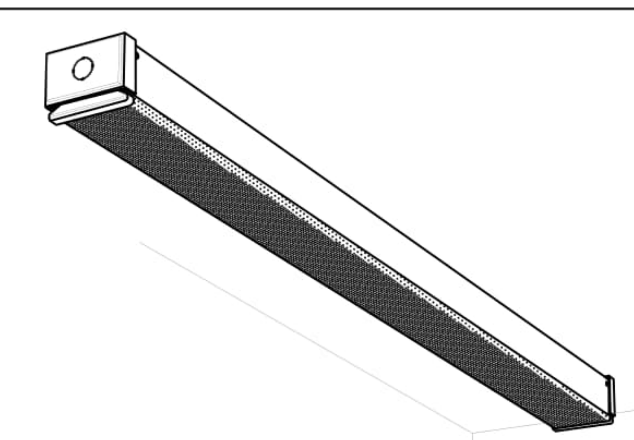
NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING – COLD-ROLLED STEEL WITH HEAT SINK. APERTURE SIZE AND SHAPE AS INDICATED IN LUMINAIRE SCHEDULE.
2. LIGHT SOURCE – SOLID STATE LEDS, 3500K CCT UON, MINIMUM 80 CRI UON, AND MINIMUM EFFICACY OF 70 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
3. DRIVER – REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DIMMABLE DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON/OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE.
4. CERTIFICATION – UL LISTED FOR DRY AND WET LOCATIONS, ROHS COMPLIANT. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
5. MOUNTING – RECESSED IN HARD AND ACOUSTICAL TILE CEILING. PROVIDE T-BAR HANGERS FOR INSTALLATION IN ACOUSTICAL TILE CEILINGS OR TABS WHEN MOUNTING IN HARD CEILINGS.
6. OPTIONS – EMERGENCY BATTERY BACK-UP, VARIOUS ACRYLIC OR POLYCARBONATE LENSES, REFLECTORS, LOUVERS AND TRIMS. VARIOUS BEAM ANGLES. IC RATED HOUSING.

FIXED RECESSED DOWNLIGHT

REVISED: NOVEMBER 2020	LIGHTING PLATE: NL-12



NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING – WELDED STEEL HOUSING WITH SNAP-ON END CAPS. SIZE AS INDICATED IN LUMINAIRE SCHEDULE.
2. OPTICS – DIFFUSE ACRYLIC LENS.
3. LIGHT SOURCE – SOLID STATE LEDS, 3500K CCT UON, MINIMUM 80 CRI UON, AND MINIMUM EFFICACY OF 90 LUMENS/WATT UON. INITIAL LUMEN OUTPUT AS INDICATED IN LUMINAIRE SCHEDULE.
4. DRIVER – REPLACEABLE, INTEGRAL, HIGH-EFFICIENCY DIMMABLE DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120-277V, THERMAL MANAGEMENT, AND < 20% THD. ON/OFF CONTROL AND FULLY DIMMABLE DOWN TO 10% MINIMUM OR AS INDICATED IN LUMINAIRE SCHEDULE.
5. CERTIFICATION – UL LISTED FOR DAMP LOCATION, ROHS COMPLIANT. DLC QUALIFIED. COMPLIES WITH IES LM79, LM80 AND TM21 TESTING STANDARDS.
6. MOUNTING – PENDANT MOUNTED WITH STAINLESS STEEL MOUNTING HARDWARE.
7. OPTIONS – EMERGENCY BATTERY BACK-UP, WIREGUARD, VARIOUS PROFILE DIMENSIONS AND RUN LENGTHS, AND VARIOUS CLEAR OR FROSTED POLYCARBONATE LENSES.

LED INDUSTRIAL LIGHT

REVISED: NOVEMBER 2020	LIGHTING PLATE: NL-23

NOTE: THIS SKETCH IS A NON-PROPRIETARY GRAPHIC REPRESENTATION OF A LUMINAIRE THAT MAY MEET THE SPECIFICATION REQUIREMENTS. IT IS NOT INTENDED TO INDICATE A CERTAIN MANUFACTURER OR PREFERENCE.

LUMINAIRE REQUIREMENTS:

1. HOUSING – HIGH-IMPACT, UV-STABILIZED, INJECTION-MOLDED THERMOPLASTIC.
2. LIGHT SOURCE – SOLID STATE LEDS.
3. DRIVER – INTEGRAL, HIGH-EFFICIENCY DRIVER WITH MINIMUM 0.9 PF, OPERATING VOLTAGE OF 120/277V, THERMAL MANAGEMENT, AND < 20% THD.
4. CERTIFICATION – NFPA 101, UL LISTED FOR DAMP LOCATION, AND ROHS COMPLIANT.
5. MOUNTING – SURFACE MOUNTED ON CEILING AND/OR WALL.
6. OPTIONS – RED LETTERING, ONE- OR TWO-SIDED. BATTERY BACKUP.

EXIT SIGN

REVISED: NOVEMBER 2020	LIGHTING PLATE: NL-28

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

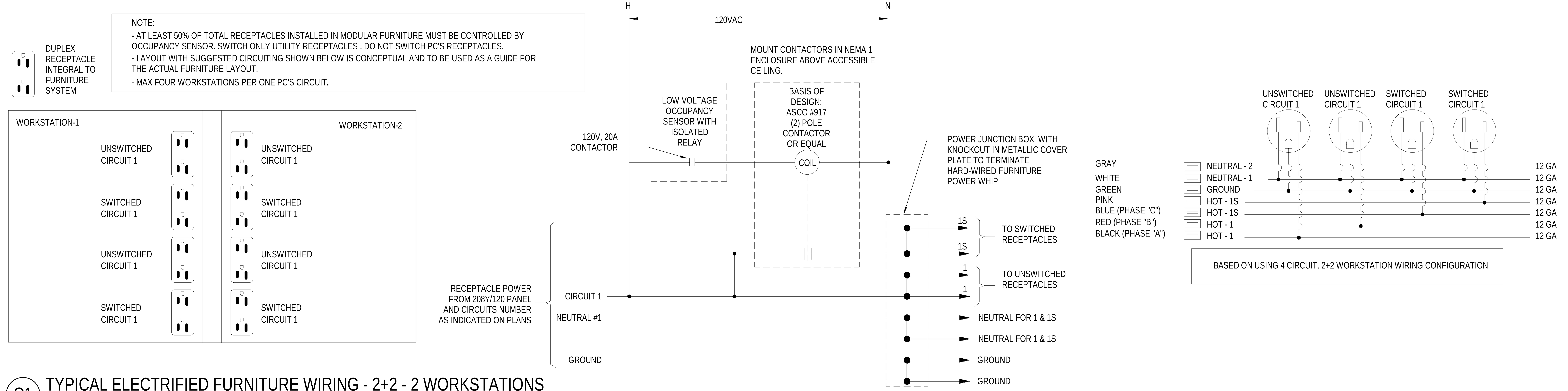
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES JHP	DRW JHP	CHK KB
PM/DM		SMS/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

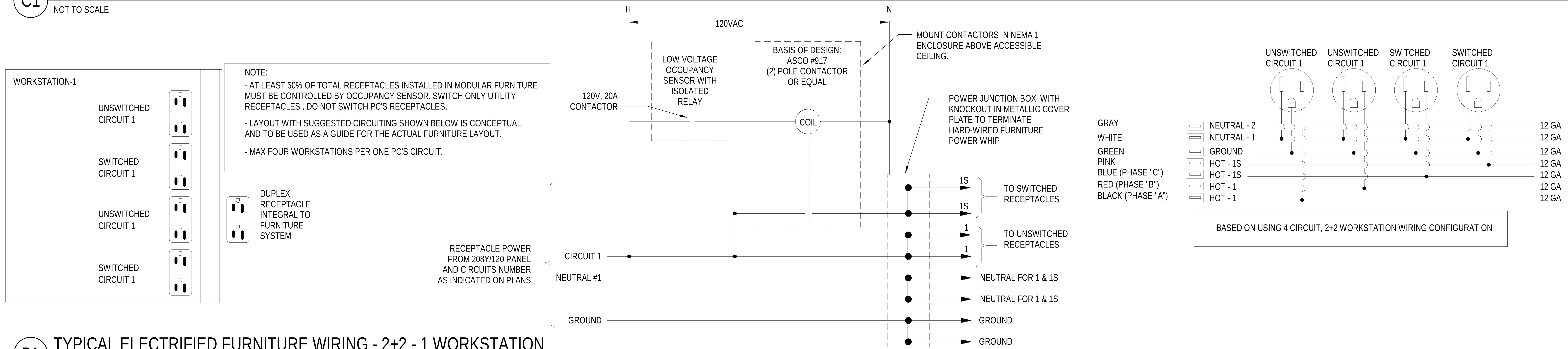
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894048
SHEET 98 OF 140
E-503



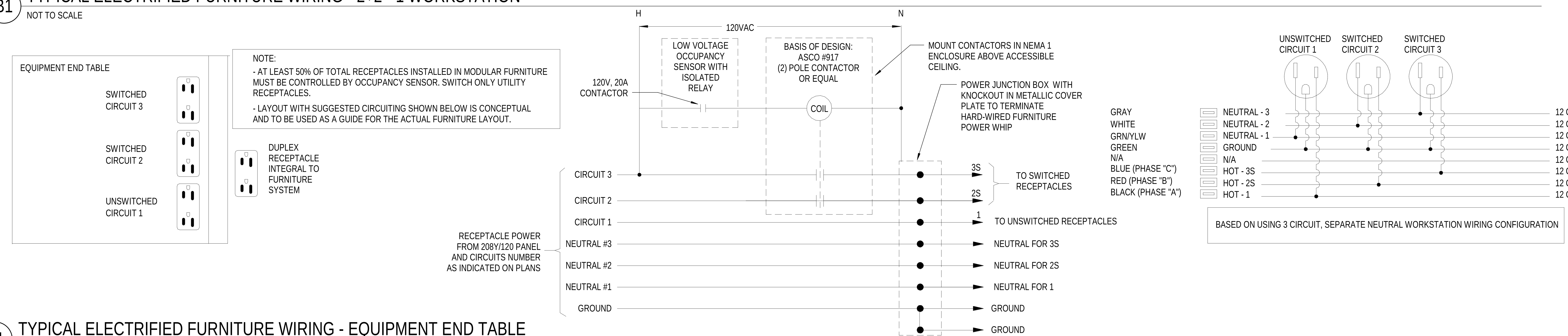
C1 TYPICAL ELECTRIFIED FURNITURE WIRING - 2+2 - 2 WORKSTATIONS

NOT TO SCALE



B1 TYPICAL ELECTRIFIED FURNITURE WIRING - 2+2 - 1 WORKSTATION

NOT TO SCALE



A1 TYPICAL ELECTRIFIED FURNITURE WIRING - EQUIPMENT END TABLE

NOT TO SCALE

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

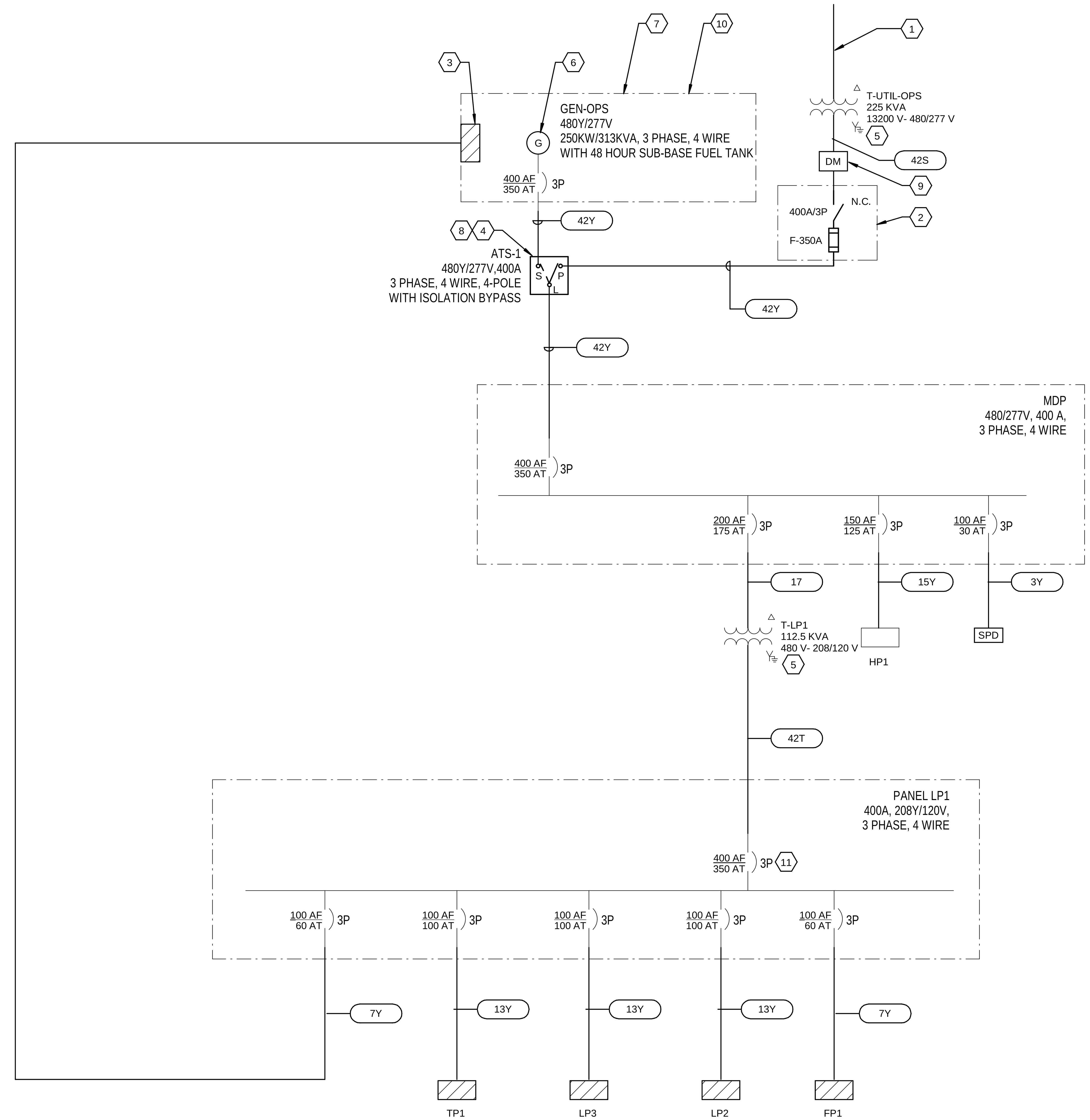
DES	JHP	DRW	JHP	CHK	KB
PM/DM	SMS/CWJ				
BRANCH MANAGER	WRB				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL DETAILS

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894049
SHEET 99 OF 140

E-504

6/20/2025 12:26:54 PM E-504
\\dc1vs01\fedprj\FDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt



A2 SINGLE LINE DIAGRAM - OPS BUILDING

NTS

GENERAL SHEET NOTES

A. REFER TO SHEETS E-001 THROUGH E-004 FOR ADDITIONAL ELECTRICAL INFORMATION.

B. REFER TO SHEET E-704 FOR FEEDER SCHEDULE.

C. THE STANDBY POWER SYSTEMS MUST MEET THE REQUIREMENTS OF NEC ARTICLE 702, OPTIONAL STANDBY.

SHEET KEYNOTES

1. MV FEEDERS TO UTILITY TRANSFORMER. REFER TO ELECTRICAL SITE PACKAGE FOR ADDITIONAL INFORMATION.
2. PROVIDE SERVICE ENTRANCE RATED FUSED SWITCH, RATING AS SHOWN.
3. PROVIDE 60A MCB, 208Y/120V, 3-PHASE, 4-WIRE PANELBOARD "GEN" INSIDE GENERATOR ENCLOSURE AND FEED FROM PANELBOARD LP1. REFER TO SHEET E-703 FOR ADDITIONAL INFORMATION REGARDING GENERATOR APPURTENANCE CIRCUITS. COORDINATE WITH GENERATOR MANUFACTURER FOR FINAL SIZE REQUIRED OF APPURTENANCES AND ADJUST CIRCUIT SIZE ACCORDINGLY.
4. PROVIDE CONTROL AND ANNUNCIATOR WIRING IN 1"C BETWEEN AUTOMATIC TRANSFER SWITCH AND GENERATOR FOR GENERATOR AUTO-ON SIGNAL.
5. PROVIDE #1/0 GEC..
6. BOND GENERATOR NEUTRAL TO GROUND BUS TO CREATE SEPARATELY DERIVED SYSTEM FOR USE WITH 4-POLE ATS. GENERATOR MUST BE SOLIDLY GROUNDED.
7. PROVIDE REMOTE EMERGENCY OFF BUTTON AT GENERATOR ENCLOSURE.
8. PROVIDE SPD AT ATS.
9. PROVIDE 600V, 400A, 3-PHASE C/T CABINET IN NEMA 1 ENCLOSURE AND EIG SHARK 270 DIGITAL METER. PROVIDE NECESSARY CT'S AND ACCESSORIES FOR PROPER FUNCTION.
10. REFER TO SHEET ES101 ON THE ELECTRICAL SITE PACKAGE FOR LOCATION OF GENERATOR.
11. PROVIDE 100% RATED CIRCUIT BREAKER.

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	JHP	DRW	JHP	CHK	KB
PM/DM	SMS/CWJ				
BRANCH MANAGER	WRB				
CHIEF ENG/ARCH	JWJ				
FIRE PROTECTION	HPN				

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL POWER SINGLE LINE DIAGRAM

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894050
SHEET 100 OF 140
E-601

PLATE	TAG	DESCRIPTION	LAMP TYPE	WATTS	LUMENS	MINIMUM EMERGENCY LUMENS	Lamp Color	Lamp Color Render Index	LUMINAIRE MOUNTING METHOD	VOLTAGE	DIMMING
NL-1	B2	RECESSED DIRECT/INDIRECT 2X2 FIXTURE, WHITE HOUSING	LED	31 W	3000 lm	0 lm	3500 K	80	RECESSED	277 V	0-10V
NL-2	B2A	RECESSED 2X2 FIXTURE WITH PRISMATIC LENS, WHITE HOUSING	LED	29 W	3300 lm	0 lm	3000 K	80	RECESSED	277 V	0-10V
NL-2	B2B	RECESSED 2X2 FIXTURE WITH PRISMATIC LENS, WHITE HOUSING	LED	18 W	2000 lm	0 lm	3000 K	80	RECESSED	277 V	0-10V
NL-1	B2E	RECESSED DIRECT/INDIRECT 2X2 FIXTURE WITH INTEGRAL BATTERY RATED FOR 90 MIN., WHITE HOUSING	LED	31 W	3000 lm	1400 lm	3500 K	80	RECESSED	277 V	0-10V
NL-1	C2	RECESSED DIRECT/INDIRECT 2X4 FIXTURE, WHITE HOUSING	LED	41 W	4000 lm	0 lm	3500 K	80	RECESSED	277 V	0-10V
NL-1	C2E	RECESSED DIRECT/INDIRECT 2X4 FIXTURE WITH INTEGRAL BATTERY RATED FOR 90 MIN., WHITE HOUSING	LED	41 W	4000 lm	1400 lm	3500 K	80	RECESSED	277 V	0-10V
NL-12	D6	6" RECESSED DOWNLIGHT, WET LOCATION RATED, WHITE HOUSING	LED	20 W	2000 lm	0 lm	3500 K	85	RECESSED	277 V	0-10V
NL-12	D6A	6" RECESSED DOWNLIGHT, WET LOCATION RATED, WHITE HOUSING	LED	10 W	1000 lm	0 lm	3500 K	85	RECESSED	277 V	0-10V
NL-12	D6AE	6" RECESSED DOWNLIGHT, WET LOCATION RATED, WITH INTEGRAL BATTERY RATED FOR 90 MIN., WHITE HOUSING	LED	10 W	1000 lm	1000 lm	3500 K	85	RECESSED	277 V	0-10V
NL-12	D6E	6" RECESSED DOWNLIGHT, WET LOCATION RATED, WITH INTEGRAL BATTERY RATED FOR 90 MIN., WHITE HOUSING	LED	20 W	2000 lm	1200 lm	3500 K	85	RECESSED	277 V	0-10V
NL-23	I1	4' INDUSTRIAL STRIP FIXTURE WITH WIREGUARD, WHITE HOUSING	LED	32 W	4000 lm	0 lm	3500 K	80	PENDANT	277 V	0-10V
NL-23	I1E	4' INDUSTRIAL STRIP FIXTURE WITH INTEGRAL BATTERY RATED FOR 90 MIN., AND WIREGUARD, WHITE HOUSING	LED	32 W	4000 lm	1500 lm	3500 K	80	PENDANT	277 V	0-10V
NL-23	I2E	4' INDUSTRIAL STRIP FIXTURE WITH INTEGRAL BATTERY RATED FOR 90 MIN., AND WIREGUARD, WHITE HOUSING	LED	44 W	7000 lm	1500 lm	3500 K	80	PENDANT	277 V	0-10V
NL-6	P8	8' PENDANT DIRECT/INDIRECT FIXTURE, WHITE HOUSING	LED	50 W	5000 lm	0 lm	3500 K	80	PENDANT	277 V	0-10V
NL-6	P8E	8' PENDANT DIRECT/INDIRECT FIXTURE WITH INTEGRAL BATTERY RATED FOR 90 MIN. WHITE HOUSING	LED	50 W	5000 lm	1500 lm	3500 K	80	PENDANT	277 V	0-10V
NL-4	R1	RECESSED 4" X 7'-6" LINEAR LED FIXTURE, WET LOCATION RATED, WHITE HOUSING	LED	30 W	2400 lm	0 lm	3500 K	80	RECESSED	277 V	0-10V
XL-10	W1E	WALL PACK LIGHTING FIXTURE WITH INTEGRAL BATTERY RATED FOR 90 MIN., WET RATED, AND MOUNTED AT 8'-0" AFF. DARK BRONZE HOUSING	LED	12 W	1500 lm	1000 lm	4000 K	70	SURFACE	277 V	0-10V
NL-28	X2	EXIT LIGHT - CEILING MOUNTED WITH INTEGRAL BATTERY RATED FOR 90 MIN. RED LETTERING	LED	3 W	270 lm	0 lm	0 K	0	SURFACE	277 V	NON-DIMMING
NL-28	X4	EXIT LIGHT - WALL MOUNTED WITH INTEGRAL BATTERY RATED FOR 90 MIN. RED LETTERING	LED	3 W	270 lm	0 lm	0 K	0	SURFACE	277 V	NON-DIMMING

GENERAL NOTES: LIGHTING FIXTURE SCHEDULE:

1. LIGHTING FIXTURES, INSTALLATION, SUPPORT, WIRING AND RELATED WORK MUST COMPLY WITH GENERAL NOTES, FIXTURE SCHEDULE, LAMP/DRIVER REQUIREMENT TABLES, FIXTURE DETAILS, SPECIFICATIONS AND CONTRACT DRAWINGS.
2. PROVIDE ACCESSORIES REQUIRED FOR A COMPLETE INSTALLATION AND COORDINATE WITH WORK BY OTHER TRADES. RECESSED FIXTURES MUST BE COMPLETELY SERVICEABLE FROM THE FIXTURE OPENING. PROVIDE REQUIRED MOUNTING HARDWARE, SUPPORTS, ACCESSORIES, AND TRIM REQUIRED FOR A COMPLETE INSTALLATION. COORDINATE FIXTURE MOUNTING WITH CEILING AND WALL CONSTRUCTION WHERE INSTALLED. PROVIDE SEPARATION BETWEEN RECESSED FIXTURES AND SURROUNDING INSULATION TO COMPLY WITH UL LISTING REQUIREMENTS AND MANUFACTURER INSTRUCTIONS.
3. FIXTURES MUST BE CBM APPROVED, LISTED AND LABELED BY ETL OR UL FOR DRY, DAMP, WET OR HAZARDOUS/CLASSIFIED LOCATIONS FOR THE INSTALLATION, UNLESS MORE STRINGENT REQUIREMENTS ARE INDICATED OR SPECIFIED.
4. DRIVERS MUST BE CBM APPROVED AND LISTED BY ETL OR UL. PROVIDE INTERNAL GREEN GROUNDING SCREW FOR FIXTURES.
5. WHERE EMERGENCY FIXTURES ARE UNSWITCHED, CONNECT EMERGENCY LIGHTING UNITS AND FIXTURES WITH EMERGENCY BATTERY PACKS TO PROVIDE ILLUMINATION AUTOMATICALLY UPON THE LOSS OF NORMAL POWER. WHERE EMERGENCY FIXTURES ARE SWITCHED, UPON THE LOSS OF NORMAL UNSWITCHED POWER, THE EMERGENCY BATTERY PACK MUST PROVIDE ILLUMINATION AUTOMATICALLY REGARDLESS OF SWITCH POSITION.
6. THE MANUFACTURERS AND MODELS LISTED ON THE LIGHTING FIXTURE SCHEDULE ARE LISTED TO INDICATE THE OVERALL LEVEL OF QUALITY REQUIRED, AND LISTING OF A MANUFACTURER OR MODEL DOES NOT AMEND OR SUPERCEDE LIGHTING FIXTURE REQUIREMENTS SPECIFIED OR INDICATE. LIGHTING FIXTURES MUST COMPLY WITH DRAWING AND SPECIFICATION REQUIREMENTS. REFER TO SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.
7. EMERGENCY LIGHT FIXTURES MUST BE DESIGNED SUCH THAT FAILURE OF (1) LED DOES NOT CAUSE ENTIRE FIXTURE TO NOT ILLUMINATE, PER NEC 700.16.

LIGHTING CONTROL MATRIX

SPACE TYPE	MANUAL ON	MANUAL OFF	MANUAL ADJUST	MANUAL OVERRIDE	TIMER ON	TIMER OFF	OCCUPANCY ON TO 100%	OCCUPANCY ON TO 50%	VACANCY OFF	VACANCY TO 50%	DAYLIGHT ADJUST	PHOTOCELL ON	PHOTOCELL OFF	SEQUENCE OF OPERATION	NOTES
BREAK ROOM	X	X	X						X					SOO A	COFFEE/MISC 106, LACTATION 118
CONFERENCE ROOM	X	X	X						X					SOO B	TRAINING/CONFERENCE 131
LIVING QUARTERS	X	X	X											SOO L	BUNKS 113, 115, 122, 125
CORRIDOR							X		X					SOO C	VESTIBULE 101, CORRIDOR 102, CORRIDOR 120
ELECTRICAL / MECHANICAL / TELECOM ROOM	X	X												SOO D	ELECTRICAL 103, MECHANICAL 104, TELECOM 105, ELEC 124
JANITOR CLOSET	X	X							X					SOO F	JANITOR 117
OPEN OFFICE	X	X	X					X	X					SOO G	SIPR OIS 107, OPERATION SUPPORT 126
LOCKER ROOMS	X	X	X					X	X					SOO E	LOCKER 114, 123
PRIVATE OFFICE	X	X	X						X					SOO H	OFFICE 108, 109, 127, 128, 129, 130
RESTROOM	X	X					X		X					SOO I	TOILETS/SHOWER 116, 121
STORAGE ROOM	X	X							X					SOO F	STORAGE 112
LAUNDRY	X	X					X		X					SOO K	WASHER/DRYER 119
EXTERIOR LIGHTING					X							X	X	SOO M	EXTERIOR BUILDING LIGHTING
ARMS/ARMORY ROOM	X	X	X						X					SOO J	ARMS ROOM 110

NOTES:
1) PROVIDE FIRE ALARM OVERRIDE OF OCCUPANCY/VACANCY SENSORS/SWITCHES PER FIRE ALARM DRAWINGS.
2) WHERE PLUG LOAD CONTROLS ARE USED FOR MODULAR FURNITURE, PROVIDE SCHEDULED/PROGRAMMED TIME OF DAY SHUT-OFF, WITH 2 HOUR OVERRIDE CAPABILITY BY OCCUPANT. CLEARLY DIFFERENTIATE MARKING BETWEEN CONTROLLED/UNCONTROLLED RECEPTACLES.

SOO A	LIGHTS CONTROLLED BY DIMMING SWITCH WITH MANUAL ON/OFF BUTTONS. OCCUPANCY SENSOR MUST TURN LIGHTS OFF WHEN SPACE IS VACANT MORE THAN 15 MINUTES.
SOO B	LIGHTS CONTROLLED BY LOCAL 3 BUTTON DIMMING SWITCH PROGRAMMED TO 0% / 50% / 100% LIGHT OUTPUT WITH RAISE LOWER OPTIONS. OCCUPANCY SENSOR MUST TURN LIGHTS OFF WHEN SPACE IS VACANT MORE THAN 15 MINUTES. LIGHTS MAY BE CONTROLLED BY AV EQUIPMENT IN SPACE. LIGHTS IN SPACE MUST BE CAPABLE OF DIMMING DOWN TO 10%.
SOO C	OCCUPANCY SENSORS MUST TURN LIGHTS ON TO 100% LIGHTING LEVEL UPON SENSING MOTION. OCCUPANCY SENSORS MUST TURN LIGHTS OFF WHEN SPACE IS VACANT MORE THAN 15 MINUTES.
SOO D	LINE VOLTAGE MANUAL ON/OFF CONTROLS ONLY.
SOO E	LIGHTS MUST TURN AUTOMATICALLY ON TO 50% LIGHT OUTPUT. IN ADDITION LIGHTS CAN BE CONTROLLED BY LOCAL ON/OFF SWITCH WITH RAISE LOWER OPTIONS. OCCUPANCY SENSOR MUST TURN LIGHTS OFF WHEN SPACE IS VACANT MORE THAN 15 MINUTES.
SOO F	LIGHTS MUST BE CONTROLLED BY MANUAL ON/OFF SWITCH. OCCUPANCY SENSORS WILL TURN OFF LIGHTS WHEN SPACE IS VACANT MORE THAN 10 MINUTES.
SOO G	LIGHTS MUST TURN AUTOMATICALLY ON TO 50% LIGHT OUTPUT. IN ADDITION LIGHTS CAN BE CONTROLLED BY LOCAL 3 BUTTON DIMMING SWITCH PROGRAMMED TO 0% / 50%/ 100% LIGHT OUTPUT WITH RAISE LOWER OPTIONS. OCCUPANCY SENSOR MUST TURN LIGHTS OFF WHEN SPACE IS VACANT MORE THAN 15 MINUTES.
SOO H	LIGHTS CONTROLLED BY LOCAL 3 BUTTON DIMMING SWITCH PROGRAMMED TO 0% / 50% / 100% LIGHT OUTPUT WITH RAISE LOWER OPTIONS. OCCUPANCY SENSOR MUST TURN LIGHTS OFF WHEN SPACE IS VACANT MORE THAN 15 MINUTES.
SOO I	LIGHTS WILL TURN ON TO 100% LIGHT OUTPUT BY OCCUPANCY SENSOR AND TURN OFF WHEN SPACE IS VACANT MORE THAN 15 MINUTES. LIGHTS MAY ALSO BE CONTROLLED BY MANUAL ON/OFF SWITCH.
SOO J	LIGHTS MUST BE CONTROLLED BY MANUAL ON/OFF SWITCH WITH RAISE/LOWER OPTIONS. OCCUPANCY SENSORS WILL TURN OFF LIGHTS WHEN SPACE IS VACANT MORE THAN 15 MINUTES.
SOO K	LIGHTS WILL TURN ON TO 100% LIGHT OUTPUT BY OCCUPANCY SENSOR AND TURN OFF WHEN SPACE IS VACANT MORE THAN 15 MINUTES. LIGHTS MAY ALSO BE CONTROLLED BY MANUAL ON/OFF SWITCH.
SOO L	LIGHTS CONTROLLED BY MANUAL ON/OFF SWITCH WITH RAISE/LOWER OPTIONS.
SOO M	LIGHTS MUST TURN ON AT DUSK. AUTOMATICALLY REDUCE LIGHTING POWER BY A MINIMUM OF 30% FROM 12 MIDNIGHT OR WITHIN ONE HOUR OF NORMAL CLOSING, WHICHEVER IS LATER, UNTIL 6AM OR NORMAL OPENING, WHICHEVER IS EARLIER. AUTOMATICALLY TURN OFF VIA PHOTOCELL WHEN SUFFICIENT DAYLIGHT IS AVAILABLE.

SYM | DESCRIPTION | DATE | APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. ..., RA
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES	JHP
DRW	JHP
CHK	KB
PM/DM	SMS/CWJ
BRANCH MANAGER	WRB
CHIEF ENG/ARCH	JWJ
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA

NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS

LIGHTING FIXTURE AND CONTROLS SCHEDULES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894051
SHEET 101 OF 140

E-701

6/20/2025 12:26:58 PM
\\dc1vs01\fwdpr\FDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt
E-701

1	2	3	4	5

SYM	DESCRIPTION	DATE	APPR

PANELBOARD: MDP

LOCATION: ELECTRICAL 103 | VOLTAGE: 480/277V,3P,4W,60Hz | INTERRUPTING RATING (kA SYMMETRICAL): 14 kA
SUPPY FROM: | BUS AMPS: 400 A | GROUND BUS: EQUIP. ONLY
MOUNTING: Surface | MCB FRAME / MCB TRIP: 400 A / 350 A | NEUTRAL BUS RATING: 100%
ENCLOSURE: Type 1 | ACCESSORY LUGS: NONE

NOTES	LOAD CLASS	DESCRIPTIONS	FRAME	TRIP AMPS	POLES	CKT NO.	A	B	C	A	B	C	CKT NO.	POLES	TRIP AMPS	FRAME	DESCRIPTIONS	LOAD CLASS	NOTES
	Spare; LTG; EQN	PANEL HP1	150 A	125 A	3	1	22.659			44.72			2	3	175 A	200 A	PANELBOARD LP1 VIA 112.5KVA XFMR	Spare; LTG; REC;...	
						3		22.73			41.753		4						
						5			20.761			38.606	6						
	REC	SPD	100 A	30 A	3	7	0			0			8	3	100 A	100 A	SPARE	--	
						9		0			0		10						
						11			0			0	12						
	--	SPACE	--	--	3	13	--			--			14	3	--	--	SPACE	--	
						15		--			--		16						
						17			--			--	18						
						19							20						
						21							22						
						23							24						
						25							26						
						27							28						
						29							30						
						31							32						
						33							34						
						35							36						
						37							38						
						39							40						
						41							42						
		TOTAL CONNECTED LOAD / PHASE:					67.378 kVA		64.482 kVA		59.366 kVA								

LOAD CLASSIFICATIONS	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTAL LOAD SUMMARY			
				LOAD (kVA)		CURRENT (AMPS)	
LTG	5.467 kVA	100.00%	5.467 kVA	CONNECTED LOAD:	191.227 kVA	CONNECTED CURRENT:	230.0 A
REC	64.73 kVA	57.72%	37.365 kVA	ESTIMATED DEMAND LOAD:	163.862 kVA	ESTIMATED DEMAND CURRENT:	197.1 A
EQC	13 kVA	100.00%	13 kVA	ESTIMATED DEMAND FACTOR:	85.69%		
EQN	108.03 kVA	100.00%	108.03 kVA	SPARE CAPACITY:	15%		
				ESTIMATED LOAD WITH SPARE:	188.441 kVA	ESTIMATED DEMAND CURRENT W/ SPARE:	227 A

PANELBOARD NOTES: UNLESS OTHERWISE NOTED, PROVIDE 2#12 + 1#12G IN 3/4"C FOR 20A/1P CIRCUITS

PANELBOARD: HP1

LOCATION: ELECTRICAL 103 | VOLTAGE: 480/277V,3P,4W,60Hz | INTERRUPTING RATING (kA SYMMETRICAL): 14 kA
SUPPY FROM: MDP | BUS AMPS: 150 | GROUND BUS: EQUIP. ONLY
MOUNTING: Surface | MCB FRAME / MCB TRIP: 150 A / 125 A | NEUTRAL BUS RATING: 100%
ENCLOSURE: Type 1 | ACCESSORY LUGS: NONE

NOTES	LOAD CLASS	DESCRIPTIONS	FRAME	TRIP AMPS	POLES	CKT NO.	A	B	C	A	B	C	CKT NO.	POLES	TRIP AMPS	FRAME	DESCRIPTIONS	LOAD CLASS	NOTES
	EQN	DHWH-1	100 A	40 A	3	1	8			8.713			2	3	40 A	100 A	DOAS-1	EQN	
						3		8			8.713		4						
						5			8			8.713	6						
	EQN	CU-10	100 A	20 A	3	7	3.547			1.766			8	1	20 A	100 A	LIGHTING - SOUTH HALF	LTG	
						9		3.547			2.469		10	1	20 A	100 A	LIGHTING - NORTH HALF	LTG	
						11			3.547			0.5	12	1	20 A	100 A	SITE LIGHTING (OPS BLDG AREA)	LTG	
	LTG	SITE LIGHTING (OPS BLDG AREA)	100 A	20 A	1	13	0.5			0.132			14	1	20 A	100 A	EXTERIOR BUILDING LIGHTING	LTG	
	--	SPARE	100 A	20 A	1	15		0			0		16	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	17			0			0	18	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	19	0			0			20	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	21		0			0		22	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	23			0			0	24	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	25	0			0			26	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	27		0			0		28	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	29			0			0	30	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	31	0			0			32	1	20 A	100 A	SPARE	--	
	--	SPACE	--	--	1	33		--			--		34	1	--	--	SPACE	--	
	--	SPACE	--	--	1	35			--			--	36	1	--	--	SPACE	--	
	--	SPACE	--	--	1	37	--			--			38	1	--	--	SPACE	--	
	--	SPACE	--	--	1	39		--			--		40	1	--	--	SPACE	--	
	--	SPACE	--	--	1	41			--			--	42	1	--	--	SPACE	--	
		TOTAL CONNECTED LOAD / PHASE:					22.659 kVA		22.73 kVA		20.761 kVA								

LOAD CLASSIFICATIONS	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTAL LOAD SUMMARY			
				LOAD (kVA)		CURRENT (AMPS)	
LTG	5.367 kVA	100.00%	5.367 kVA	CONNECTED LOAD:	66.149 kVA	CONNECTED CURRENT:	79.6 A
EQN	60.782 kVA	100.00%	60.782 kVA	ESTIMATED DEMAND LOAD:	66.149 kVA	ESTIMATED DEMAND CURRENT:	79.6 A
				ESTIMATED DEMAND FACTOR:	100.00%		
				SPARE CAPACITY:	15%		
				ESTIMATED LOAD WITH SPARE:	76.071 kVA	ESTIMATED DEMAND CURRENT W/ SPARE:	91 A

PANELBOARD NOTES: UNLESS OTHERWISE NOTED, PROVIDE 2#12 + 1#12G IN 3/4"C FOR 20A/1P CIRCUITS

PANELBOARD: LP1

LOCATION: ELECTRICAL 103 | VOLTAGE: 208/120V,3P,4W,60Hz | INTERRUPTING RATING (kA SYMMETRICAL): 14 kA
SUPPY FROM: T-LP1 | BUS AMPS: 400 A | GROUND BUS: EQUIP. ONLY
MOUNTING: Surface | MCB FRAME / MCB TRIP: 400 A / 350 A | NEUTRAL BUS RATING: 100%
ENCLOSURE: Type 1 | ACCESSORY LUGS: NONE

NOTES	LOAD CLASS	DESCRIPTIONS	FRAME	TRIP AMPS	POLES	CKT NO.	A	B	C	A	B	C	CKT NO.	POLES	TRIP AMPS	FRAME	DESCRIPTIONS	LOAD CLASS	NOTES
	REC	REC ELEC 103	100 A	20 A	1	1	0.54			0.36			2	1	20 A	100 A	REC MECH 104	REC	
	REC	REC EWC CORRIDOR 102	100 A	20 A	1	3		0.18			0.72		4	1	20 A	100 A	REC EXTERIOR BLDG	REC	
	REC	REC WORKSTATION OFFICE 127	100 A	20 A	1	5			0.72			0.72	6	1	20 A	100 A	REC WORKSTATION OFFICE 128	REC	
	REC	REC OFFICE 127, 128	100 A	20 A	1	7	1.08			0.8			8	1	20 A	100 A	REC REFRIGERATOR RM 108	REC	
	REC	REC WATER/ICE DISP. RM 108	100 A	20 A	1	9		0.8			1		10	1	20 A	100 A	REC COFFEE MACHINE RM 108	REC	
	REC	REC COFFEE/MISC 106	100 A	20 A	1	11			0.54			1	12	1	20 A	100 A	REC MICROWAVE RM 106	REC	
	REC	REC MICROWAVE RM 106	100 A	20 A	1	13	1			0.72			14	1	20 A	100 A	REC WORKSTATION OFFICE 129	REC	
	REC	REC TRAINING/CONF. 131	100 A	20 A	1	15		0.9			0.9		16	1	20 A	100 A	REC TRAINING/CONF. 131	REC	
	REC	REC COPIER RM 126	100 A	20 A	1	17			1			0.5	18	1	20 A	100 A	REC AV RACK TRAINING/CONF....	REC	
	REC	REC AV MONITORS RM 131	100 A	20 A	1	19	0.72			0.36			20	1	20 A	100 A	REC TRAINING/CONF 131	REC	
	REC	REC RMS 129, 130	100 A	20 A	1	21		0.36			0.72		22	1	20 A	100 A	REC RM 101, 102, 112	REC	
	REC	REC RM 126	100 A	20 A	1	23			0.72			0.8	24	1	20 A	100 A	REC SHREDDER RM 126	REC	
	EQN	CU-3 / AC-3	100 A	30 A	2	25	1.997			1			26	1	20 A	100 A	POWERPOLE RM 126	REC	
						27		1.997			1		28	1	20 A	100 A	POWERPOLE RM 126	REC	
	REC	POWERPOLE RM 126	100 A	20 A	1	29			1			1	30	1	20 A	100 A	SYSTEMS FURN RM 107	REC	
	REC	POWERPOLE RM 126	100 A	20 A	1	31	1			1.997			32	2	30 A	100 A	CU-5 / AC-5	EQN	
	EQN	UH-1	100 A	30 A	2	33		2.496			1.997		34						
						35			2.496			1.997	36	2	30 A	100 A	CU-4 / AC-4	EQN	
2	Spare; EQN	PANELBOARD FP1	100 A	60 A	3	37	1			1.997			38						
						39		0			1.581		40	2	25 A	100 A	CU-1 / AC-1	EQN	
						41			0			1.581	42						
	Spare; LTG; REC;...	PANELBOARD GEN	100 A	60 A	3	43	3.6			10.885			44	3	100 A	100 A	PANELBOARD LP2	Spare; REC; EQN	
						45		2.6			8.53		46						
						47			0.28			8.184	48						
	Spare; REC; EQC;...	PANELBOARD TP1	100 A	100 A	3	49	6.4			9.265			50	3	100 A	100 A	PANELBOARD LP3	Spare; REC; EQN	
						51		6.081			9.892		52						
						53			5.081			10.987	54						
		TOTAL CONNECTED LOAD / PHASE:					44.72 kVA		41.753 kVA		38.606 kVA								

LOAD CLASSIFICATIONS	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTAL LOAD SUMMARY			
				LOAD (kVA)		CURRENT (AMPS)	
LTG	0.1 kVA	100.00%	0.1 kVA	CONNECTED LOAD:	125.078 kVA	CONNECTED CURRENT:	347.2 A
REC	64.73 kVA	57.72%	37.365 kVA	ESTIMATED DEMAND LOAD:	97.713 kVA	ESTIMATED DEMAND CURRENT:	271.2 A
EQC	13 kVA	100.00%	13 kVA	ESTIMATED DEMAND FACTOR:	78.12%		
EQN	47.248 kVA	100.00%	47.248 kVA	SPARE CAPACITY:	15%		
				ESTIMATED LOAD WITH SPARE:	112.37 kVA	ESTIMATED DEMAND CURRENT W/ SPARE:	312 A

PANELBOARD NOTES: 1) UNLESS OTHERWISE NOTED, PROVIDE 2#12 + 1#12G IN 3/4"C FOR 20A/1P CIRCUITS 2) PROVIDE RED "LOCK-ON" CIRCUIT BREAKER.

PANELBOARD: LP2

LOCATION: ELECTRICAL 103 | VOLTAGE: 208/120V,3P,4W,60Hz | INTERRUPTING RATING (kA SYMMETRICAL): 14 kA
SUPPY FROM: LP1 | BUS AMPS: 100 A | GROUND BUS: EQUIP. ONLY
MOUNTING: Surface | MCB FRAME / MCB TRIP: 100 A / 100 A | NEUTRAL BUS RATING: 100%
ENCLOSURE: Type 1 | ACCESSORY LUGS: NONE

NOTES	LOAD CLASS	DESCRIPTIONS	FRAME	TRIP AMPS	POLES	CKT NO.	A	B	C	A	B	C	CKT NO.	POLES	TRIP AMPS	FRAME	DESCRIPTIONS	LOAD CLASS	NOTES
	REC	REC OFFICE 108, 109	100 A	20 A	1	1	0.9			0.72			2	1	20 A	100 A	REC WORKSTATION RM 108	REC	
	REC	SECURITY PANEL, RM 107	100 A	20 A	1	3		0.5			0.12		4	1	20 A	100 A	RF-3	EQN	
	EQN	DHWH-2	100 A	20 A	2	5			0.75			0.1	6	1	20 A	100 A	TRAP PRIMER RM 108	EQN	
						7	0.75			0.5			8	1	20 A	100 A	REC FAX MACHINE RM 107	REC	
	REC	REC PRINTER RM 107	100 A	20 A	1	9		0.8			0.5		10	1	20 A	100 A	REC SCANNER RM 107	REC	
	REC	REC RADIO CHARGER RM 107	100 A	20 A	1	11			0.5			0.5	12	1	20 A	100 A	REC FAX MACHINE RM 107	REC	
	REC	REC PRINTER RM 107	100 A	20 A	1	13	0.8			0.36			14	1	20 A	100 A	REC SIPR OIS 107	REC	
	REC	REC WORKSTATIONS RM 109	100 A	20 A	1	15		0.72			1		16	1	20 A	100 A	POWERPOLE RM 126	REC	
	REC	REC SHREDDER RM 107	100 A	20 A	1	17			0.8			1	18	1	20 A	100 A	POWERPOLE RM 126	REC	
	EQN	CU-6 / AC-6	100 A	20 A	2	19	0.915			0.915			20	2	20 A	100 A	CU-7 / AC-7	EQN	
						21		0.915			0.915		22						
	EQN	UH-2	100 A	20 A	2	23			0.998			0.54	24	1	20 A	100 A	FLOOR RECEPTS RM 131	EQN	
						25	0.998			0.54			26	1	20 A	100 A	FLOOR RECEPTS RM 131	EQN	
	EQN	MECH CONTROL PANEL, RM 104	100 A	20 A	1	27		0.2			0.36		28	1	20 A	100 A	REC MECH RM 104	REC	
	REC	REC TV MONITOR, RM 129, 130	100 A	20 A	1	29			0.72			0.416	30	2	20 A	100 A	F-1	EQN	
	REC	REC RM 131	100 A	20 A	1	31	0.9			0.416			32						
	EQN	LIGHTING CONTROLS, RM 103	100 A	20 A	1	33		0.5			0		34	1	20 A	100 A	SPARE	--	
	EQN	CDP-1, CDP-2	100 A	20 A	1	35			0.36			0	36	1	20 A	100 A	SPARE	--	
	REC	REC REFRIGERATOR RM 112	100 A	20 A	1	37	0.65			0			38	1	20 A	100 A	SPARE	--	
	REC	POWERPOLE RM 126	100 A	20 A	1	39		1			0		40	1	20 A	100 A	SPARE	--	
	REC	POWERPOLE RM 126	100 A	20 A	1	41			1			0	42	1	20 A	100 A	SPARE	--	
	REC	REC IDS MONITORING 130	100 A	20 A	1	43	0.72			0			44	1	20 A	100 A	SPARE	--	
	REC	POWERPOLE RM 126 FURN.	100 A	20 A	1	45		0.5			0		46	1	20 A	100 A	SPARE	--	
	REC	POWERPOLE RM 126 FAX	100 A	20 A	1	47			0.5			0	48	1	20 A	100 A	SPARE	--	
	REC	POWERPOLE RM 126 PRINTER	100 A	20 A	1	49	0.8			0			50	1	20 A	100 A	SPARE	--	
	REC	POWERPOLE RM 126 RADIO	100 A	20 A	1	51		0.5			0		52	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	53			0			0	54	1	20 A	100 A	SPARE	--	
		TOTAL CONNECTED LOAD / PHASE:					10.885 kVA		8.53 kVA		8.184 kVA								

LOAD CLASSIFICATIONS	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTAL LOAD SUMMARY			
				LOAD (kVA)		CURRENT (AMPS)	
REC	17.25 kVA	78.99%	13.625 kVA	CONNECTED LOAD:	27.599 kVA	CONNECTED CURRENT:	76.6 A
EQN	10.349 kVA	100.00%	10.349 kVA	ESTIMATED DEMAND LOAD:	23.974 kVA	ESTIMATED DEMAND CURRENT:	66.5 A
				ESTIMATED DEMAND FACTOR:	86.87%		
				SPARE CAPACITY:	15%		
				ESTIMATED LOAD WITH SPARE:	27.57 kVA	ESTIMATED DEMAND CURRENT W/ SPARE:	77 A

PANELBOARD NOTES: 1) UNLESS OTHERWISE NOTED, PROVIDE 2#12 + 1#12G IN 3/4"C FOR 20A/1P CIRCUITS.

GENERAL NOTE:

A. PANEL SCHEDULE TOTAL CONNECTED LOAD / PHASE VALUES ARE SUMMATIONS FOR PHASES A, B, AND C FROM LEFT TO RIGHT. (TYPICAL FOR ALL PANEL SCHEDULES SHOWN ON THIS SHEET).

MDP	HP1
LP1	LP2

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
James E. Donahue, RA
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025

DES JHP	DRW JHP	CHK KB
PM/DM		SMS/CWJ
BRANCH MANAGER		WRB
CHIEF ENG/ARCH		JWJ
FIRE PROTECTION		HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL PANEL SCHEDULES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894052
SHEET 102 OF 140

E-702

E-702
6/20/2025 12:27:00 PM
\\dc1vs01\wdp\jFDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt

PANELBOARD: TP1

LOCATION: TELECOM 105
SUPPY FROM: LP1
MOUNTING: Surface
ENCLOSURE: Type 1

VOLTAGE: 208/120V,3P,4W,60Hz
BUS AMPS: 100 A
MCB FRAME / MCB TRIP: 100 A / 100 A

INTERRUPTING RATING (kA SYMMETRICAL): 14 kA
GROUND BUS: EQUIP. ONLY
NEUTRAL BUS RATING: 100%
ACCESSORY LUGS: NONE

NOTES	LOAD CLASS	DESCRIPTIONS	FRAME	TRIP AMPS	POLES	CKT NO.	A	B	C	A	B	C	CKT NO.	POLES	TRIP AMPS	FRAME	DESCRIPTIONS	LOAD CLASS	NOTES
	REC	REC TELECOM 105	100 A	20 A	1	1	0.54			0.36			2	1	20 A	100 A	REC TELECOM 105	REC	
	EQC	RACK L5-20R TELECOM 105	100 A	20 A	1	3		1			1		4	1	20 A	100 A	RACK L5-20R TELECOM 105	EQC	
	EQC	RACK L6-30R TELECOM 105	100 A	30 A	2	5			1.5			1.5	6	2	30 A	100 A	SIPR CABINET RM 107	EQC	
						7	1.5			1.5			8						
	EQN	CU-2 / AC-2	100 A	25 A	2	9		1.581			0		10	1	20 A	100 A	SPARE	--	
						11			1.581			--	12	1	--	--	SPACE	--	
	EQC	RACK L6-30R TELECOM 105	100 A	30 A	2	13	1.5			--			14	1	--	--	SPACE	--	
						15		1.5			--		16	1	--	--	SPACE	--	
	REC	SECURITY PANEL	100 A	20 A	1	17			0.5			--	18	1	--	--	SPACE	--	
	EQC	RACK L5-20R TELECOM 105	100 A	20 A	1	19	1			--			20	1	--	--	SPACE	--	
	EQC	RACK L5-20R TELECOM 105	100 A	20 A	1	21		1			--		22	1	--	--	SPACE	--	
	--	SPARE	100 A	20 A	1	23			0			--	24	1	--	--	SPACE	--	
	--	SPARE	100 A	20 A	1	25	0			--			26	1	--	--	SPACE	--	
	--	SPARE	100 A	20 A	1	27		0			--		28	1	--	--	SPACE	--	
	--	SPARE	100 A	20 A	1	29			0			--	30	1	--	--	SPACE	--	

TOTAL CONNECTED LOAD / PHASE: 6.4 kVA | 6.081 kVA | 5.081 kVA

LOAD CLASSIFICATIONS	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTAL LOAD SUMMARY			
				LOAD (kVA)		CURRENT (AMPS)	
REC	1.4 kVA	100.00%	1.4 kVA	CONNECTED LOAD:	17.562 kVA	CONNECTED CURRENT:	48.7 A
EQC	13 kVA	100.00%	13 kVA	ESTIMATED DEMAND LOAD:	17.562 kVA	ESTIMATED DEMAND CURRENT:	48.7 A
EQN	3.162 kVA	100.00%	3.162 kVA	ESTIMATED DEMAND FACTOR:	100.00%		
				SPARE CAPACITY:	15%		
				ESTIMATED LOAD WITH SPARE:	20.196 kVA	ESTIMATED DEMAND CURRENT W/ SPARE:	56 A

PANELBOARD NOTES: UNLESS OTHERWISE NOTED, PROVIDE 2#12 + 1#12G IN 3/4"C FOR 20A/1P CIRCUITS

PANELBOARD: GEN

LOCATION:
SUPPY FROM: LP1
MOUNTING: Surface
ENCLOSURE: Type 1

VOLTAGE: 208/120V,3P,4W,60Hz
BUS AMPS: 100 A
MCB FRAME / MCB TRIP: 100 A / 60 A

INTERRUPTING RATING (kA SYMMETRICAL): 14 kA
GROUND BUS: EQUIP. ONLY
NEUTRAL BUS RATING: 100%
ACCESSORY LUGS: NONE

NOTES	LOAD CLASS	DESCRIPTIONS	FRAME	TRIP AMPS	POLES	CKT NO.	A	B	C	A	B	C	CKT NO.	POLES	TRIP AMPS	FRAME	DESCRIPTIONS	LOAD CLASS	NOTES
	EQN	GENERATOR JACKET HEATER	100 A	30 A	2	1	2.1			1.5			2	1	20 A	100 A	GENERATOR BATTERY CHARGER	EQN	
						3		2.1			0.5		4	1	20 A	100 A	GENERATOR LUBE OIL HEATER	EQN	
	REC	REC GENERATOR ENCLOSURE	100 A	20 A	1	5			0.18			0.1	6	1	20 A	100 A	GEN. ENCLOSURE LIGHTING	LTG	
	--	SPARE	100 A	20 A	1	7	0			0			8	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	9		0			0		10	1	20 A	100 A	SPARE	--	
	--	SPACE	--	--	1	11			--			--	12	1	--	--	SPACE	--	
	--	SPACE	--	--	1	13	--			--			14	1	--	--	SPACE	--	
	--	SPACE	--	--	1	15		--			--		16	1	--	--	SPACE	--	
	--	SPACE	--	--	1	17			--			--	18	1	--	--	SPACE	--	

TOTAL CONNECTED LOAD / PHASE: 3.6 kVA | 2.6 kVA | 0.28 kVA

LOAD CLASSIFICATIONS	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTAL LOAD SUMMARY			
LTG	0.1 kVA	100.00%	0.1 kVA	LOAD (kVA)		CURRENT (AMPS)	
REC	0.18 kVA	100.00%	0.18 kVA	CONNECTED LOAD:	6.480 kVA	CONNECTED CURRENT:	18.0 A
EQN	6.2 kVA	100.00%	6.2 kVA	ESTIMATED DEMAND LOAD:	6.480 kVA	ESTIMATED DEMAND CURRENT:	18.0 A
				ESTIMATED DEMAND FACTOR:	100.00%		
				SPARE CAPACITY:	15%		
				ESTIMATED LOAD WITH SPARE:	7.452 kVA	ESTIMATED DEMAND CURRENT W/ SPARE:	21 A

PANELBOARD NOTES: UNLESS OTHERWISE NOTED, PROVIDE 2#12 + 1#12G IN 3/4"C FOR 20A/1P CIRCUITS

PANELBOARD: LP3

LOCATION: ELEC 124
SUPPLY FROM: LP1
MOUNTING: Surface
ENCLOSURE: Type 1

VOLTAGE: 208/120V,3P,4W,60Hz
BUS AMPS: 100 A
MCB FRAME / MCB TRIP: 100 A / 100 A

INTERRUPTING RATING (kA SYMMETRICAL): 14 kA
GROUND BUS: EQUIP. ONLY
NEUTRAL BUS RATING: 100%
ACCESSORY LUGS: NONE

NOTES	LOAD CLASS	DESCRIPTIONS	FRAME	TRIP AMPS	POLES	CKT NO.	A	B	C	A	B	C	CKT NO.	POLES	TRIP AMPS	FRAME	DESCRIPTIONS	LOAD CLASS	NOTES
	REC	REC E6 BUNKS 125	100 A	20 A	1	1	0.54			0.8			2	1	20 A	100 A	REC SHREDDER RM 126	REC	
	REC	REC COPIER RM 126	100 A	20 A	1	3		1			0.54		4	1	20 A	100 A	REC RM 122, 123	REC	
	REC	REC TOILETS/SHOWER 121	100 A	20 A	1	5			0.54			0.54	6	1	20 A	100 A	REC TOILETS/SHOWER 116	REC	
	REC	REC RM 114, 115	100 A	20 A	1	7	0.54			0.54			8	1	20 A	100 A	REC E6 BUNKS 113	REC	
	REC	REC BLDG EXTERIOR	100 A	20 A	1	9		0.54			0.5		10	1	20 A	100 A	REC UNDERCOUNT. FRIDGE 118	REC	
	REC	REC 14-30R DRYER RM 119	100 A	30 A	2	11			2.15			1.32	12	1	20 A	100 A	REC WASHING MACH. RM 119	REC	
						13	2.15			0.18			14	1	20 A	100 A	REC EWC RM 126	REC	
	EQN	DCP-1 RM 117	100 A	20 A	1	15		0.24			0.72		16	1	20 A	100 A	REC RM 110, 111	REC	
	REC	REC WORKSTATION RM 111	100 A	20 A	1	17			0.36			0.54	18	1	20 A	100 A	REC RM 112, 117, 126	REC	
	EQN	CU-8 / AC-8	100 A	20 A	2	19	0.915			0.36			20	1	20 A	100 A	CORD REEL RM 111	REC	
						21		0.915			0.54		22	1	20 A	100 A	REC RM 124, 126	REC	
	REC	REC RM 118, 119	100 A	20 A	1	23			0.54			1	24	1	20 A	100 A	POWERPOLE RM 126	REC	
	REC	POWERPOLE RM 126	100 A	20 A	1	25	1			1			26	1	20 A	100 A	POWERPOLE RM 126	REC	
	REC	POWERPOLE RM 126	100 A	20 A	1	27		1			1		28	1	20 A	100 A	POWERPOLE RM 126	REC	
	REC	POWERPOLE RM 126	100 A	20 A	1	29			1			0	30	1	20 A	100 A	SPARE	--	
	EQN	RF-1	100 A	20 A	1	31	0.12			0.12			32	1	20 A	100 A	RF-2	EQN	
	EQN	TRAP PRIMER RM 117	100 A	20 A	1	33		0.1			1.997		34	2	30 A	100 A	CU-9 / AC-9	EQN	
	REC	POWERPOLE RM 126 FURN.	100 A	20 A	1	35			0.5			1.997	36						
	REC	POWERPOLE RM 126 FAX	100 A	20 A	1	37	0.5			0.5			38	1	20 A	100 A	SECURITY PANEL, RM 111	REC	
	REC	POWERPOLE RM 126 PRINTER	100 A	20 A	1	39		0.8			0		40	1	20 A	100 A	SPARE	--	
	REC	POWERPOLE RM 126 RADIO	100 A	20 A	1	41			0.5			0	42	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	43	0			0			44	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	45		0			0		46	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	47			0			0	48	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	49	0			0			50	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	51		0			0		52	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	53			0			0	54	1	20 A	100 A	SPARE	--	

TOTAL CONNECTED LOAD / PHASE: 9.265 kVA | 9.892 kVA | 10.987 kVA

LOAD CLASSIFICATIONS	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTAL LOAD SUMMARY			
REC	23.74 kVA	71.06%	16.87 kVA	LOAD (kVA)		CURRENT (AMPS)	
EQN	6.403 kVA	100.00%	6.403 kVA	CONNECTED LOAD:	30.143 kVA	CONNECTED CURRENT:	83.7 A
				ESTIMATED DEMAND LOAD:	23.273 kVA	ESTIMATED DEMAND CURRENT:	64.6 A
				ESTIMATED DEMAND FACTOR:	77.21%		
				SPARE CAPACITY:	15%		
				ESTIMATED LOAD WITH SPARE:	26.764 kVA	ESTIMATED DEMAND CURRENT W/ SPARE:	74 A

PANELBOARD NOTES: UNLESS OTHERWISE NOTED, PROVIDE 2#12 + 1#12G IN 3/4"C FOR 20A/1P CIRCUITS

PANELBOARD: FP1

LOCATION: ELECTRICAL 103
SUPPY FROM: LP1
MOUNTING: Surface
ENCLOSURE: Type 1

VOLTAGE: 208/120V,3P,4W,60Hz
BUS AMPS: 100 A
MCB FRAME / MCB TRIP: 100 A / 60 A

INTERRUPTING RATING (kA SYMMETRICAL): 14 kA
GROUND BUS: EQUIP. ONLY
NEUTRAL BUS RATING: 100%
ACCESSORY LUGS: NONE

NOTES	LOAD CLASS	DESCRIPTIONS	FRAME	TRIP AMPS	POLES	CKT NO.	A	B	C	A	B	C	CKT NO.	POLES	TRIP AMPS	FRAME	DESCRIPTIONS	LOAD CLASS	NOTES
	EQN	FIRE ALARM EQ. RM 103	100 A	20 A	1	1	0.5			0.5			2	1	20 A	100 A	RADIO TRANS. RM 103	EQN	
	--	SPARE	100 A	20 A	1	3		0			0		4	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	5			0			0	6	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	7	0			0			8	1	20 A	100 A	SPARE	--	
	--	SPARE	100 A	20 A	1	9		0			0		10	1	20 A	100 A	SPARE	--	
	--	SPACE	--	--	1	11			--			--	12	1	--	--	SPACE	--	
	--	SPACE	--	--	1	13	--			--			14	1	--	--	SPACE	--	
	--	SPACE	--	--	1	15		--			--		16	1	--	--	SPACE	--	
	--	SPACE	--	--	1	17			--			--	18	1	--	--	SPACE	--	

TOTAL CONNECTED LOAD / PHASE: 1 kVA | 0 kVA | 0 kVA

LOAD CLASSIFICATIONS	CONNECTED LOAD	DEMAND FACTOR	ESTIMATED DEMAND	TOTAL LOAD SUMMARY			
EQN	1 kVA	100.00%	1 kVA	LOAD (kVA)		CURRENT (AMPS)	
				CONNECTED LOAD:	1.000 kVA	CONNECTED CURRENT:	2.8 A
				ESTIMATED DEMAND LOAD:	1.000 kVA	ESTIMATED DEMAND CURRENT:	2.8 A
				ESTIMATED DEMAND FACTOR:	100.00%		
				SPARE CAPACITY:	15%		
				ESTIMATED LOAD WITH SPARE:	1.15 kVA	ESTIMATED DEMAND CURRENT W/ SPARE:	3 A

PANELBOARD NOTES: 1) UNLESS OTHERWISE NOTED, PROVIDE 2#12 + 1#12G IN 3/4"C FOR 20A/1P CIRCUITS. 2) PROVIDE LABEL INDICATING "FIRE PROTECTION/LIFE SAFETY EQUIPMENT"

GENERAL NOTE:

A. PANEL SCHEDULE TOTAL CONNECTED LOAD / PHASE VALUES ARE SUMMATIONS FOR PHASES A, B, AND C FROM LEFT TO RIGHT. (TYPICAL FOR ALL PANEL SCHEDULES SHOWN ON THIS SHEET.)

TP1	LP3
GEN	FP1

SYM	DESCRIPTION	DATE	APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600
A/E INFO

APPROVED
James E. D[illegible], RA
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES JHP DRW JHP CHK KB
PM/DM SMS/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON
GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL PANEL SCHEDULES

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894053
SHEET 103 OF 140
E-703

6/20/2025 12:27:03 PM
\\dc1vs01\fedpr\FDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt
E-703

FEEDER SCHEDULE

TAG	CONDUCTORS AND CONDUIT	MAX CURRENT
2	(3) #12 + #12G IN 3/4"C	20 A
2K	(3) #12 + #8N + #12G IN 3/4"C	20 A
2Y	(3) #12 + #12N + #12G IN 3/4"C	20 A
3	(3) #10 + #10G IN 3/4"C	30 A
3K	(3) #10 + #4N + #10G IN 1"C	30 A
3Y	(3) #10 + #10N + #10G IN 3/4"C	30 A
4	(3) #8 + #10G IN 3/4"C	40 A
4K	(3) #8 + #1N + 10G IN 1"C	40 A
4Y	(3) #8 + #8N + #10G IN 3/4"C	40 A
5	(3) #6 + #10G IN 3/4"C	55 A
5K	(3) #6 + #1N + #10G IN 1 1/4"C	55 A
5Y	(3) #6 + #6N + #10G IN 1"C	55 A
7	(3) #4 + #8G IN 1"C	70 A
7K	(3) #4 + #1/0N + #8G IN 1 1/4"C	70 A
7Y	(3) #4 + #4N + #8G IN 1 1/4"C	70 A
9	(3) #2 + #8G IN 1 1/4"C	95 A
9K	(3) #2 + #3/0N + #8G IN 1 1/2"C	95 A
9Y	(3) #2 + #2N + #8G IN 1 1/4"C	95 A
13	(3) #1 + #6G IN 1 1/2"C	130 A
13K	(3) #1 + (2) #1/0N + #6G IN 2"C	130 A
13Y	(3) #1 + #1N + #6G IN 1 1/2"C	130 A
15	(3) #1/0 + #6G IN 1 1/2"C	150 A
15K	(3) #1/0 +(2) #1/0N + #6G IN 2"C	150 A
15Y	(3) #1/0 + #1/0N + #6G IN 2"C	150 A
17	(3) #2/0 + #6G IN 2"C	175 A
17K	(3) #2/0 + (2) #2/0N + #6G IN 2"C	175 A
17Y	(3) #2/0 + #2/0N + #6G IN 2"C	175 A
20	(3) #3/0 + #6G IN 2"C	200 A
20K	(3) #3/0 + (2) #3/0N + #6G IN 2 1/2"C	200 A
20Y	(3) #3/0 + #3/0N + #6G IN 2"C	200 A

FEEDER SCHEDULE

TAG	CONDUCTORS AND CONDUIT	MAX CURRENT
23	(3) #4/0 + #4G IN 2"C	230 A
23K	(3) #4/0 + (2) #4/0N + #4G IN 2 1/2"C	230 A
23Y	(3) #4/0 + #4/0N + #4G IN 2 1/2"C	230 A
25	(3) 250 kcmil + #4G IN 2 1/2"C	255 A
25K	(3) 250 kcmil + (2) 250 kcmil N + #4G IN 2 1/2"C	255 A
25Y	(3) 250 kcmil + 250 kcmil N + #4G IN 2 1/2"C	255 A
28	(3) 300 kcmil + #4G IN 2 1/2"C	285 A
28K	(3) 300 kcmil + (2) 300 kcmil N + #4G IN 3"C	285 A
28Y	(3) 300 kcmil + 300 kcmil N + #4G IN 2 1/2"C	285 A
31	(3) 350 kcmil + #1G IN 2 1/2"C	310 A
31K	(3) 350 kcmil + (2) 350 kcmil N + #1G IN 3"C	310 A
31Y	(3) 350 kcmil + 350 kcmil N + #1G IN 3"C	310 A
33	(3) 400 kcmil + #1G IN 2 1/2"C	335 A
33K	(3) 400 kcmil + (2) 400 kcmil N + #1G IN 3"C	335 A
33Y	(3) 400 kcmil + 400 kcmil N + #1G IN 3"C	335 A
38	(3) 500 kcmil + #1G IN 3"C	380 A
38K	(3) 500 kcmil + (2) 500 kcmil N + #1G IN 4"C	380 A
38Y	(3) 500 kcmil + 500 kcmil N + #1G IN 3"C	380 A
42S	(3) 600 kcmil + (1) 600 kcmil N IN 4"C	420 A
42T	(3) 600 kcmil + (1) 600 kcmil N + #1/0G IN 4"C	420 A
42Y	(3) 600 kcmil + 600 kcmil N + #1G IN 4"C	420 A
46	2 SETS OF EACH: (3) #4/0 + #2G IN (2) 2"C	460 A
46K	2 SETS OF EACH: (3) #4/0 + (2) #4/0N + #2G IN (2) 2 1/2"C	460 A
46Y	2 SETS OF EACH: (3) #4/0 + #4/0N + #2G IN (2) 2 1/2"C	460 A
51	2 SETS OF EACH: (3) 250 kcmil + #2G IN (2) 2 1/2"C	510 A
51K	2 SETS OF EACH: (3) 250 kcmil + (2) 250 kcmil N + #2G IN (2) 2 1/2"C	510 A
51Y	2 SETS OF EACH: (3) 250 kcmil + 250 kcmil N + #2G IN (2) 2 1/2"C	510 A
62	2 SETS OF EACH: (3) 350 kcmil + #1G IN (2) 2 1/2"C	620 A
62K	2 SETS OF EACH: (3) 350 kcmil + (2) 350 kcmil N + #1G IN (2) 3"C	620 A
62Y	2 SETS OF EACH: (3) 350 kcmil + 350 kcmil N + #1G IN (2) 3"C	620 A

FEEDER SCHEDULE

TAG	CONDUCTORS AND CONDUIT	MAX CURRENT
76	2 SETS OF EACH: (3) 500 kcmil + #1/0G IN (2) 3"C	760 A
76K	2 SETS OF EACH: (3) 500 kcmil + (2) 500 kcmil N + #1/0G IN (2) 4"C	760 A
76Y	2 SETS OF EACH: (3) 500 kcmil + 500 kcmil N + #1/0G IN (2) 3"C	760 A
84	2 SETS OF EACH: (3) 600 kcmil + #1/0G IN (2) 3"C	840 A
84K	2 SETS OF EACH: (3) 600 kcmil + (2) 600 kcmil N + #1/0G IN (2) 4"C	840 A
84Y	2 SETS OF EACH: (3) 600 kcmil + 600 kcmil N + #1/0G IN (2) 4"C	840 A
100	3 SETS OF EACH: (3) 400 kcmil + #2/0G IN (3) 2 1/2"C	1005 A
100K	3 SETS OF EACH: (3) 400 kcmil + (2) 400 kcmil N + #2/0G IN (3) 3"C	1005 A
100Y	3 SETS OF EACH: (3) 400 kcmil + 400 kcmil N + #2/0G IN (3) 3"C	1005 A
120	3 SETS OF EACH: (3) 600 kcmil + #3/0G IN (3) 3"C	1260 A
120K	3 SETS OF EACH: (3) 600 kcmil + (2) 600 kcmil N + #3/0G IN (3) 4"C	1260 A
120Y	3 SETS OF EACH: (3) 600 kcmil + 600 kcmil N + #3/0G IN (3) 4"C	1260 A
160	4 SETS OF EACH: (3) 600 kcmil + #4/0G IN (4) 3"C	1680 A
160K	4 SETS OF EACH: (3) 600 kcmil + (2) 600 kcmil N + #4/0G IN (4) 4"C	1680 A
160Y	4 SETS OF EACH: (3) 600 kcmil + 600 kcmil N + #4/0G IN (4) 4"C	1680 A
210	5 SETS OF EACH: (3) 600 kcmil + 250 kcmil G IN (5) 3"C	2100 A
210K	5 SETS OF EACH: (3) 600 kcmil + (2) 600 kcmil N + 250 kcmil G IN (5) 4"C	2100 A
210Y	5 SETS OF EACH: (3) 600 kcmil + 600 kcmil N + 250 kcmil G IN (5) 4"C	2100 A
250	6 SETS OF EACH: (3) 600 kcmil + 250 kcmil G IN (6) 3"C	2520 A
250K	6 SETS OF EACH: (3) 600 kcmil + (2) 600 kcmil N + 250 kcmil G IN (6) 4"C	2520 A
250Y	6 SETS OF EACH: (3) 600 kcmil + 600 kcmil N + 250 kcmil G IN (6) 4"C	2520 A
300	8 SETS OF EACH: (3) 500 kcmil + 400 kcmil G IN (8) 3"C	3040 A
300K	8 SETS OF EACH: (3) 500 kcmil + (2) 500 kcmil N + 400 kcmil G IN (8) 4"C	3040 A
300Y	8 SETS OF EACH: (3) 500 kcmil + 500 kcmil N + 400 kcmil G IN (8) 4"C	3040 A
420	10 SETS OF EACH: (3) 600 kcmil + 500 kcmil G IN (10) 3"C	4200 A
420K	10 SETS OF EACH: (3) 600 kcmil + (2) 600 kcmil N + 500 kcmil G IN (10) 4"C	4200 A
420Y	10 SETS OF EACH: (3) 600 kcmil + 600 kcmil N + 500 kcmil G IN (10) 4"C	4200 A
500	12 SETS OF EACH: (3) 600 kcmil + (2) 500 kcmil G IN (12) 4"C	5040 A
500K	12 SETS OF EACH: (3) 600 kcmil + (2) 600 kcmil N + (2) 500 kcmil G IN (12) 4"C	5040 A
500Y	12 SETS OF EACH: (3) 600 kcmil + 600 kcmil N + (2) 500 kcmil G IN (12) 4"C	5040 A

FEEDER SCHEDULE NOTES:

1. CONDUCTOR SIZES ARE BASED ON 60°C RATED COPPER CONDUCTORS WITH 60°C RATED TERMINATIONS FOR FEEDERS RATED 100A AND LESS. CONDUCTOR SIZES FOR FEEDERS RATED MORE THAN 100A ARE BASED ON 75°C RATED COPPER CONDUCTORS WITH 75°C RATED TERMINATIONS.
2. CONDUIT SIZES FOR INDOOR FEEDERS ARE MINIMUM SIZES FOR THHN/THWN IN EMT.
3. WHERE MULTIPLE SETS ARE IDENTIFIED, PROVIDE EACH SET OF PHASE, NEUTRAL, AND GROUNDING CONDUCTORS IN SEPARATE CONDUITS EQUAL TO THE NUMBER OF SETS UNLESS NOTED OTHERWISE.
4. NOT ALL FEEDER SIZES ARE USED IN THIS DRAWING SET.

SYM DESCRIPTION DATE APPR

NAVFAC

COMMONWEALTH OF VIRGINIA
KELLEY BOURASSA
Lic. No. 0402061759
6/23/2025
PROFESSIONAL ENGINEER
SEAL

Jacobs
1100 NORTH GLEBE ROAD
SUITE 500
ARLINGTON, VIRGINIA 22201
571.218.1000
FAX: 571.218.1600

A/E INFO

APPROVED
FOR COMMANDER NAVFAC 8/14/2025
ACTIVITY
APPROVED BY LCDR ELIZABETH OLOKODE NAVFAC NLON FEAD DIRECTOR NEW LONDON, GROTON, CT
SATISFACTORY TO DATE 05/27/2025
DES JHP DRW JHP CHK KB
PM/DM SMS/CWJ
BRANCH MANAGER WRB
CHIEF ENG/ARCH JWJ
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
MID-ATLANTIC NAVAL STATION - NORFOLK, VA
NAVAL SUBMARINE BASE NEW LONDON GROTON, CONNECTICUT
P-1102 WEAPONS MAGAZINE & ORDNANCE OPERATIONS FACILITY
OPERATIONS
ELECTRICAL FEEDER SCHEDULE

SCALE: AS NOTED
EPROJECT NO.: 1667770
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12894054
SHEET 104 OF 140

E-704

6/20/2025 12:27:05 PM E-704
\\dc1vs01\fwdpr\FDZE1628\ELEC\BLDG-1667770-E-BIM_v21.rvt